南方科技大學

CHAPTER II

Solid-State Chemistry

2.1 Amorphous vs crystalline solids

A SOLID is a material that retains both its shape and volume over time.

Crystalline material

Amorphous material

The atoms, ions, or molecules are organized into regular arrangements (lattice)

No long-range structural order throughout the solid

Crystalline Solids	Amorphous solids
They have a definite shape and geometrical form.	They do not have a definite geometrical shape.
They have a sharp (definite) melting point.	They melt over a wide range of temperatures.
They are rigid and incompressible.	They too are usually rigid and cannot be compressed to any appreciable extent. However graphite is soft because of its unusual structure.
They give a clean clevage, i.e, break into pieces with plane surfaces.	They give irregular clevage.
They have a definite heat of fusion.	They do not have a definite heat of fusion.
Anisotropic, i.e. their mechanical properties and electrical properties depend on the direction along which they are measured.	Isotropic, i.e. they have similar physical properties in all directions because the constituents are arranged in random manner.

Galena

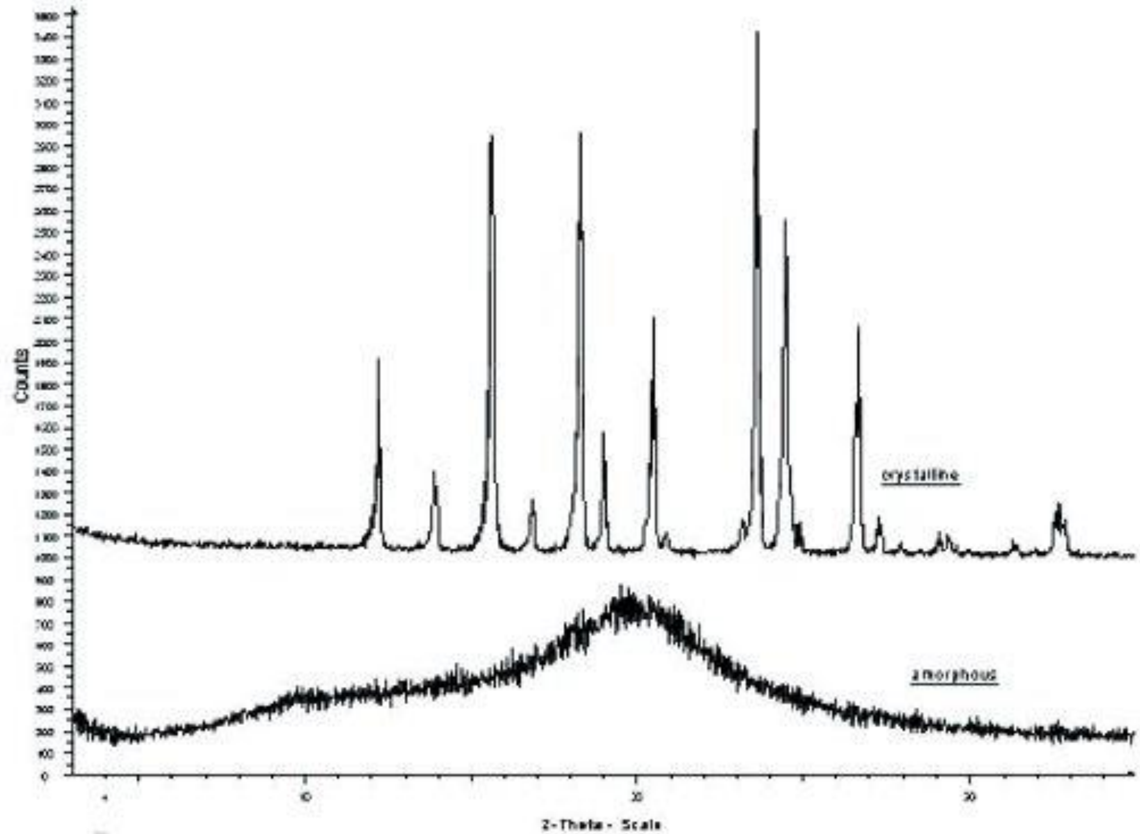
Quartz

Obsidian

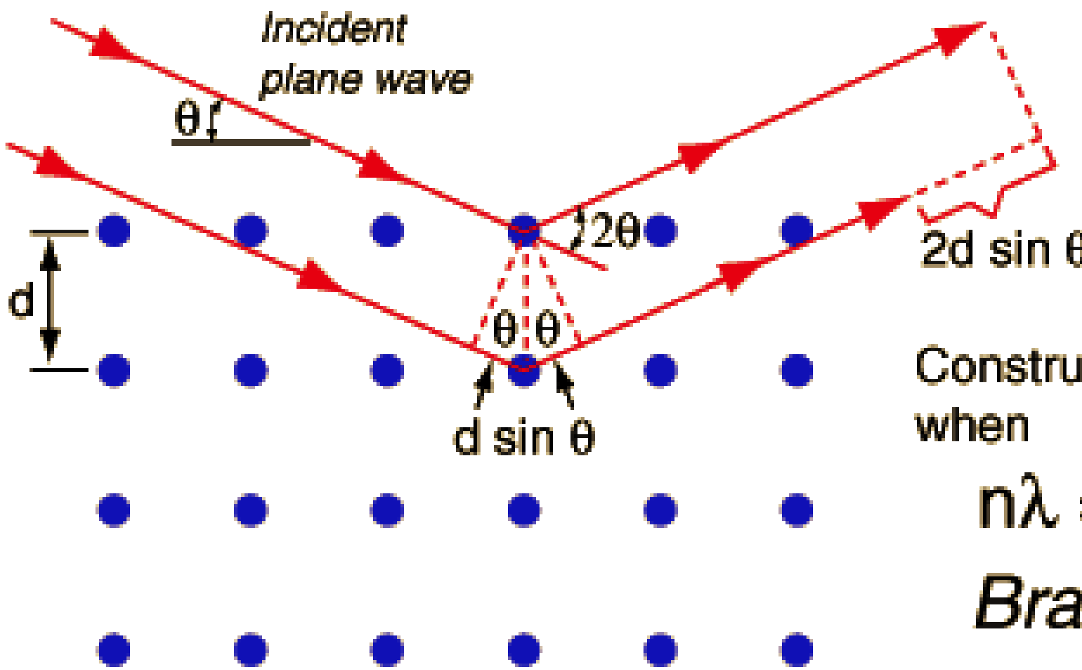
Pyrite

(Left) Crystalline faces. The faces of crystals can intersect at right angles, as in galena (PbS) and pyrite (FeS_2), or at other angles, as in quartz. **(Right) Cleavage surfaces of an amorphous solid.** Obsidian, a volcanic glass with the same chemical composition as granite (typically $KAlSi_3O_8$), tends to have curved, irregular surfaces when cleaved.

X-ray Powder Diffraction (XRD)



X-ray Powder Diffraction (XRD)

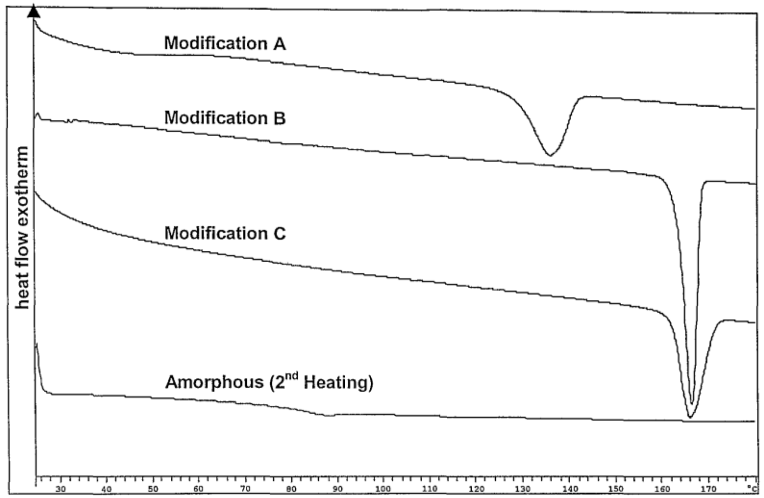


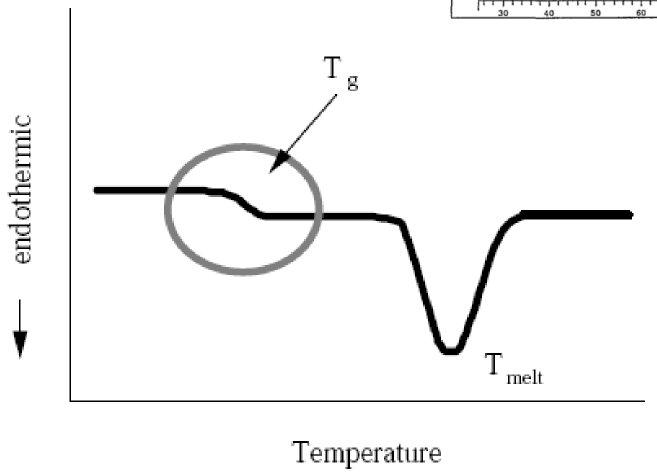
Constructive interference when

$$n\lambda = 2d \sin\theta$$

Bragg's Law

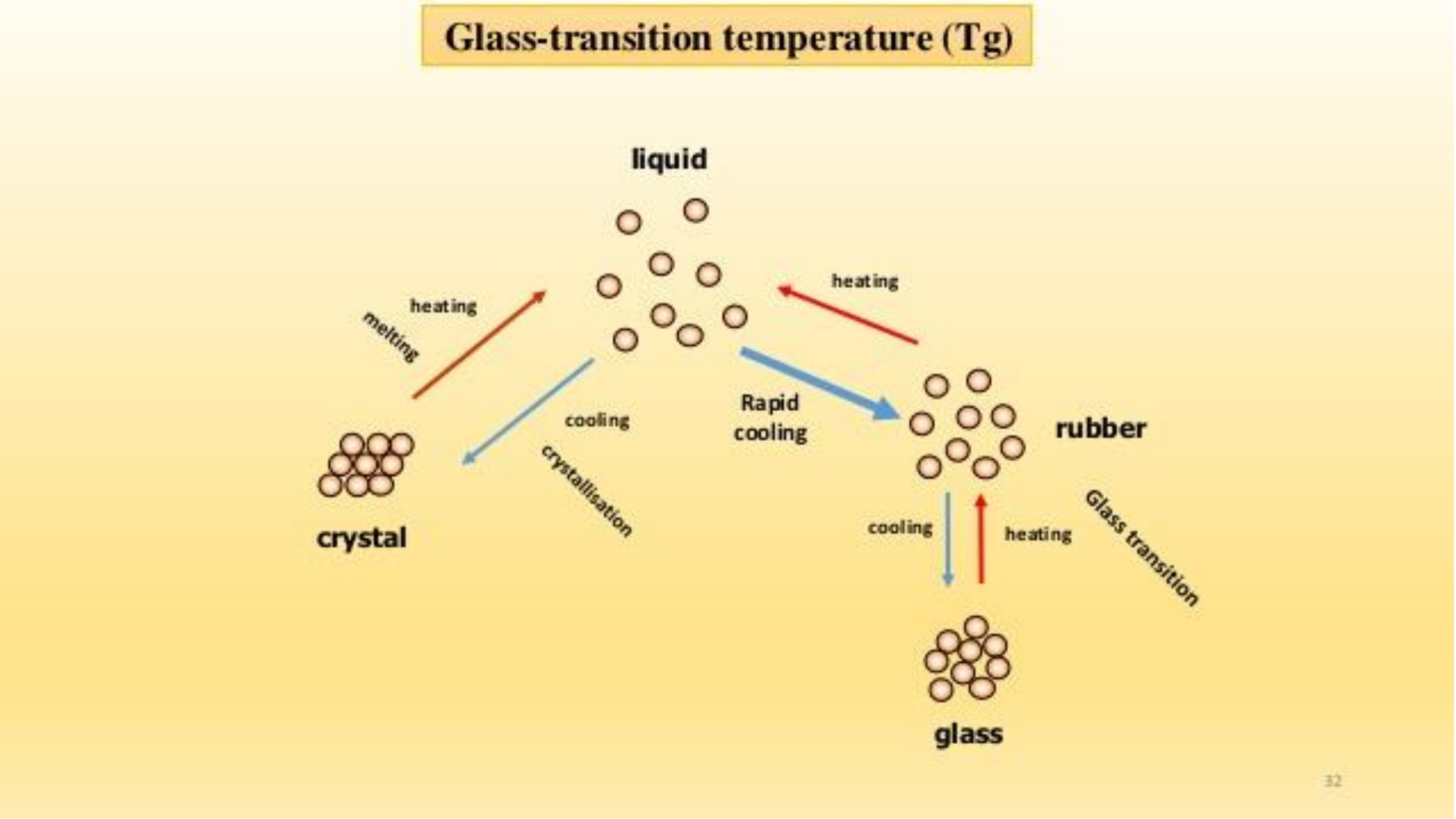
Differential scanning calorimetry (DSC)





The glass transition temperature (T_g) is the reversible transition in amorphous materials (or in amorphous regions within semi-crystalline materials) from a hard and relatively brittle state into a molten or rubber-like state. The T_g is a function of chain flexibility.

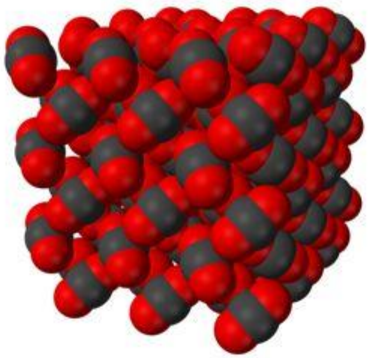
T_g is always lower than the melting temperature (Tm) of the crystalline state of the material, if one exists.


Glass-transition temperature (Tg)
liquid
heating
melting
cooling
crystallisation
crystal
Rapid cooling
heating
rubber
cooling
heating
Glass transition
glass
32

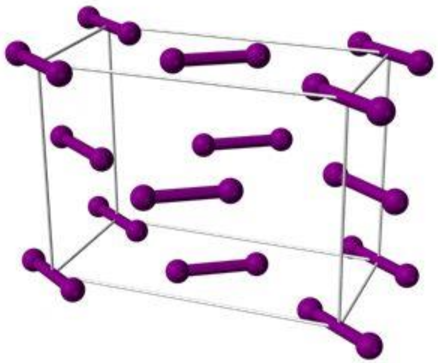
2.2 Types of bonding in solids

Types of Solid	Units	Forces	Properties
Ionic	Positive and negative ions	Electrostatic attractions	Hard and brittle; High Tm; Poor conductivity.
Metallic	atoms	Metallic bonds	Soft to very hard; Low to very high Tm; Excellent conductivity; Malleable and ductile.
Molecular	Atoms or molecules	London dispersion, dipole-dipole, hydrogen bonding	Fairly soft; Low to moderately high Tm; Poor conductivity.
Covalent network	Atoms connected in a network of covalent bonds	Covalent bonds	Very hard; Very high Tm; Often poor conductivity.

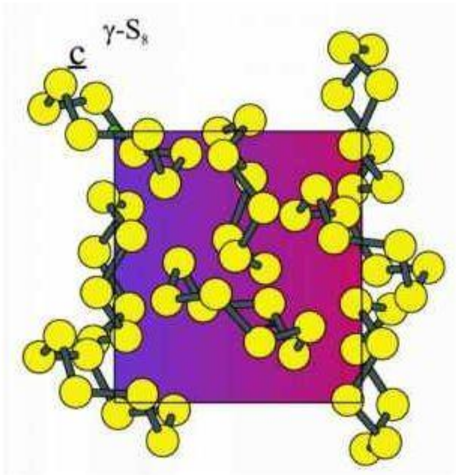
Molecular Solids



Carbon Dioxide

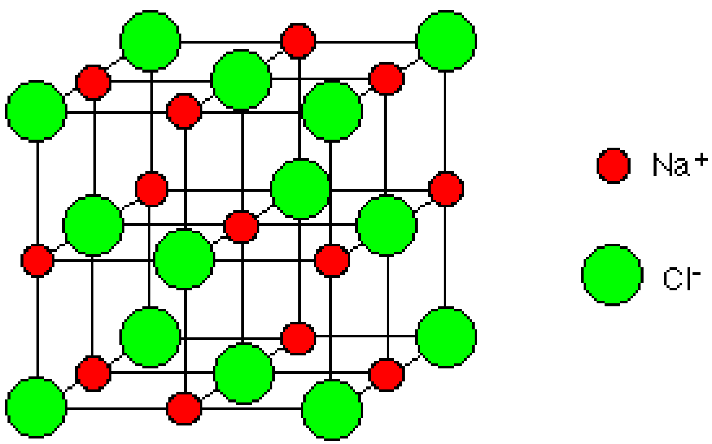


Iodine (I_2)



Sulfur (S_8)

2.2.1 Ionic in solids



Cationic and anionic species are associated through electrostatic interactions.

The most common forms:
Group 1-17 MX (LiF, KCl, NaCl etc.)
Group 2-17 MX_2 ($MgCl_2$, $CaCl_2$ etc.)
Group 1: alkali metals
Group 2: alkali earth metals
Group 17: halogens

Strong electrostatic attraction → Lattice energy U → High T_m

$$U = \frac{NMZ_{\text{cation}}Z_{\text{anion}}}{r_0}\left(\frac{e^2}{4\pi\varepsilon_0}\right)\left(1-\frac{\rho}{r_0}\right),$$

where: N = Avagadro's number (6.02×10^{23} molecules mol^{-1})

$Z_{\text{cation, anion}}$= magnitude of ionic charges;

r_0= average ionic bond length

e= electronic charge (1.602×10^{-19} C)

$4\pi\varepsilon_0$, permittivity of a vacuum (1.11×10^{-10} C^2 J^{-1} m^{-1})

M= the Madelung constant

ρ= the Born exponent

The Madelung constant is related to the specific arrangement of ions in the crystal lattice.

- sizes of ions;
- charges of ions;
- the specific arrangement of ions.

Na^+	0.102 nm
K^+	0.138 nm
Mg^{2+}	0.072 nm
Cl^-	0.181 nm

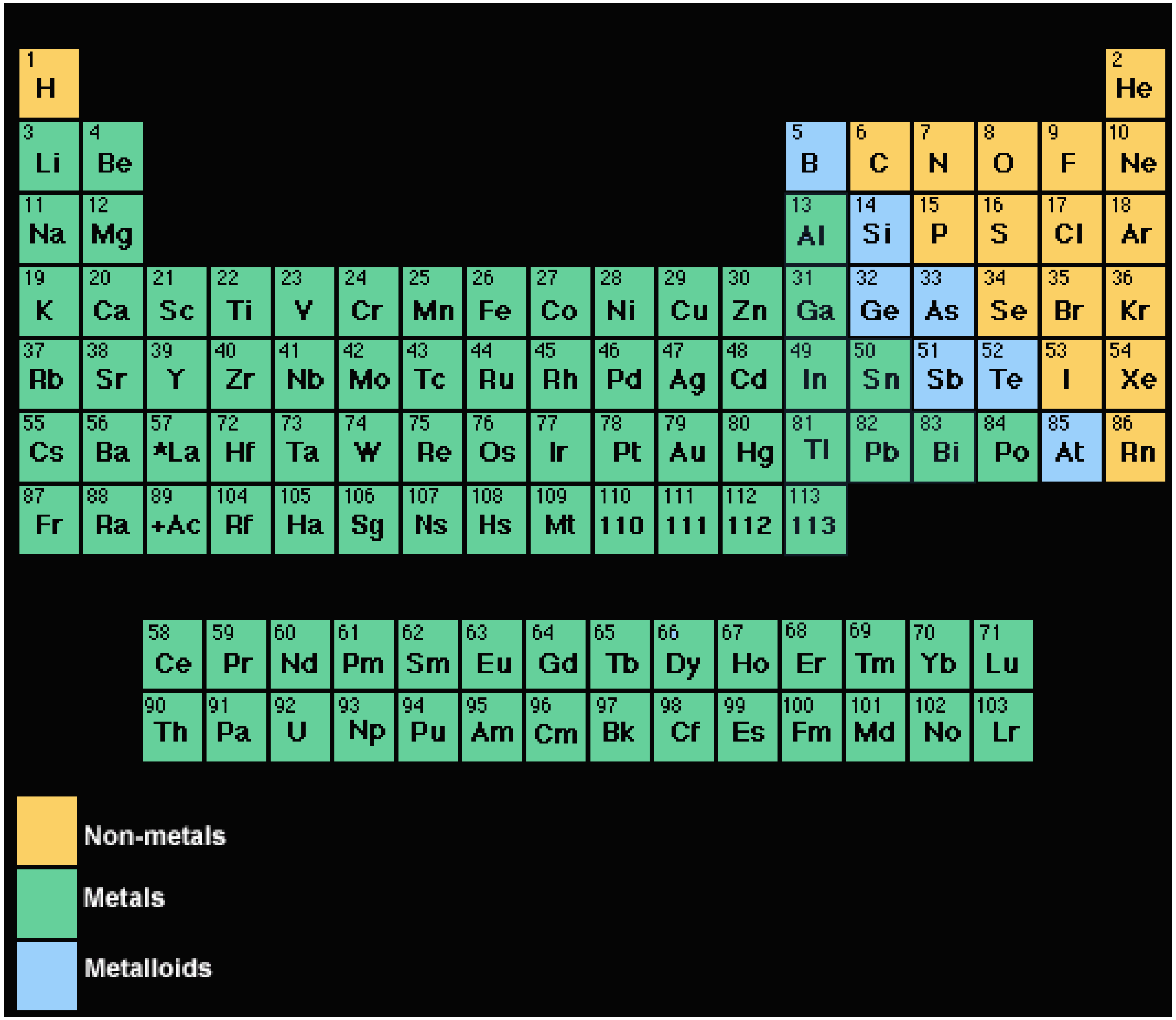
Q: Compare the Tm of NaCl with KCl.

NaCl (800.8℃) > KCl (770 ℃)

Q: Compare the Tm of NaCl with $MgCl_2$.

NaCl (800.8℃) > $MgCl_2$ **(708 ℃)**

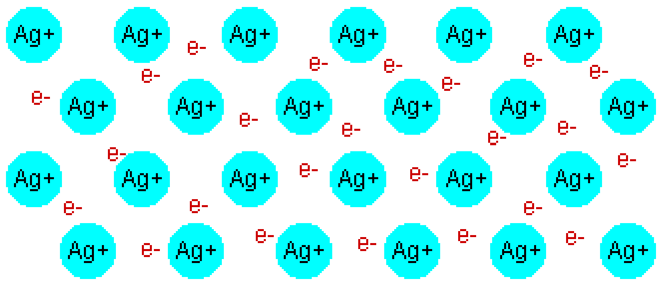
2.2.2 Metallic solids

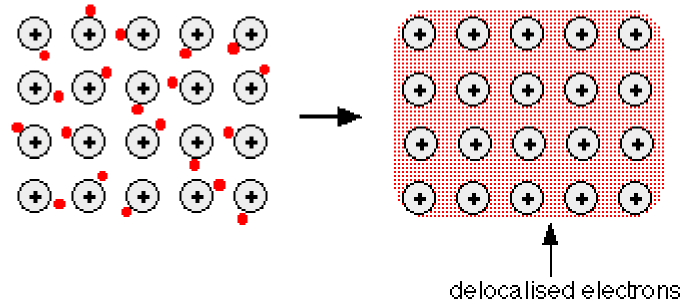


2.2.2 Metallic solids

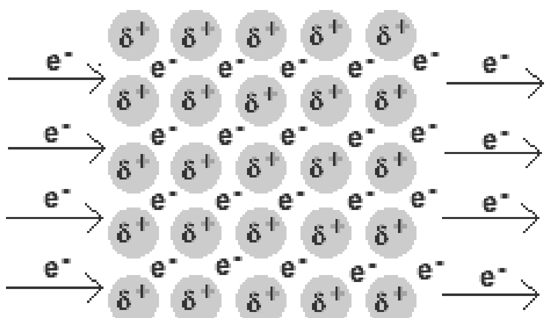
- A metal is a lattice of positive metal “ion” in a “sea” of delocalized electrons
- Metallic bonding refers to the interaction between the delocalized electrons and the metal nuclei.
- The physical properties of metals are the result of the delocalization of the electrons involved in metallic bonding.
- The physical properties of solid metals are:
  - ✓ Conduct heat
  - ✓ Conduct electricity
  - ✓ Generally high melting and boiling points
  - ✓ Strong
  - ✓ Malleable (can be hammered or pressed out of shape)
  - ✓ Ductile (able to be drawn into a wire)
  - ✓ Metallic luster
  - ✓ Opaque (reflect light)
  - ✓ Easily oxidized by the surrounding environment

Metallic bonding



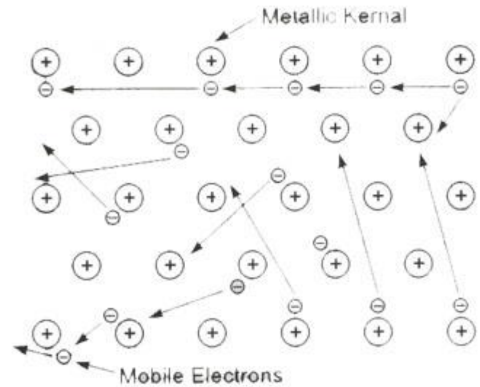


Electricity conductivity



Under the influence of an external potential difference across a metal piece the free electron start moving in one particular direction.

Thermal conductivity



Thermal conductivity in metallic crystals

When a part of a metal is heated, the electrons of that part get large amount of energy and start to move in the lattice in zig-zag motion.

Malleability and Ductility

Shifting of metal ions layer in metallic crystals

In these properties one layer of atoms slides over another without a series disturbance at any stage on applying the force. Now, since the atoms are arrange in a particular fashion. When the form is applied one layer of atoms may slide over the next and we can get fine sheet or wire of the metal.

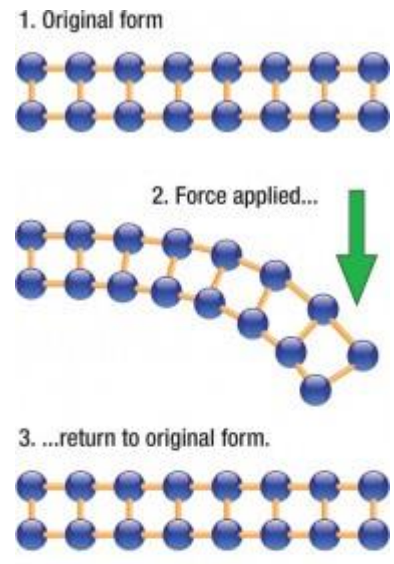
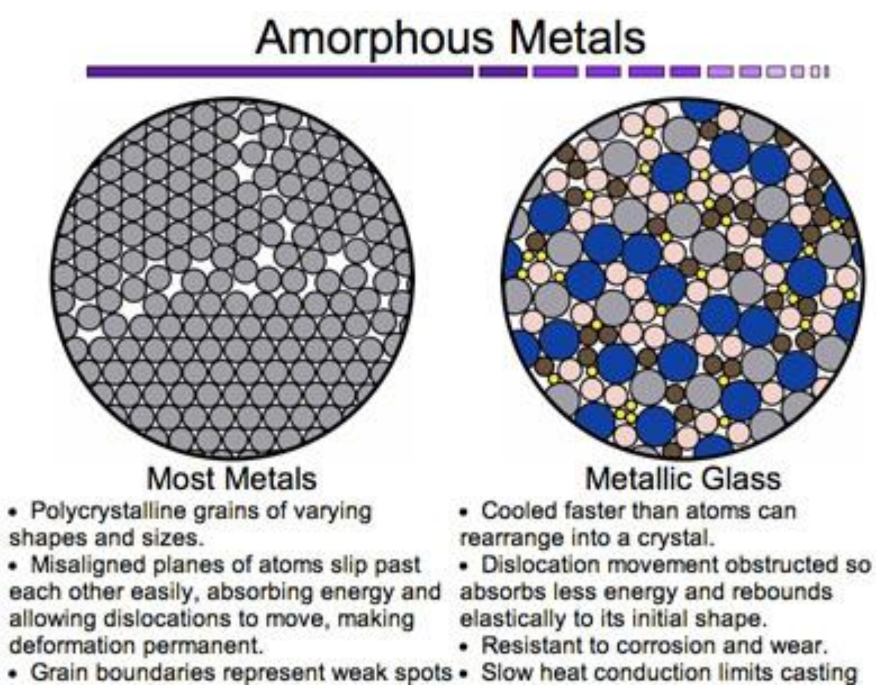


Fig. 4 Elastic Deformation

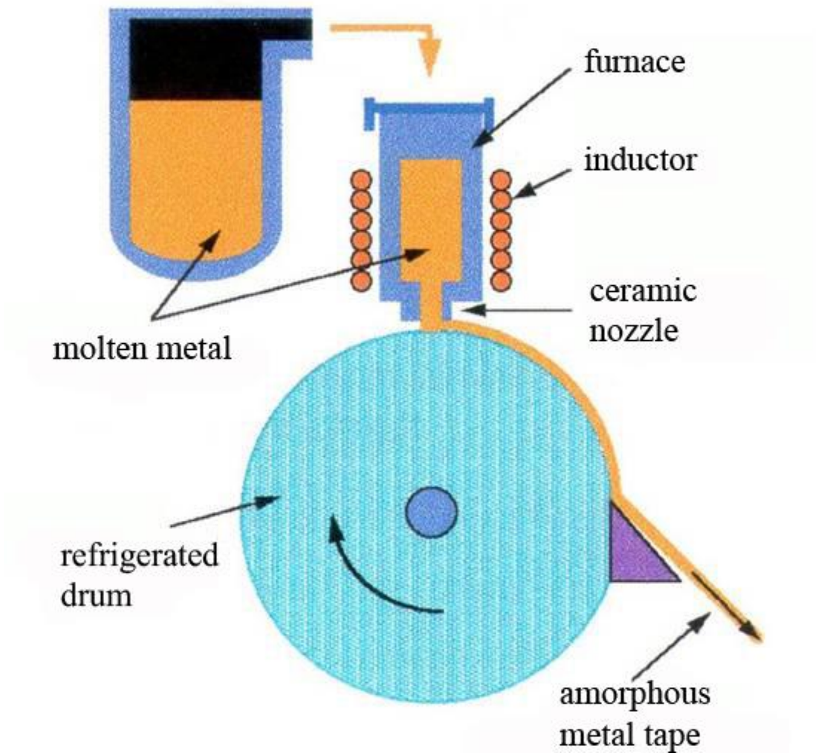
An **amorphous metal** (also known metallic glass) is a solid metallic material, usually an alloy ($Zr_{41}Ti_{14}Ni_{12}Cu_{10}Be_{23}$), with a disordered atomic-scale structure.

Many traditional metals are relatively easy to deform, or bend permanently out of shape, because their crystal lattices are riddled with defects. A metallic glass, in contrast, will spring back to its original shape much more readily.



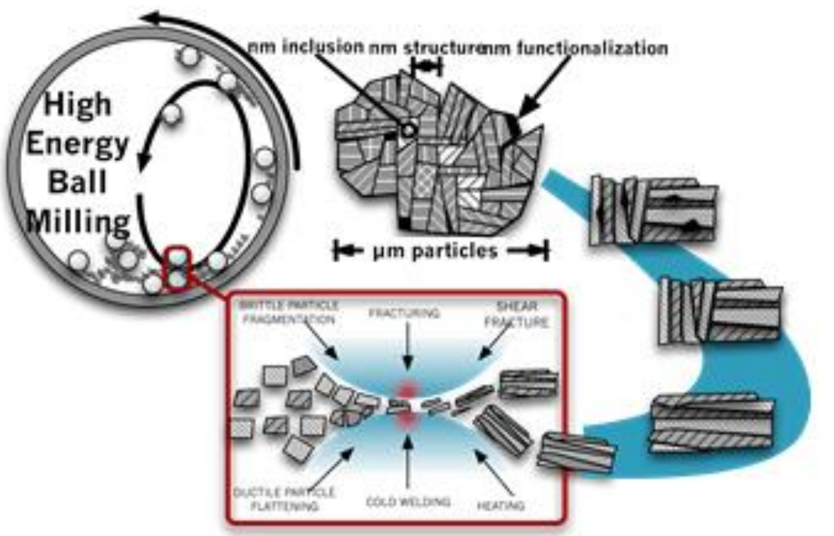
- resistance to plastic deformation.
- The absence of grain boundaries, the weak spots of crystalline materials, leads to better resistance to wear and corrosion.
- Tougher and less brittle than oxide glasses and ceramics.

The production of amorphous metallic materials



Melt Spinning or Melt Extraction

Providing a very high cooling rate ($>10^6$ K s^{-1})



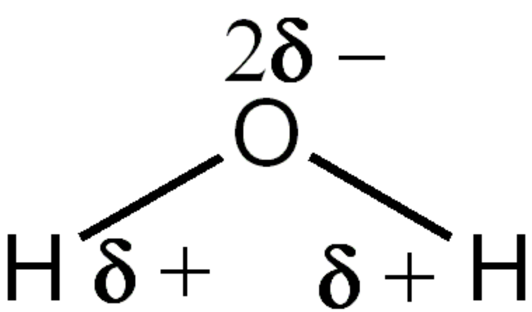
Ball-milling

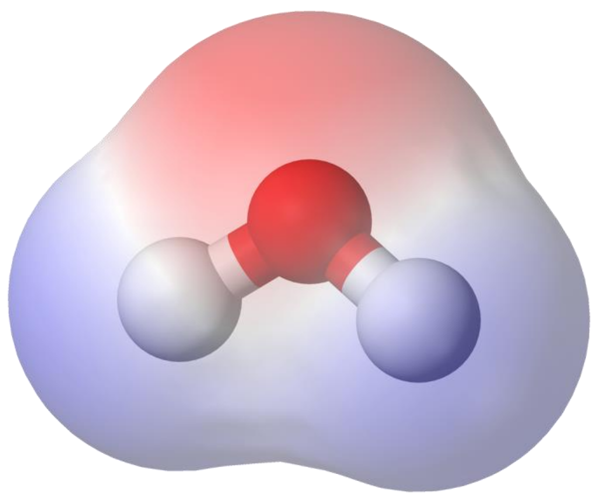
High pressure causes local melting of the crystalline powders, breaking apart metallic bonds, and facilitating atomic diffusion along preferential crystalline interfaces.

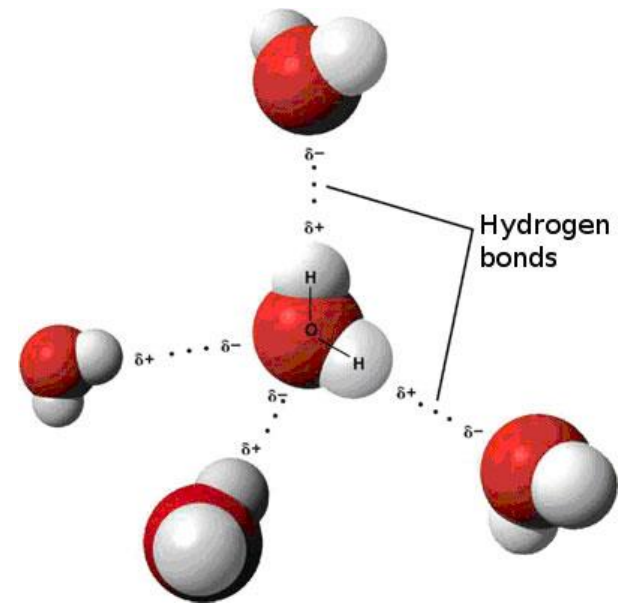
2.2.3 Molecular solids

Intermolecular Forces are weaker than covalent or ionic bonds

Are caused by molecule polarity







3 Types of Intermolecular Forces:

Listed in order of increasing strength:

London dispersion forces

Dipole-dipole interactions

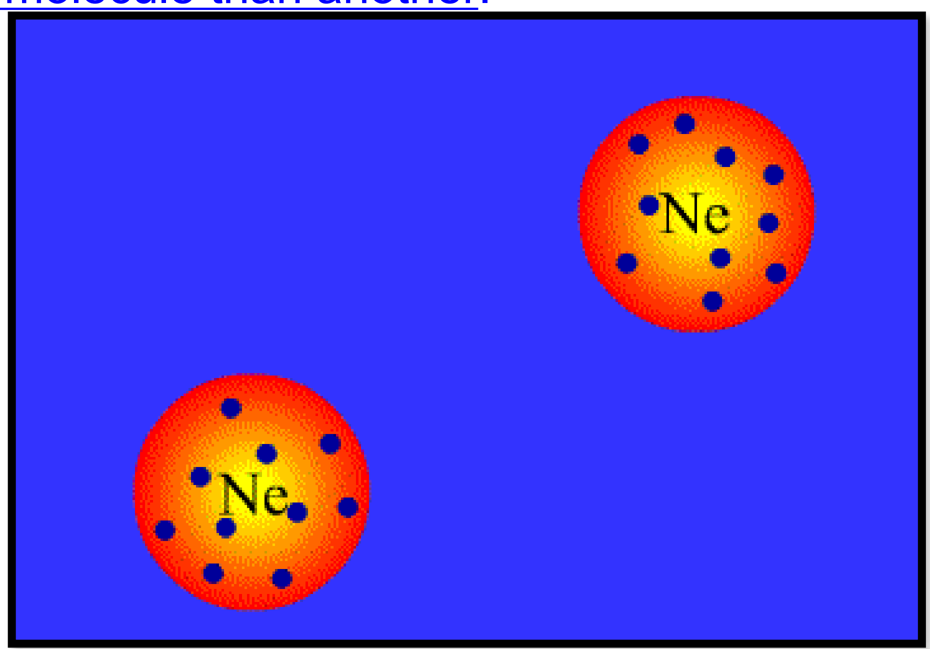
Hydrogen bonding

Note: London dispersion forces and dipole-dipole interactions are collectively called **van der Waals forces**.

London Dispersion Forces

- Occur in ALL molecules; the only type of intermolecular force found in nonpolar molecules
- The weakest of molecular interactions
- Caused by the motion of electrons

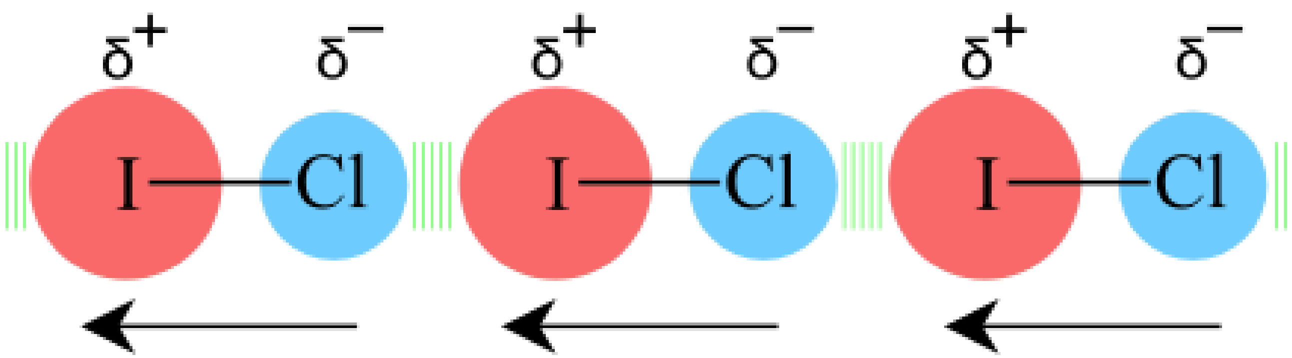
  Electrons are in constant motion. At times, more electrons can be on one side of a molecule than another.



Dipole-dipole Forces

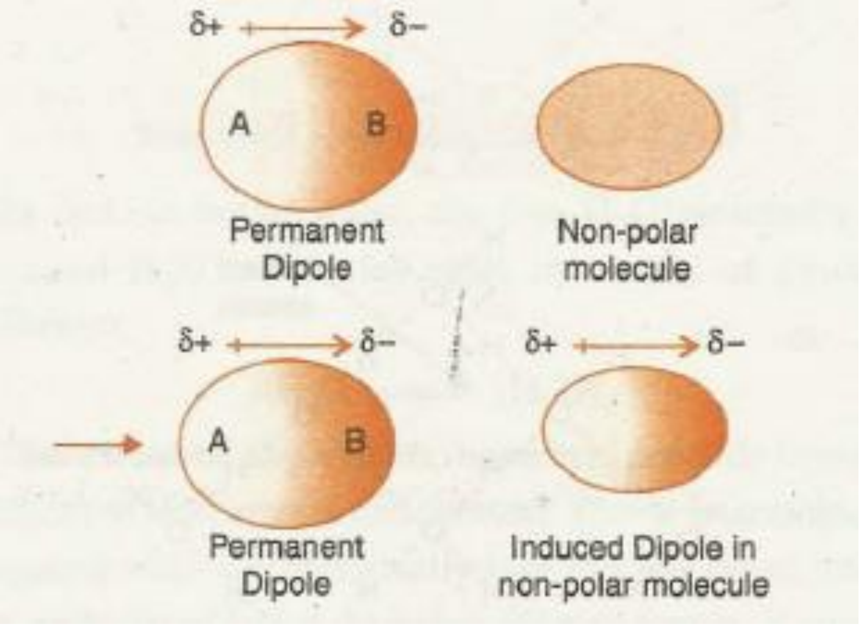
- The slightly negative region of a polar molecule is weakly attracted to the slightly positive region of another polar molecule.

- Dipole-dipole interactions are *similar to* but *much weaker* than ionic bonds.



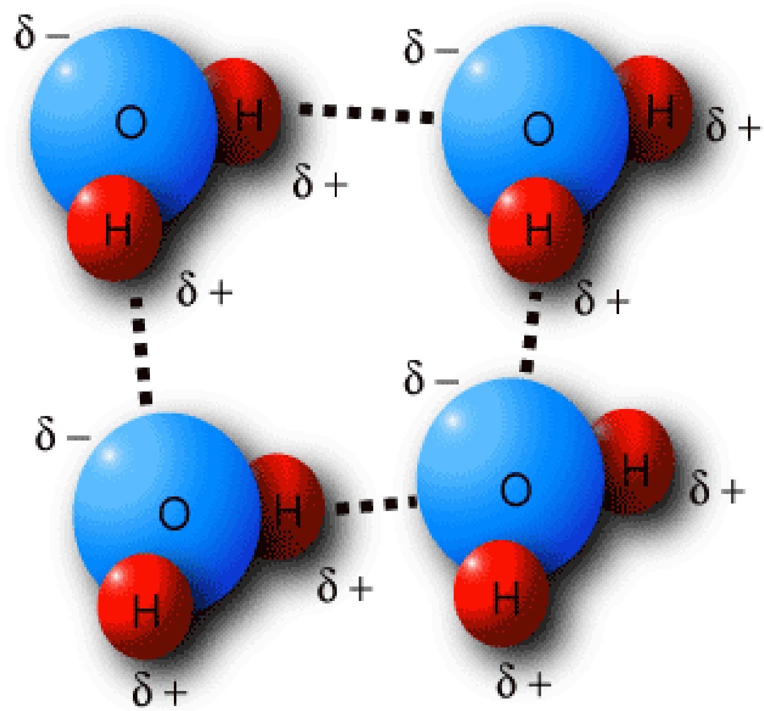
Dipole-induced dipole Forces

If both polar and nonpolar molecules are present, a dipole-induced dipole interaction may occur.



Hydrogen Bonding

- Strongest type of intermolecular force (12-30 kJ mol^{-1});
- Hydrogen bonding always involves **hydrogen;**
- Hydrogen is bonded to a very electronegative atom **(oxygen, nitrogen, or fluorine).**



Dipole-Dipole Forces	$\text{dipole–dipole} = \dfrac{\mu^4}{kTR^6}$
London Dispersion Forces	$\text{London Dispersion} = \dfrac{I\alpha^2}{R^6}$
Dipole-Induced Dipole Force	$\text{Dipole–induced dipole} = \dfrac{\mu\alpha^2}{R^6}$

μ is the molecular dipole moment (Debyes);
I is the ionization potential of the molecule;
α is the polarizability of the molecules
R is the average distance of separation (A˚);
T is the temperature (K);
k is the Boltzmann constant (1.38065×10^{-23} J K^{-1})

Properties of molecular solids

The dipole forces are weaker than ionic or covalent bonds. The relatively weak intermolecular forces cause molecular solids to have relatively low melting points, typically less than 300 °C. Most molecular solids are relatively soft, electrical insulators and have low density.

Examples of Molecular Solids:

water ice	solid carbon dioxide	sucrose or table sugar
Hydrocarbons	fullerenes	sulfur
white phosphorus	yellow arsenic	solid halogens

halogen compound with hydrogen (e.g., HCl)

2.2.4 Covalent Network solids

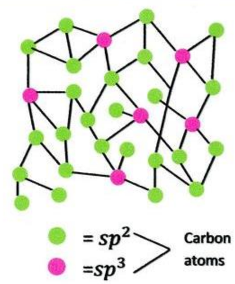
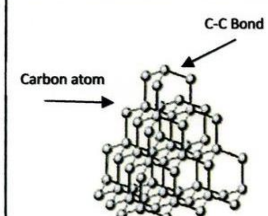
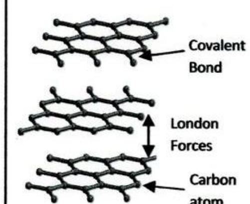
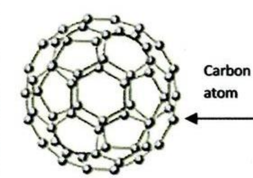
A network solid or covalent network solid is a chemical compound (or element) in which the atoms are bonded by covalent bonds in a continuous network extending throughout the material. In a network solid there are no individual molecules, and the entire crystal may be considered a macromolecule.

- High T_m
- Bulk hardness
- Allotropes

Macroscopic Observations

Nontetrahedral Amorphous	Diamond	Graphite	Fullerene (Soot)
*Glassy *Crumbles *Black *Lump *Not very dense *Not very hard	*Crystal-like *Clear *Repels water *Reflects light *Very dense *Very Hard	*Crystal-like *Grayish/Black *High conductivity *Properties of both metals and nonmetals *Very Soft	*Black *Chalky *No Odor *Sticks to the can *Showed up on my finger once rubbed

Micromodels

Nontetrahedral Amorphous	Diamond	Graphite	Fullerene (Soot)
			
*Also called charcoal *Random/jungle like *No pattern *Low conductivity *More sp^2 than sp^3 *More trigonal planar than tetrahedral	*Also called ice *4 cloud (sp^3) *Tetrahedral (109.5°) *All single bonds *Makes hexagonal chairs *Looks like a honeycomb	*Also called pencil lead *3 clouds (sp^2) *Layers are offset *Hexagonal layers *Trigonal planar (120°) *One double bond and two single bonds -The double bond has a sigma and a pi. The pi bond has high energy. It also has the ability to move, which allows it to conduct electricity.	*Also called Buckyball *3 cloud (sp^2) *Trigonal planar (120°) *One double bond and two single bonds *Cage-like *32 faces-looks like a soccer ball *20 hexagons around 12 pentagons

2.3 The Crystalline State

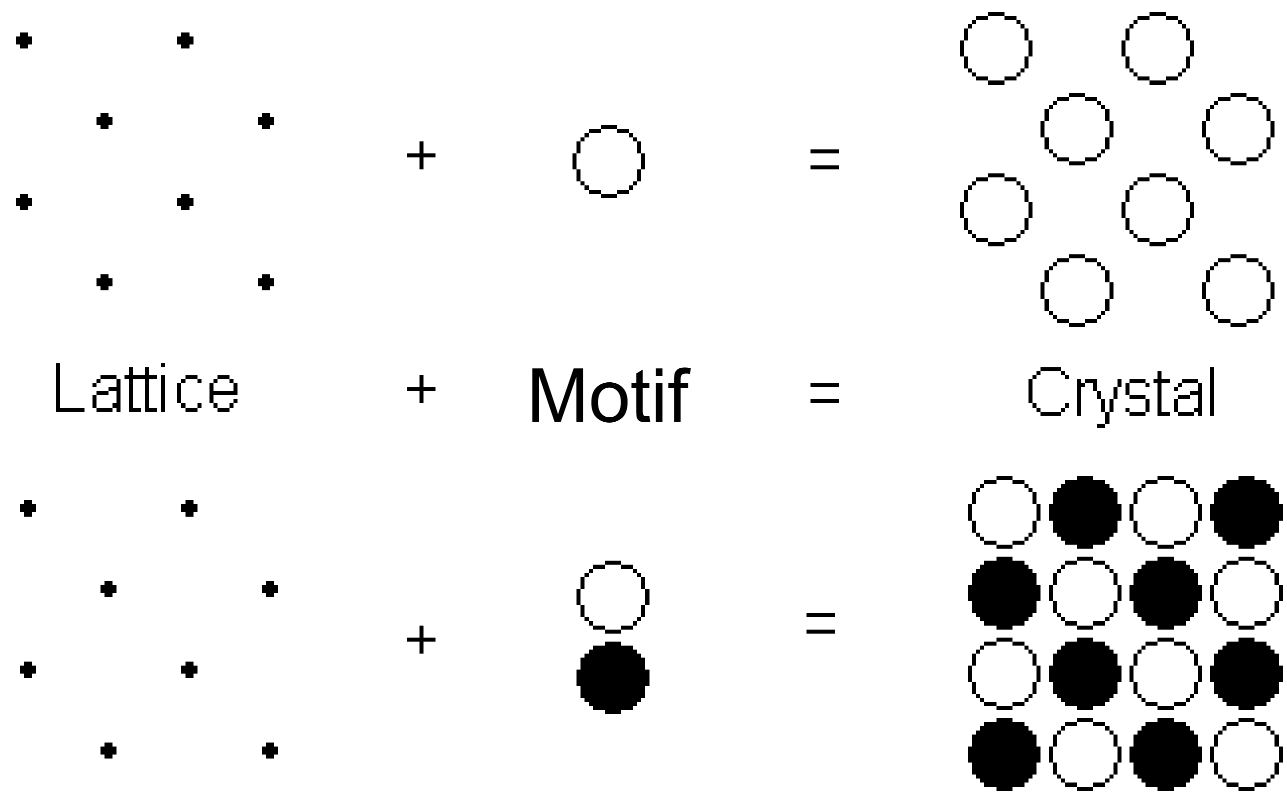
Lattice	Crystal
Translationally periodic arrangement of **points**	Translationally periodic arrangement of **motifs**

Crystal = Lattice + Motif

- Lattice ➢ the underlying periodicity of the crystal
- Motif ➢ atom/ion/molecule or a group of atoms/ions/molecules associated with each lattice points

- Lattice ➢ how to repeat
- Motif ➢ what to repeat

Crystal



Crystal stability----- Lattice energy

The strength of the interactions between the species comprising the crystal, which is dependent on the nature and degree of interaction between adjacent species.

Ionic and covalent crystals	700-900 kJ mol^{-1}
Metallic crystals	400-500 kJ mol^{-1}
Molecular crystals	5-20 kJ mol^{-1}

Crystal arrangement

The ions, molecules, or atoms pack in an arrangement that minimizes the total free energy of the crystal lattice.

- Charge balance
- Hydrogen bonding
- Polarity arrangement
- Hydrophobic and hydrophilic

The residual solvent in crystals

Depending on how strongly a solvent is contained within the crystal lattice.

Efflorescence (Weak association)

The encapsulated solvent is lost

Deliquescence (Strong association)

If the solid contains ions with a high charge density and is soluble in water, the crystals will readily adsorb water from the atmosphere and may even be transformed to a solution. For example, $CaCl_2$ is employed as a dehydrating agent.

Crystal Morphology--- Isomorphism vs polymorphism

The overall shape or form of a crystal is known as the morphology.

Isomorphism（类质同晶） is when two different compounds have the same crystal structure. May also have isostructural groups that are related by a common anion or anion group, such as the carbonates, sulphates, and phosphates.

Polymorphism（同质多晶） is when a chemical compound may crystallize in more than one structure.

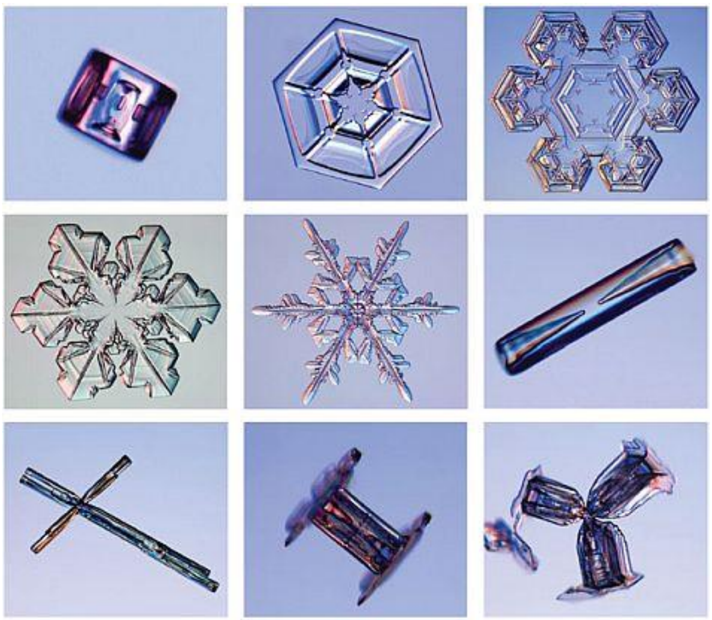
The interconversion is easy between different polymorphs.

Table 4.3 Calcite Isostructural Group[a]

Mineral	Formula
Calcite	$CaCO_3$
Magnesite	$MgCO_3$
Siderite	$FeCO_3$
Rhodochrosite	$MnCO_3$
Smithsonite	$ZnCO_3$

[a] All crystallize in the $\bar{3}2/m$ point group and differ only in the cation in the structure.

Polymorphism

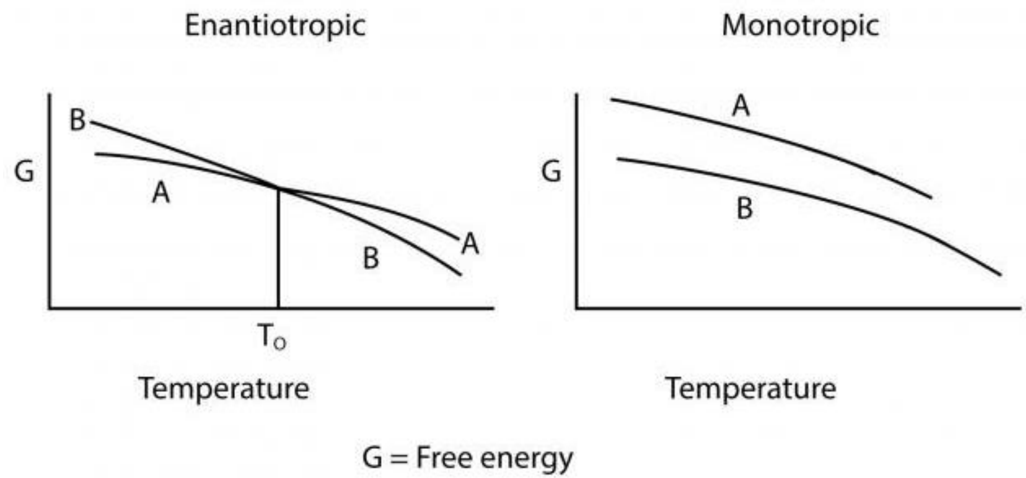


It is the ability of the compounds to crystallize as more than one distinct crystalline species with different internal lattice.

When a substance exists in more than one crystalline form, the different forms are designated as polymorphs and the phenomenon as polymorphism.

In pseudopolymorphism the different crystal types are the result of hydration or solvation. This is more correctly referred to as solvomorphism as different solvates have different chemical formulae.

Enantiotropic and monotropic polymorphs



- The polymorph which can be changed from one form into another by varying temp. or pressure is called as enantiotropic（双变性）polymorph.
  e.g. Sulfur.

- One polymorph which is unstable at all temp. & pressure is called as monotropic（单变性）polymorph.
  e.g. Glyceryl stearate.

The conversion between polymorphs can be observed by differential scanning calorimetry (DSC)

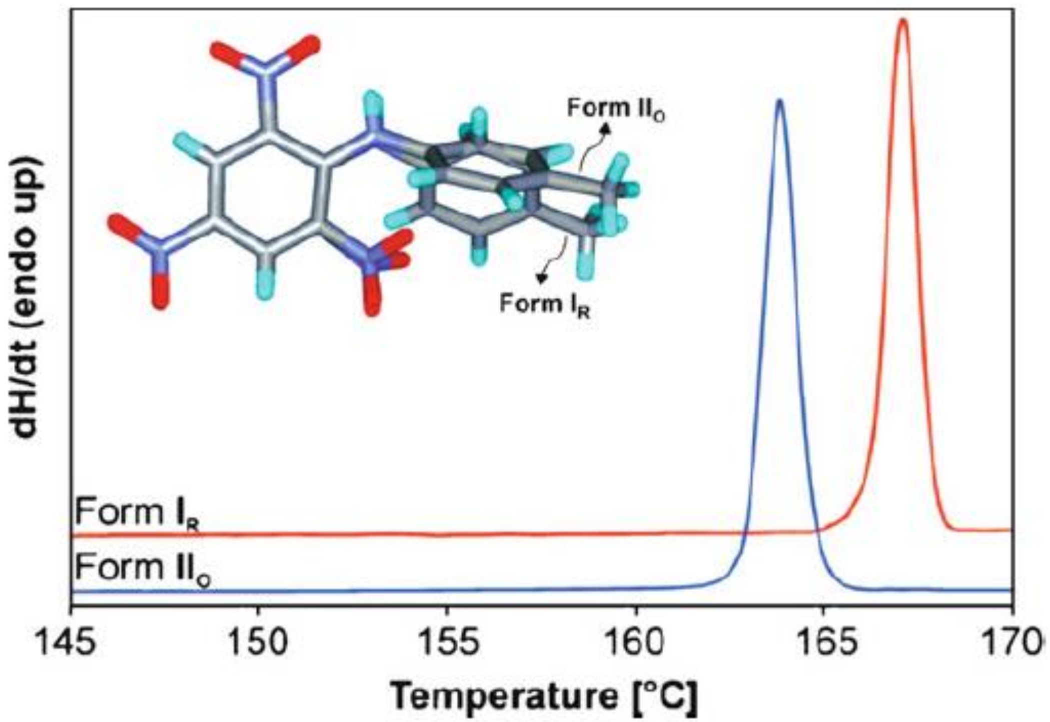
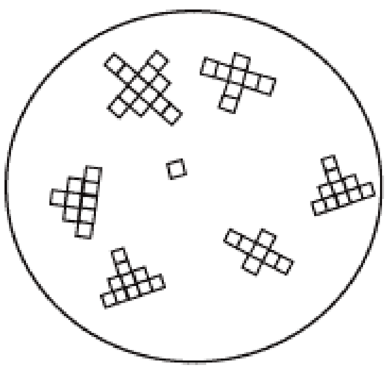


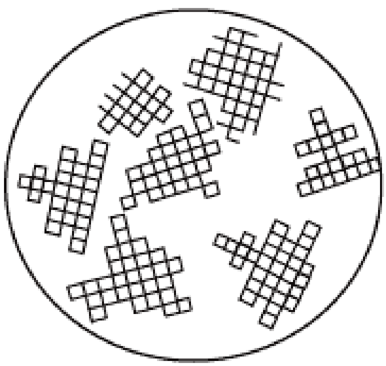
Figure 2.5. Differential scanning calorimetry (DSC) curves of two polymorphs of picryltoluidine. Crystals of form $\mathbf{I}_R$ were obtained from methanol, whereas form $\mathbf{II}_o$ was obtained from an acetone-water solution. Whereas form $\mathbf{I}_R$ exhibits an endothermic melting peak at 166°C, form $\mathbf{II}_o$ melts at 163°C. Further, the enthalpies of fusion for $\mathbf{I}_R$ and $\mathbf{II}_o$ forms are determined as 31.3 and 28.6 kJ/mol, respectively. Reproduced with permission from *Cryst. Growth Des.* **2008**, *8*, 1977. Copyright 2008 American Chemical Society.

2.3.1 Crystal Growth Technique

Crystal growth involves a phase change from liquid (precipitation) or gas (sublimation) to a solid.



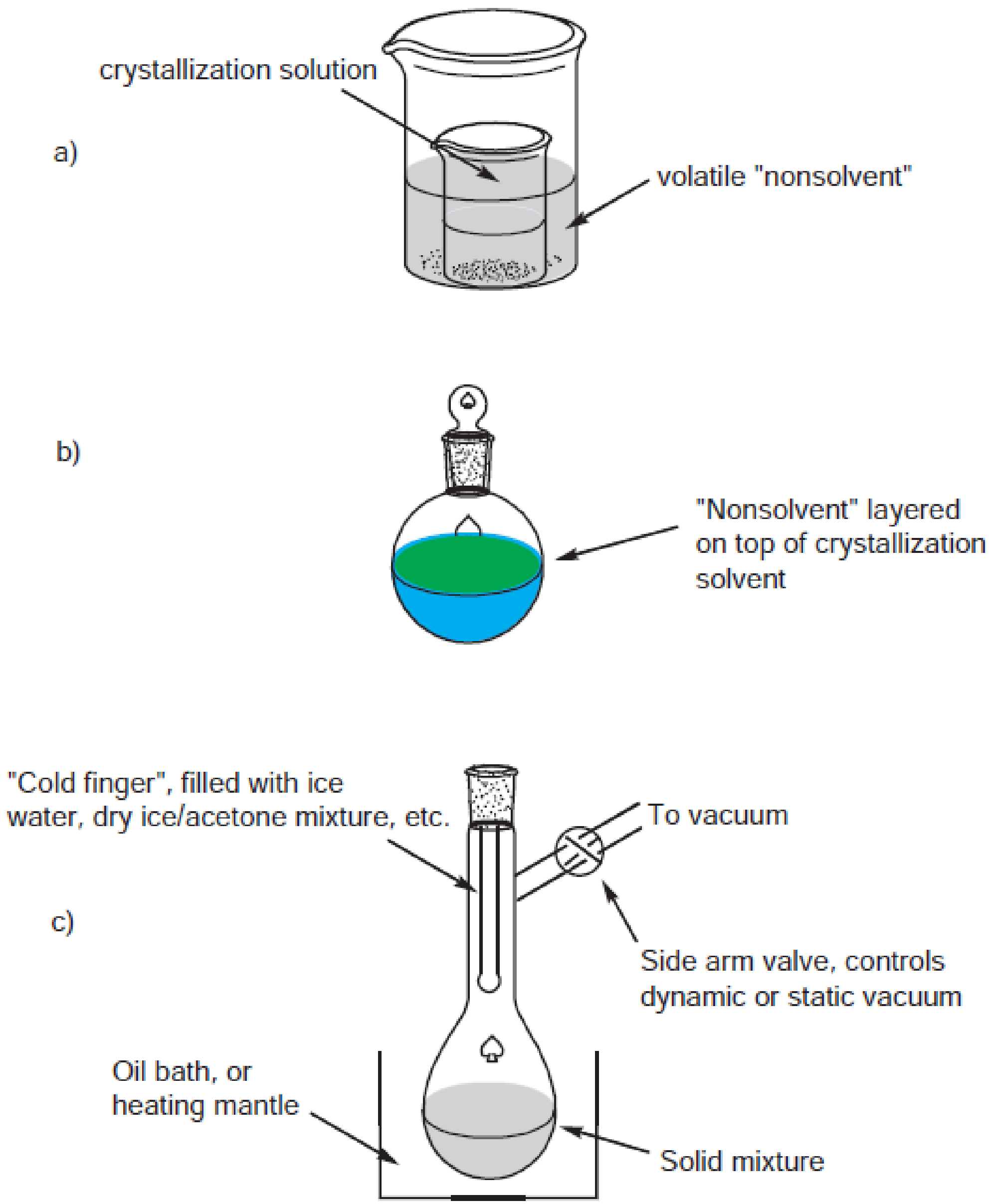
Nucleation



Growth

For high quality single crystal

- Relatively few nuclei should be formed rather than multiple sites of nucleation that will yield microcrystalline solids.
- The growth should be kept at a rate sufficiently low to allow oriented growth. A high growth rate may lead to defects in the crystal, forming multi-branched or dendritic crystallites.



(a) Diffusion where vapors from a volatile "nonsolvent" meet the crystallization solvent;

(b) Interfacial where the nonsolvent is layered on top of the crystallization solvent;

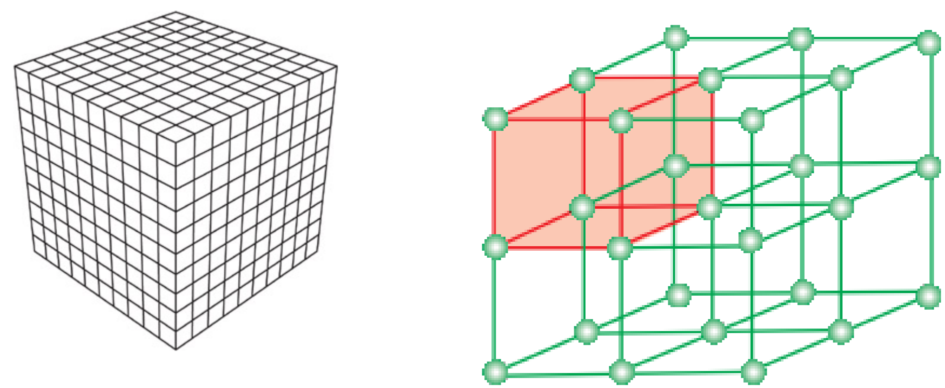
(c) Sublimation where the solid mixture is heated and the vapors form crystallites on the surface of a cold finger.

2.3.2 The Unit Cell

A **unit cell** is the smallest repeating unit in space lattice, which governs the symmetry and structure of the overall bulk crystal.

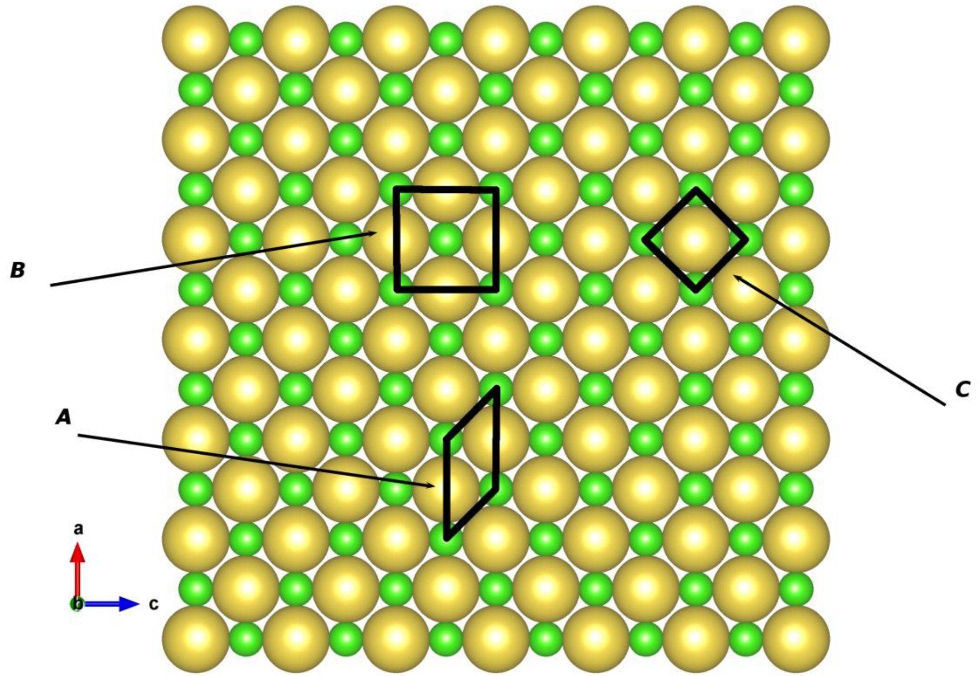
A unit cell will generate the entire extended crystal lattice via translations along the unit cell axes.

Two dimensional Lattice



Three dimensional Lattice

Choosing a cell unit

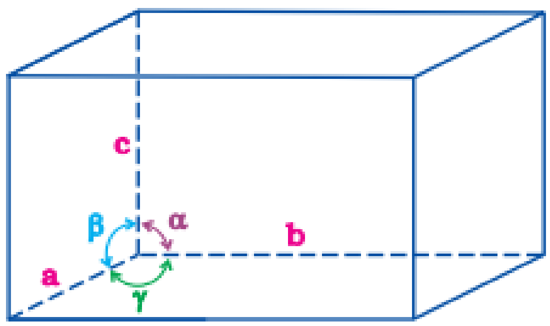


The whole space lattice may be generated by placing any one of these unit cells side by side in space. But usually a single unit cell is chosen. The real unit cell representation may be found out by considering:

- **the volume of the unit cell;**
- **the symmetry of the unit cell.**

- The unit cell is the simplest repeating unit in the crystal.
- Opposite faces of a unit cell are parallel.
- The edge of the unit cell connects equivalent points.

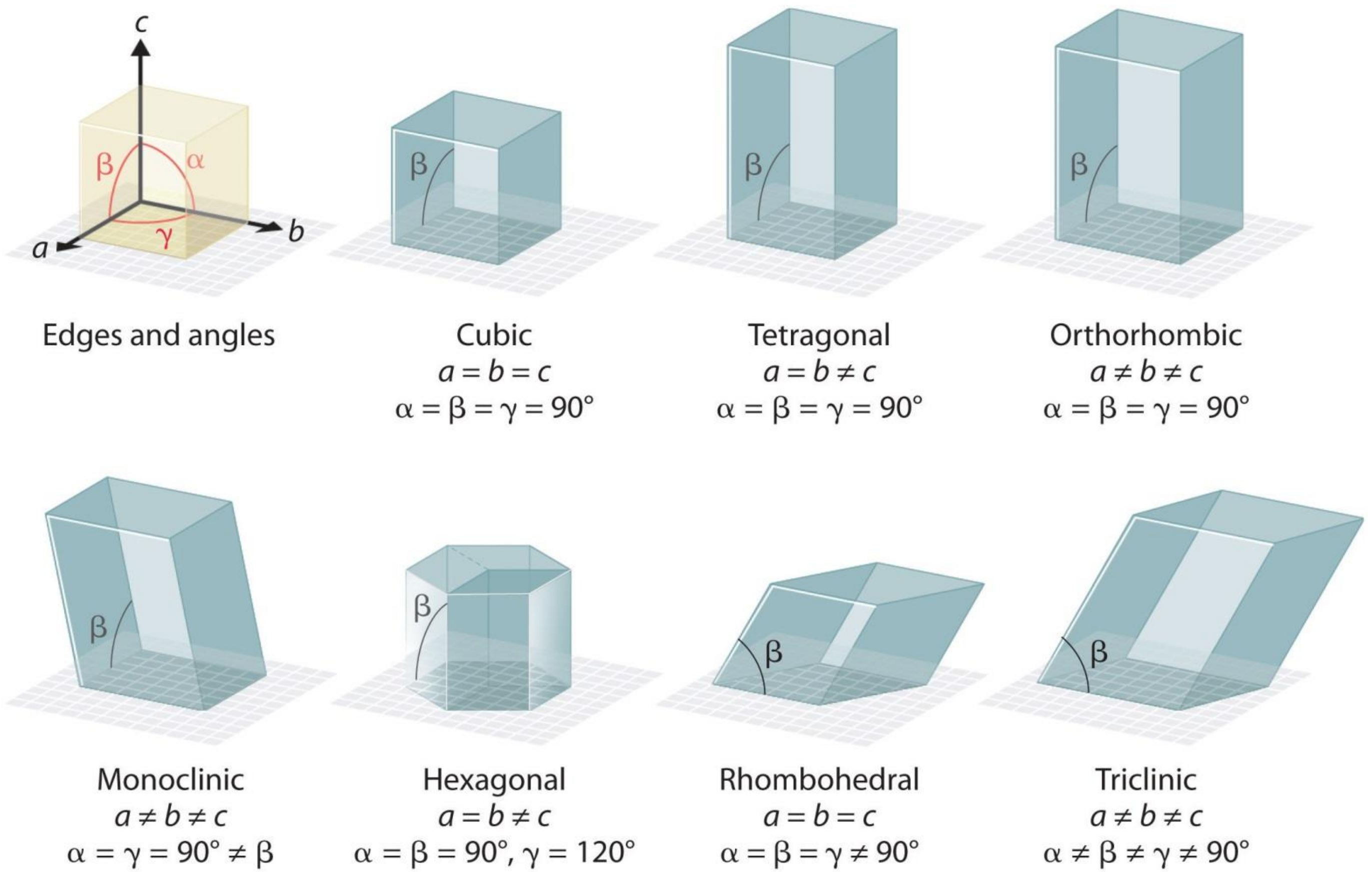
Coordinates for unit cells



A unit cell is characterized by:

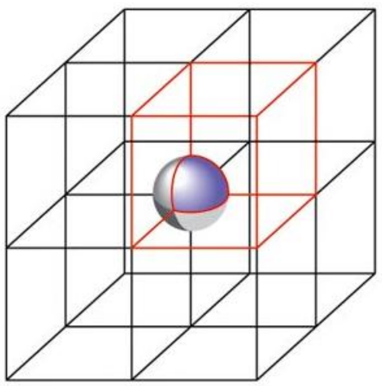
- its dimensions along the three edges, a, b and c.
- angles between the edges,
  - α (between b and c)
  - β (between a and c)
  - γ (between a and b).

The General Features of the Seven Basic Unit Cells

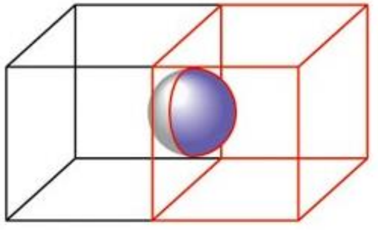


motifs/Unit Cell

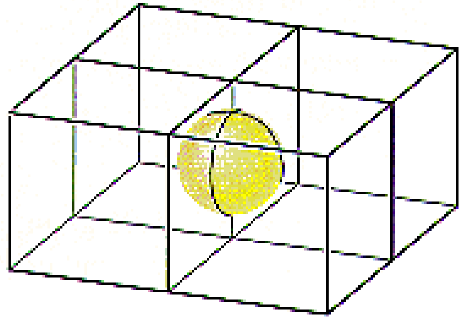
For atoms in a cubic unit cell:



motifs in corners are ⅛ within the cell



motifs on faces are ½ within the cell



motifs on edges are ¼ within the cell

motifs/Unit Cell

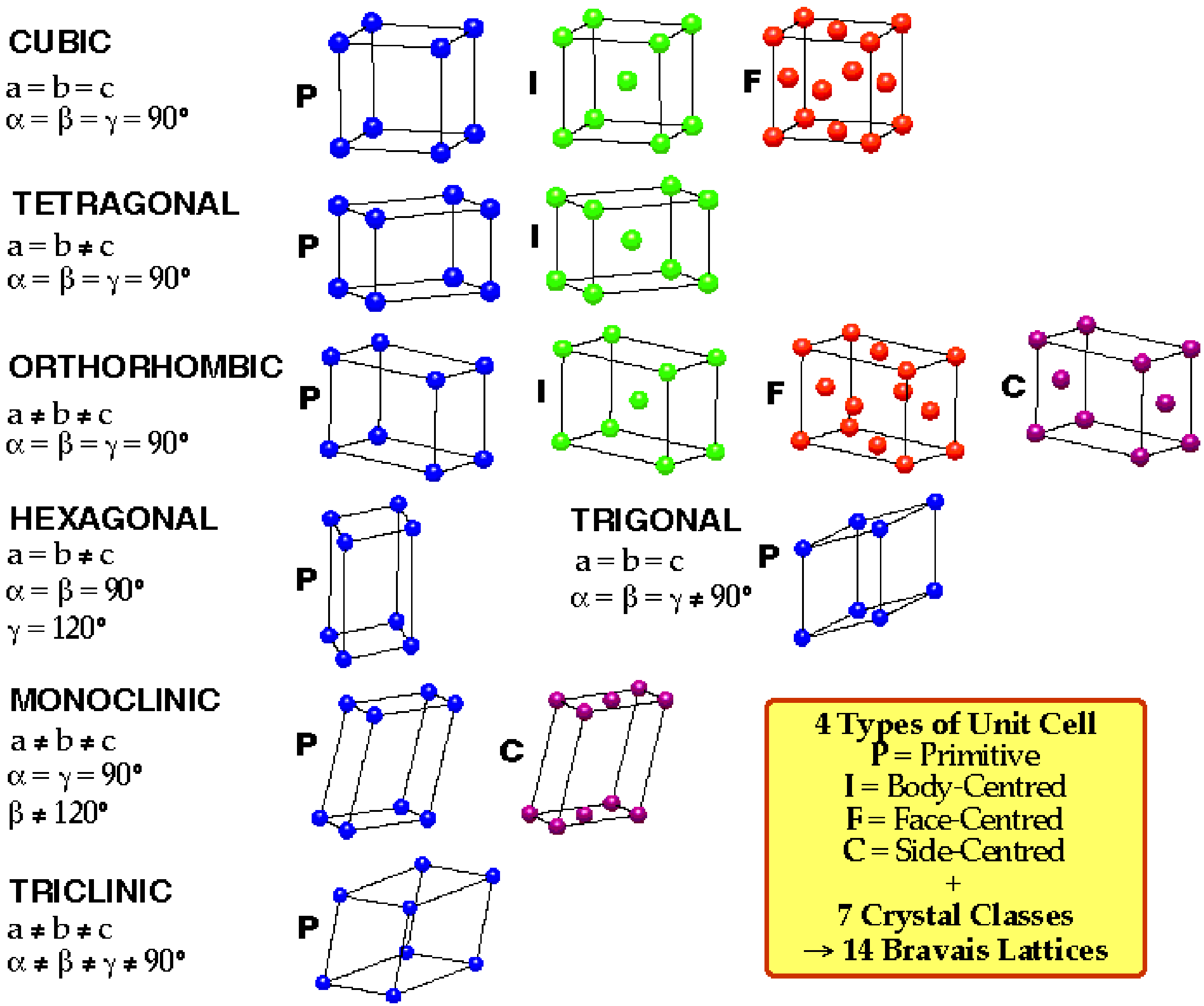
Position in unit cell	Fraction in unit cell
Center	1
Face	½
Edge	¼
Corner	⅛

Types of unit cells

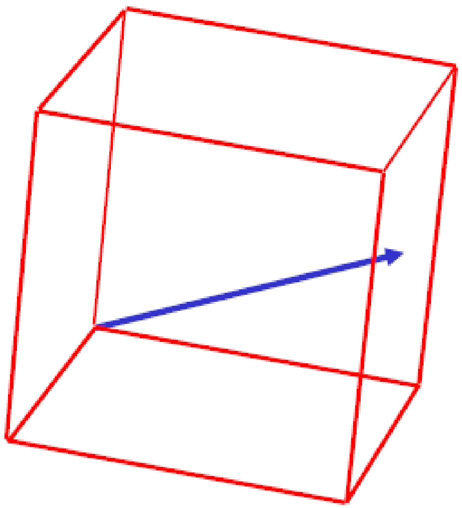
Unit Cell

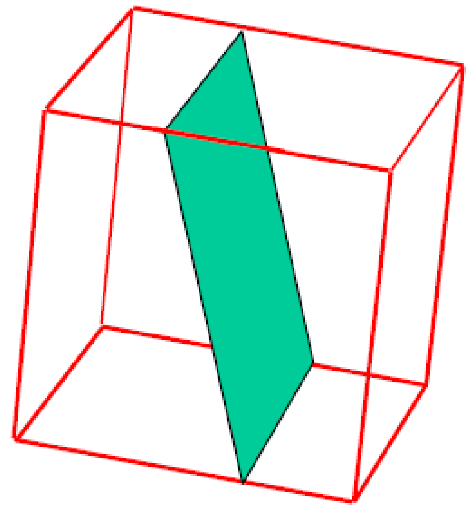
- Primitive Unit Cell (P)
  - Particles are only present at corners
- Centered Unit Cell
  - Particles are present at corners and other positions
  - Body-Centered (I): Contains one particle at body center
  - Face-Centered (F): Contains one particle at the center of each face.
  - Side-Centered (C): Contains one particle at the center of any two opposing faces.

The 14 Bravais unit cells



Crystallographic Planes & Directions





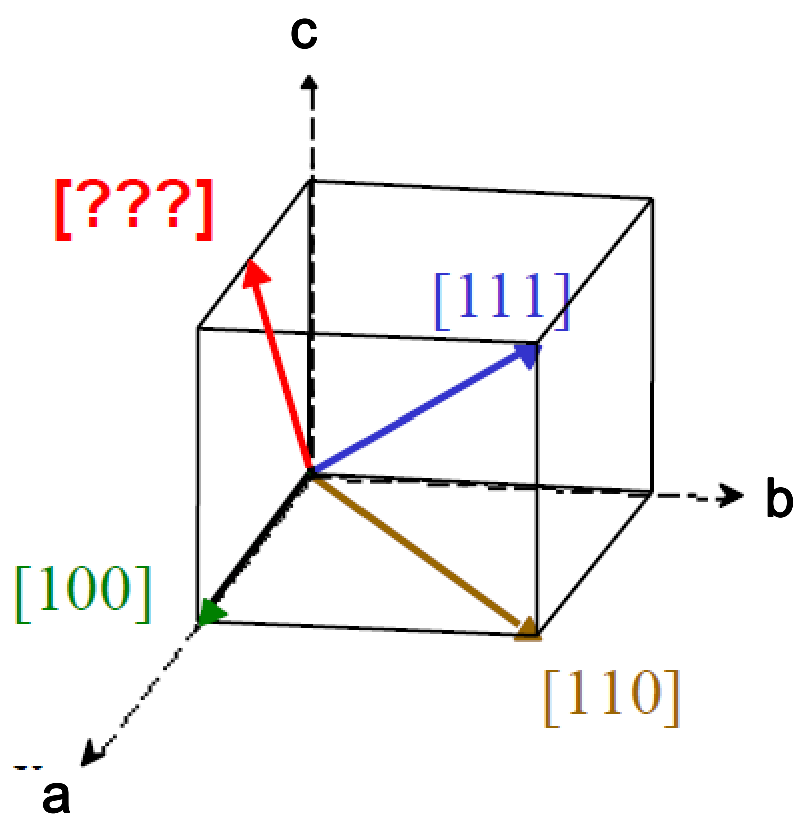
- It is often necessary to be able to specify certain directions and planes in crystals.
- Many material properties and processes vary with direction in the crystal.
- Directions and planes are described using three integers - **Miller Indices**

General Rules for Lattice Directions, Planes & Miller Indices

- Miller indices used to express lattice *planes* and *directions*
- a, b, c are the axes (on arbitrarily positioned origin)
- |a|, |b|, |c| are lattice parameters (length of unit cell along a side)
- h, k, l are the Miller indices for planes and directions - expressed as **planes: (hkl)** and **directions: [hkl]**

- Conventions for naming
– There are NO COMMAS between numbers
– Negative values are expressed with a bar over the number
- Example: -2 is expressed $\bar{2}$

Miller Indices for Directions



Method

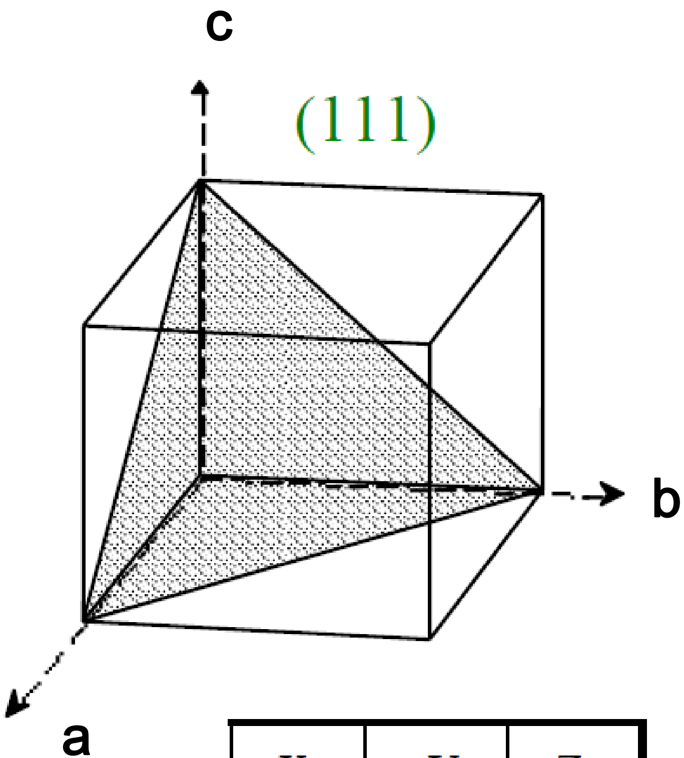
– Draw vector, and find the coordinates of the head, a_1,b_1,c_1 and the tail a_2,b_2,c_2.

– subtract coordinates of tail from coordinates of head.

– Remove fractions by multiplying by smallest possible factor.

– Enclose in square brackets.

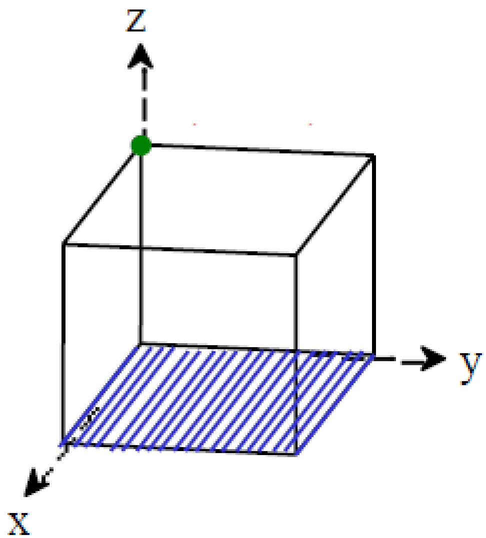
Miller Indices for Planes

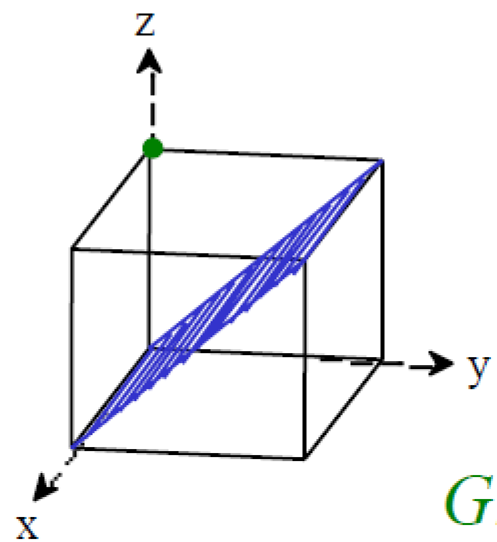


	x	y	z
Intercepts	1	1	1
Reciprocals	1	1	1

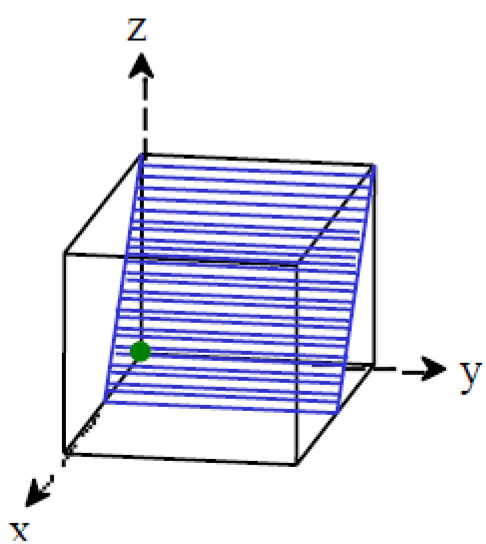
Method

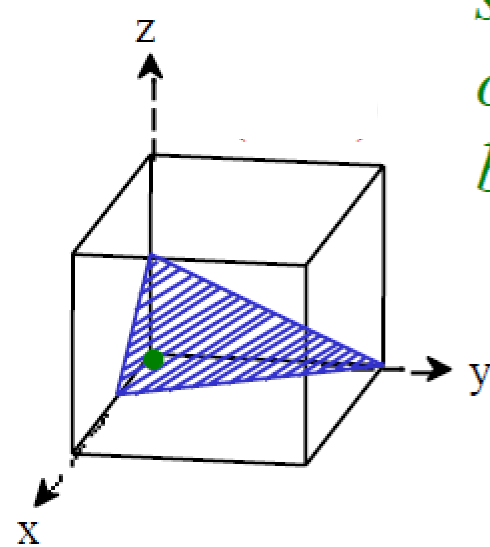
- If the plane passes through the origin, select an equivalent plane or move the origin.
- Determine the intersection of the plane with the axes in terms of a, b, and c
- Take the reciprocal (1/∞ = 0)
- Convert to smallest integers (optional)
- Enclose by parentheses

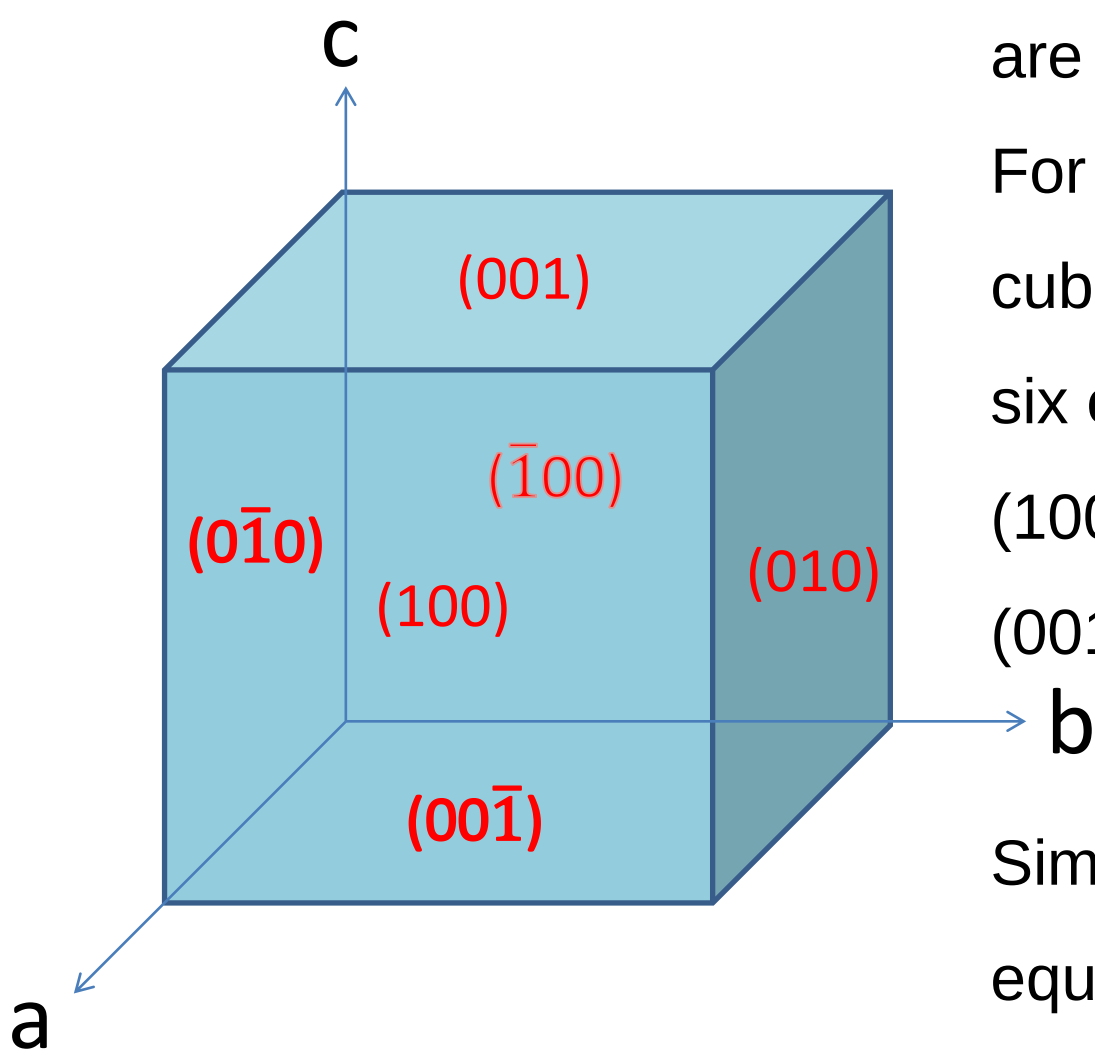




Green circles show where the origins have been placed.





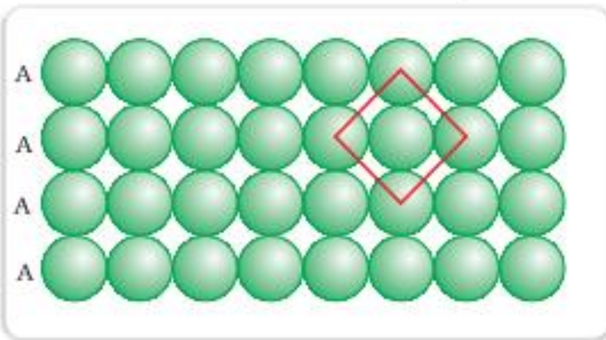


Symmetry-equivalent planes are indicated by $\{hkl\}$.

For example: $\{100\}$ for a cubic unit cell would have six equivalent plances: (100), ($\bar{1}$00), (010), (0$\bar{1}$0), (001), and (00$\bar{1}$)

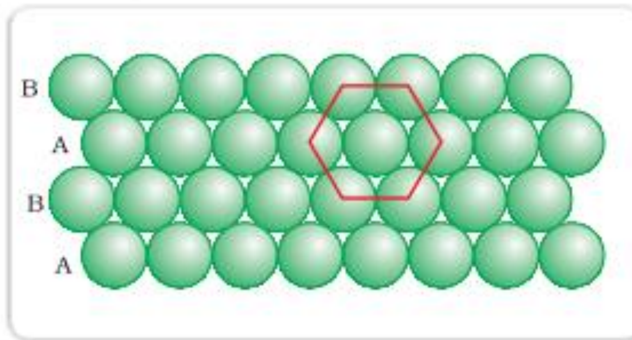
Similarly, the family of equivalent directions are indicated by $\langle hkl \rangle$

2.3.3 Crystal Lattices and close packing



(a)

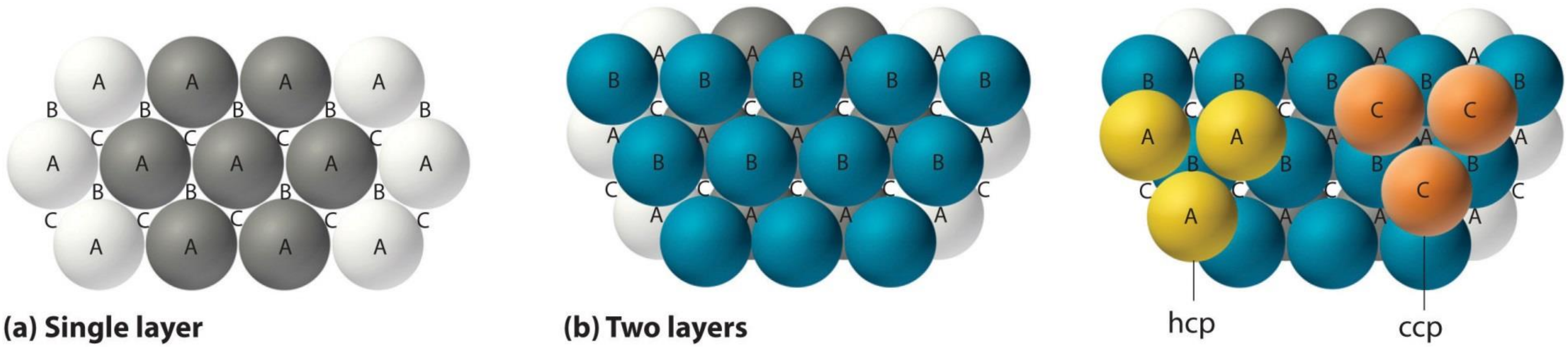
Simple cubic packing



(b)

Close packing

the coordination number (*i.e.*, number of nearest neighbors) of individual species for cubic packing is 6, the coordination number for close-packing is 12.



(a) Single layer (b) Two layers (c) Three layers

There are now two ways for placing the third layer:

- If the atoms/ions in the third layer sit directly above the spheres in the first layer to form a ABABABA….pattern, the arrangement has a hexagonal unit cell, or is said to be hexagonal close packed (hcp).

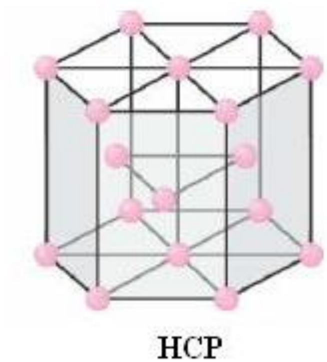
- If the atoms/ions in the third layer are placed in C sites to form a ABCABC….pattern, the arrangement is called cubic close packing (ccp) or face centered cubic packing (fcc).

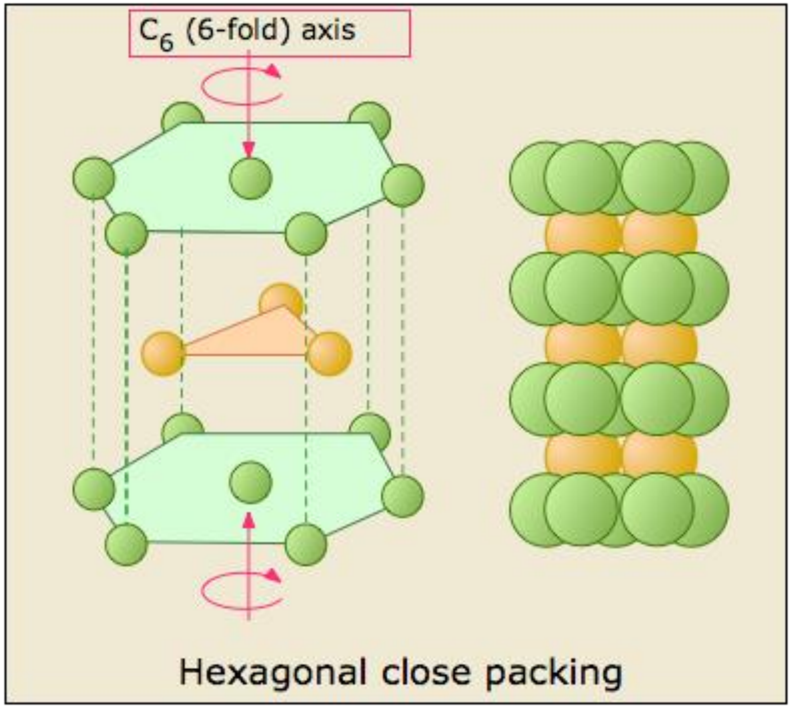
H hcp																	**He** hcp
Li bcc	**Be** hcp											**B** tetrag	**C** hcp	**N** hcp	**O** cubic	**F**	**Ne** fcc
Na bcc	**Mg** hcp											**Al** fcc	**Si** diam	**P** orthor	**S** orthor	**Cl** tetrag	**Ar** fcc
K bcc	**Ca** fcc	**Sc** hcp	**Ti** hcp	**V** bcc	**Cr** bcc	**Mn** bcc	**Fe** bcc	**Co** hcp	**Ni** fcc	**Cu** fcc	**Zn** hcp	**Ga** orthor	**Ge** diam	**As** rhomb	**Se** hcp	**Br** orthor	**Kr** fcc
Rb bcc	**Sr** fcc	**Y** hcp	**Zr** hcp	**Nb** bcc	**Mo** bcc	**Tc** hcp	**Ru** hcp	**Rh** fcc	**Pd** fcc	**Ag** fcc	**Cd** hcp	**In** tetrag	**Sn** tetrag	**Sb** rhomb	**Te** hcp	**I** orthor	**Xe** fcc
Cs bcc	**Ba** bcc	**La** hcp	**Hf** hcp	**Ta** bcc	**W** bcc	**Re** hcp	**Os** hcp	**Ir** fcc	**Pt** fcc	**Au** fcc	**Hg** rhomb	**Tl** hcp	**Pb** fcc	**Bi** rhomb	**Po** monoc	**At**	**Rn** fcc
Fr bcc	**Ra** bcc	**Ac** fcc	**Rf**	**Db**	**Sg**	**Bh**	**Hs**	**Mt**									

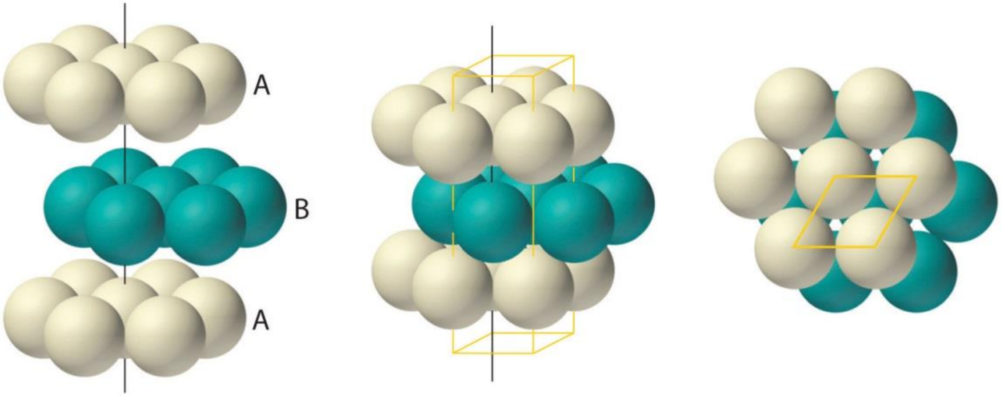
Ce fcc	**Pr** hcp	**Nd** hcp	**Pm**	**Sm** rhomb	**Eu** bcc	**Gd** hcp	**Tb** hcp	**Dy** hcp	**Ho** hcp	**Er** hcp	**Tm** hcp	**Yb** fcc	**Lu** hcp
Th fcc	**Pa** tetrag	**U** orthor	**Np** orthor	**Pu** monoc	**Am** hcp	**Cm**	**Bk**	**Cf**	**Es**	**Fm**	**Md**	**No**	**Lr**

Figure 2.13. *Preferred* crystal structures of the elements. Shown are the ordinary forms of each element in its solid state (*e.g.*, graphite, α-iron, *etc.*). The following abbreviations are used: fcc, bcc (body-centered cubic), hcp, cubic (simple cubic), diam (diamond or zinc blende), monoc (monoclinic), rhomb (rhombohedral), tetrag (tetragonal), and orthor (orthorhombic).

Hexagonal close packing (hcp)



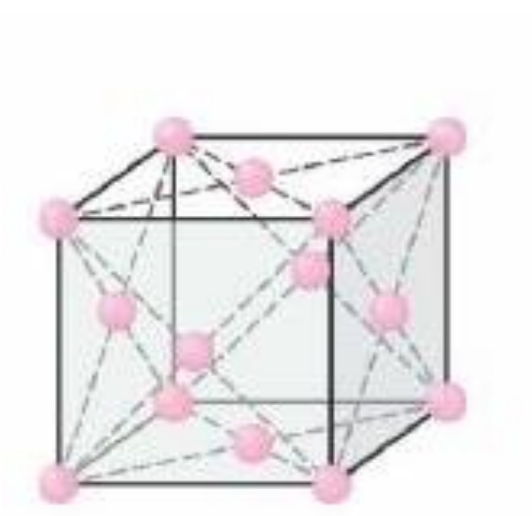




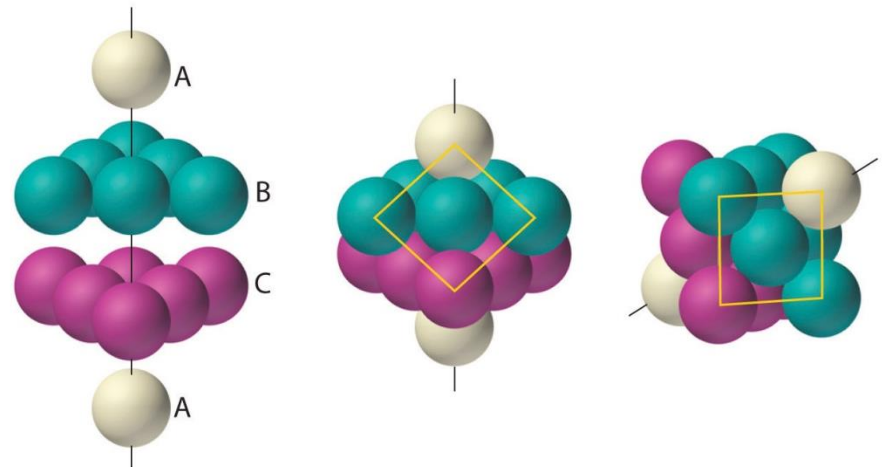
(a) Hexagonal close-packed (hcp)

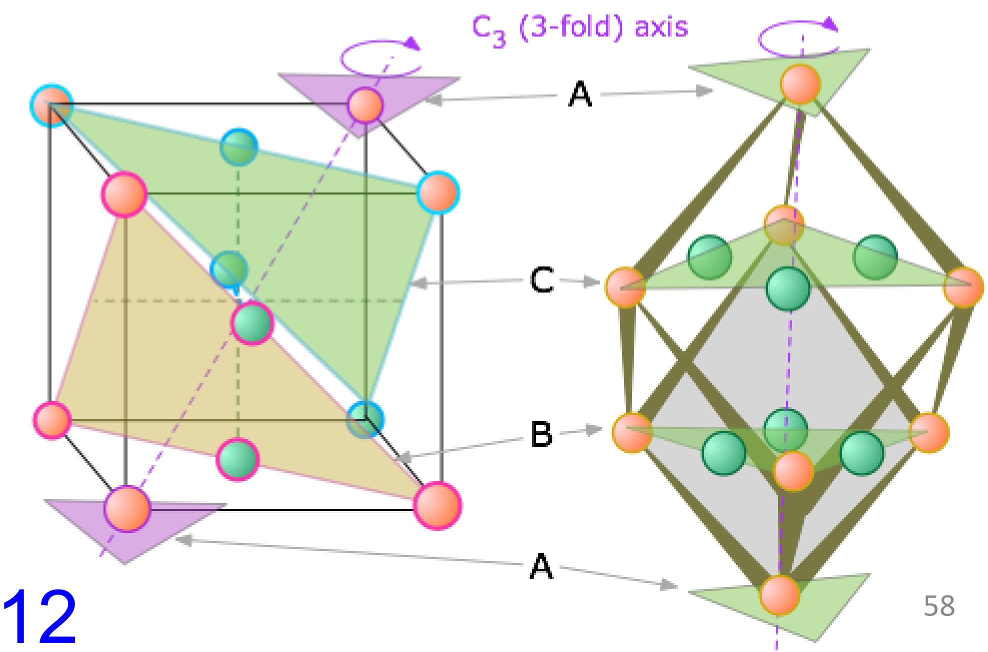
Coordination number 12

Face centered cubic close packing (fcc)



FCC





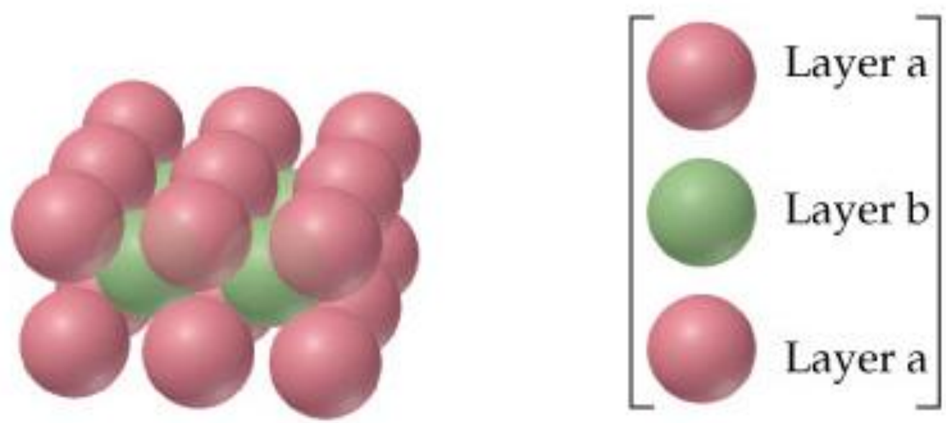
Coordination number 12

TABLE 10.10 Summary of the Four Kinds of Packing for Spheres

Structure	Stacking Pattern	Coordination Number	Space Used (%)	Unit Cell
Simple cubic	*a-a-a-a-*	6	52	Primitive cubic
Body-centered cubic	*a-b-a-b-*	8	68	Body-centered cubic
Hexagonal closest-packed	*a-b-a-b-*	12	74	(Noncubic)
Cubic closest-packed	*a-b-c-a-b-c-*	12	74	Face-centered cubic

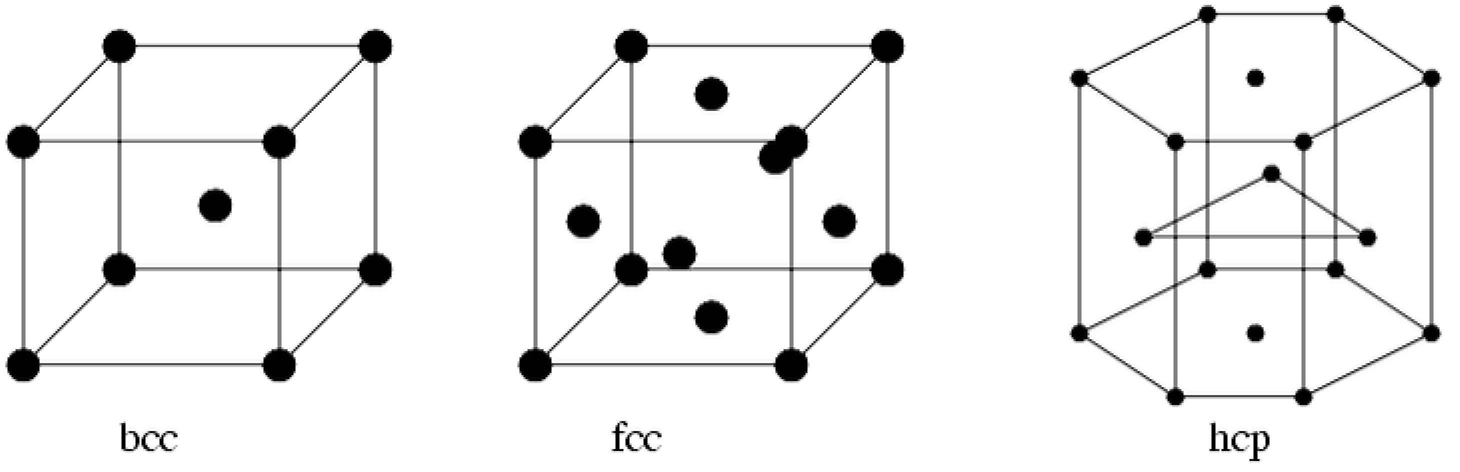
Simple cubic and Body-centered cubic

Simple cubic
(a)

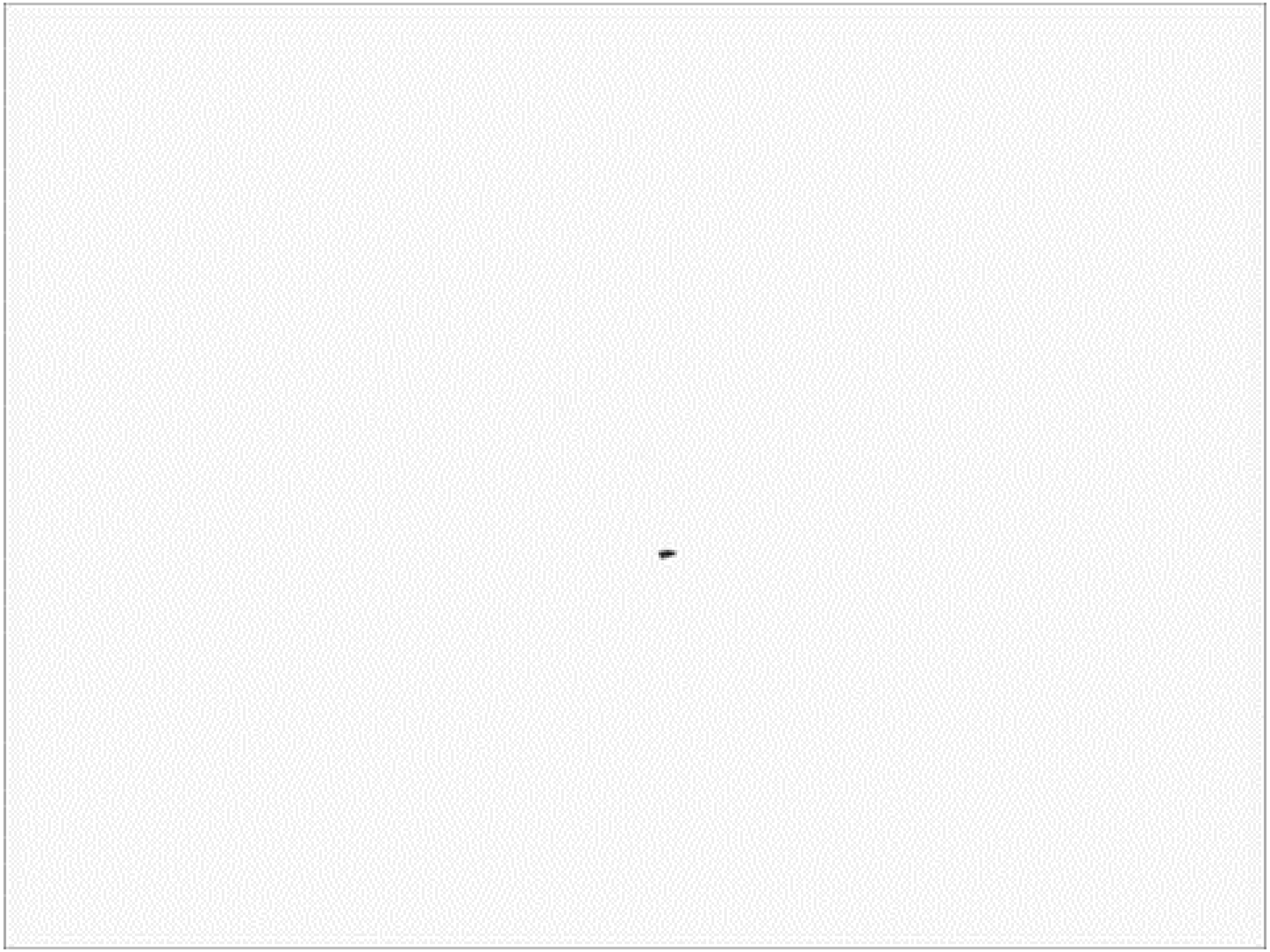


Body-centered cubic
(b)

Number of particles per unit cell



Particles/Cell	2	4	6



Interstitial crystal lattices

Either arrangement utilizes 74% of the available space, producing a dense arrangement of atoms. Small holes make up the other 26% of the unit cell.

Only for ab type structure

Table 2.3. Ionic Radii Ratios Corresponding to Interstitial Sites

Radii ratio range (r_+/r_-)	Geometry of interstitial site (coordination number)
<0.225	Trigonal (3)
<0.414	Tetrahedral (4)
<0.732	Octahedral (6)
>0.732	Cubic (8)

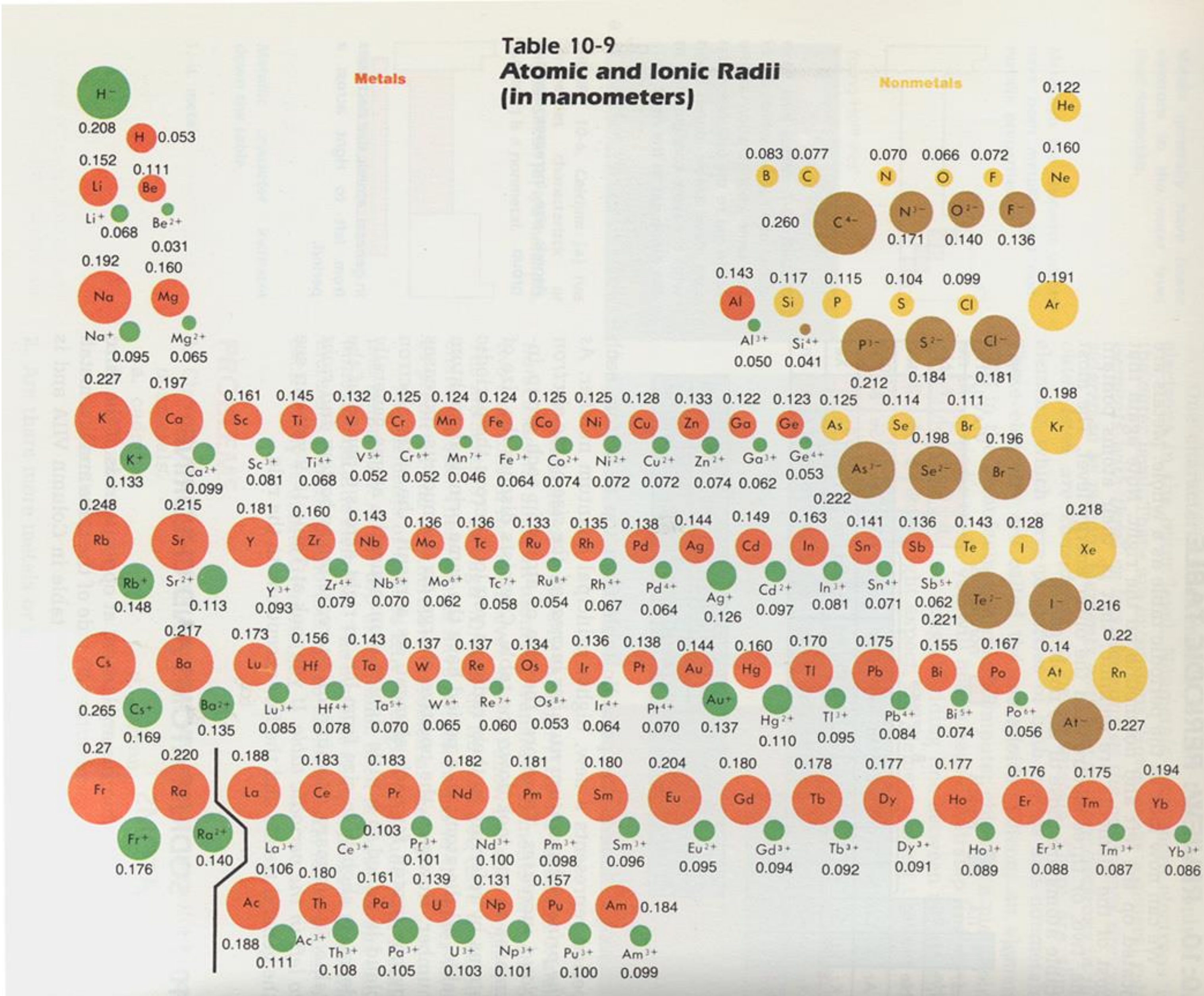
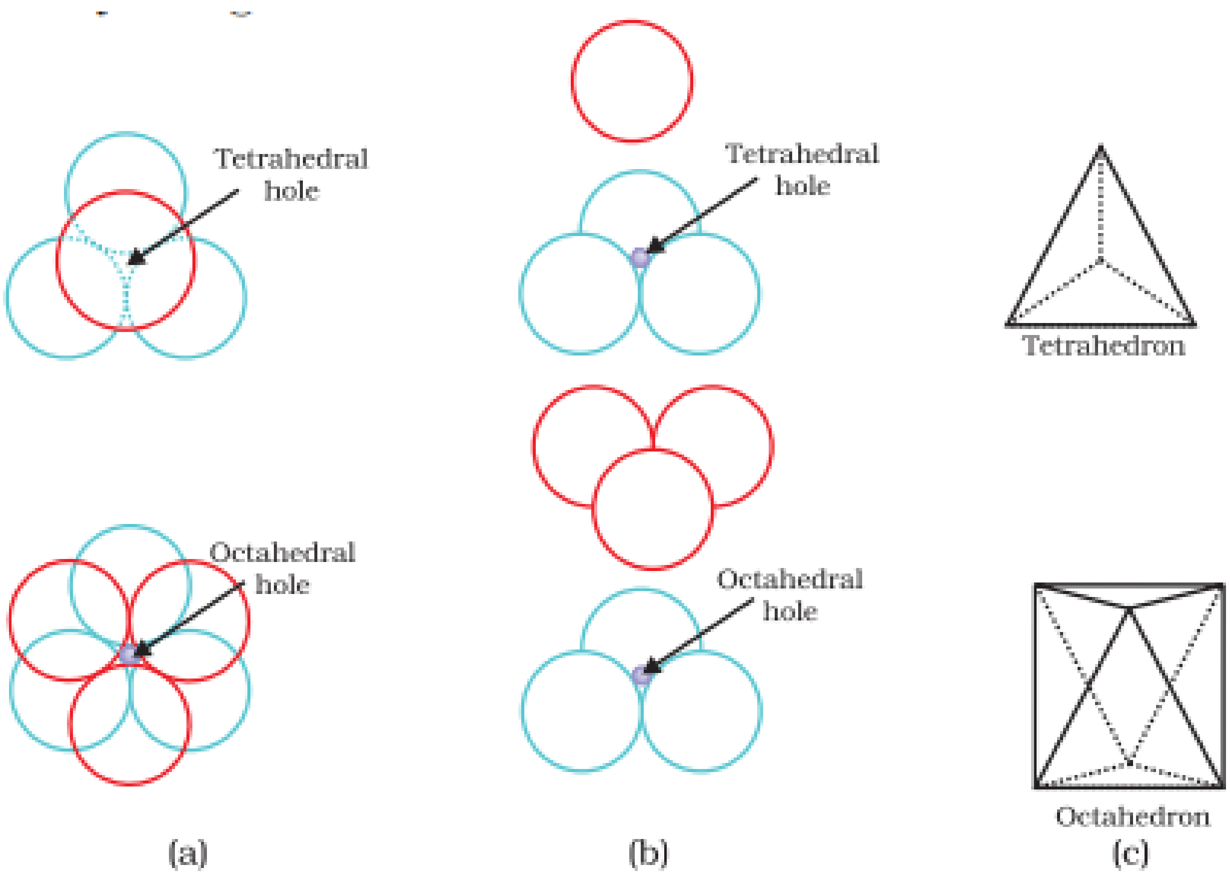
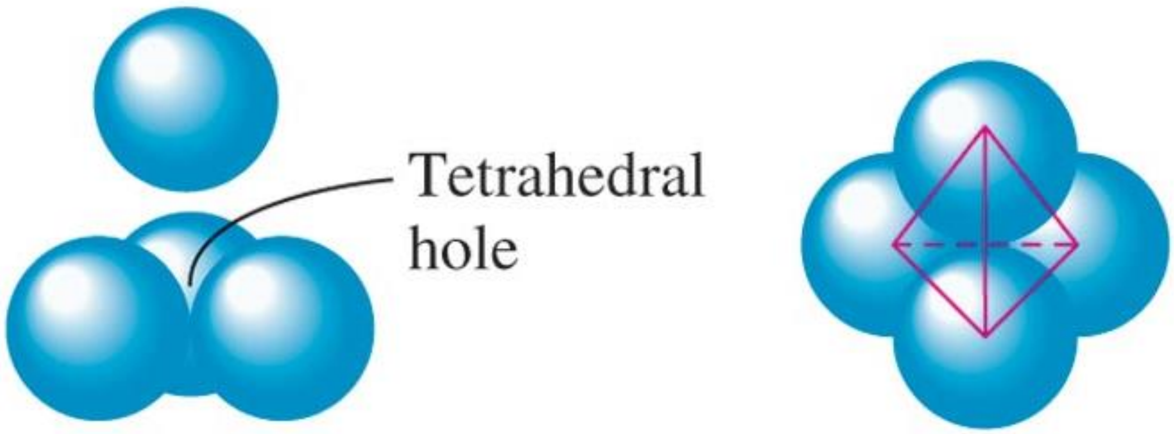


Table 10-9
Atomic and Ionic Radii (in nanometers)
Metals
Nonmetals
H^- 0.208
H 0.053
Li 0.152 Li^+ 0.068
Be 0.111 Be^{2+} 0.031
Na 0.192 Na^+ 0.095
Mg 0.160 Mg^{2+} 0.065
K 0.227 K^+ 0.133
Ca 0.197 Ca^{2+} 0.099
Rb 0.248 Rb^+ 0.148
Sr 0.215 Sr^{2+} 0.113
Cs 0.265 Cs^+ 0.169
Ba 0.217 Ba^{2+} 0.135
Fr 0.27 Fr^+ 0.176
Ra 0.220 Ra^{2+} 0.140
Sc 0.161 Sc^{3+} 0.081
Ti 0.145 Ti^{4+} 0.068
V 0.132 V^{5+} 0.052
Cr 0.125 Cr^{6+} 0.052
Mn 0.124 Mn^{7+} 0.046
Fe 0.124 Fe^{3+} 0.064
Co 0.125 Co^{2+} 0.074
Ni 0.125 Ni^{2+} 0.072
Cu 0.128 Cu^{2+} 0.072
Zn 0.133 Zn^{2+} 0.074
Ga 0.122 Ga^{3+} 0.062
Ge 0.123 Ge^{4+} 0.053
As 0.125 As^{3-} 0.222
Se 0.114 Se^{2-} 0.198
Br 0.111 Br^- 0.196
Kr 0.198
He 0.122
B 0.083
C 0.077 C^{4-} 0.260
N 0.070 N^{3-} 0.171
O 0.066 O^{2-} 0.140
F 0.072 F^- 0.136
Ne 0.160
Al 0.143 Al^{3+} 0.050
Si 0.117 Si^{4+} 0.041
P 0.115 P^{3-} 0.212
S 0.104 S^{2-} 0.184
Cl 0.099 Cl^- 0.181
Ar 0.191
Y 0.181 Y^{3+} 0.093
Zr 0.160 Zr^{4+} 0.079
Nb 0.143 Nb^{5+} 0.070
Mo 0.136 Mo^{6+} 0.062
Tc 0.136 Tc^{7+} 0.058
Ru 0.133 Ru^{8+} 0.054
Rh 0.135 Rh^{4+} 0.067
Pd 0.138 Pd^{4+} 0.064
Ag 0.144 Ag^+ 0.126
Cd 0.149 Cd^{2+} 0.097
In 0.163 In^{3+} 0.081
Sn 0.141 Sn^{4+} 0.071
Sb 0.136 Sb^{5+} 0.062
Te 0.143 Te^{2-} 0.221
I 0.128 I^- 0.216
Xe 0.218
Lu 0.173 Lu^{3+} 0.085
Hf 0.156 Hf^{4+} 0.078
Ta 0.143 Ta^{5+} 0.070
W 0.137 W^{6+} 0.065
Re 0.137 Re^{7+} 0.060
Os 0.134 Os^{8+} 0.053
Ir 0.136 Ir^{4+} 0.064
Pt 0.138 Pt^{4+} 0.070
Au 0.144 Au^+ 0.137
Hg 0.160 Hg^{2+} 0.110
Tl 0.170 Tl^{3+} 0.095
Pb 0.175 Pb^{4+} 0.084
Bi 0.155 Bi^{5+} 0.074
Po 0.167 Po^{6+} 0.056
At 0.14 At^- 0.227
Rn 0.22
La 0.188 La^{3+} 0.106
Ce 0.183 Ce^{3+} 0.103
Pr 0.183 Pr^{3+} 0.101
Nd 0.182 Nd^{3+} 0.100
Pm 0.181 Pm^{3+} 0.098
Sm 0.180 Sm^{3+} 0.096
Eu 0.204 Eu^{2+} 0.095
Gd 0.180 Gd^{3+} 0.094
Tb 0.178 Tb^{3+} 0.092
Dy 0.177 Dy^{3+} 0.091
Ho 0.177 Ho^{3+} 0.089
Er 0.176 Er^{3+} 0.088
Tm 0.175 Tm^{3+} 0.087
Yb 0.194 Yb^{3+} 0.086
Ac 0.188 Ac^{3+} 0.111
Th 0.180 Th^{3+} 0.108
Pa 0.161 Pa^{3+} 0.105
U 0.139 U^{3+} 0.103
Np 0.131 Np^{3+} 0.101
Pu 0.157 Pu^{3+} 0.100
Am 0.184 Am^{3+} 0.099

Interstitial crystal lattices

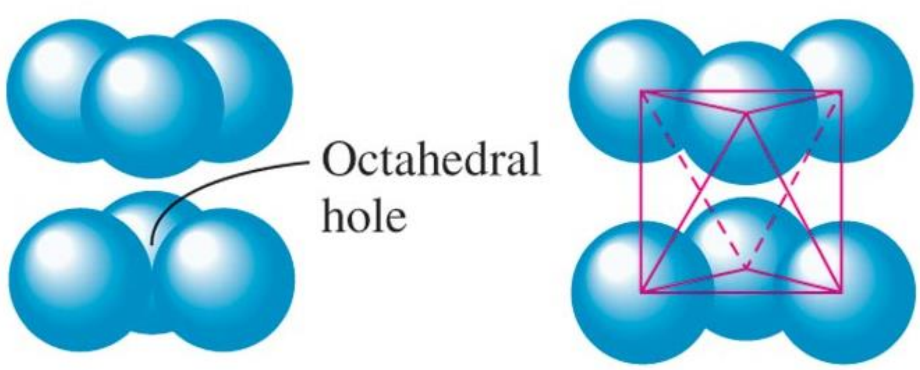


Tetrahedral voids

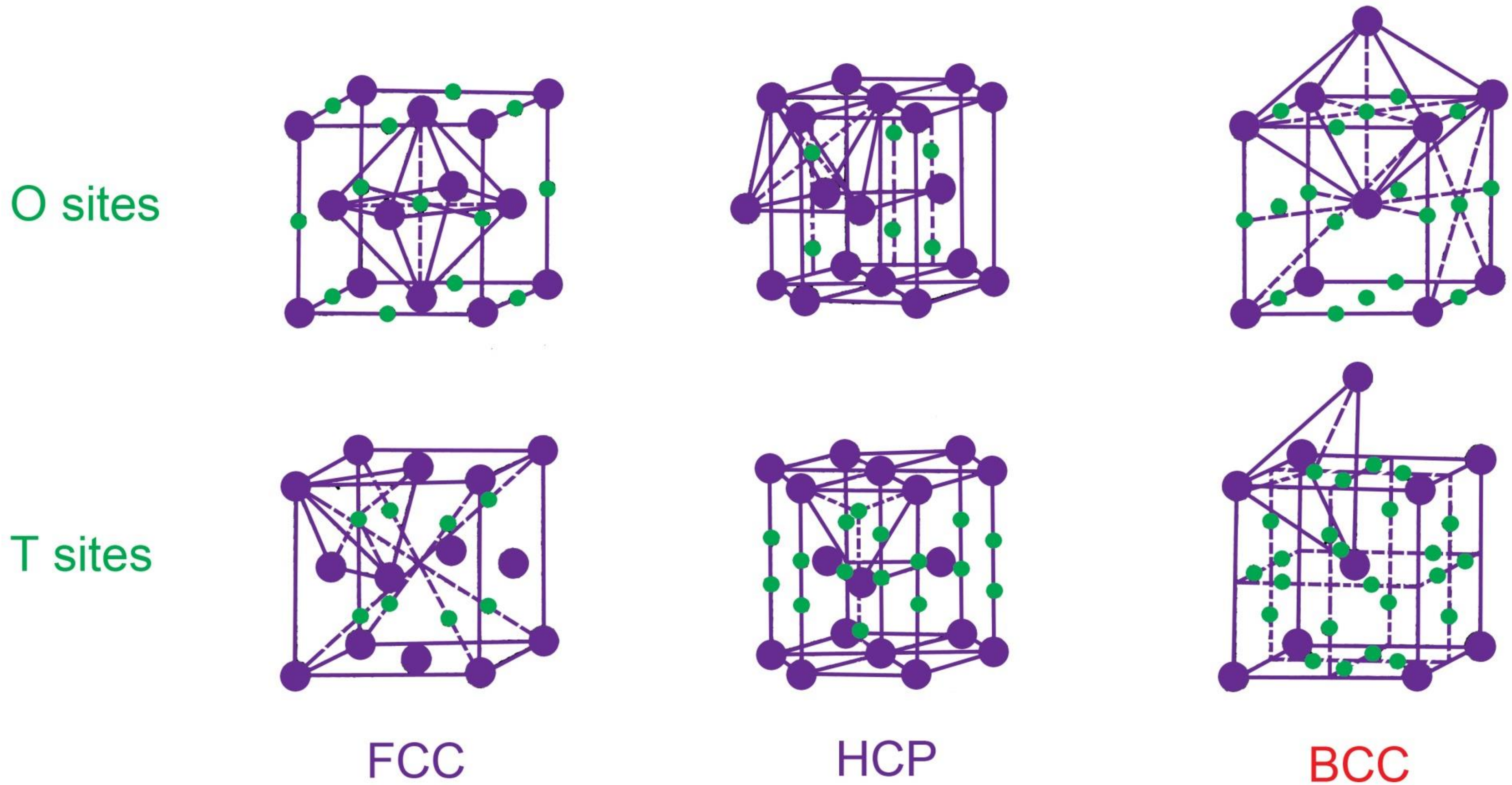


Tetrahedral voids are formed by a planar triangle of atoms, with a 4th atom covering the indentation in the center. The resulting hole has a coordination number of 4.

Octahedral voids

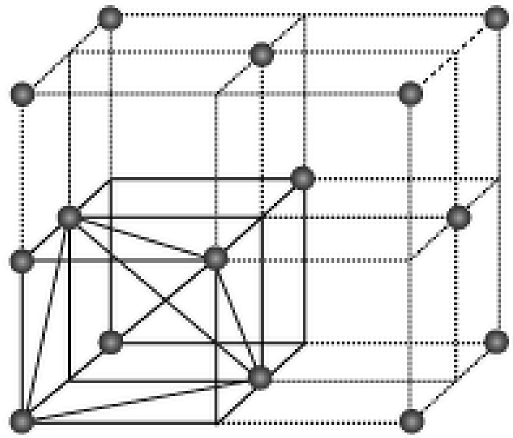


- The interstitial void, formed by combination of two triangular voids of the first and second layer is called octahedral void because this is enclosed between six spheres centers of which occupy corners of a regular octahedron.

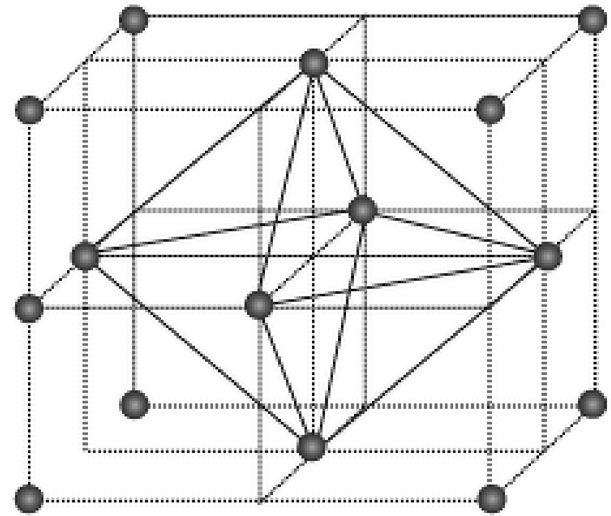


Voids/cell

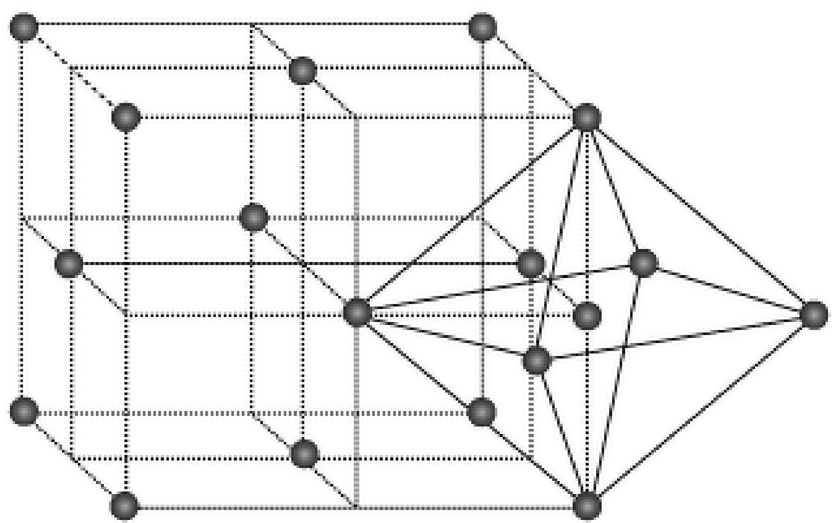
	Octahedral voids	Tetrahedral voids
FCC	4	8
HCP	6	12
BCC	6	12



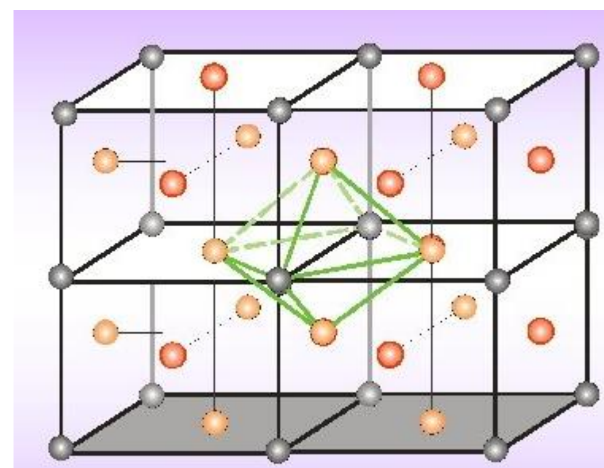
Tetrahedral



Octahedral



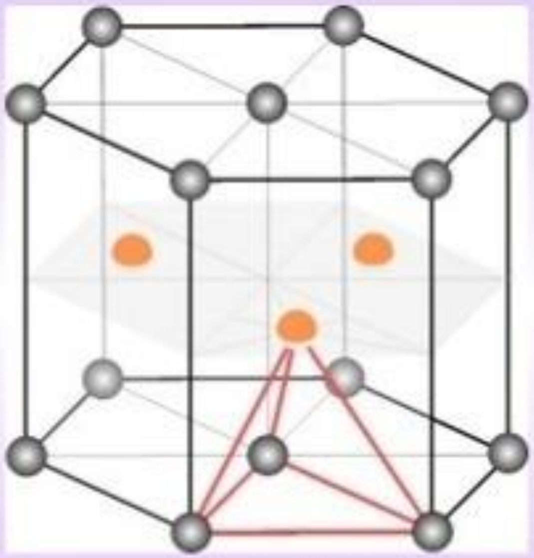
Octahedral



HCP

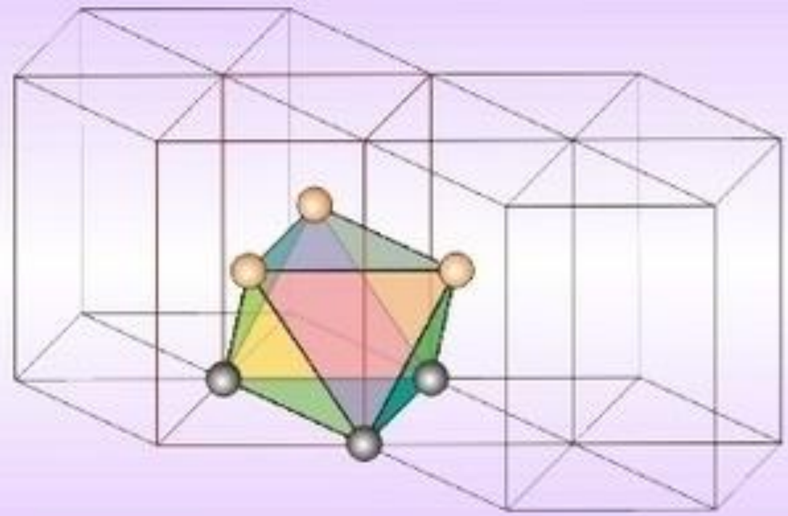
VOIDS

TETRAHEDRAL



Coordinates: $(0,0,\frac{3}{8}),(0,0,\frac{5}{8}),(\frac{2}{3},\frac{1}{3},\frac{1}{8}),(\frac{2}{3},\frac{1}{3},\frac{7}{8})$

OCTAHEDRAL



Coordinates: $(\frac{1}{3}\ \frac{2}{3},\frac{1}{4}), (\frac{1}{3},\frac{2}{3},\frac{3}{4})$

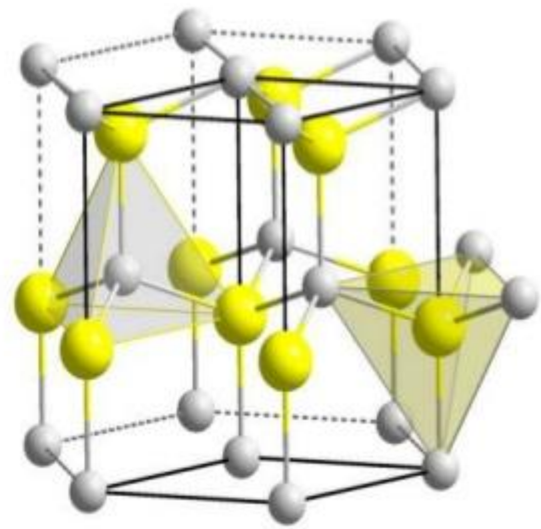
- These voids are identical to the ones found in FCC *(for ideal c/a ratio)*
- When the c/a ratio is non-ideal then the octahedra and tetrahedra are distorted *(non-regular)*

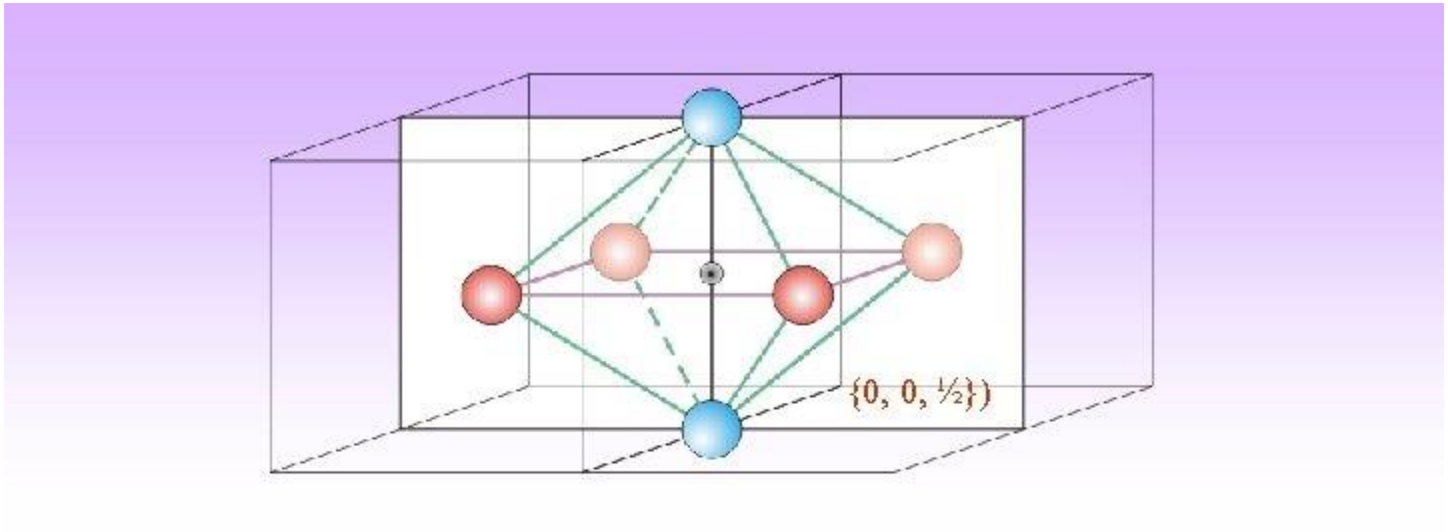
Important Note: often in these discussions an ideal c/a ratio will be assumed (without stating the same explicitly)

Note: Atoms are coloured differently but are the same

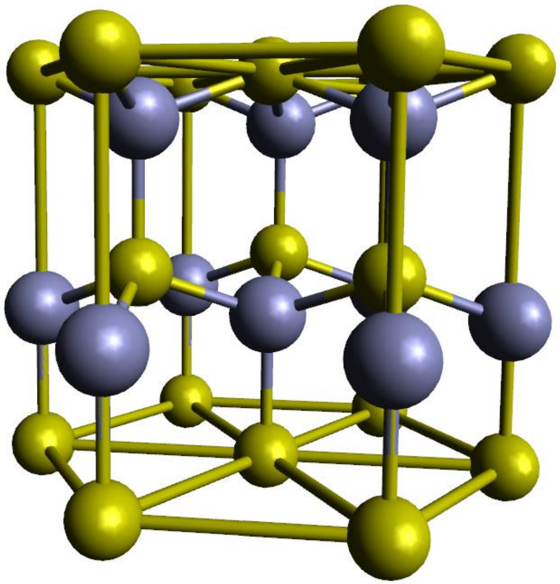
Wurtzite (ZnS)

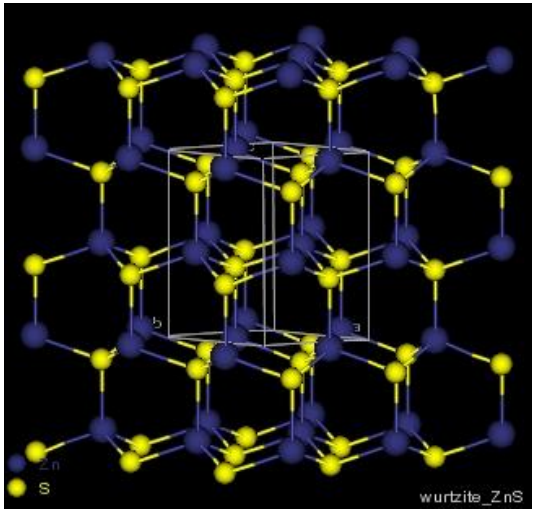
- HCP w.r.t anion.
- Cations occupy tetrahedral sites, either T_+ or T_- rest are empty.
- c/a ratio=1.633 (assuming anions are in contact)
- Tetrahedral site at a distance 3/8 above anion.
- 12 tetrahedral voids are present, only 6 are occupied.




{0, 0, ½})

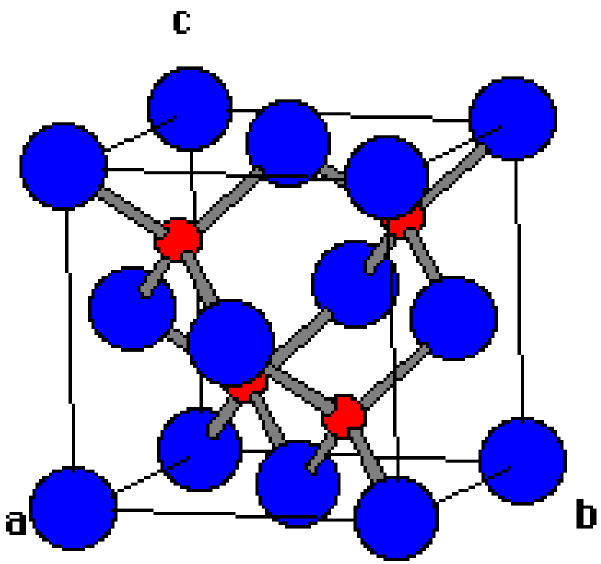
The Wurtzite structure

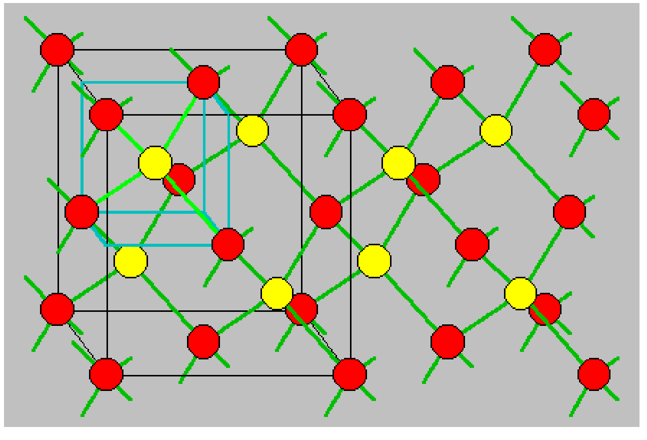




- $rZn^{2+}/ rS^{2-} = 0.40$.
- HCP type of arrangement.
- Zn^{2+} 1/2 tetrahedral voids.
- Each Zn^{2+} is surrounded by 4 S^{2-} and in turn each S^{2-} is also surrounded by 4 Zn^{2+} . The coordination of Wurtizite structure is described as 4:4.

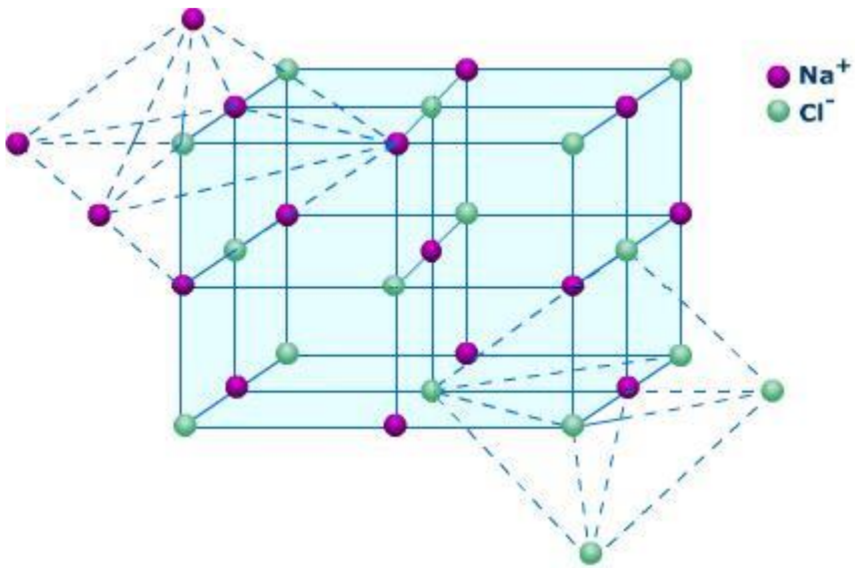
The Zinc Blende structure





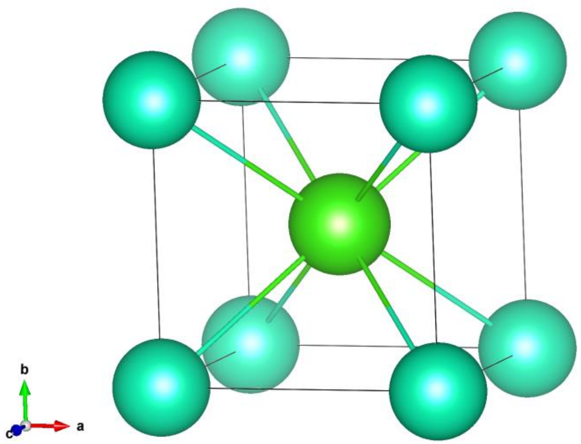
- $rZn^{2+}/ rS^{2-} = 0.40$.
- FCC type of arrangement.
- Zn^{2+} in ½ tetrahedral voids.
- Each Zn^{2+} is surrounded by 4 S^{2-} and in turn each S^{2-} is also surrounded by 4 Zn^{2+} . The coordination of zinc blende structure is described as 4:4.
- It may be concluded that the structure of Wurtzite is very similar to the structure of zinc blende. The only difference is the S^{2-} arrangements.

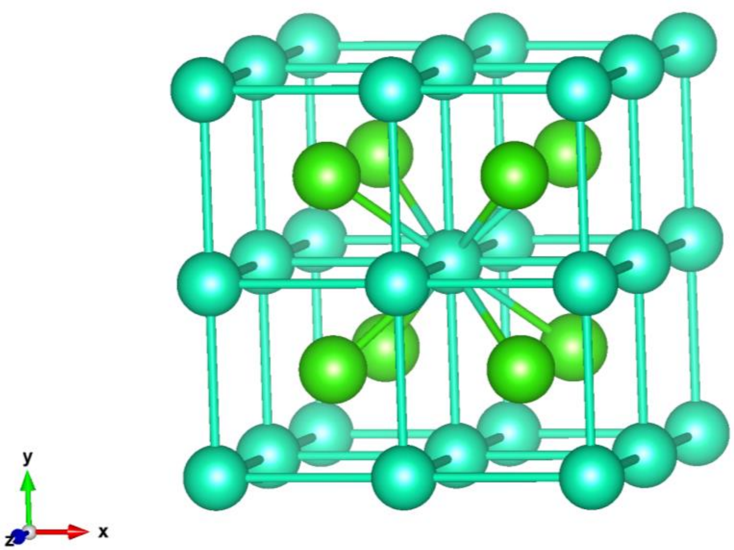
The Sodium Chloride structure



- $rNa^+/ rCl^- = 0.525$.
- FCC type of arrangement.
- Na^+ in all the octahedral voids.
- Each Na^+ is surrounded by 6 Cl^- and in turn each Cl^- is also surrounded by 6 Na^+. The coordination of sodium chloride structure is described as 6:6.

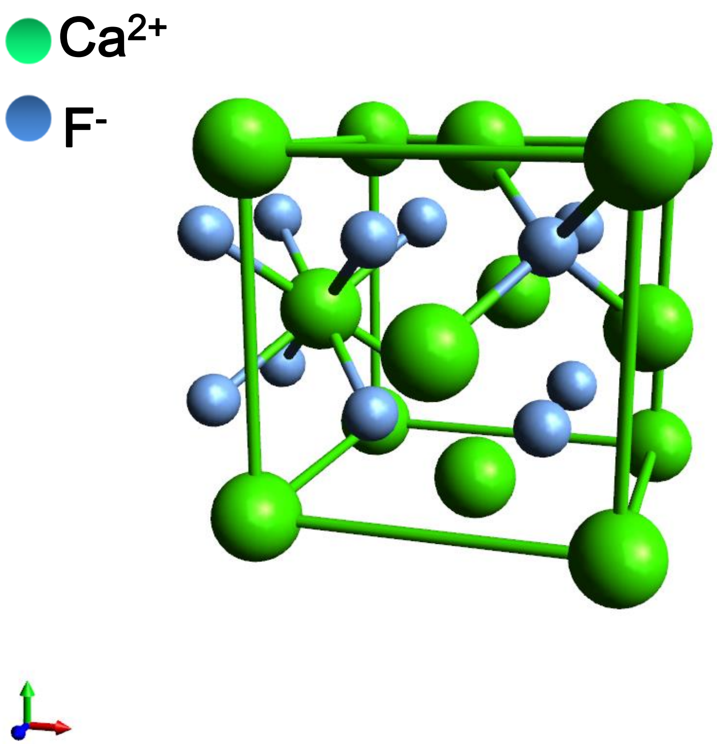
Cubic voids---the Cesium Chloride structure





- $rSc^+/ rCl^- = 1.08$.
- Simple cubic arrangement.
- Cl^- in all the cubic voids.
- Each Sc^+ is surrounded by 8 Cl^- and in turn each Cl^- is also surrounded by 8 Sc^+. The coordination of sodium chloride structure is described as 8:8.

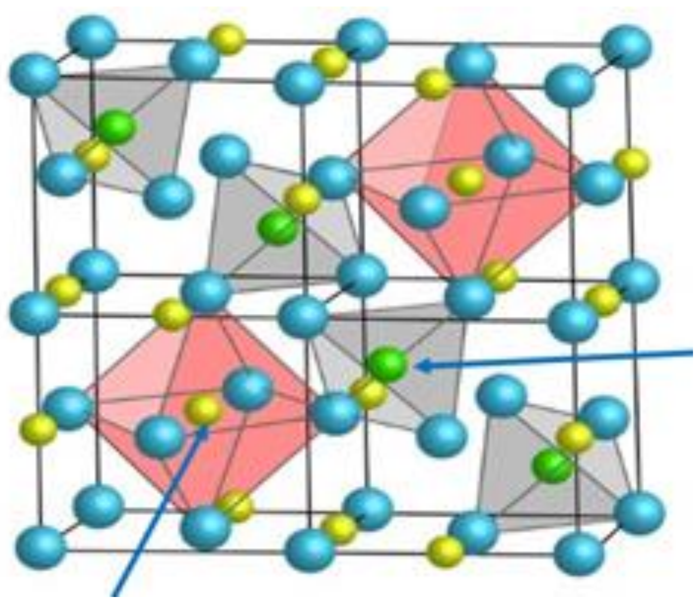
The CaF_2 (fluorite) structure



- $rCa^{2+}/ rF^{-} = 0.75$.
- FCC type of arrangement.
- F^- in all tetrahedral voids.
- Each Ca^{2+} is surrounded by 8 F^- and in turn each F^- is surrounded by 4 Ca^{2+} . The coordination of fluorite structure is described as 8:4.

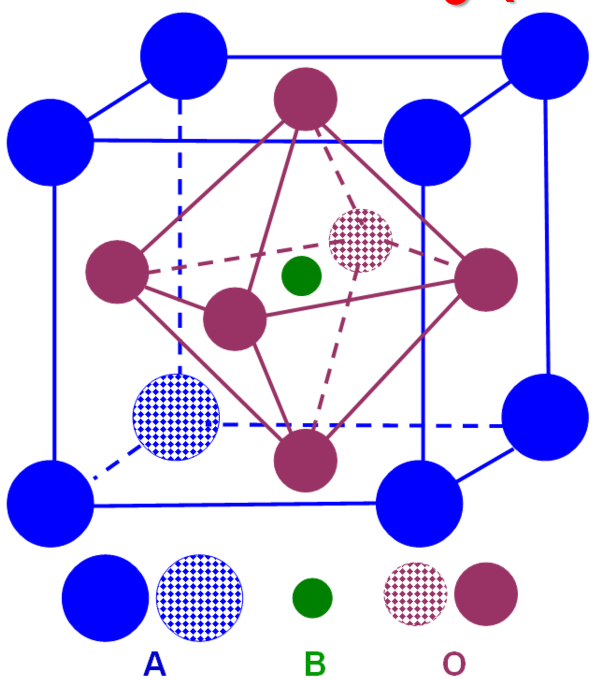
Metal Oxide Lattices

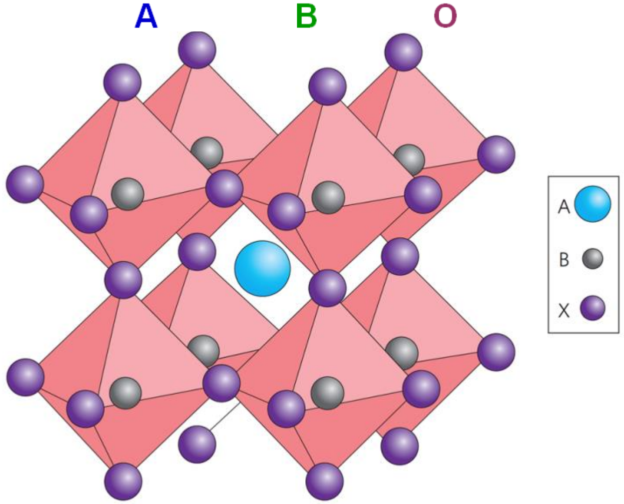
Normal spinel lattice AB_2O_4



- A FCC array of oxide ions (blue).
- A^{2+} ions occupy 1/8 of the tetrahedral holes (green).
- B^{3+} ions occupy 1/2 of the octahedral holes (yellow).

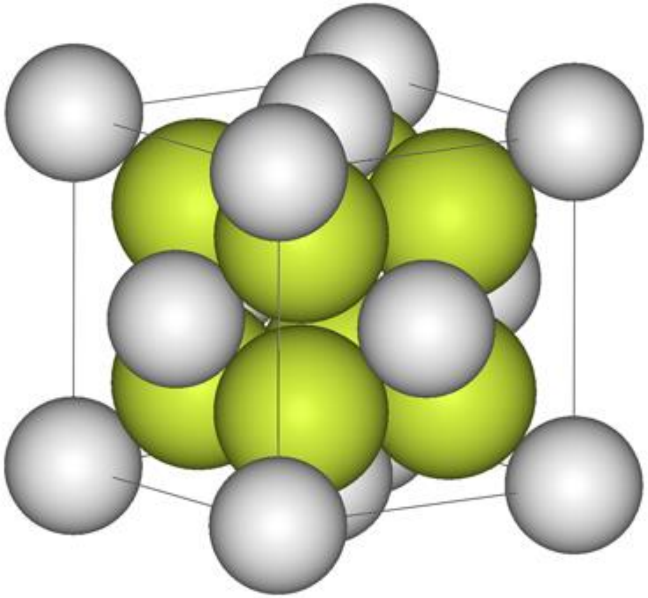
Perovskite ABO_3 (A is a cation of larger size than B)





- A FCC arrangement of both oxide and the larger cation.
- The smaller cation occupies the octahedral hole at the position (½, ½, ½).
- High temperature super-conductors

What is the empirical formula of the unit cell shown below? (Explain your work) (10 points)
The grey balls are Ca^{2+} and the yellow balls are F^{-}.



There are 8 F ions (yellow green) inside and
(8 x 1/8) = 1 Ca ions (grey) at the vertices plus
(6 x ½) = 3 Ca at the face centers
so the ratio is 4 Ca” 8 F which is CaF_2 (or the mineral Fluorite)

2.3.4 Crystal symmetry and space groups

The 32 point groups

The point groups are made up from point symmetry operations and their combinations. A point group is defined as a group of point symmetry operations whose operation leaves at least one point unmoved. Any operation involving translation is excluded.

Symmetry operation

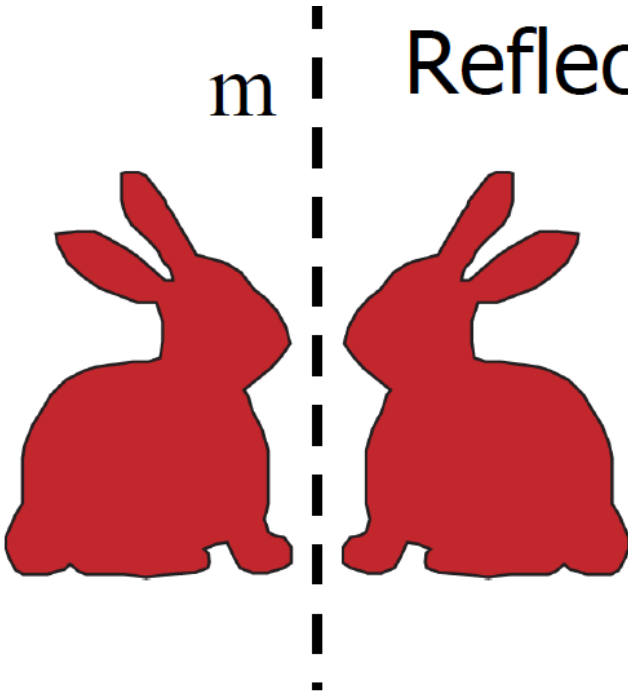
A Symmetry operation is an operation that can be performed either physically or imaginatively that results in no change in the appearance of an object.

In crystallography, a crystallographic point group is a set of symmetry operations, like rotations or reflections, that leave a central point fixed while moving other directions and faces of the crystal to the positions of features of the same kind.

Point Symmetry Elements
Symmetry elements fixed at a point

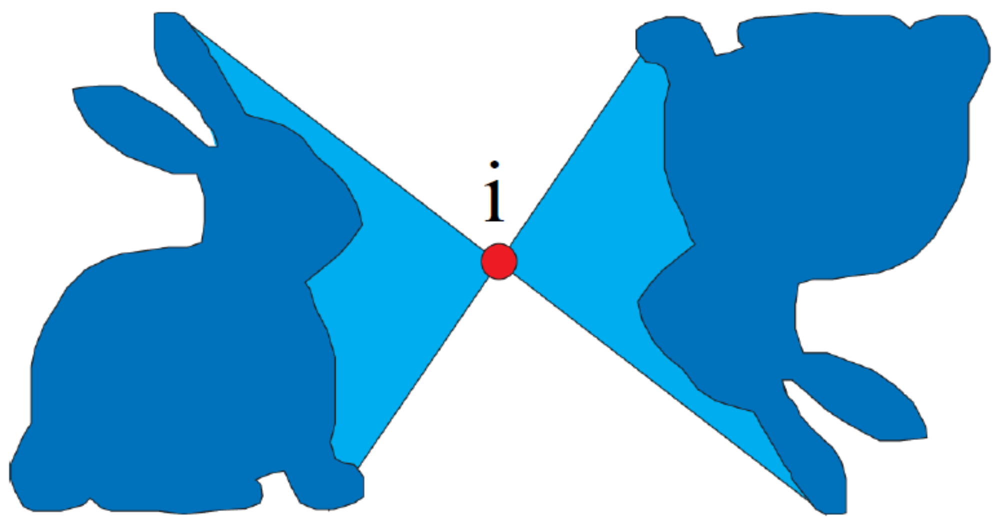
1. **center of symmetry** (or inversion):	point	$\bar{1}$		inversion
2. **rotation** (or proper) **axis**	: line	**n**		
3. **mirror**	: plane	*m*		reflection
4. **rotation–inversion axis**	: line	$\bar{n}$		
5. **identity**	: no element	**1**		

Reflection versus Inversion



Let us look at this purely in 2 dimensions.

Reflection of a 2-dimensional object occurs across a plane (m)



After inversion everything is an equal and opposite distance through a single point *i*.

Results in congruent pairs.

Rotation Axis: n

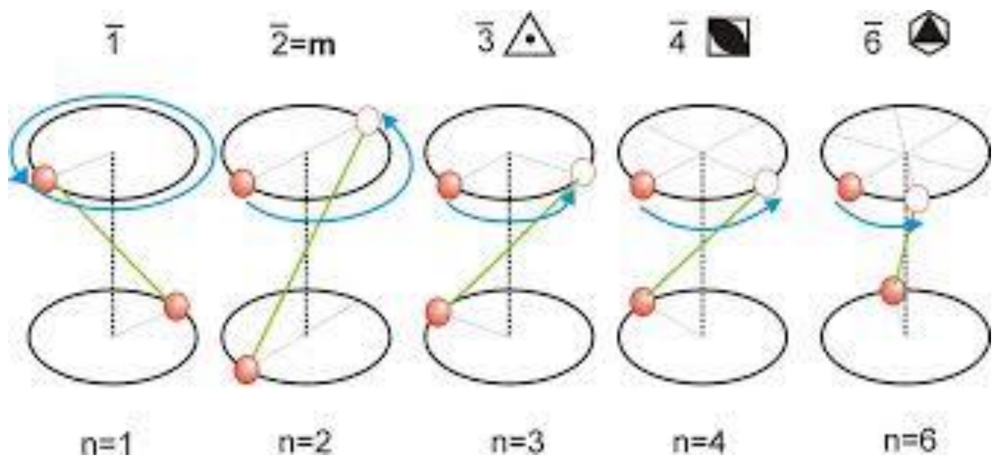
n is an integer which gives the degrees of rotation:
Each time stopping at an identical appearance.

n is the number of times rotated,
n is the foldness of the rotation axis.

Only 2, 3, 4, and 6-fold axes allowed in crystal symmetry

Rotation-Inversion $\bar{n}$

Rotation followed by **inversion**



Representation of Symmetry

point symmetry often represented symbolically in the form of points on a circle (projection of a sphere):

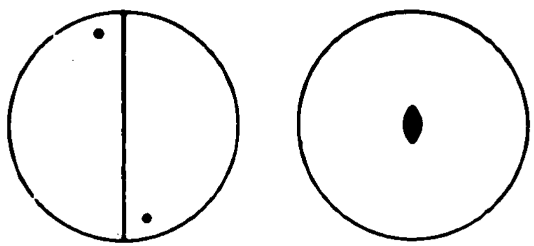
a point above plane is a **filled** circle: ●

a point below plane is an **open** circle: ○

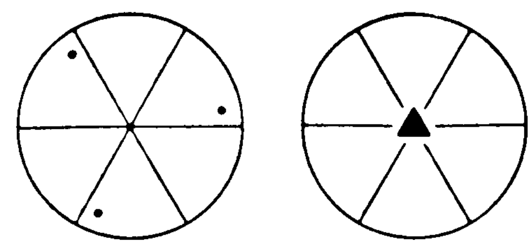
two points directly on top of each other: ◉

Cyclic Point Groups

Single operation

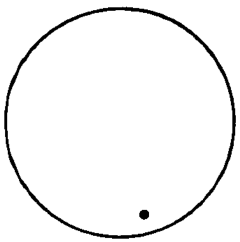


$2(C_2)$

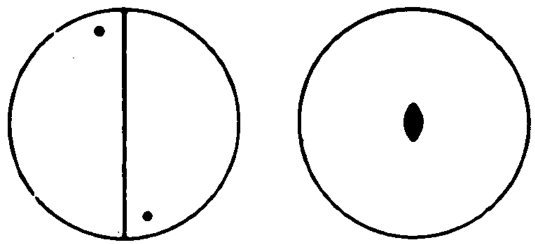


$3(C_3)$

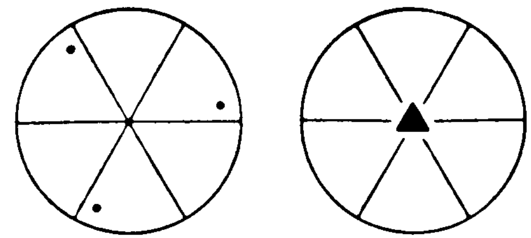
◆ Cyclic Point Groups



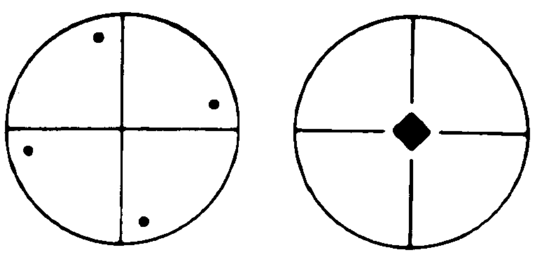
$1(C_1)$



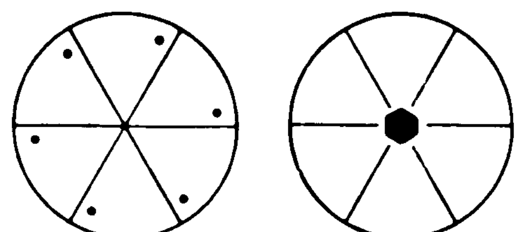
$2(C_2)$



$3(C_3)$



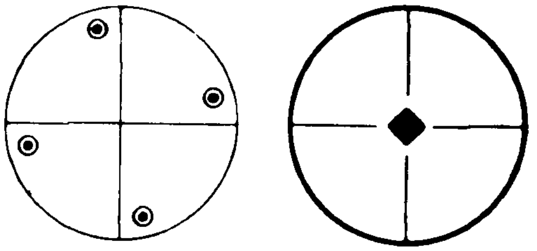
$4(C_4)$



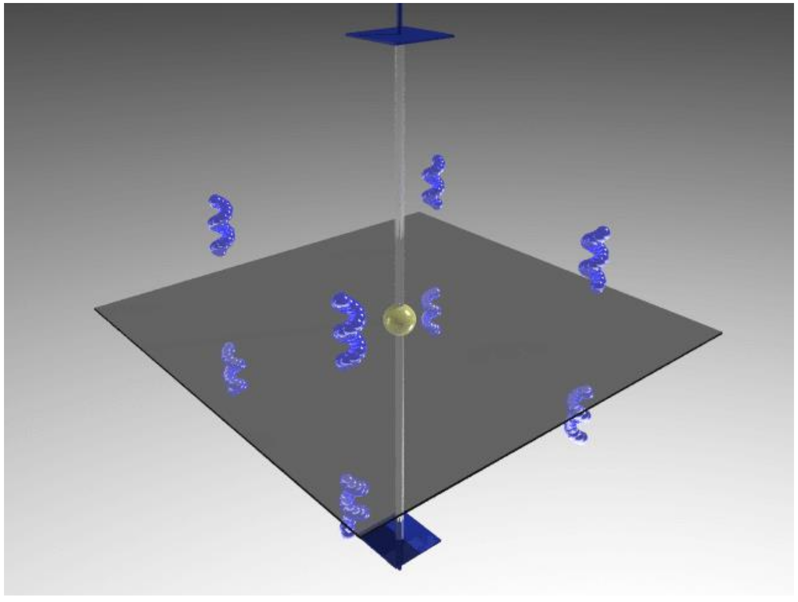
$6(C_6)$

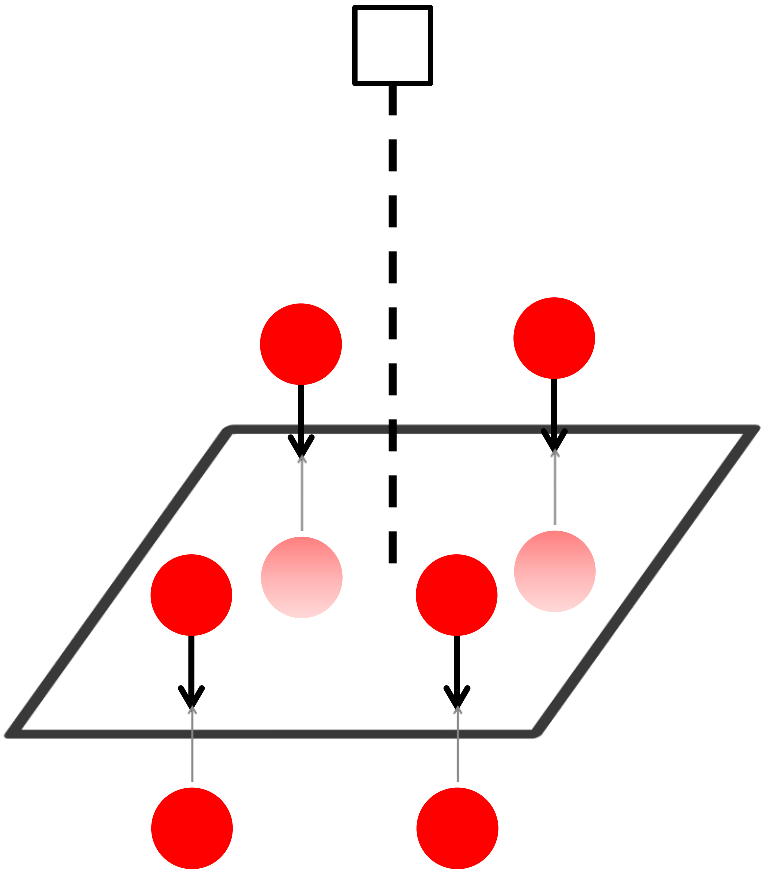
Sum = 5

◆ Cyclic + Horizontal Mirror Groups

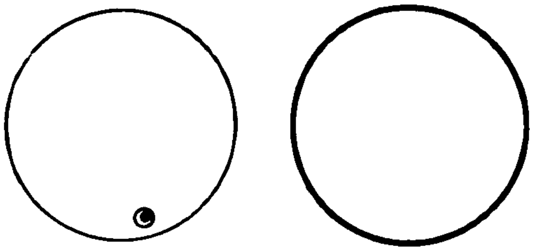


$$\frac{4}{m}(C_{4h})$$

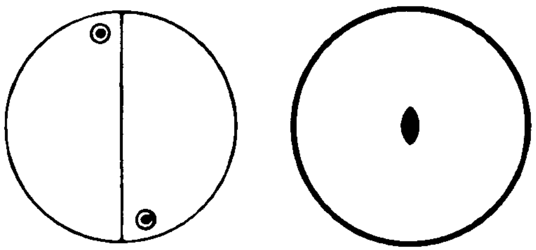




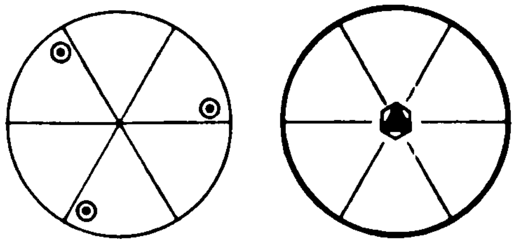
◆ Cyclic + Horizontal Mirror Groups



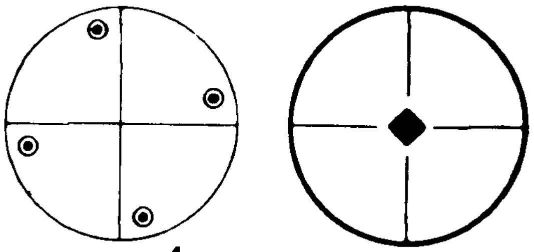
$m(C_{1h})$



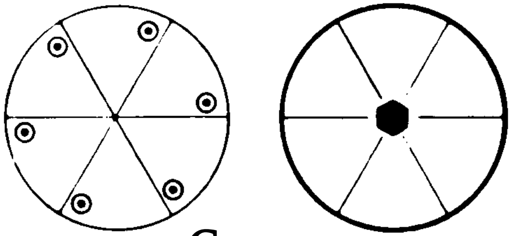
$\frac{2}{m}(C_{2h})$



$\frac{3}{m}(C_{3h})$



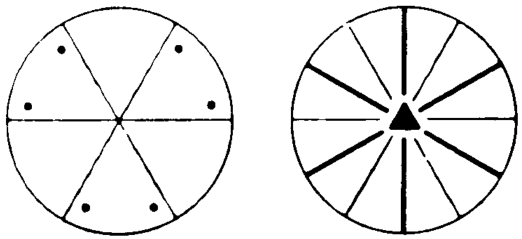
$\frac{4}{m}(C_{4h})$



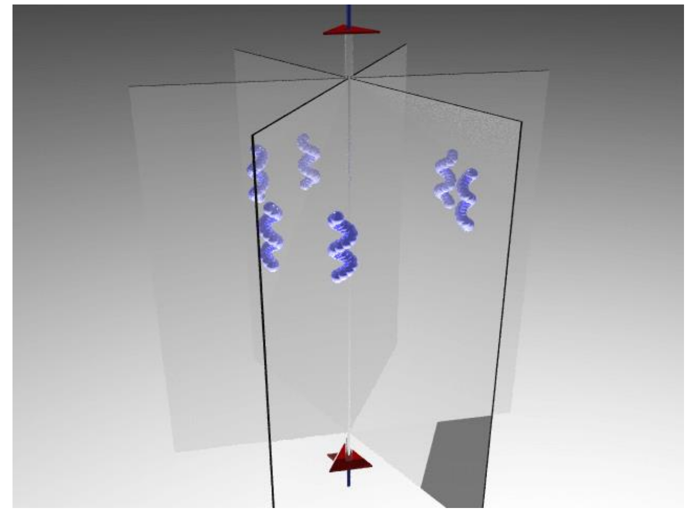
$\frac{6}{m}(C_{6h})$

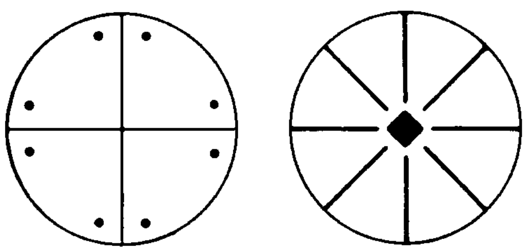
Sum = 5+5=10

◆ Cyclic + Vertical Mirror Groups

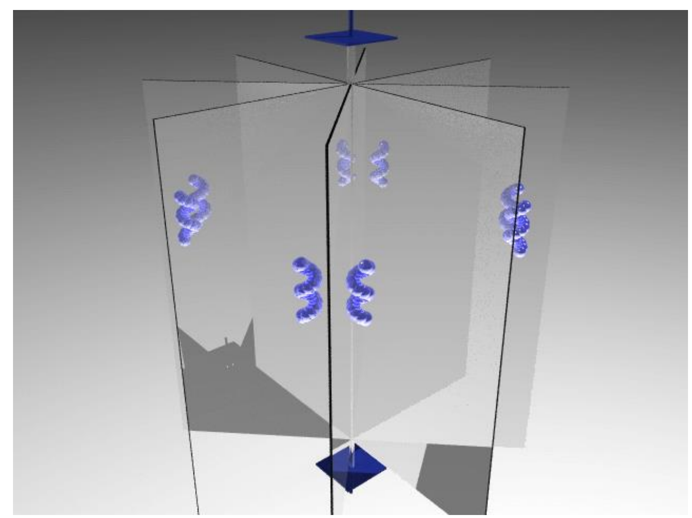


$3m(C_{3v})$

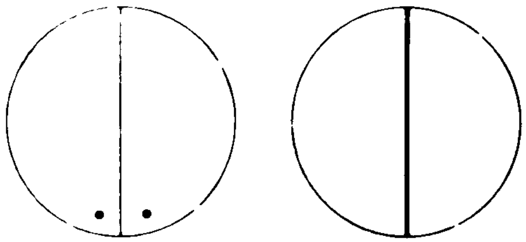




$4mm(C_{4v})$

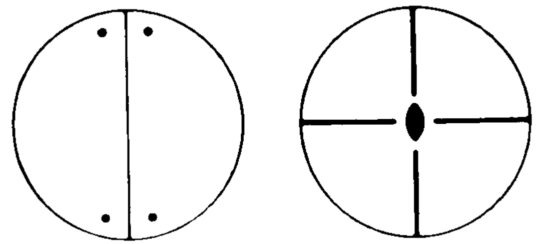


◆ Cyclic + Vertical Mirror Groups

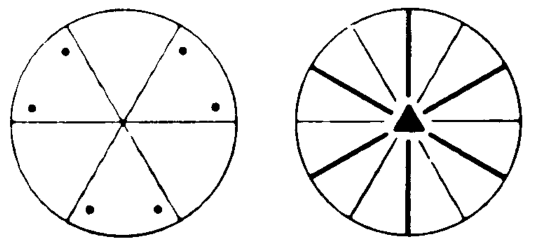


$\cancel{m(C_{1v})}$

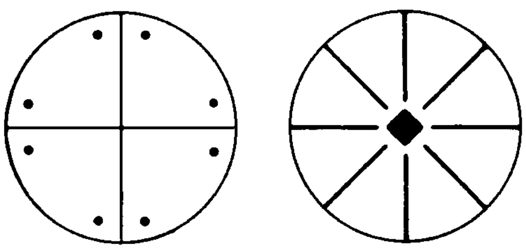
$m(C_{1h})$



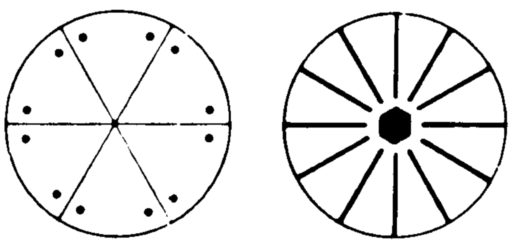
$mm2(C_{2v})$



$3m(C_{3v})$



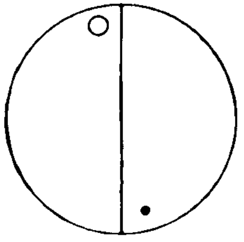
$4mm(C_{4v})$



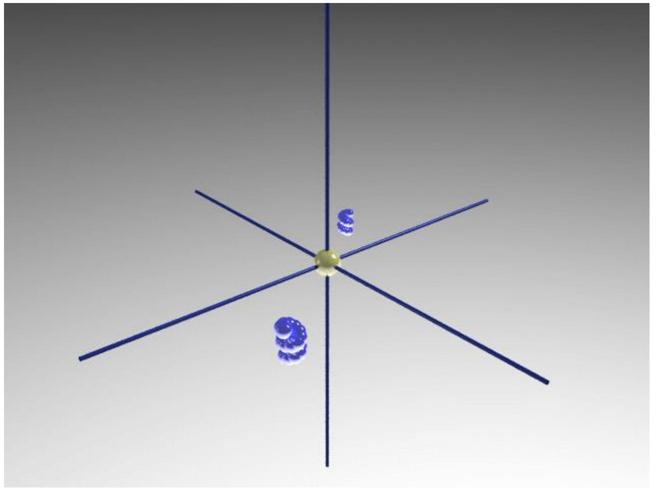
$6mm(C_{6v})$

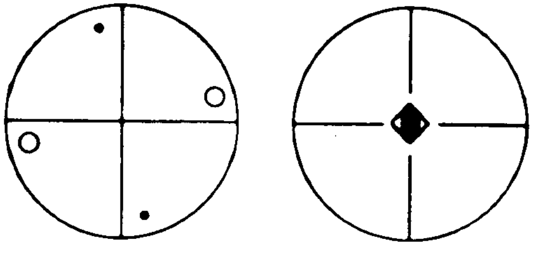
Sum = 10+4=14

◆ Rotoinversion Groups

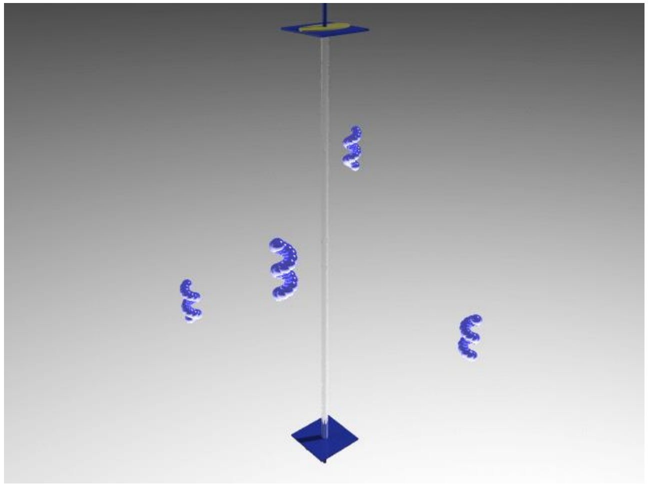


$\bar{1}(S_2)$

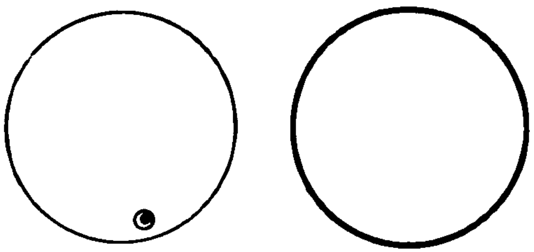




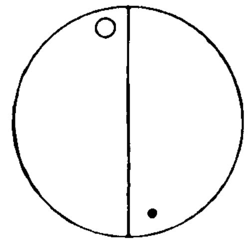
$\bar{4}(S_4)$



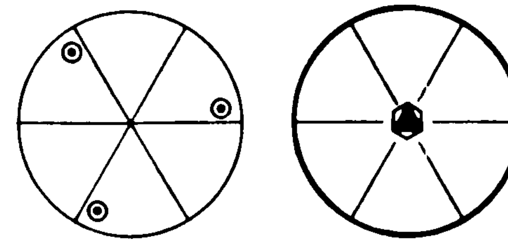
◆ Rotoinversion Groups



$\cancel{\bar{2}(S_1)}$

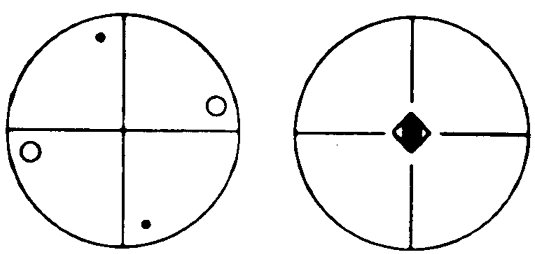


$\bar{1}(S_2)$

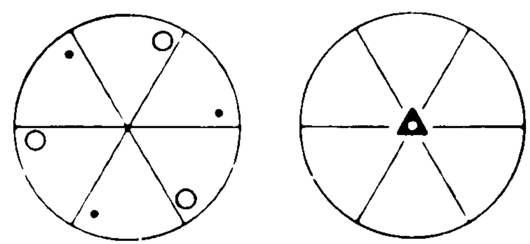


$\cancel{\bar{6}(S_3)}$

$\frac{3}{m}(C_{3h})$



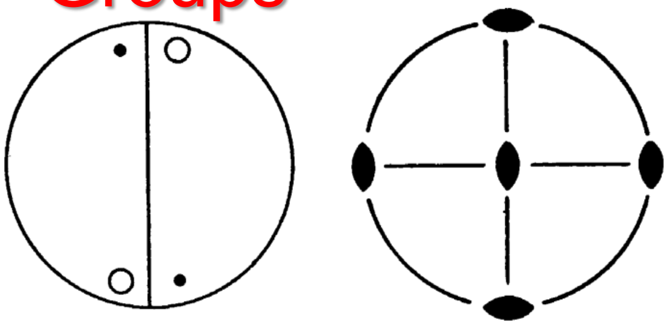
$\bar{4}(S_4)$



$\bar{3}(S_6)$

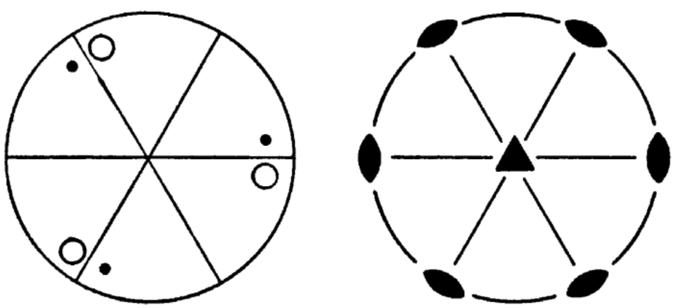
Sum = 14+3=17

◆ Rotations + Perpendicular 2-folds Dihedral (D_n) Groups



Motif is generated via
a 2-fold axis,
2 set 2-fold axes,
The notation for this combination is 422

$$222(D_2)$$

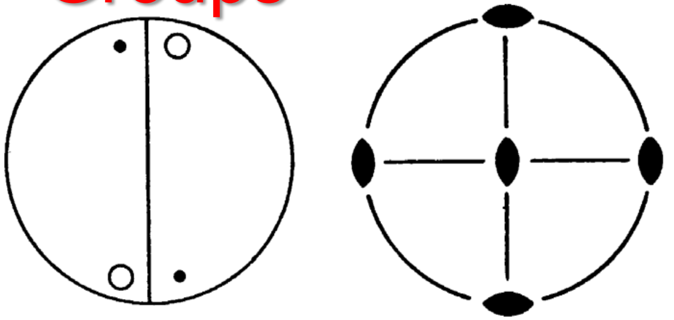


Motif is generated via
a 3-fold axis,
3 set 2-fold axes,
The notation for this combination is 32

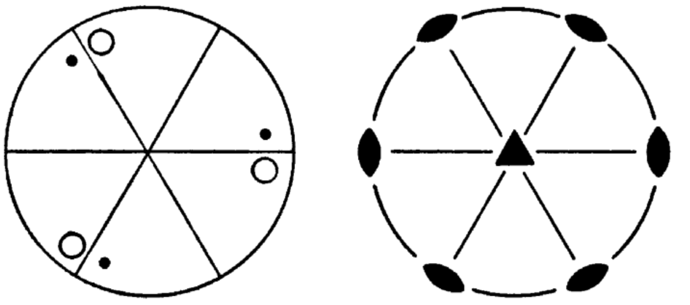
$$32(D_3)$$

http://symmetry.otterbein.edu/gallery/index.html

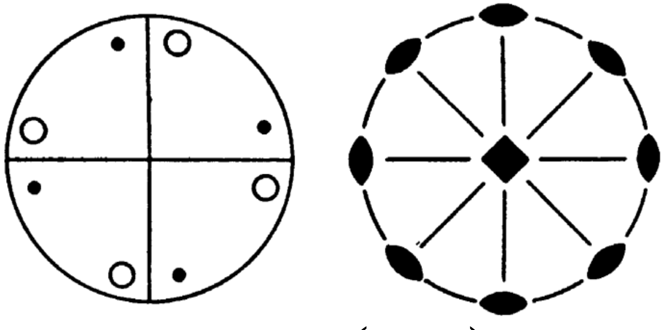
◆ Rotations + Perpendicular 2-folds Dihedral (D_n) Groups



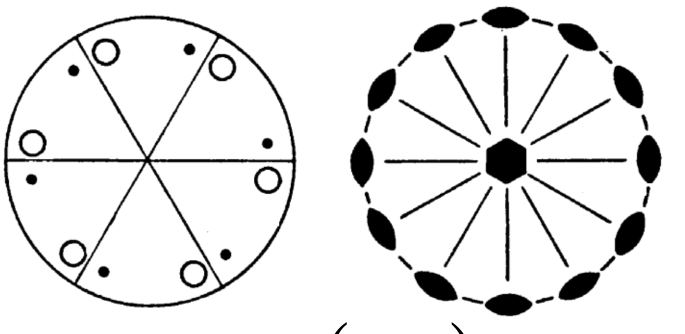
$222(D_2)$



$32(D_3)$



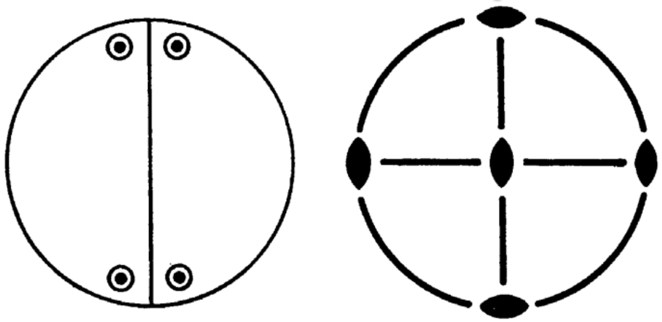
$422(D_4)$



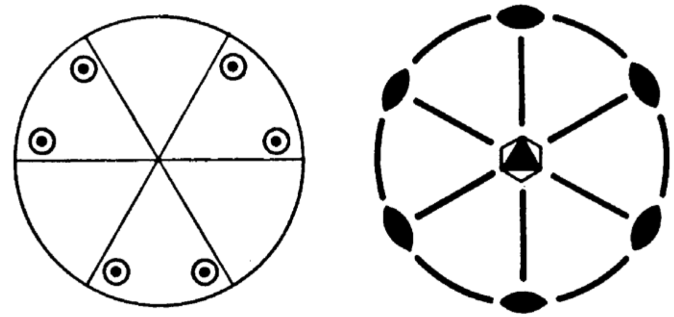
$622(D_6)$

Sum = 17+4=21

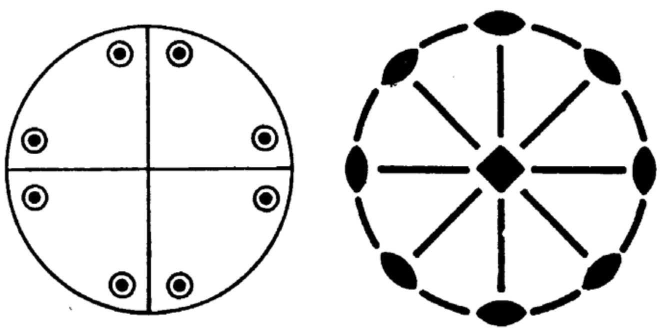
◆ Dihedral Groups + σ_h



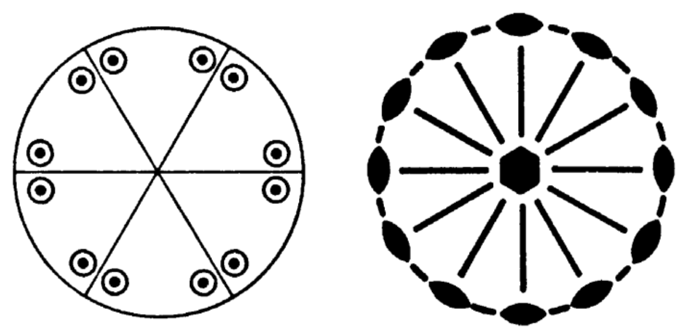
$$mmm(D_{2h})$$



$$\bar{6}m2(D_{3h})$$



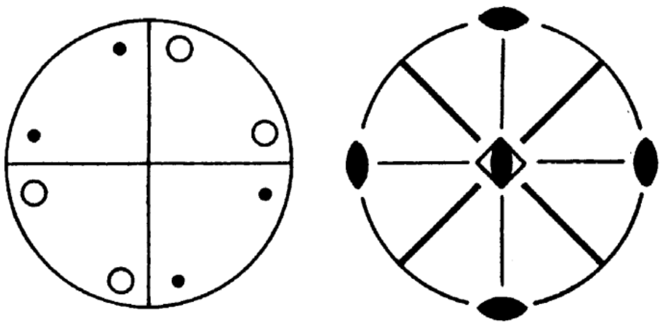
$$\frac{4}{m}mm(D_{4h})$$



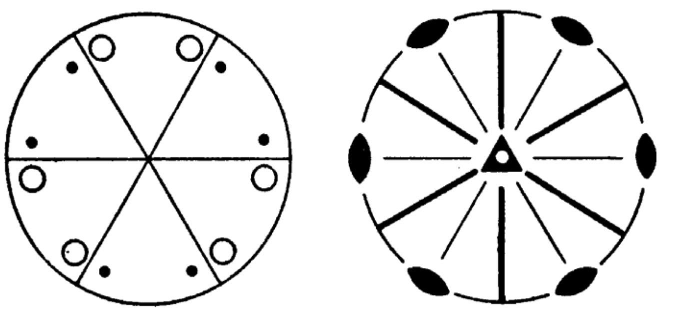
$$\frac{6}{m}mm(D_{6h})$$

Sum = 21+4=25

Dihedral Groups + σ_d



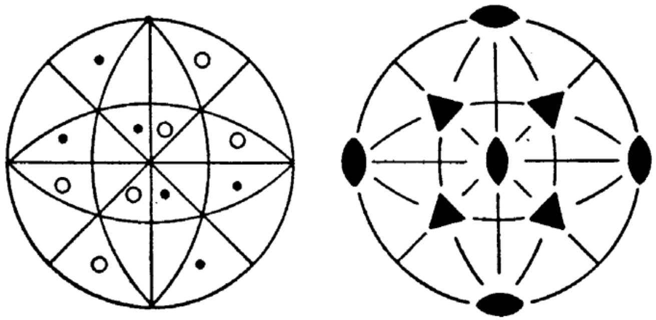
$\bar{4}2m(D_{2d})$



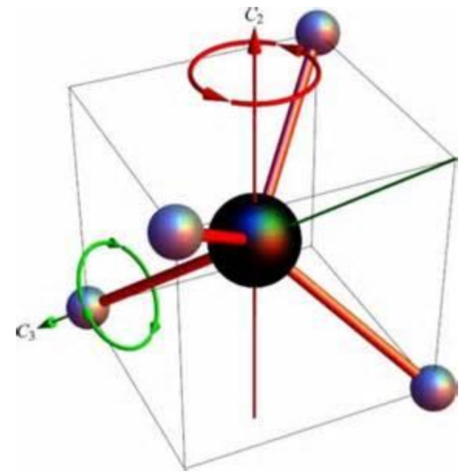
$\bar{3}m(D_{3d})$

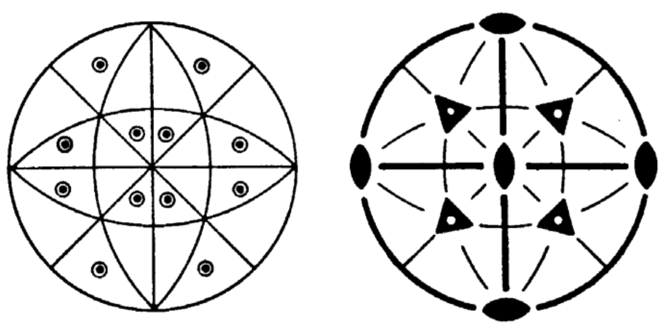
Sum = 25+2=27

◆ T (tetrahedral) Groups

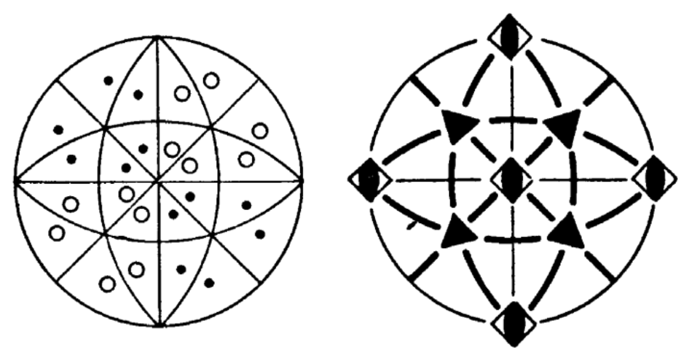


$$23(T)$$





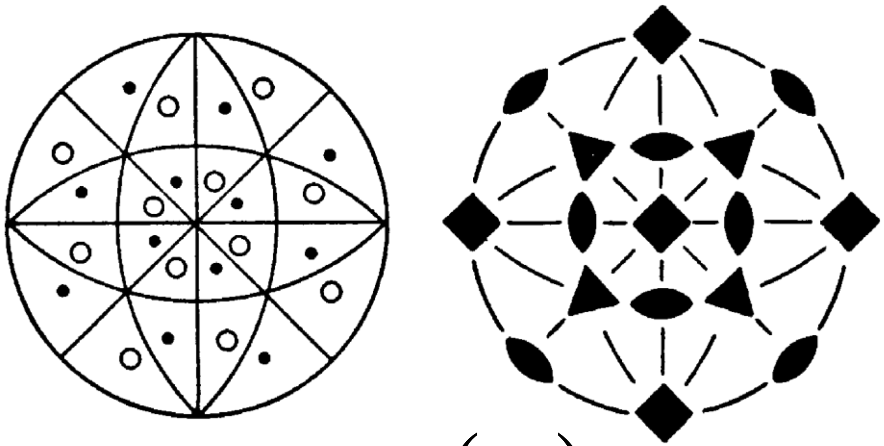
$$m3(T_h)$$



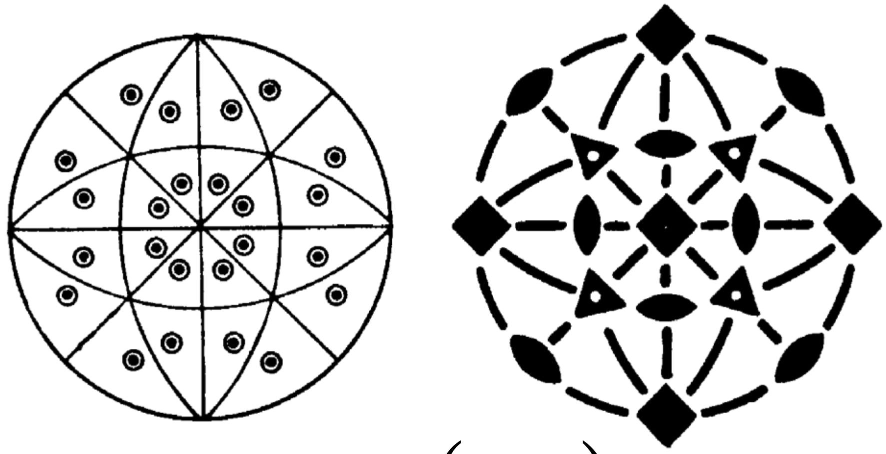
$$\bar{4}3m(T_d)$$

Sum = 27+3=30

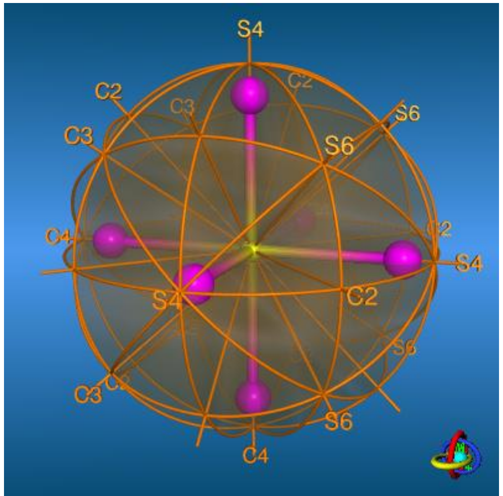
◆ O (octahedral) Groups



$432(O)$



$m3m(O_h)$



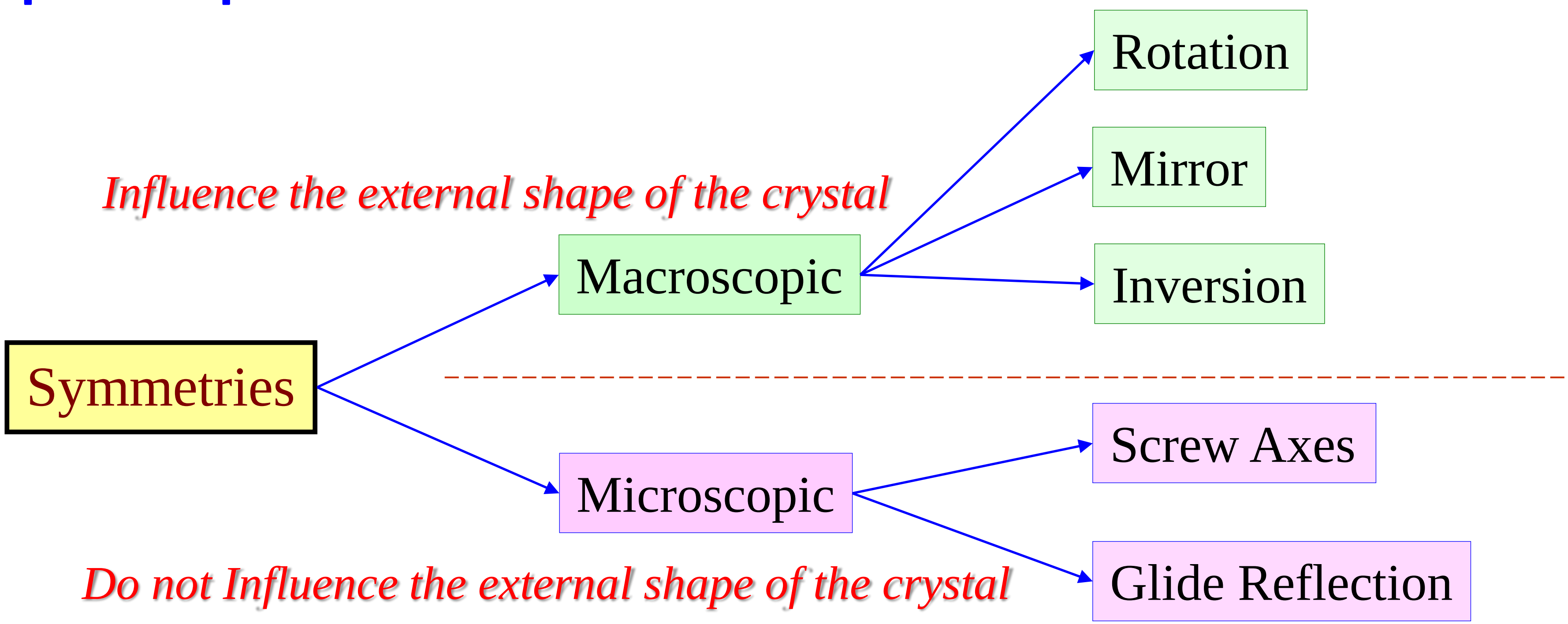
Sum = 30+2=32

32 point groups compatible with 7 crystal systems

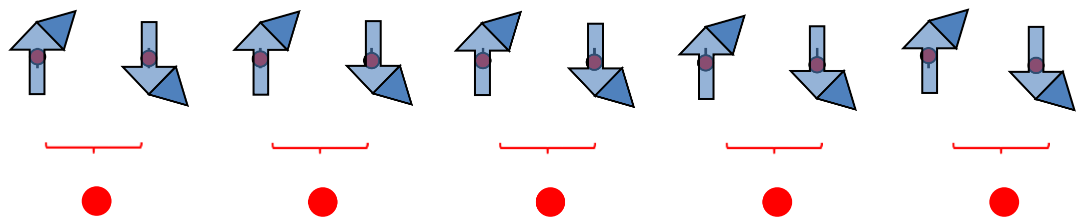
System	Class: International	Class: Schönfließ
triclinic	1	C_1
	$\bar{1}$	C_i
monoclinic	m	C_s
	2	C_2
	$2/m$	C_{2h}
orthorhombic	$2mm$	C_{2v}
	222	D_2
	mmm	D_{2h}
tetragonal	4	C_4
	$\bar{4}$	S_4
	$4/m$	C_{4h}
	$4mm$	C_{4v}
	$\bar{4}2m$	D_{2d}
	422	D_4
	$4/mmm$	D_{4h}
trigonal (rhombohedral)	3	C_3
	$\bar{3}$	S_6
	$3m$	C_{3v}
	32	D_3
	$\bar{3}m$	D_{3d}
hexagonal	$\bar{6}$	C_{3h}
	6	C_6
	$6/m$	C_{6h}
	$\bar{6}m2$	D_{3h}
	$6mm$	C_{6v}
	622	D_6
	$6/mmm$	D_{6h}
cubic	23	T
	$m3$	T_h
	$\bar{4}3m$	T_d
	432	O
	$m3m$	O_h

http://www.metafysica.nl/derivation_32.html

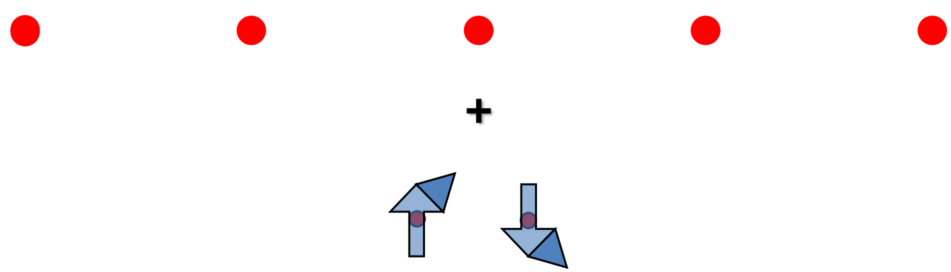
Space operation --- Micro-translations



The combination of all symmetry elements including the translation symmetry elements yields only 230 combination. These combinations are called "**space groups**".



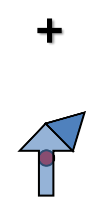
Crystal = Lattice + Motif =



OR

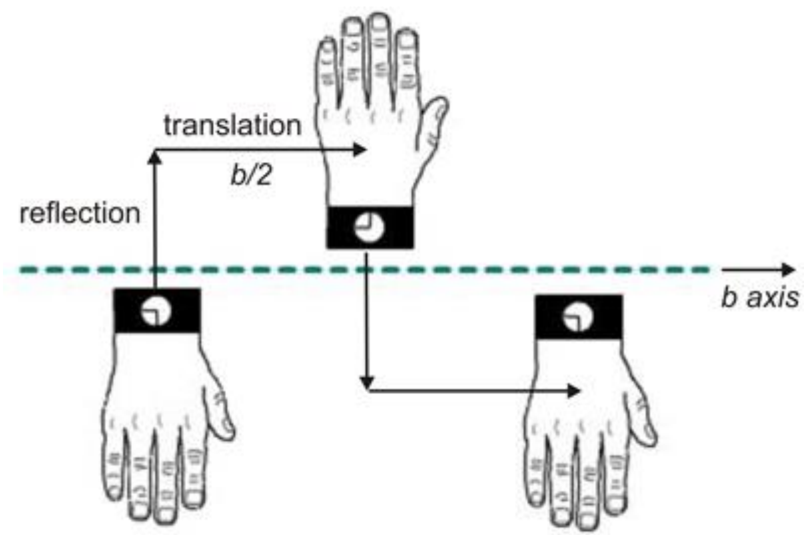
Glide reflection operation

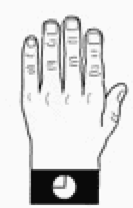
Crystal = unit cell + micro-translation=



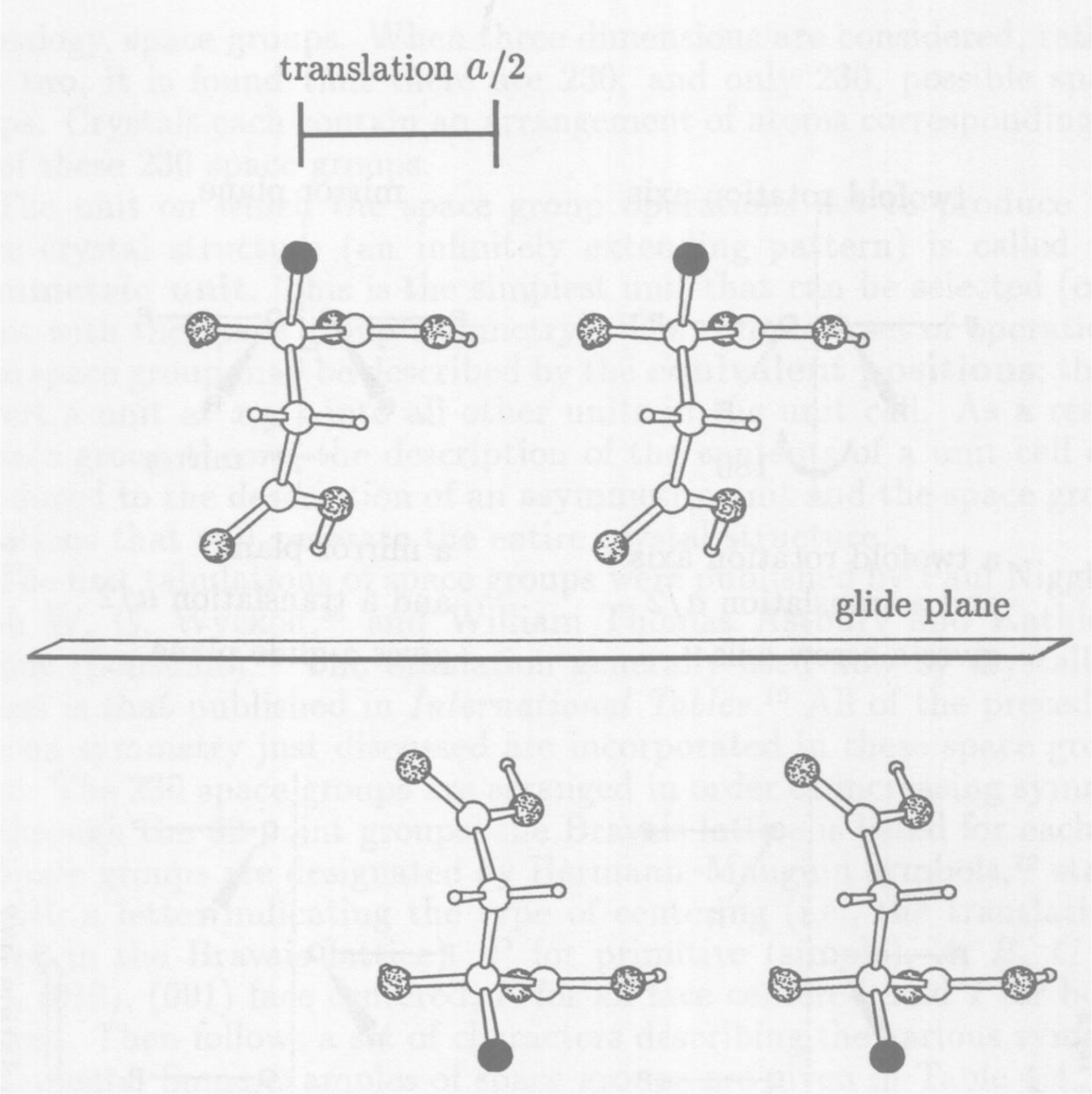
Glide Reflection = Mirror + Translation

- reflection across plane;
- followed by translation of **1/2** (usually) unit cell parallel to plane along **a**, **b**, **c**, **face diagonal (*n*)**, or **body diagonal (*d*)**





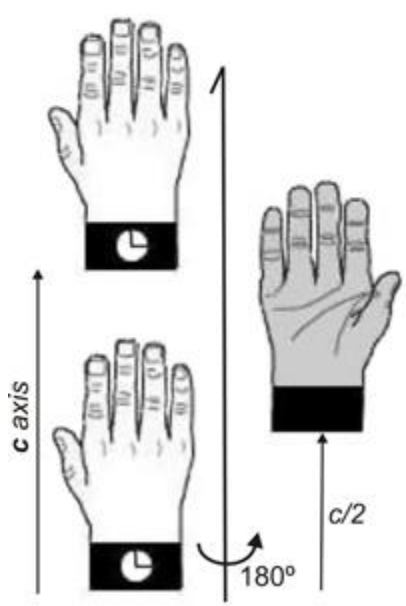
Solid-State Chemistry

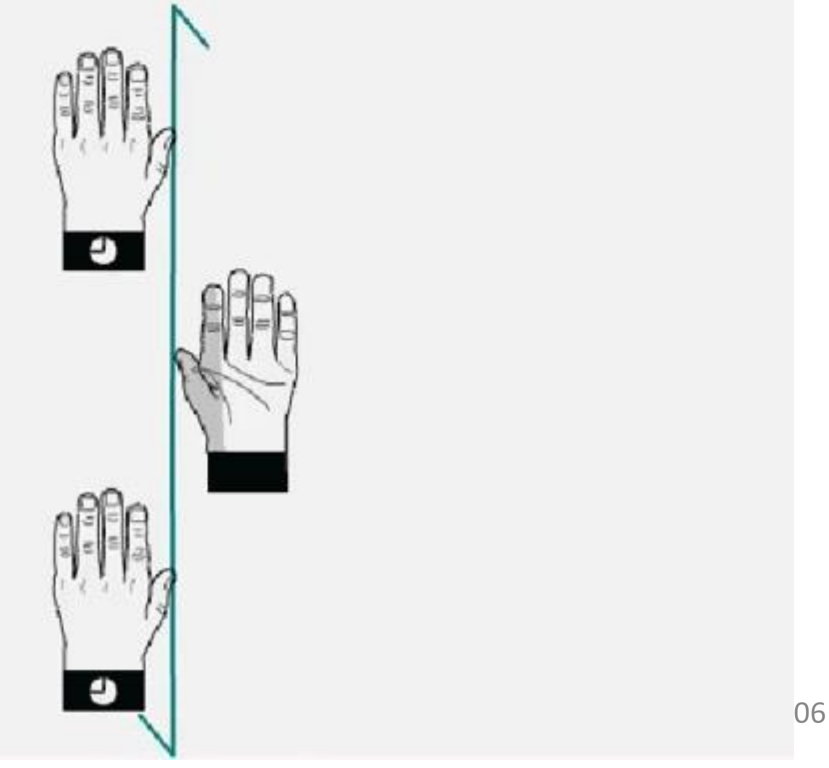


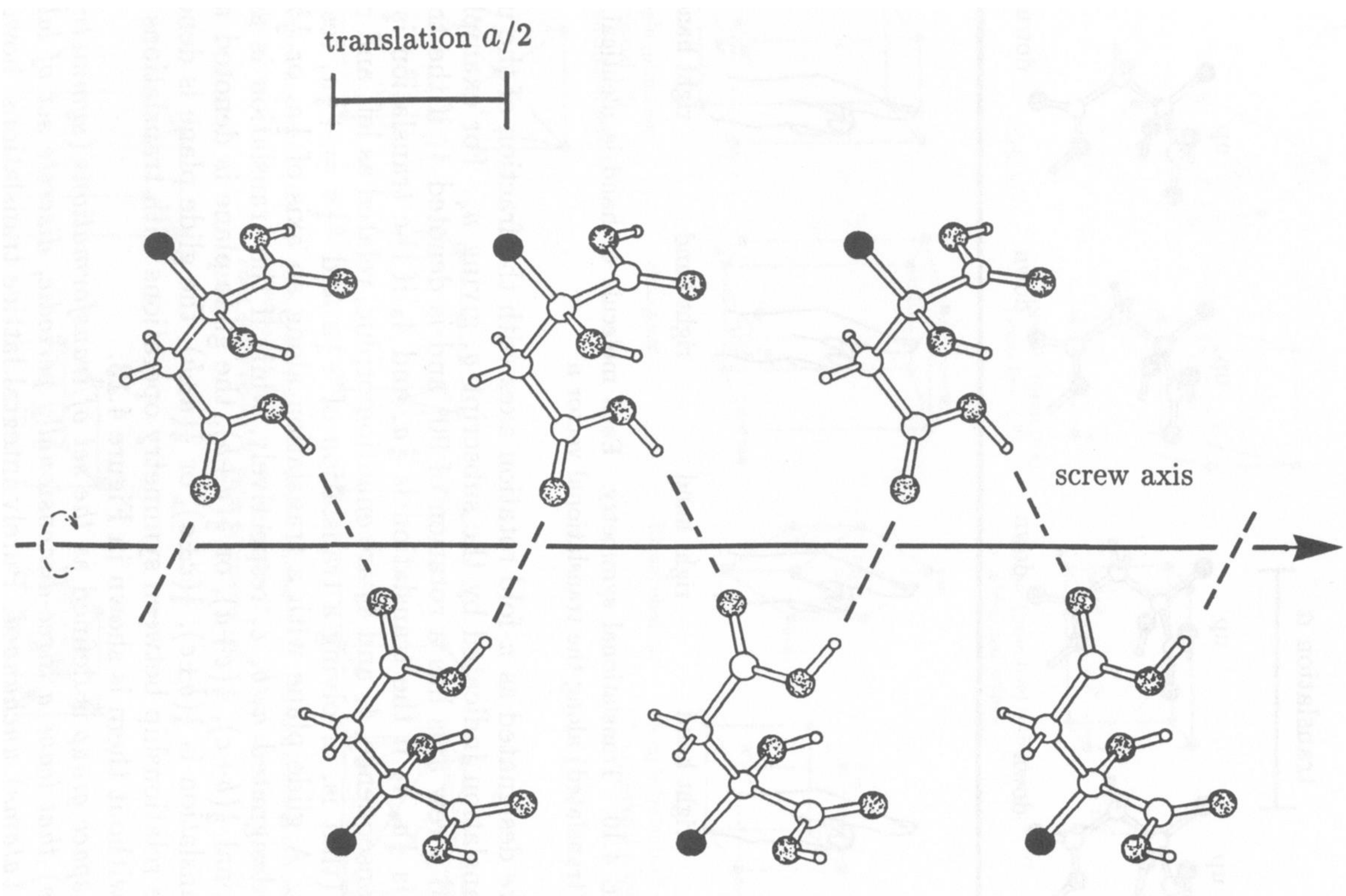
screw axis = rotation + translation:

The nomenclature n_x

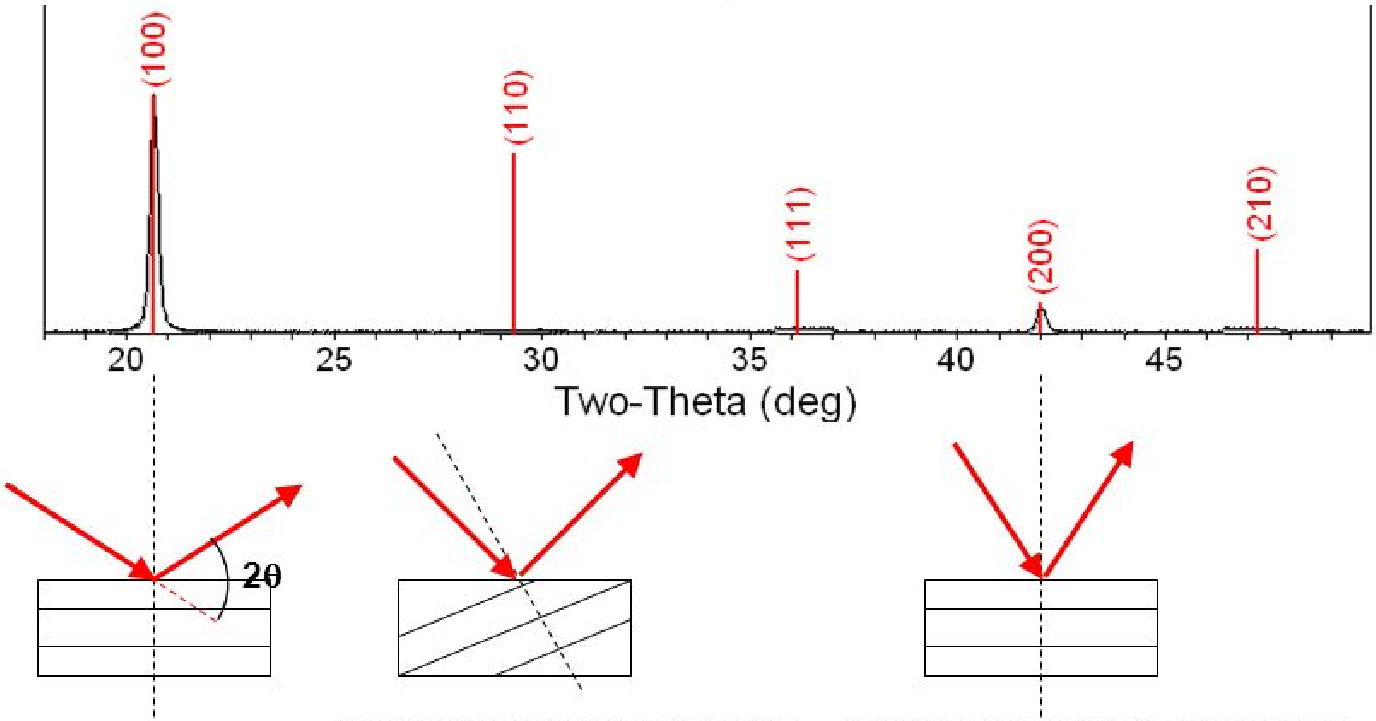
- Rotation by **360º/n**;
- Followed by a $\boldsymbol{r} \times \boldsymbol{x}/\boldsymbol{n}$ translation along one of the unit cell vectors, a, b or c.






translation $a/2$
screw axis

A single crystal specimen in a Bragg-Brentano diffractometer would produce only one family of peaks in the diffraction pattern.

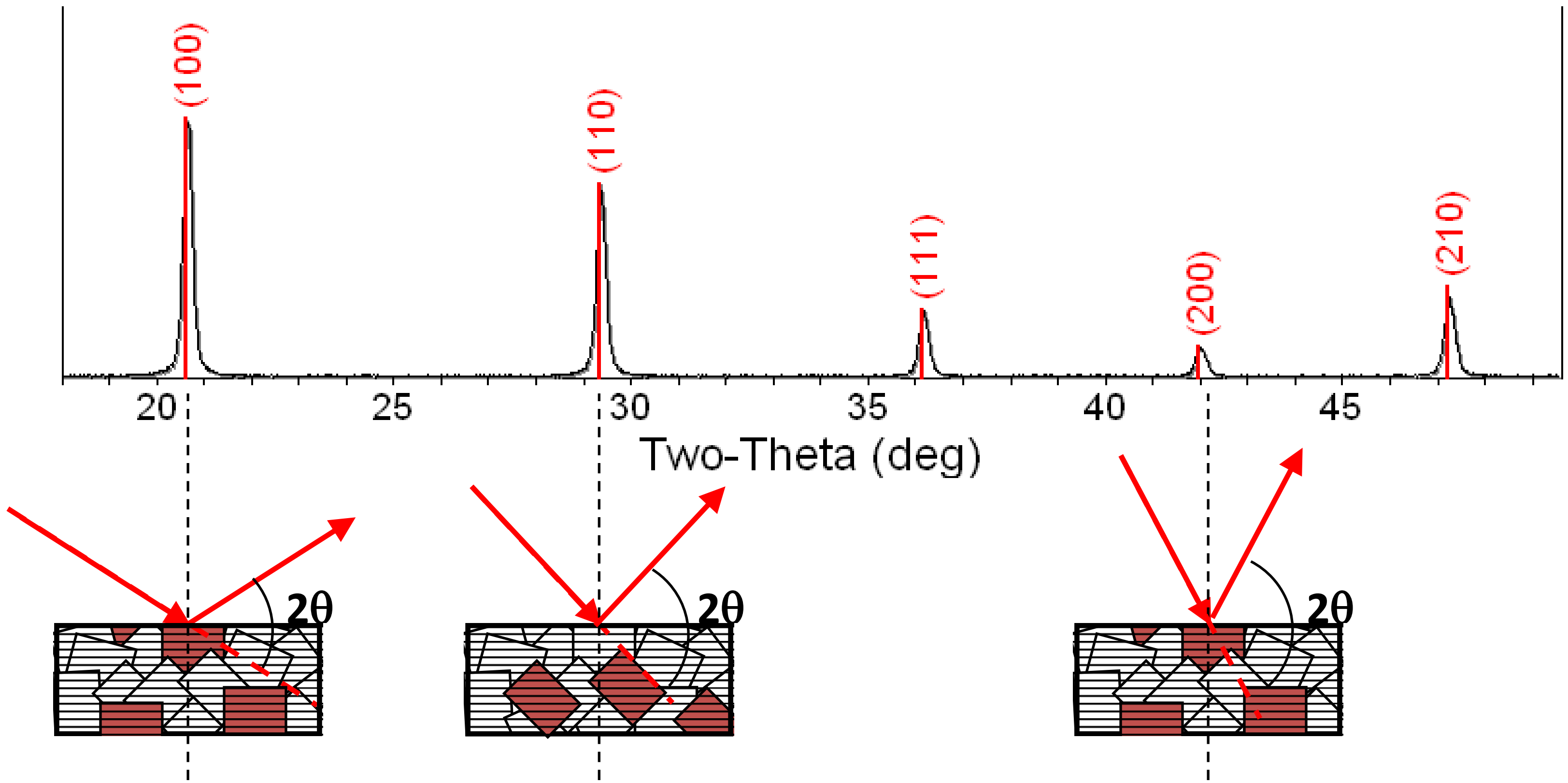


At 20.6 °2θ, Bragg's law fulfilled for the (100) planes, producing a diffraction peak.

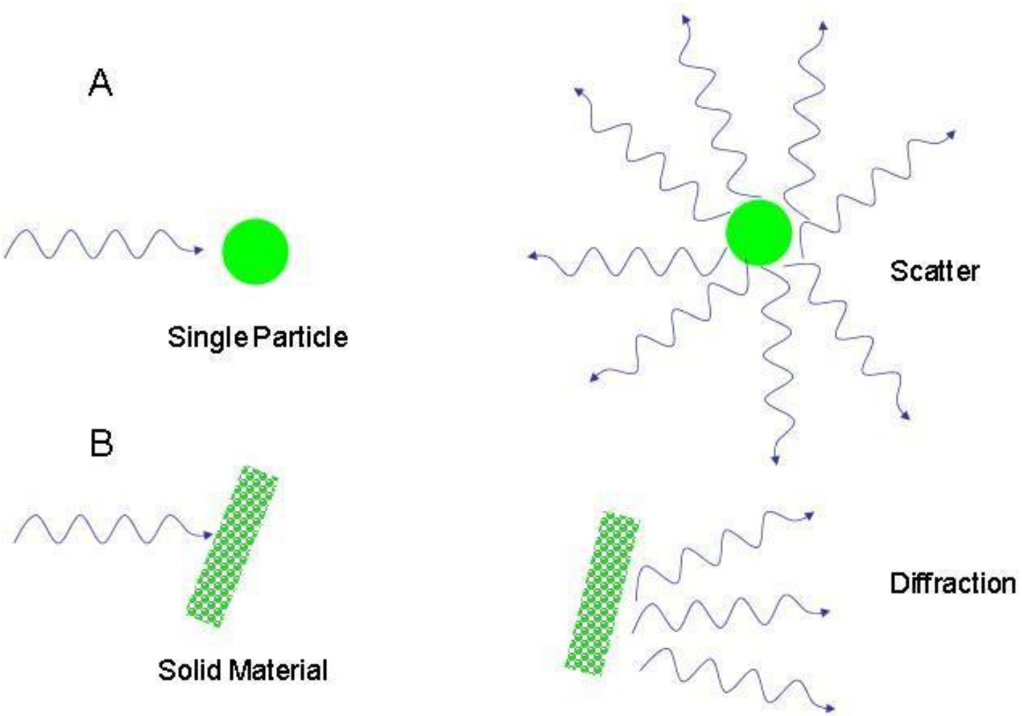
The (110) planes would diffract at 29.3 °2θ; however, they are not properly aligned to produce a diffraction peak (the perpendicular to those planes does not bisect the incident and diffracted beams). Only background is observed.

The (200) planes are parallel to the (100) planes. Therefore, they also diffract for this crystal. Since d_{200} is ½ d_{100}, they appear at 42 °2θ.

A polycrystalline sample should contain thousands of crystallites. Therefore, all possible diffraction peaks should be observed.



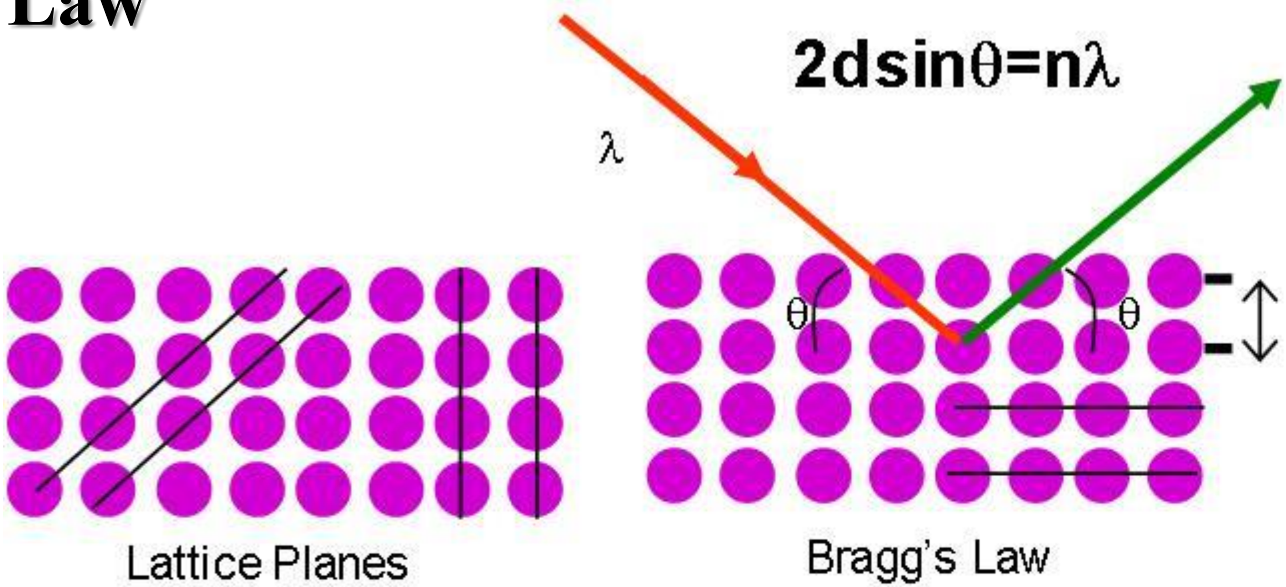
- For every set of planes, there will be a small percentage of crystallites that are properly oriented to diffract (the plane perpendicular bisects the incident and diffracted beams).
- Basic assumptions of powder diffraction are that for every set of planes there is an equal number of crystallites that will diffract and that there is a statistically relevant number of crystallites, not just one or two.



A. When X-rays interact with a single particle, it scatters the incident beam uniformly in **all** directions.

B. When X-rays interact with a solid material the scattered beams can add together in a few directions and reinforce each other to yield diffraction. The regularity of the material is responsible for the diffraction of the beams.

Bragg's Law



$$2d\sin\theta = n\lambda$$

λ = wavelength of the x-ray

θ = scattering angle

n = integer representing the order of the diffraction peak.

d = inter-plane distance of (i.e atoms, ions, molecules)

Application

In X-ray diffraction (XRD) the interplanar spacing (d-spacing) of a crystal is used for identification and characterization purposes. In this case, the wavelength of the incident X-ray is known and measurement is made of the incident angle (Θ) at which constructive interference (结构干涉) occurs. Solving Bragg's Equation gives the d-spacing between the crystal lattice planes of atoms that produce the constructive interference. A given unknown crystal is expected to have many rational planes of atoms in its structure; therefore, the collection of "reflections" of all the planes can be used to uniquely identify an unknown crystal. In general, crystals with high symmetry (e.g. isometric system) tend to have relatively few atomic planes, whereas crystals with low symmetry (in the triclinic or monoclinic systems) tend to have a large number of possible atomic planes in their structures.

2.3.5 Crystal imperfections

Crystals with a purity of 99.9999%
There are ca. 6×10^{16} impurities cm^{-3}

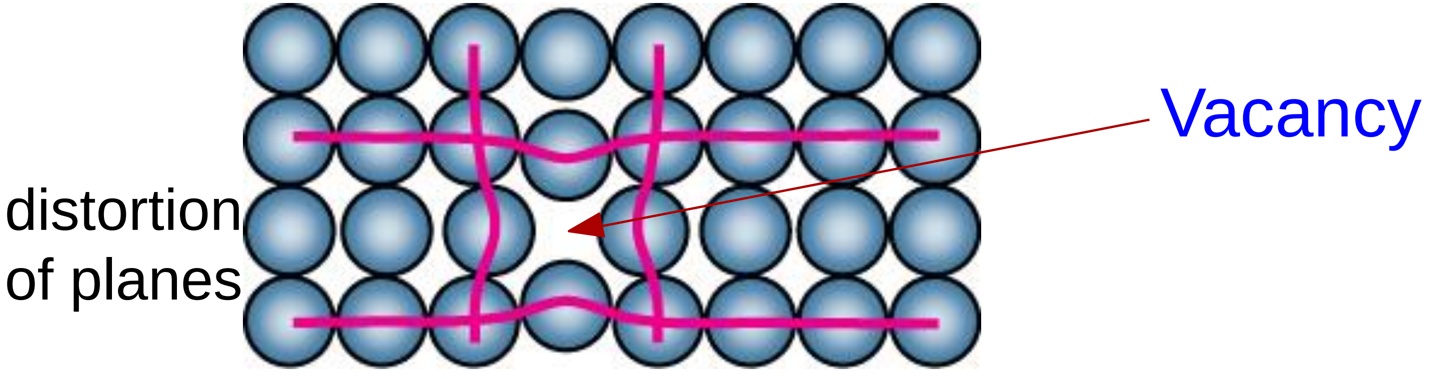
- Point defects
  - vacancies
  - interstitial/substitutional dopants
  - Schottky/Frenkel defects
- Linear defects
  - Edge and screw dislocations
- Planar defects
  - Grain boundary
- Bulk defects
  - pores, cracks

May be good
May be bad

Point defects

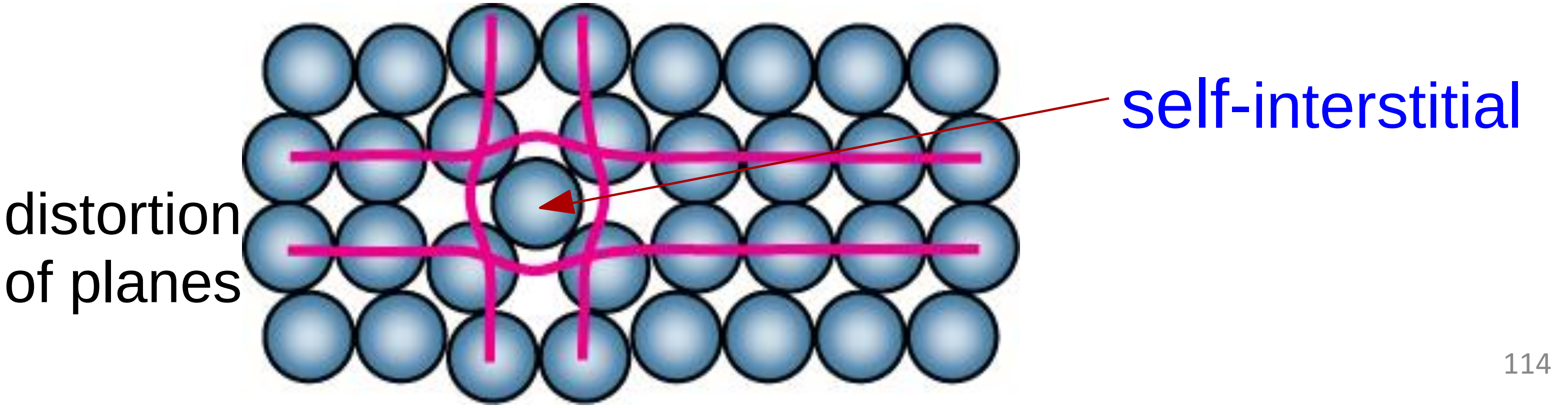
- Vacancies:

  -vacant atomic sites in a structure.



- Self-Interstitials:

  -"extra" atoms positioned between atomic sites.

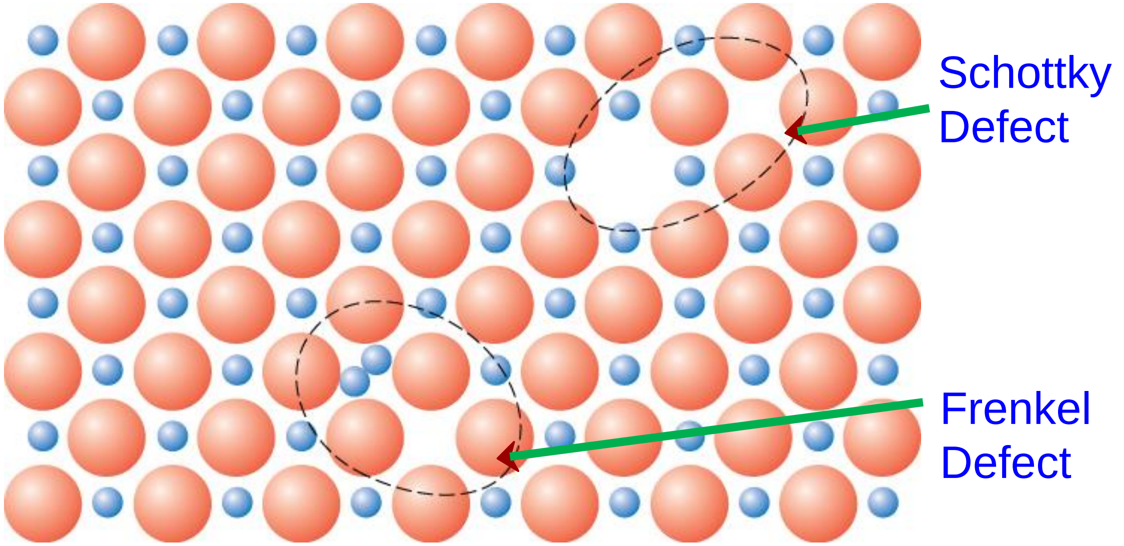


- **Frenkel Defect**

To maintain the charge neutrality, a cation vacancy-cation interstitial pair occur together. The cation leaves its normal position and moves to the interstitial site.

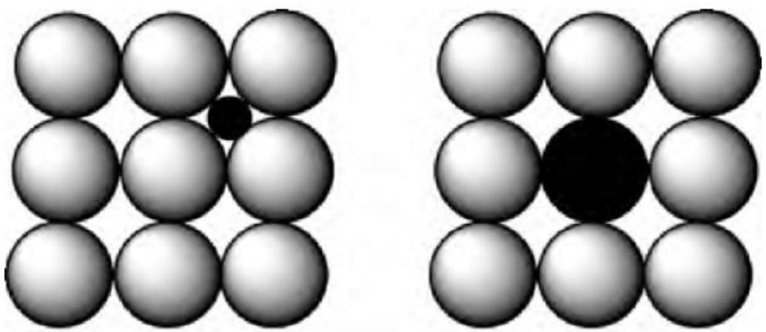
- **Schottky Defect**

To maintain the charge neutrality, remove 1 cation and 1 anion; this creates 2 vacancies.



- Interstitial/ substitutional solid solutions:

Interstitial dopant Substitutional dopant



a) b)

Cr-doping aluminum oxide

Emeralds

Rubies

- Solid solutions are formed upon the placement of foreign atoms/molecules within a host crystal lattice. If the regular crystal lattice comprises metal atoms, then this solution is referred to as an alloy.
- Solutions that contain two or more species in their crystal lattice may either be substitutional or interstitial in nature, corresponding to shared occupancy of regular lattice sites or vacancies between lattice sites, respectively.

➢ <u>The substitutional doping</u>: William Hume-Rothery Rules

- The percentage difference between solute and solvent atomic radii should be less than 15%.

$$\frac{|r_{\text{solute}} - r_{\text{solvent}}|}{r_{\text{solvent}}} \times 100\% \leq 15\%$$

- The crystal structures of the dopant and solvent atoms must be matched.

$$\text{FCC}_{\text{solute}} - \text{FCC}_{\text{solvent}} > \text{FCC}_{\text{solute}} - \text{BCC}_{\text{solvent}}$$
$$> \text{FCC}_{\text{solute}} - \text{HCP}_{\text{solvent}} > \text{BCC}_{\text{solute}} - \text{HCP}_{\text{solvent}}$$

- In order for dopant atoms to be stabilized within a host lattice, both types of atoms must have similar electro-negativities.

- The solute and solvent atoms should have similar valences in order for maximum solubility.

➢ The interstitial doping: diffusion energy

Energy is required to form a defect (endothermic process)

$$\frac{N_i}{N_T} \propto e^{-(E_a/kT)}$$

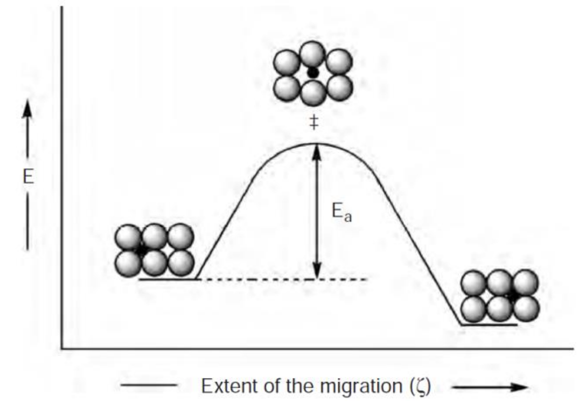


Figure 2.27. Illustration of the energetics involved for the atomic diffusion of interstitial impurities.

N_i: the number of impurities when $E > E_a$;

N_T: the total number of potential impurity atoms in the crystal lattice;

E_a: the activation energy for the diffusion process;

k: Boltzmann constant;

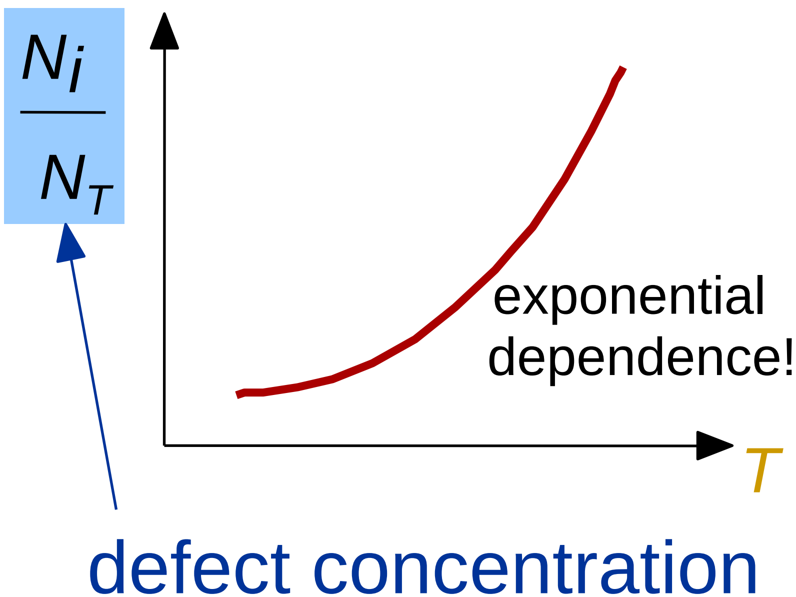
T: temperature.

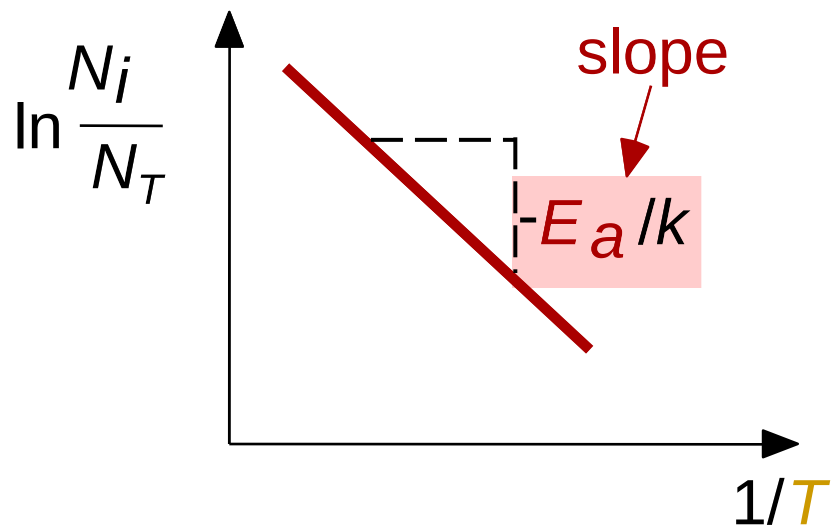
Measuring Activation Energy

- We can get E_a from an experiment.

$$\frac{N_\mathrm{i}}{N_\mathrm{T}} \propto e^{-(E_a/kT)}$$

- Measure this...
- Replot it...





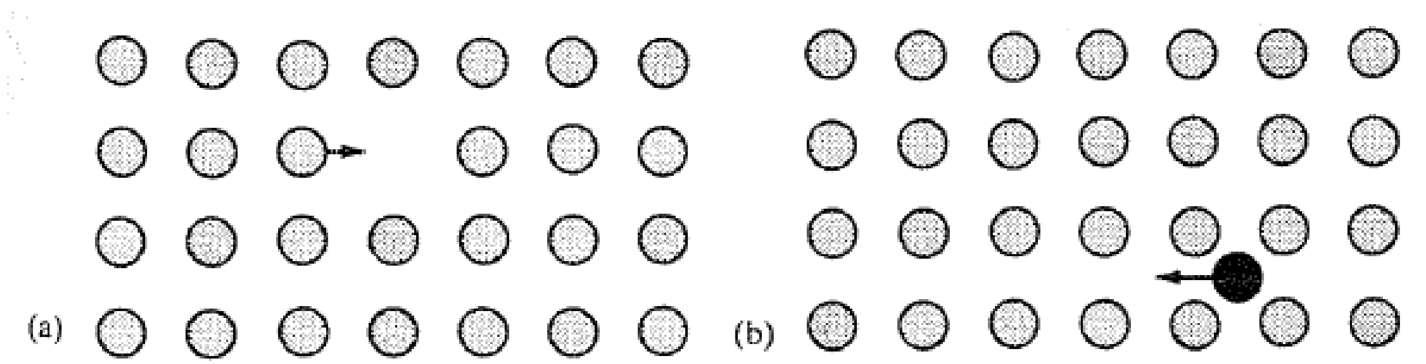
Doping with selected 'impurities' can introduce vacancies into a crystal.

- Consider incorporating $CaCl_2$ into NaCl, in which each Ca^{2+} replaces two Na^+ and creates one cation vacancy.

Defects and Ionic Conductivity in Solids

Defects make it possible for atoms or ions to move, through diffusion through the lattice or ionic conductivity (ions under the influence of an external electric field) through the structure.

Two possible mechanisms for the movement of ions through a lattice:



Vacancy mechanism　　　Interstitial mechanism

Solid Electrolytes

Ionic conductivity of solids is important towards the development of solid state batteries.

- Primary batteries are not reversible and are discarded after use.
- Secondary or storage batteries are reversible and significant research is performed to improve properties of materials.

Line defects

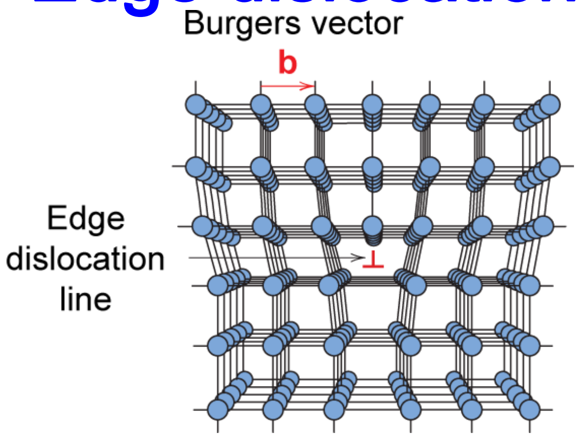
Linear defects (Dislocations) are one-dimensional defects that cause misalignment of nearby atoms.

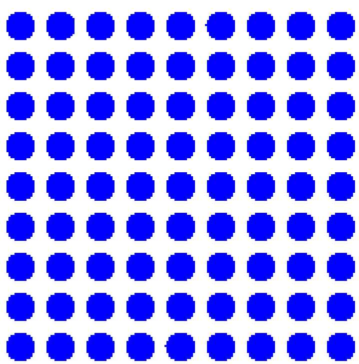
Dislocations: Defects in the order arrangement of a metal's atomic structure.

Dislocations are found along the slip plane and are commonly caused by a shearing stress. Dislocations explain why the strength of materials is less than theoretically calculated or predicted by theory.

Types of dislocations: edge, screw, mixed.

Edge dislocation



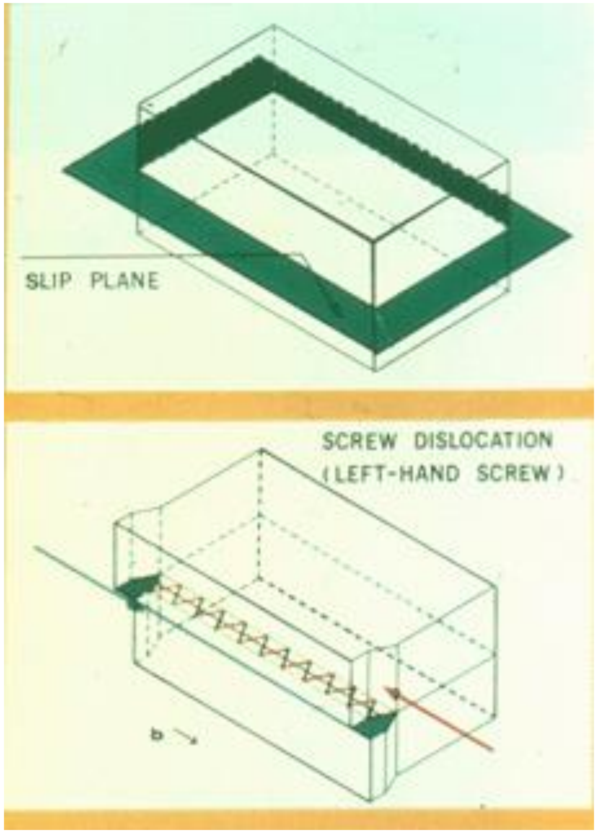


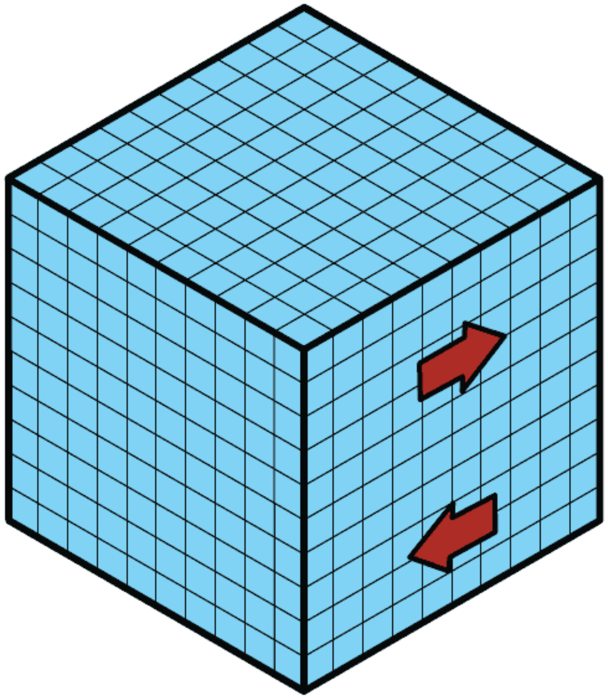
- Extra half-plane of atoms inserted in a crystal structure; the edge of the plane terminates within the crystal.
- Around the dislocation line there is some localized distortion.

The edge dislocation frequently determines whether the entire solid will deform and fail under stress.

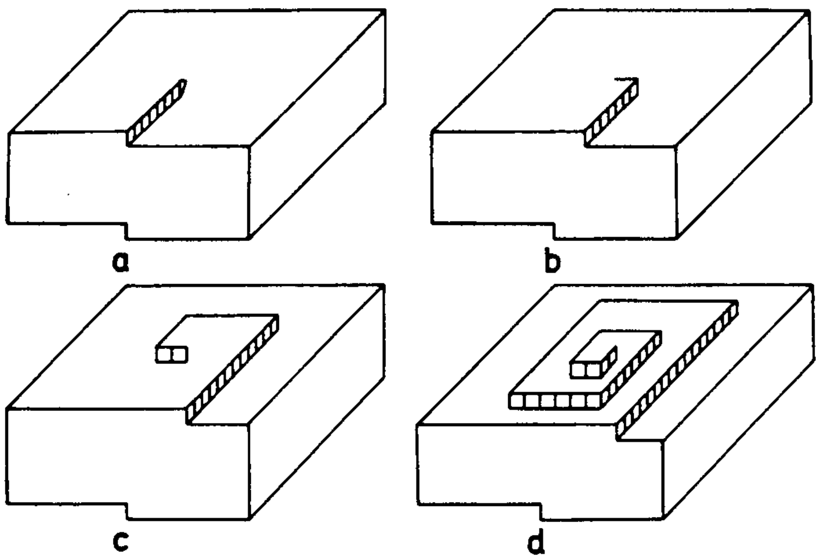
Screw dislocation

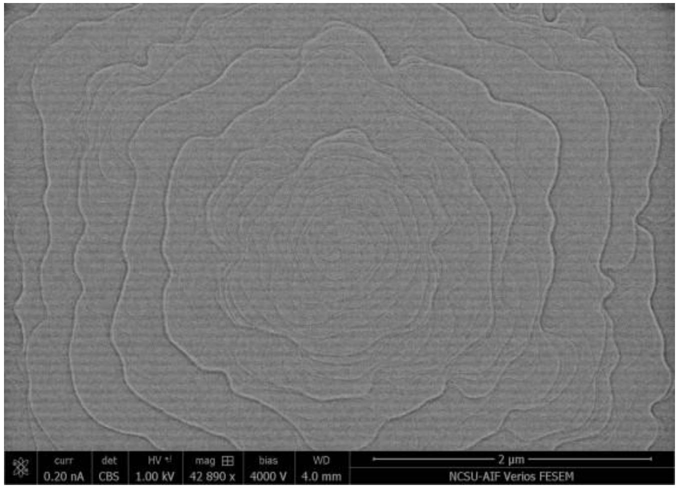
- Named for the spiral stacking of crystal planes around the dislocation line; results from shear deformation

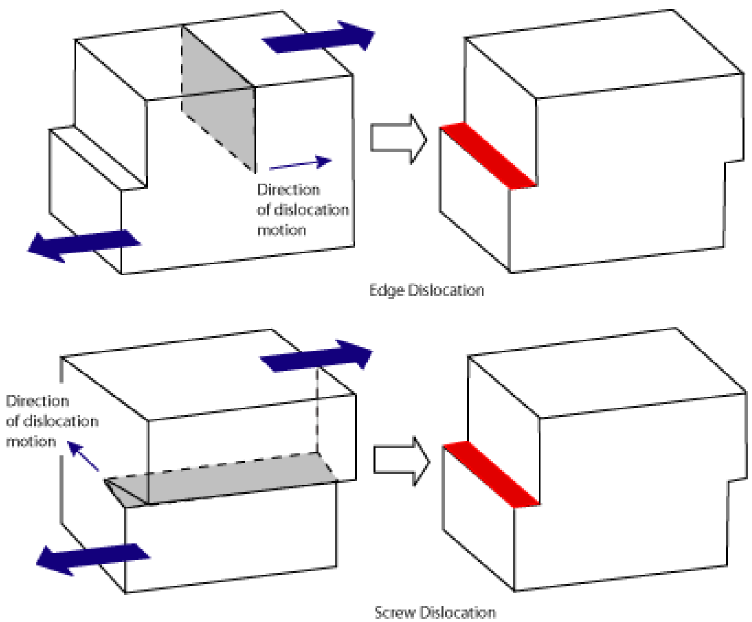


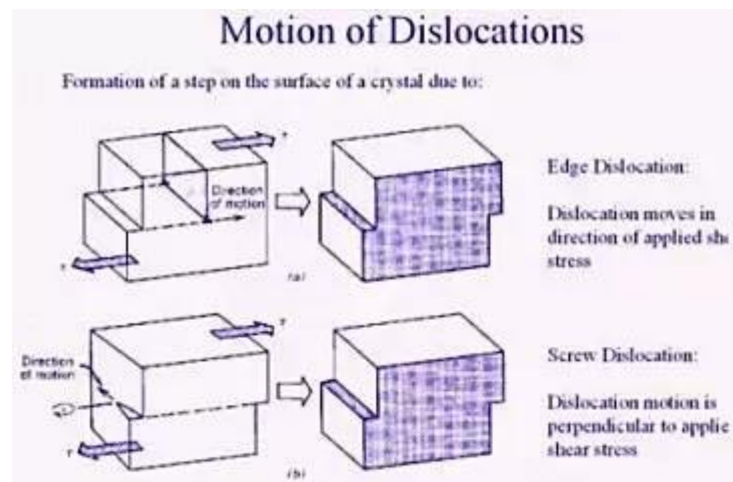


Screw dislocation

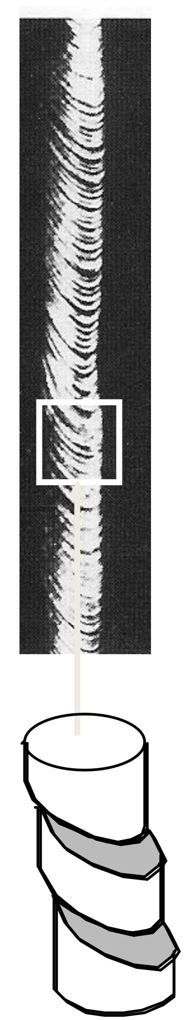





Direction
of dislocation
motion
Edge Dislocation
Direction
of dislocation
motion
Screw Dislocation


Motion of Dislocations
Formation of a step on the surface of a crystal due to:
Edge Dislocation:
Dislocation moves in
direction of applied sh
stress
Screw Dislocation:
Dislocation motion is
perpendicular to applie
shear stress

- The strength of a material with no dislocations is 20-100 times greater than the strength of a material with a high dislocation density.
- So, materials with no dislocations may be very strong, but they cannot be deformed.
- The dislocations weaken a material, but make plastic deformation possible.
- The ability of a metal to deform depends on the ability of dislocations to move.
- Restricting dislocation motion makes the material stronger



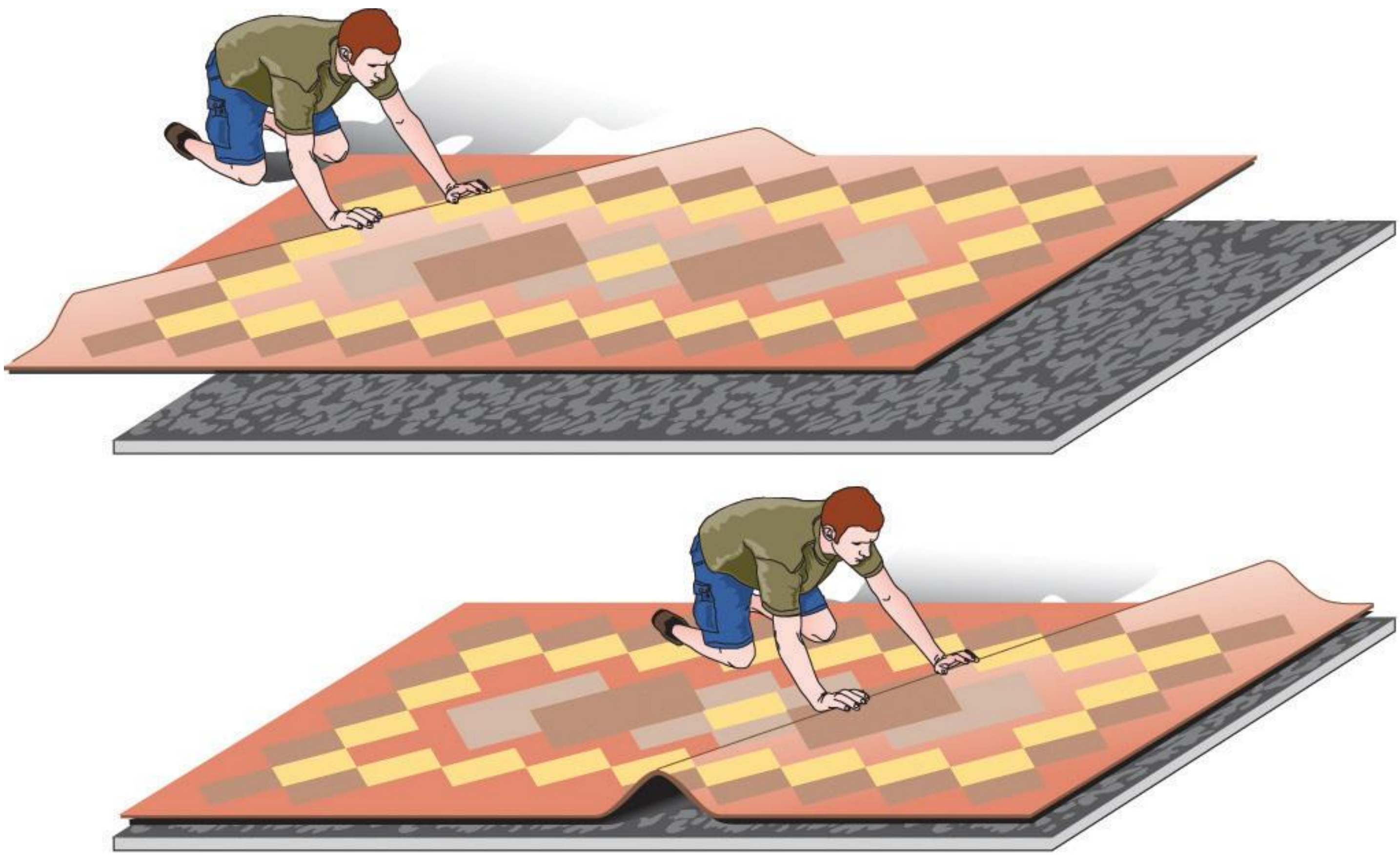
Plastically stretched zinc single crystal.

Q:

1. Why metals could be plastically deformed?

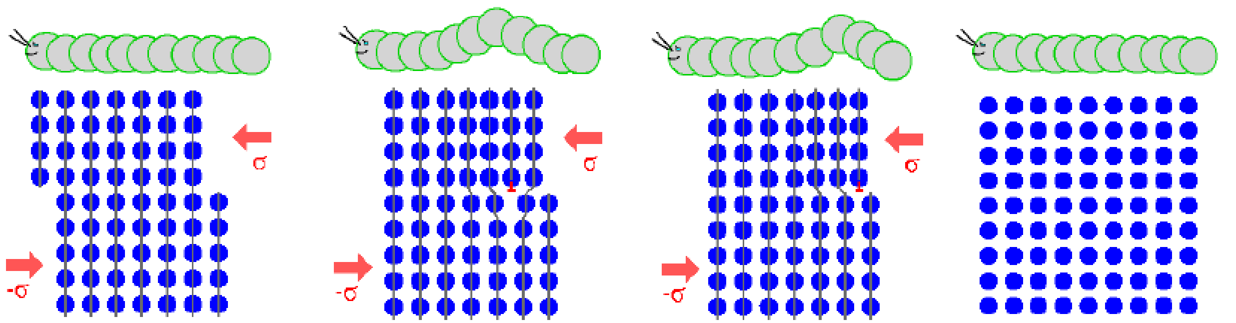
2. Why the plastic deformation properties could be changed to a very large degree by forging without changing the chemical composition?

3. Why plastic deformation occurs at stresses that are much smaller than the theoretical strength of perfect crystals?



(a) Dislocation

Dislocations allow deformation at much lower stress than in a perfect crystal



If the top half of the crystal is slipping one plane at a time then only a small fraction of the bonds are broken at any given time and this would require a much smaller force.

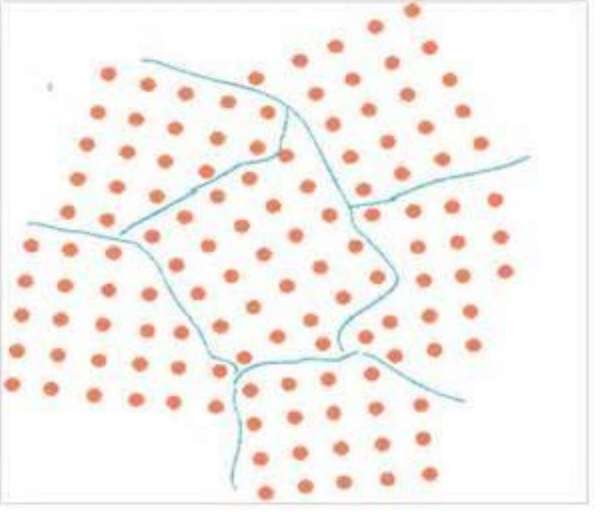
The propagation of one dislocation across the plane causes the top half of the crystal to move (**to slip**) with respect to the bottom half but we do not have to break all the bonds across the middle plane simultaneously (which would require a very large force).

Planar defects

- Most materials are polycrystalline, which means they consist of many microscopic individual crystals called **grains** that are randomly oriented with respect to one another.

- The place where two grains intersect is called a **grain boundary**.

- The movement of a deformation through a solid tends to stop at a grain boundary. Consequently, controlling the grain size in solids is critical for obtaining desirable mechanical properties; <u>fine-grained materials are usually much stronger than coarse-grained ones.</u>



2.3.6 Physical Properties of Crystals

- **Hardness**

  - The macroscopic physical properties of crystalline materials are directly related to these arrangements.
  - For instance, the overall hardness of a crystal depends on the nature of the interactions among the discrete components of the crystal lattice. Those crystals possessing covalent interactions (e.g., diamond) will have a high hardness, whereas those containing only van der Waals forces will be soft (e.g., talc).

The Mohs scale (Hardness) is generated by a qualitative assessment of how easily a surface is scratched by harder materials.

Table 2.5. Hardness Scales

Solid	Mohs	Vickers	Knoop
Talc	1	27	N/A
Graphite	1.5	37	N/A
Gypsum	2	61	N/A
Fingernail	2.5	102	117
Calcite	3	157	169
Fluorite	4	315	327
Apatite	5	535	564
Knife blade	5.5	669	705
Feldspar	6	817	839
Pyrex glass	6.5	982	929
Quartz	7	1,161	N/A
Topaz/Porcelain	8	1,567	N/A
Sapphire/Corundum	9	2,035	N/A
Diamond	10	N/A	N/A

N/A indicates the hardness value is above/below the acceptable range of the particular hardness scale. Values were obtained from the conversion site: http://www.efunda.com/units/hardness/convert_hardness.cfm?HD=HM&Cat=Steel#ConvInto

Diamond: 10

Talc: 1

Fingernail: 2.5

Glass: 5.5

Steel Nail: 5.5

Testing method--- Tensile test

Tensile strength is a measurement of the force required to pull something such as rope, wire, or a structural beam to the point where it breaks. The hardness of a material is directly proportional to its tensile strength.

Testing method--- Indentation technique

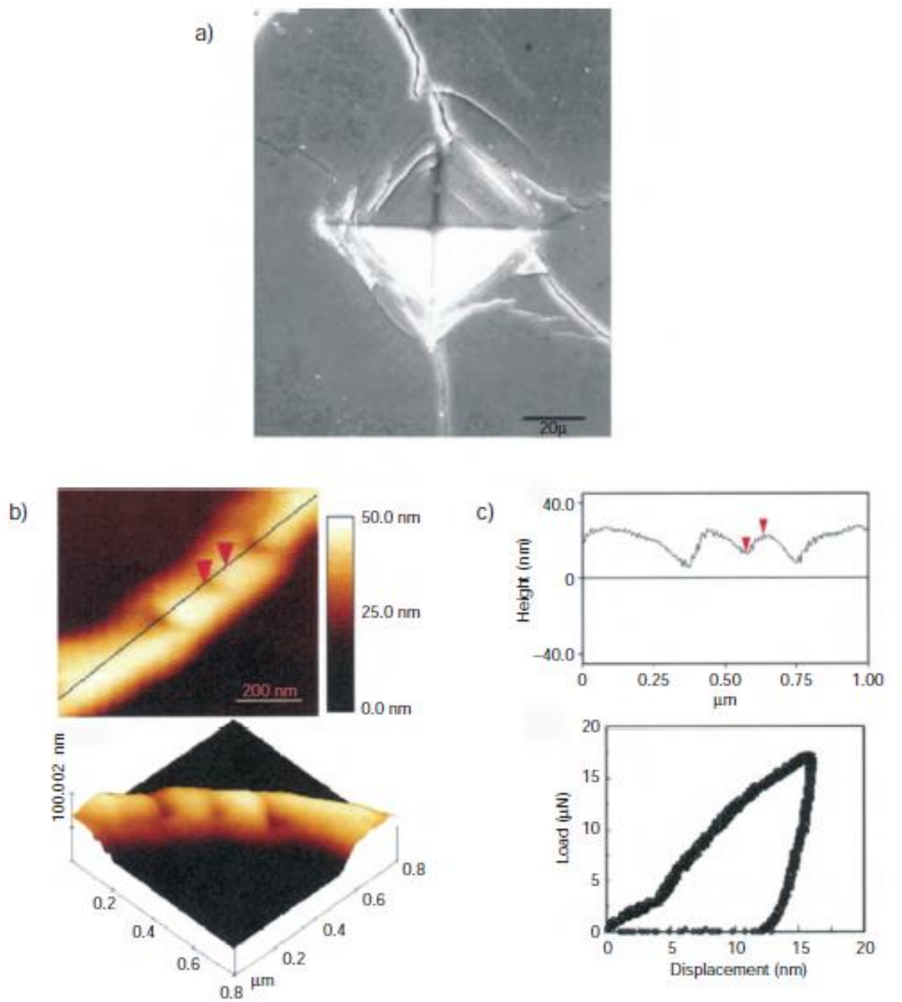


Figure 2.35. Examples of indentation processes to determine surface hardness. Shown are (a) Vickers indentation on a SiC–BN composite, (b) atomic force microscope images of the nanoindentation of a silver nanowire, and (c) height profile and load–displacement curve for an indent on the nanowire. Reproduced with permission from *Nano Lett.* **2003**, 3(11), 1495. Copyright 2003 American Chemical Society.

$$H = F\frac{L}{A_r}$$

H: hardness

L: the maximum load

A_r: the residual indentation area

F: the coefficient varies depending on which indentation method is used.

Nano-indentation technique: the hardness of a material is evaluated by the depth and symmetry of the cavity created from controlled perforation of a surface with a *nano-sized* tip.

• Cleavage and fracturing

❖ Cleavage plane

Created where little force is needed to separate the crystal into two unites

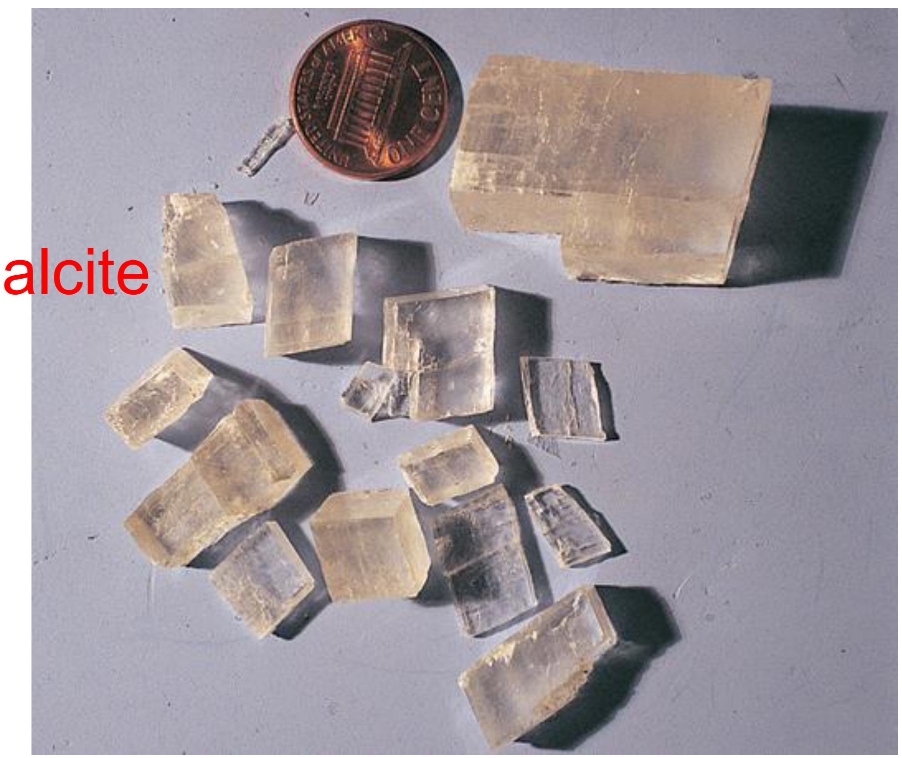
❖ Fracture

Chipping a crystal into rough, jagged pieces.

Cleavage: Orientation and number of planes of weakness within a crystal. Directly reflects the orientation of weak bonds within the crystal structure. This feature is also highly diagnostic.

Since the cleavage planes are parallel to crystal faces, the fragments formed upon cleavage will retain the symmetry exhibited by the bulk crystal.

Rhombohedral Cleavage in Calcite



Ulexite ($NaCaB_5O_6(OH)_6 \cdot 5(H_2O)$) (hydrated sodium calcium borate hydroxide), sometimes known as TV rock, is a mineral occurring in silky white rounded crystalline masses or in parallel fibers.

The crystal will preferentially cleave in directions parallel to the crystallite fibers.

a)
b)
c)

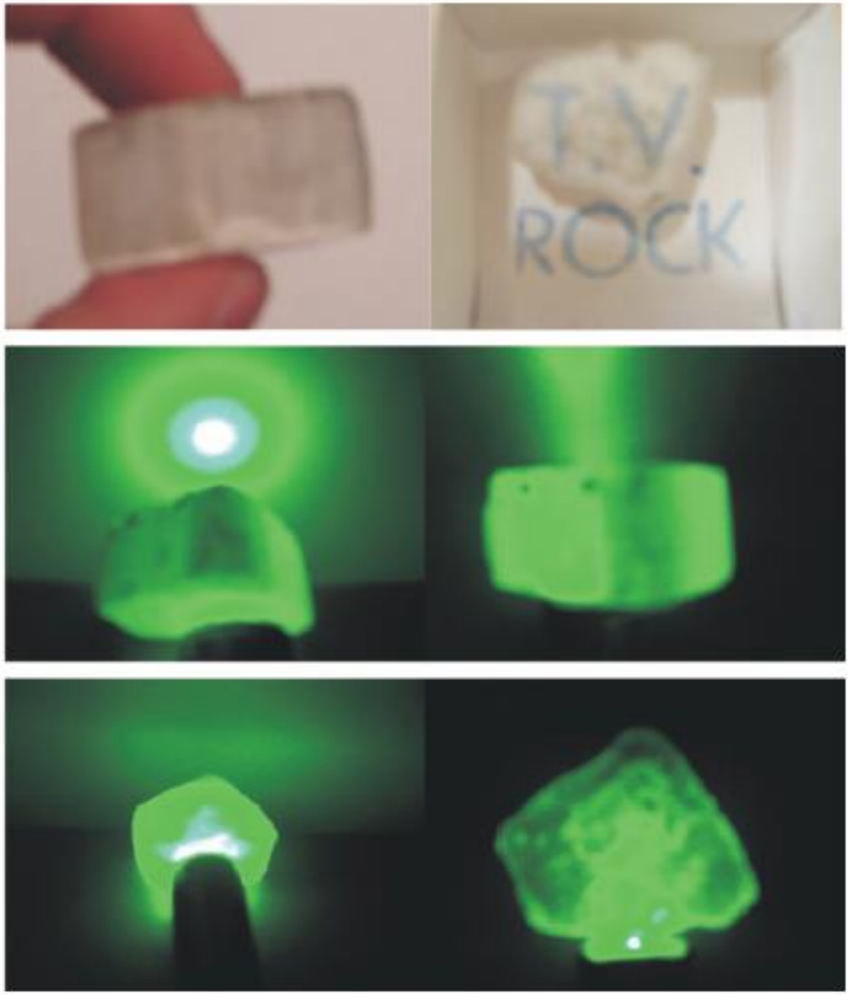


Figure 2.37. Optical properties of ulexite, the "T.V. rock." Shown are (a) the fibrous morphology of the crystal, and projection of the words "T.V." on the surface of the rock from transmission through parallel fibers,; (b) passage, and (c) blockage of a green laser beam when impinged on the crystal at angles parallel and perpendicular to the fibers, respectively.

Fracture: This describes how a mineral breaks if it is not along well defined planes. In crystals with low symmetry and highly interconnected atomic networks, irregular fracture is common.

Conchoidal Fracture in quartz

- **Color**

colorless

Since visible light (350–700 nm) is not energetically sufficient to cause bond rupturing and/or electronic transitions of the constituent metal atoms/ions, this energy is not absorbed by pure crystals, giving rise to a colorless state.

diamond

quartz

corundum

Impurities induce colors

Impurities are trapped in the anion vacancy

✓ heating

$$[(KCl)_n(KCl)]_{(s)} + K_{(g)} \xrightarrow{500\ K} [(KCl)_n(K)(e^-)]_{(s)} + KCl_{(s)},$$

Colorless KCl becomes a violet color

✓ X-ray irradiation

This high-energy radiation will cause the removal of a halide ion from the lattice and will excite some of the lattice electrons from valence to conduction bands. At this point, the electrons are free to diffuse through the crystal, where they remain mobile until they find an anion vacancy site.

Self-darkening sunglasses

✓ Transition metal doping

Pure corundum (α-alumina)

<1% Cr^{3+} doping

Gemstone owe their color from trace transition-metal ions

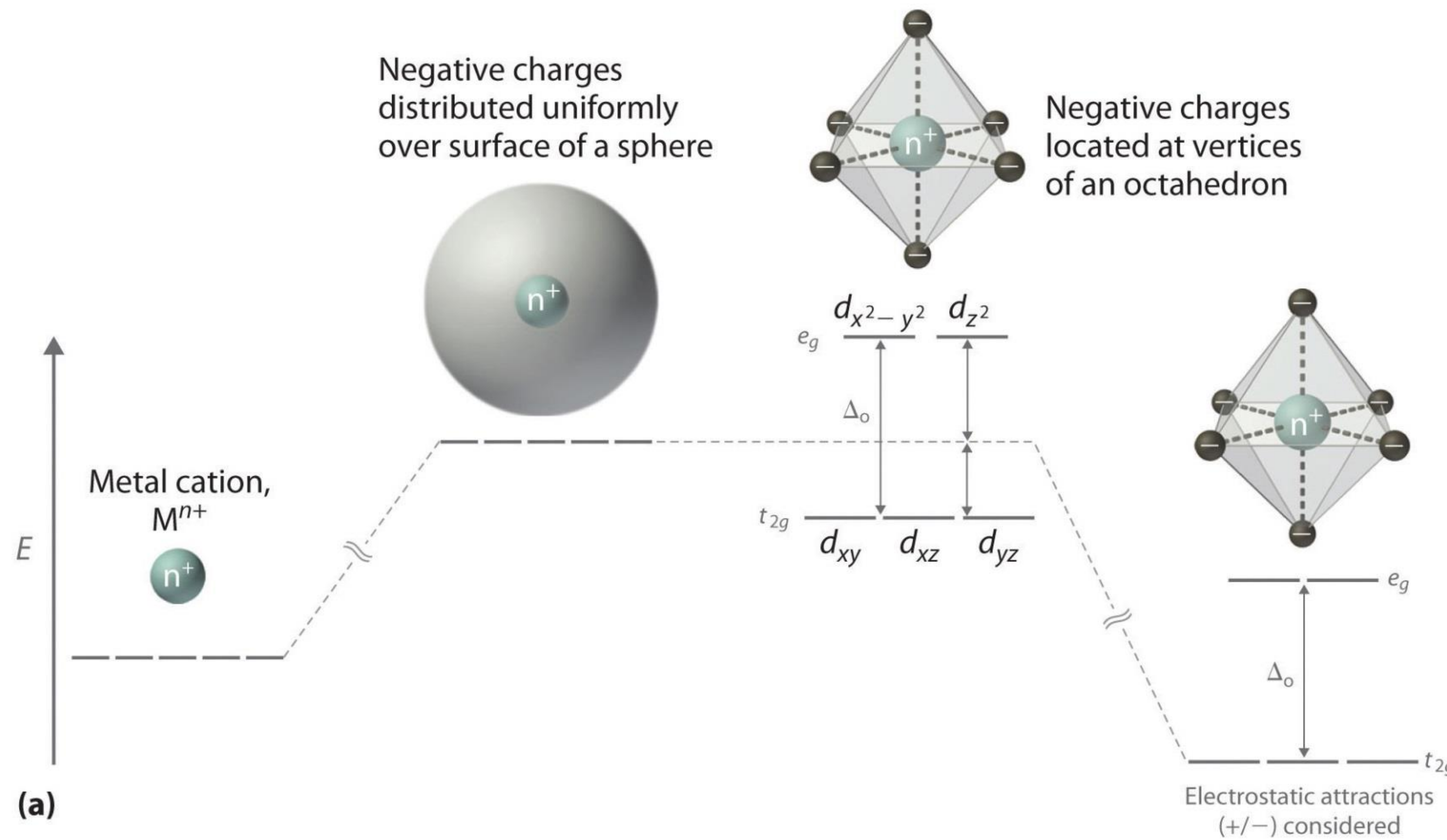
Corundum mineral, Al_2O_3: Colorless

Cr ⇒ Al :	Ruby (red)
Mn ⇒ Al:	Amethyst (purple)
Fe ⇒ Al:	Topaz (yellow)
Ti &Co ⇒ Al:	Sapphire (blue)

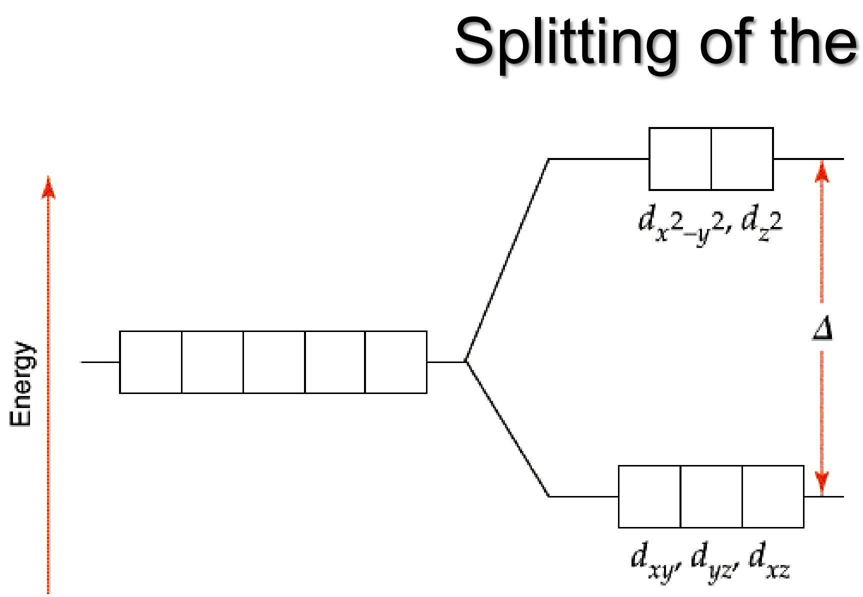
Beryl mineral, $Be_3Al_2Si_6O_{18}$: Colorless

Cr ⇒ Al :	Emerald (green)
Fe ⇒ Al :	Aquamarine (ocean blue)

Crystal Field Theory



Octahedral field Splitting Pattern:



The energy gap is referred to as Δ (10 Dq) , the crystal field splitting energy.

The d_{z2} and d_{x2-y2} orbitals lie on the same axes as negative charges.

Therefore, there is a large, unfavorable interaction between ligand (-) orbitals.

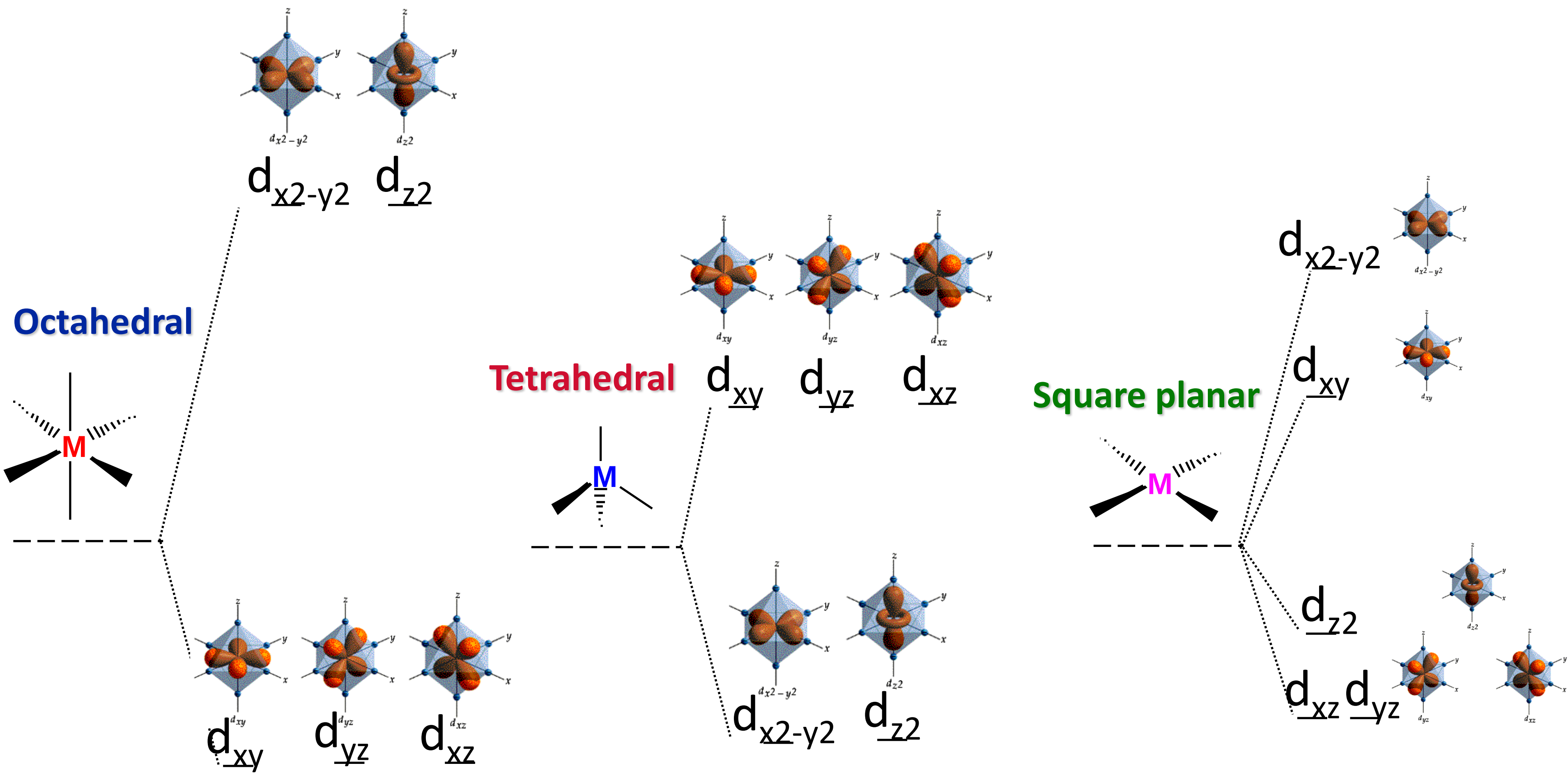
These orbitals form the degenerate high energy pair of energy levels.

The d_{xy} , d_{yx} and d_{xz} orbitals bisect the negative charges.

Therefore, there is a smaller repulsion between ligand & metal for these orbitals.

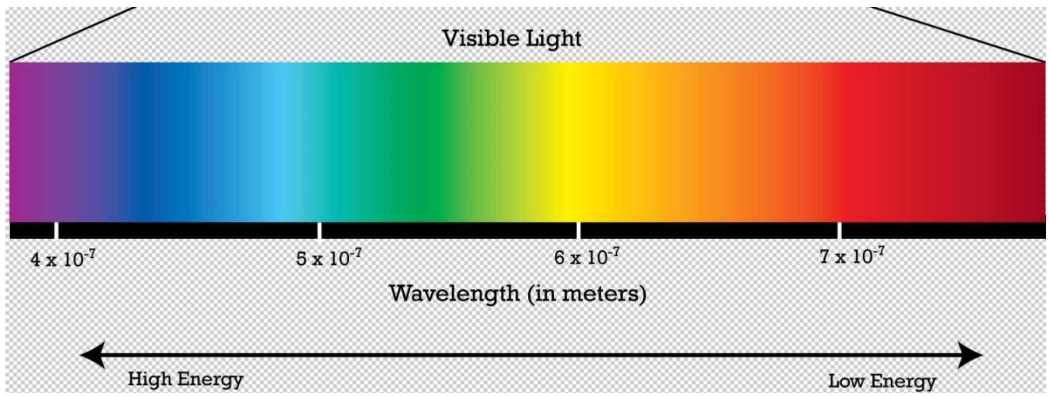
These orbitals form the degenerate low energy set of energy levels.

Octahedral, Tetrahedral & Square Planar

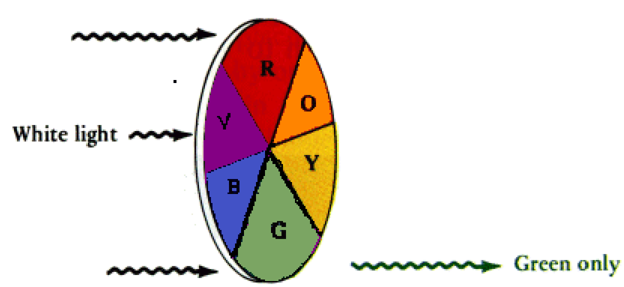


Complementary color after absorption

complementary colors lie opposite each other on the color wheel. Further, a black object absorbs all wavelengths of light, while an object will appear as white (or colorless) if no light is absorbed.



(a) Sample absorbs all but green light. Green is perceived.



(b) Sample absorbs violet, red, and orange light. Blue, green, and yellow light are transmitted. Green light is perceived.

Since the Δ (10 Dq) between d-orbitals is in the violet region of the visible spectrum, the reflected complementary red color is observed for ruby crystals.

For beryl crystals, the Be^{2+} ions will pull electron density away from the oxygen ions, which will cause less electron–electron repulsions between the Cr^{3+} d-orbitals and lone pairs of the oxygen ligands. This will correspond to a decrease in the 10 Dq value, the absorption of lower-energy wavelengths, and a shift of the reflected color from red to green.

Not only does the observed color depend on the nature of the transition metal impurity, but on the oxidation state of the dopant.

the color of a beryl-based crystal:

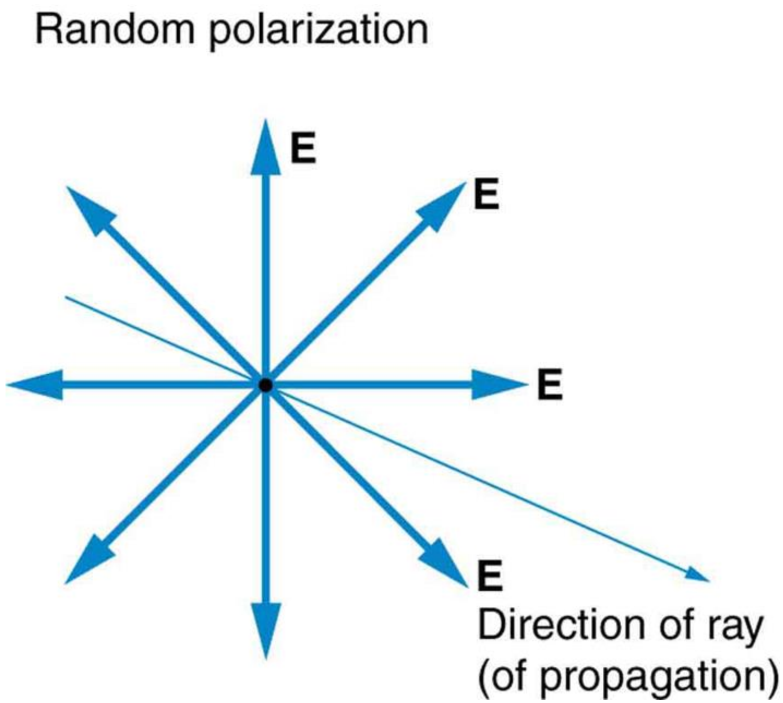
Fe^{2+} doping	Aquamarine	blue
Fe^{3+} doping	Heliodor	yellow
Mn^{2+} doping	Morganite	pink
Mn^{3+} doping	Red Beryl	Red

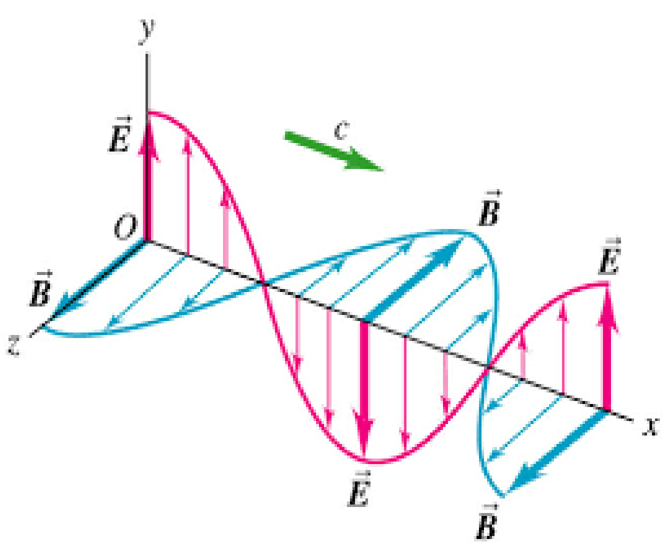
As a general rule, as the oxidation state of the transition metal ion increases, the ligand ions are drawn in closer to the metal center. This will result in more electron–electron repulsions between the metal and ligand, and a larger 10 Dq. The increase in the energy gap between d-orbitals causes the absorption of higher energy wavelengths, and a corresponding red shift for the observed/transmitted color.

- **Properties resulting from crystal anisotropy**

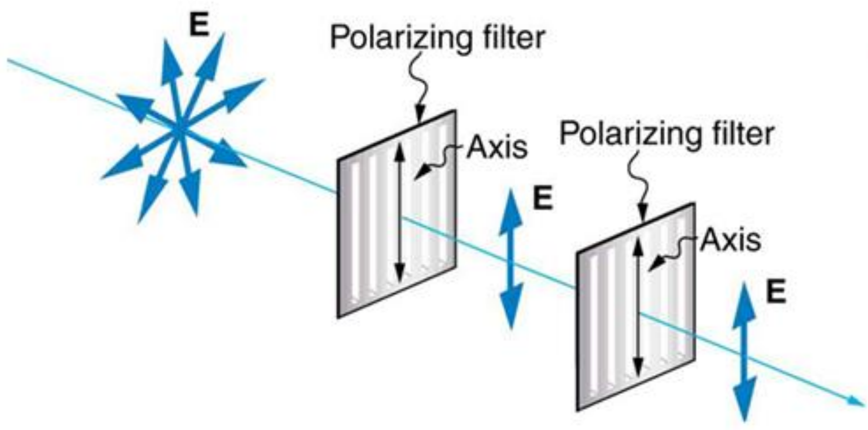
  Light

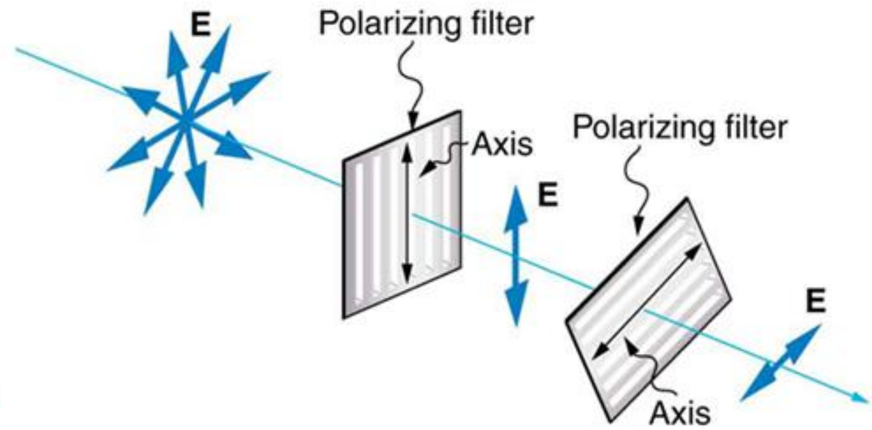
Birefringence, or **double refraction**, is the decomposition of a ray of light into two rays when it passes through anisotropic materials.





- Light is one type of electromagnetic wave.
- EM waves oscillate perpendicular to the direction of propagation.
- There are specific directions for the oscillations of the electric and magnetic fields.
- Polarization is the attribute that a wave's oscillations have a definite direction relative to the direction of propagation of the wave.


E
Polarizing filter
Axis
E
Polarizing filter
Axis
E
(a)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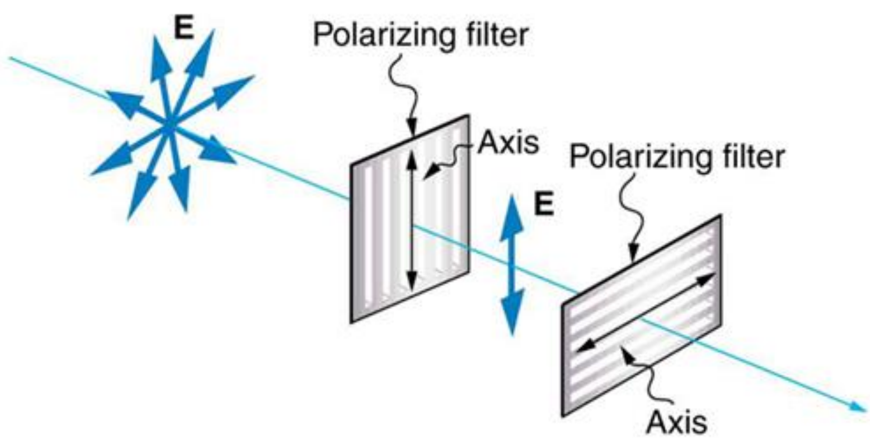
(b)


E
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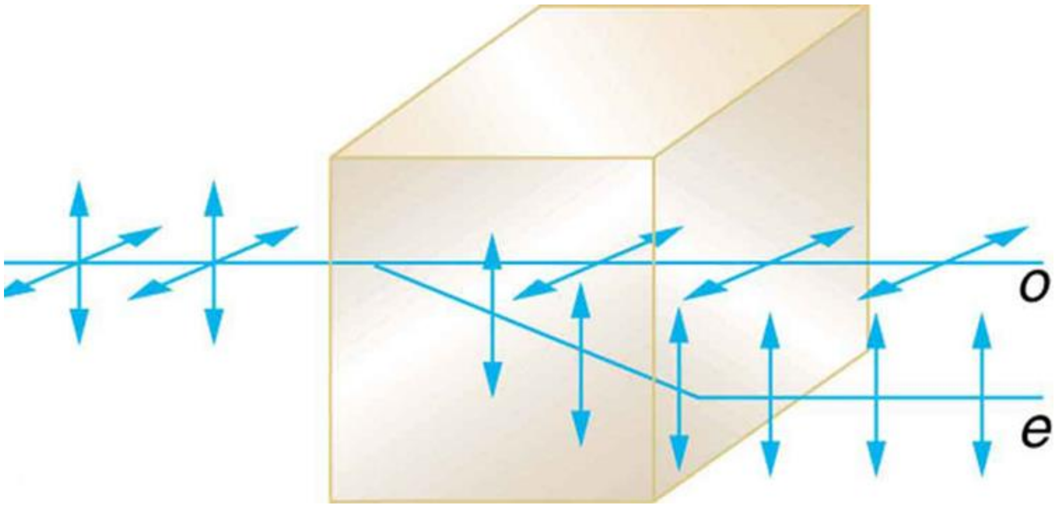
(c)


(d)



- Birefringent materials split un-polarized beams of light into two.
- The ordinary ray behaves as expected,
- but the extraordinary ray does not obey Snell’s law.

In optics the index of refraction n of an optical medium is a dimensionless number that describes how light, or any other radiation, propagates through that medium. It is defined as

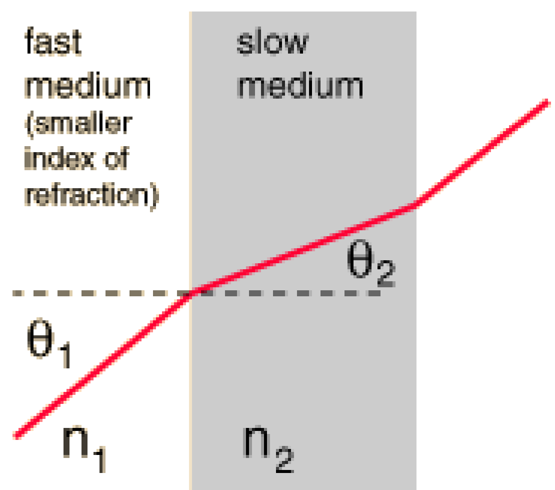
$$n = \frac{c}{v}$$

c : the speed of light in vacuum;

v : the phase velocity of light in the medium.

Snell's Law

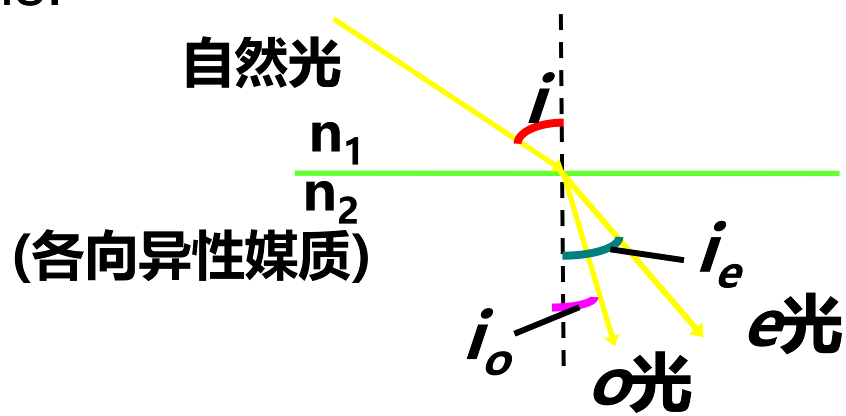
$$\frac{n_1}{n_2} = \frac{\sin\theta_2}{\sin\theta_1}$$



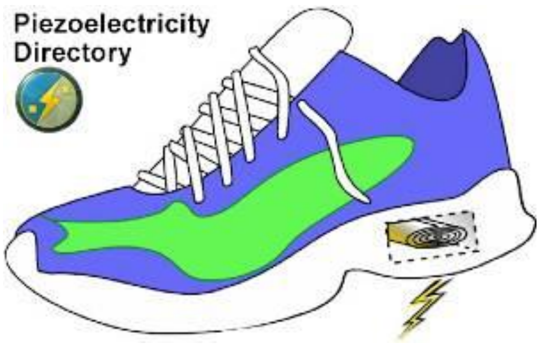
- the ordinary (O) ray, is characterized by an index of refraction that is the same in all directions.

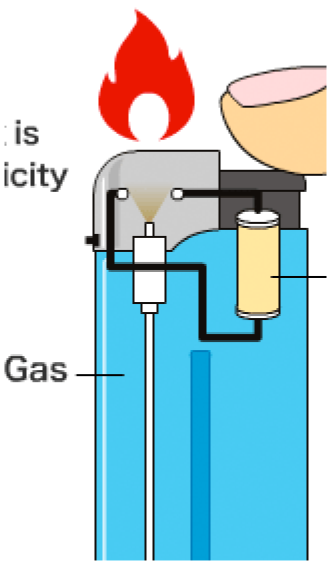
- the extraordinary (E) ray, travels with different speeds in different directions and hence is characterized by an index of refraction that varies with the direction of propagation.

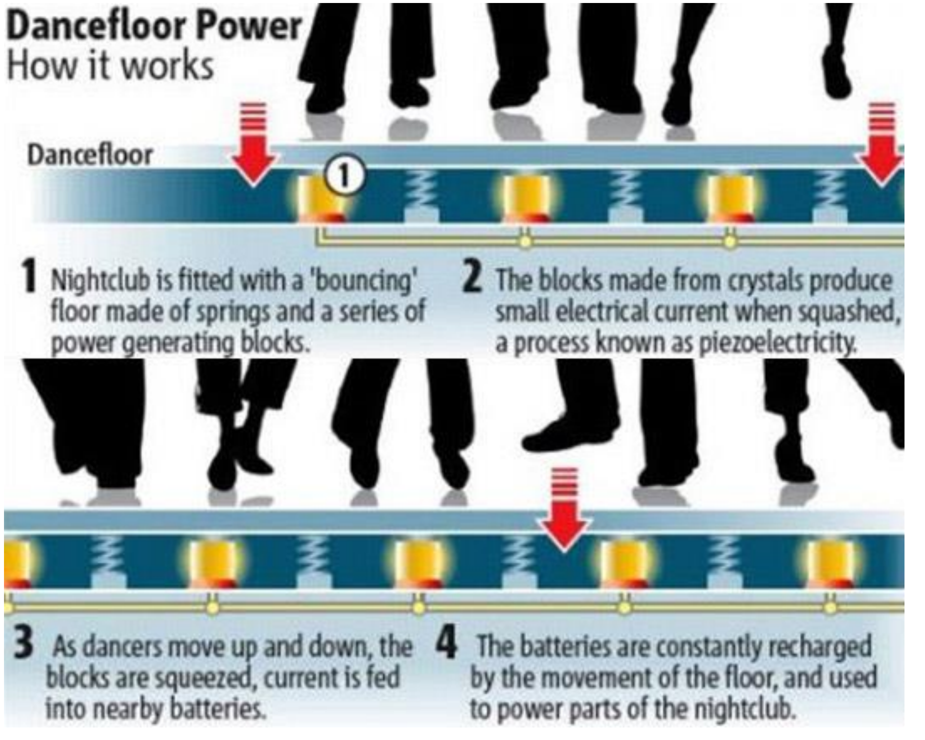
- The ordinary and extraordinary rays follow different paths inside the crystal, but on leaving the crystal, they follow parallel paths.



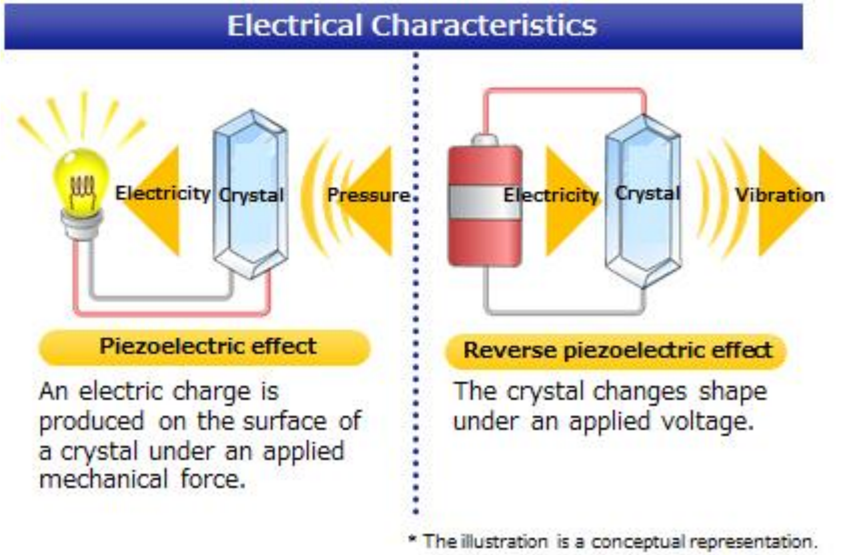
- # Pressure and electricity







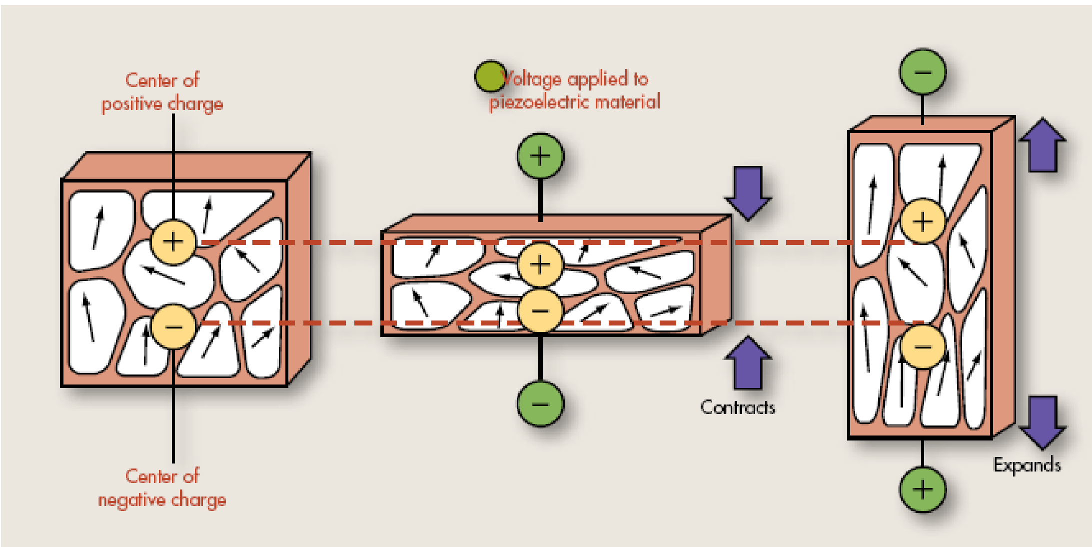
Pressure and electricity



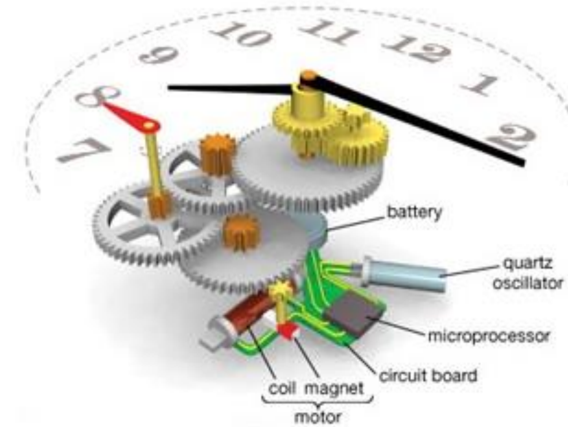
Crystals that do not possess a center of symmetry (i.e., non-centrosymmetric) exhibit interesting properties when exposed to pressure – a phenomenon known as piezoelectricity.

Conversely, if a piezoelectric crystal is placed in an electric field, the ions move toward opposite electrodes, thereby changing the shape of the crystal.

Quartz clock



1. The piezoelectric effect causes crystal materials like quartz to generate an electric charge when the crystal material is compressed, twisted, or pulled. The reverse also is true, as the crystal material compresses or expands when an electric voltage is applied.



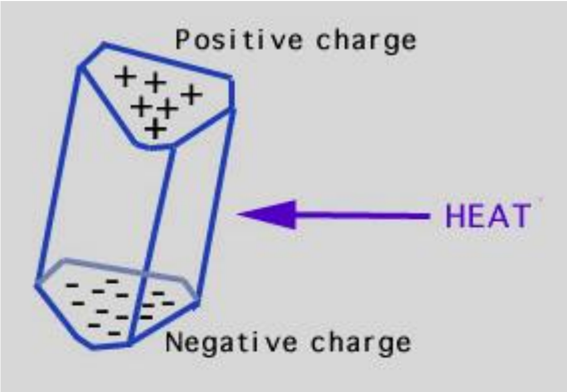
- **Heat and electricity**

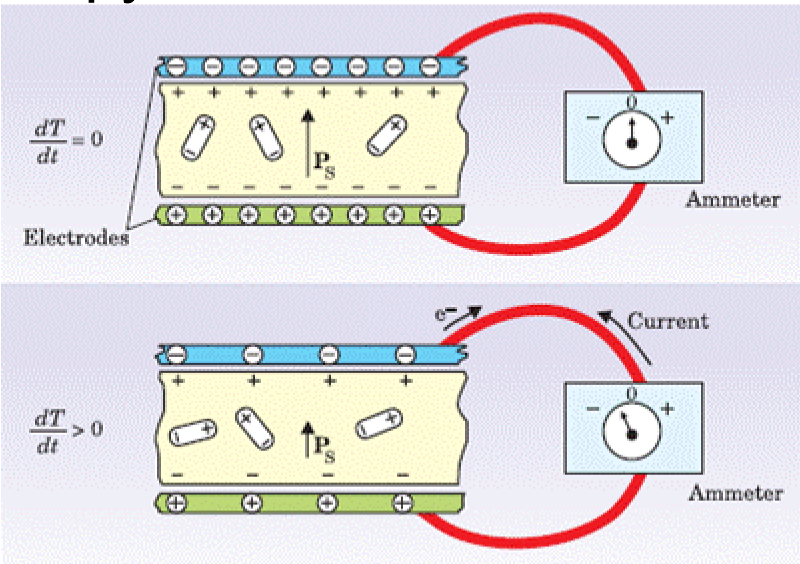
Other noncentrosymmetric crystals that alter their shape in response to changes in temperature are referred to as pyroelectric.

Infrared detectors

- Pyroelectricity is a phenomenon of certain crystals, displaying temporary voltage followed by external current, when subjected to a temperature gradient.

- The temperature gradient modifies the position of atoms within the crystal, thus the spontaneous polarization changes. Consequently, depolarization charges on the surface redistribute, compensating the change in polarization, expressed as the pyroelectric current.





2.4 The amorphous state

- Amorphous metals
- Glasses
- Cementitious materials

2.4.1 Sol-gel processing

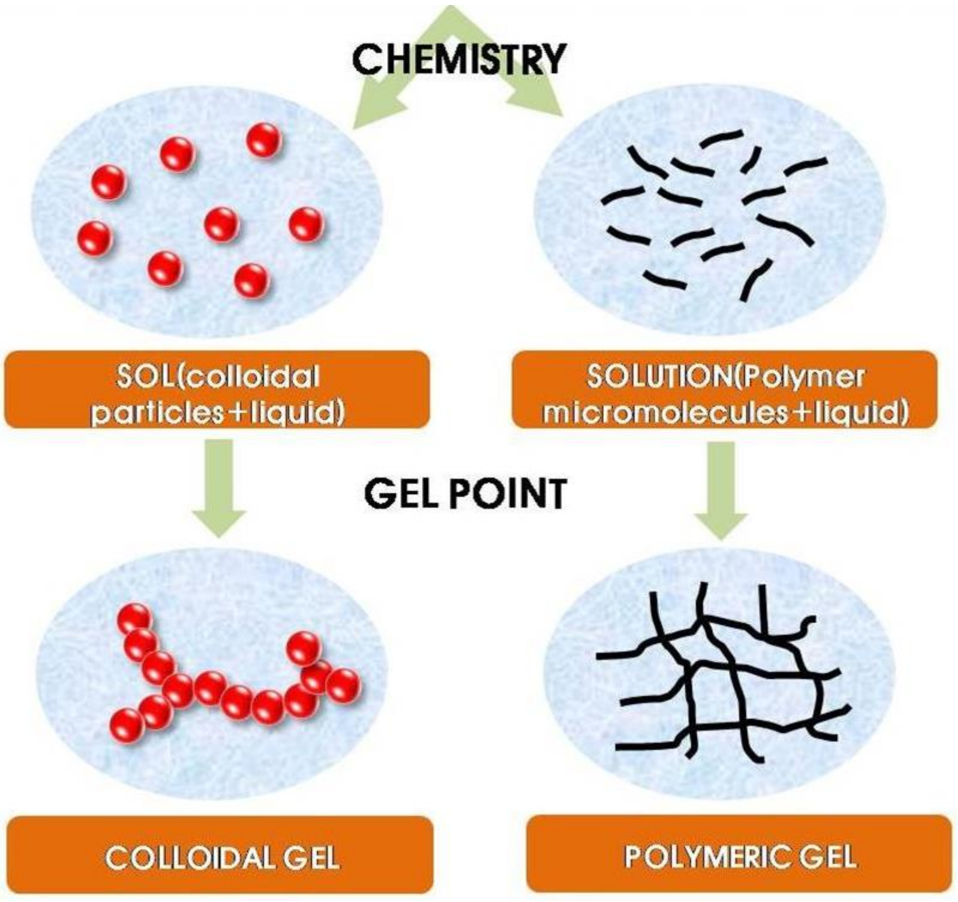
Definition

Formation of an oxide network through polycondensation reactions of a molecular precursor in a liquid.

The sol-gel method was developed in the 1960s mainly due to the need of new synthesis methods in the nuclear industry. A method was needed where dust was reduced (compared to the ceramic method) and which needed a lower sintering temperature. In addition, it should be possible to do the synthesis by remote control.

Sol: a stable suspension of colloidal solid particles (ca. > 200 nm particles) or polymers in a liquid.

Gel: porous, three-dimensional, continuous solid <u>network</u> surrounding a continuous liquid phase.



- The idea behind sol-gel synthesis is to “dissolve” the compound in a liquid in order to bring it back as a solid in a controlled manner.
- Multi component compounds may be prepared with a controlled stoichiometry by mixing sols of different compounds.
- The sol-gel method prevents the problems with co-precipitation, which may be inhomogeneous, be a gelation reaction.
- Enables mixing at an atomic level.
- Results in small particles, which are easily sinterable.

Sol

- A sol consists of a liquid with colloidal particles which are not dissolved, but do not agglomerate or sediment.
- Agglomeration of small particles are due to van der Waals forces and a tendency to decrease the total surface energy. Van der Waals forces are weak, and extend only for a few nanometers.
- In order to counter the van der Waals interactions, repulsive forces must be established.

May be accomplished by:

Electrostatic repulsion. By adsorption of charged species onto the surface of the particles, repulsion between the particles will increase and agglomeration will be prevented. Most important for colloidal systems.

Steric hindrance. By adsorbing a thick layer of organic molecules, the particles are prevented from approaching each other reducing the role of the van der Waals forces. Works best in concentrated dispersions. Branched adsorbates works best. Usual for nanomaterials.

PZC, Point of zero charge

Stabilization due to electrostatic repulsion are due to formation of a double layer at the particle.

The surface of a particle is covered by ionic groups, which determines the surface potential. Counter ions in the solution will cover this layer, shielding the rest of the solution from the surface charges.

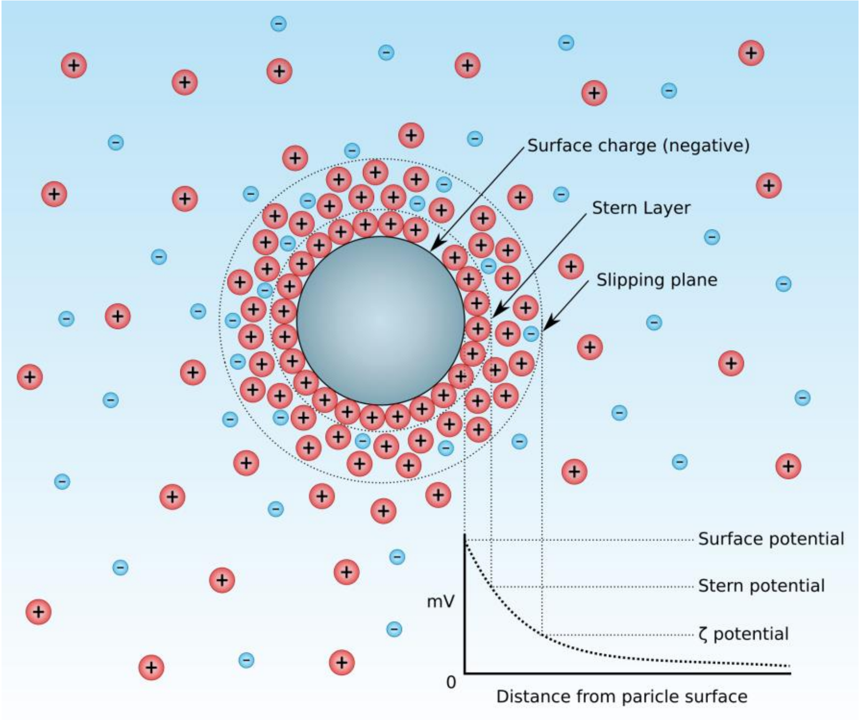
The pH where the particle is neutral is called PZC, point of zero charge.

For **pH > PZC** the surface is **negatively** charged

For **pH < PZC** the surface is **positively** charged.

Double Layer

- **Electric Double Layer** is the layer surrounding a particle of the dispersed phase and including the ions adsorbed on the particle surface and a film of the counter charged dispersion medium.
- The "slip plane" divides the part that moves with the particle and the solution. The potential at the slip plane is called the zeta (ζ) potential φ_ζ.
- The pH for which φ_ζ=0 is called the isoelectrical point (IEP).
- The stability of a colloid depends on φ_ζ; the larger the φ_ζ the more stable the colloid. Should be >30-50 mV.
- Given the same surface potential, the repulsive forces will increase with the particle size.



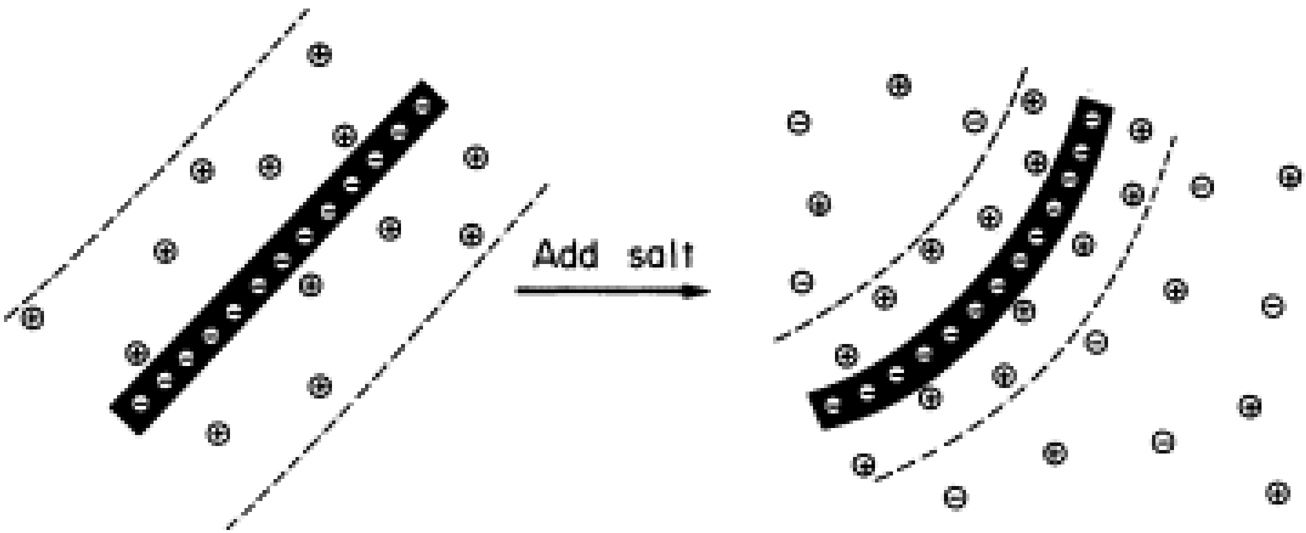
The PZC is the same as the isoelectric point (IEP) if there is no adsorption of other ions than the potential determining H^+/OH^- at the surface. This is often the case for pure ("pristine surface") oxides in water. In the presence of specific adsorption, PZC and isoelectric point generally have different values.

In the absence of positive or negative charges, the surface is best described by the PZC. If positive and negative charges are both present in equal amounts, then this is the IEP.

Coagulation/flocculation

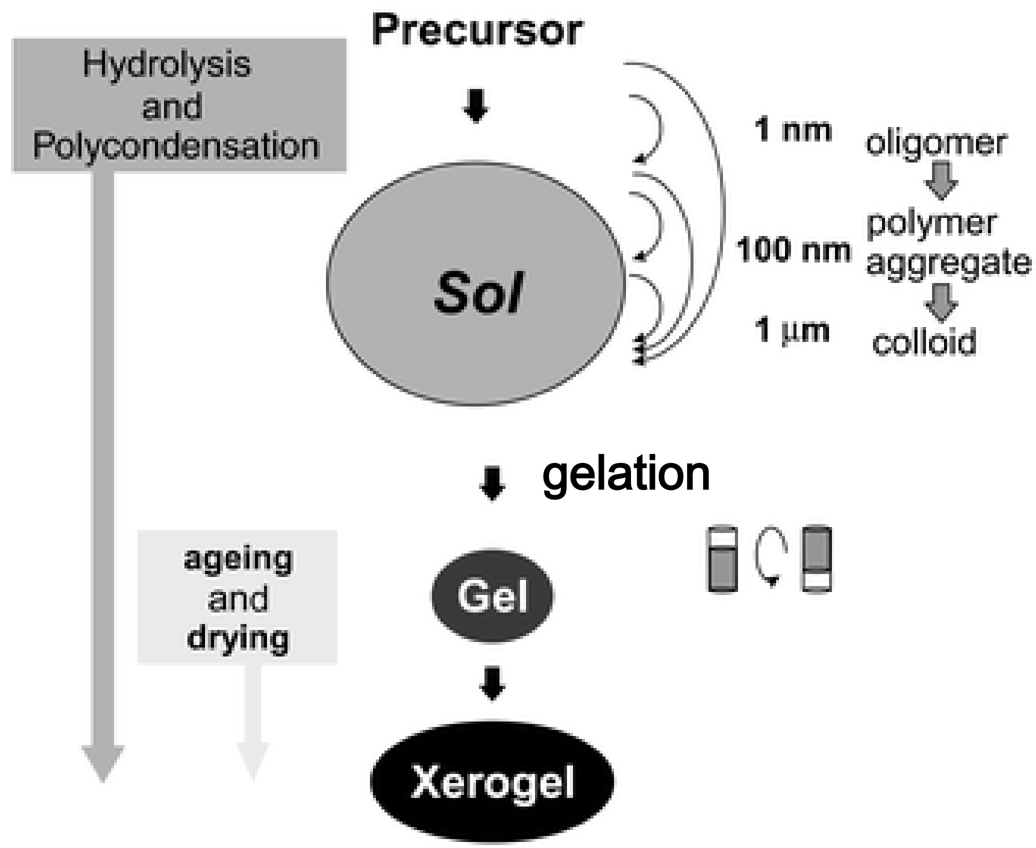
Coagulation of a sol may occur if:

- The surface potential (φ_0) is lowered (by changing pH).
- By increasing the number of counter ions. An increase in the concentration of counter ions result in a decrease of the thickness of the double layer.
- In some cases a coagulated colloid may be re-dispersed. This is called "**peptizing**". This is done e.g. by removing the surplus counter ions by washing, or by adding charged ions, so that the double layer is restored.


Add salt

Sol-gel process

- Hydrolysis
- Condensation
- Gelation
- Ageing
- Drying



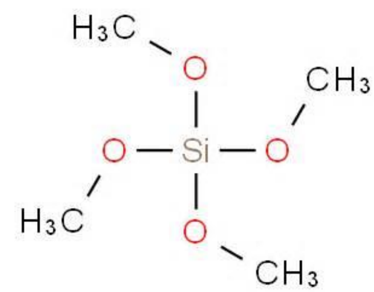
Silica gel

The starting materials for formation of silica gels are usually "water glass", Na_2SiO_3, or silicon alkoxides (e.g. $Si(OMe)_4$). The differences are:

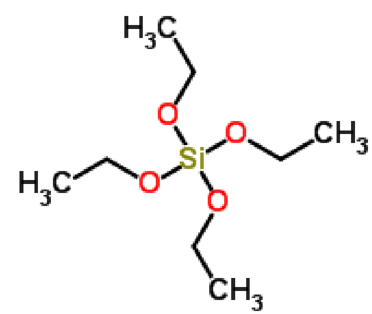
- Water glass is dissolved in water while the alkoxide is dissolved in an organic solvent, usually alcohol.
- The reactive groups in water glass are silanol. In the alkoxide system, a hydrolysis reaction occur, converting Si-OR to Si-OH. Gelation therefore starts with a change in pH (in water based systems) or by addition of water (in alcohol based systems).
- The alkoxide systems are more complex with more parameters. This may result in better control of the reactions.

Metal alkoxide complexes ($M(OR)_x$, where R = alkyl group (e.g., CH_3, C_2H_5, CF_3, etc.).

$Si(OR)_4$
tetramethoxysilane (TMOS)



tetraethoxysilane (TEOS).



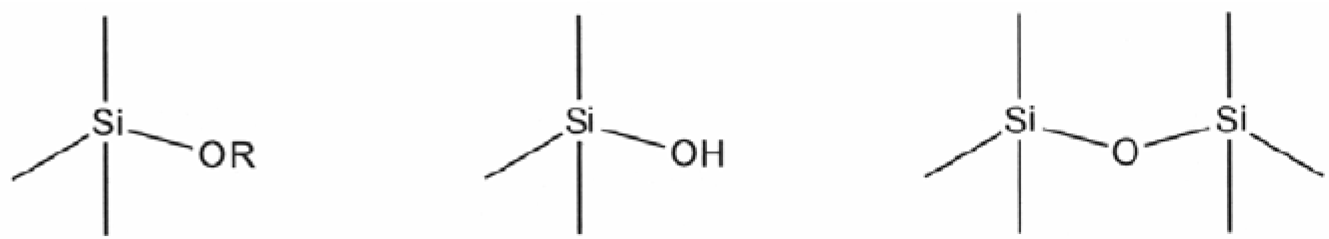
Hydrolysis and condensation

$$\equiv\text{Si–OR} + H_2O \longrightarrow \equiv\text{Si–OH} + \text{ROH} \qquad \text{Hydrolysis}$$

$$\equiv\text{Si–OH} + \equiv\text{Si–OR} \longrightarrow \equiv\text{Si–O–Si}\equiv + \text{ROH}$$

$$\equiv\text{Si–OH} + \equiv\text{Si–OH} \longrightarrow \equiv\text{Si–O–Si}\equiv + H_2O$$

Condensation



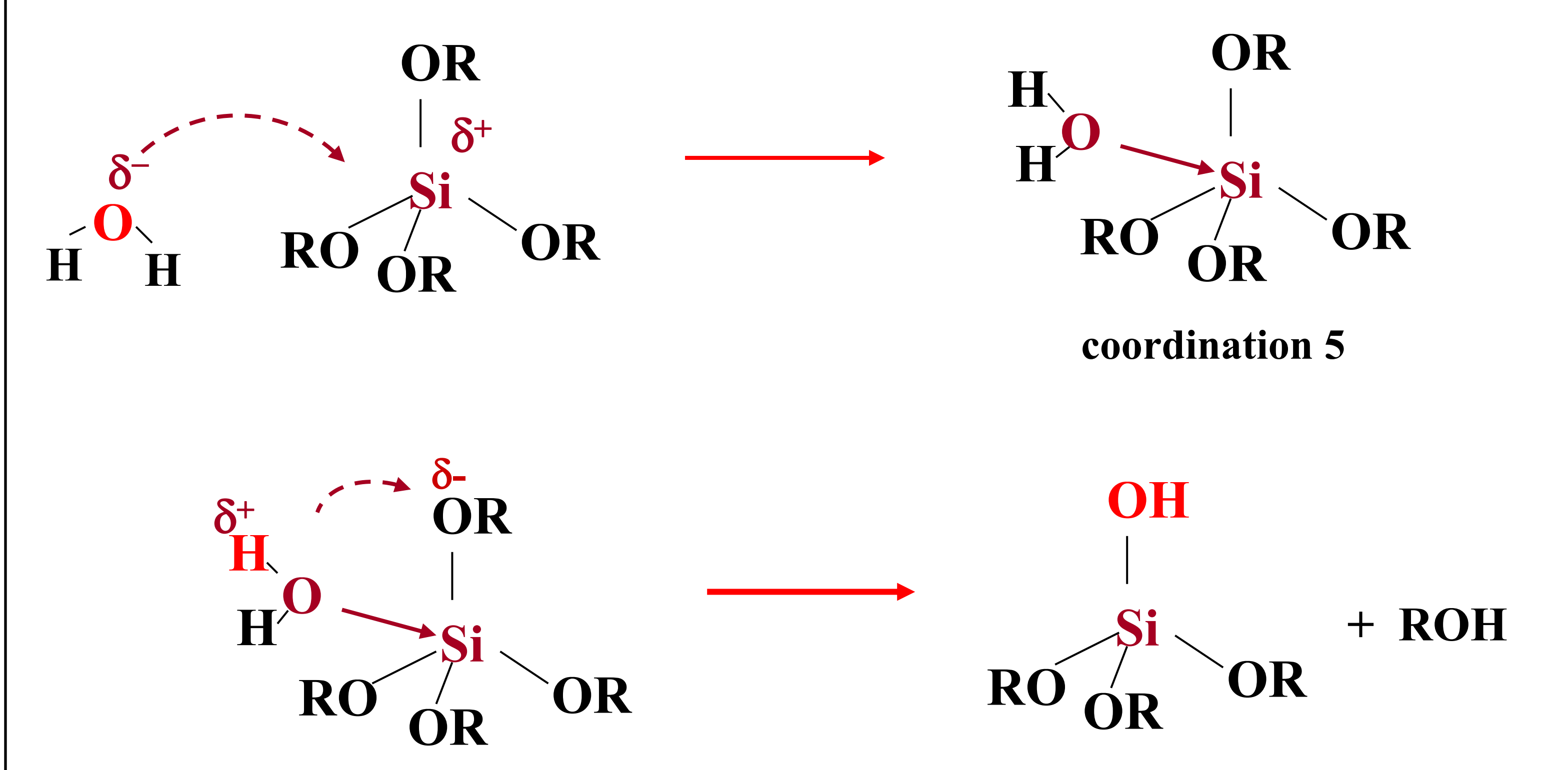
Alkoxide Silanol Siloxane

Sol–gel syntheses are typically carried out in the presence of polar solvents such as alcohol or water media, which facilitate the two primary reactions of hydrolysis and condensation.

Hydrolysis of silicon alkoxides $Si(OR)_4$

$$\text{Si - O}\boxed{\text{R} + \text{HO}}\text{- H} \longrightarrow \text{Si - OH} + \text{ROH}$$

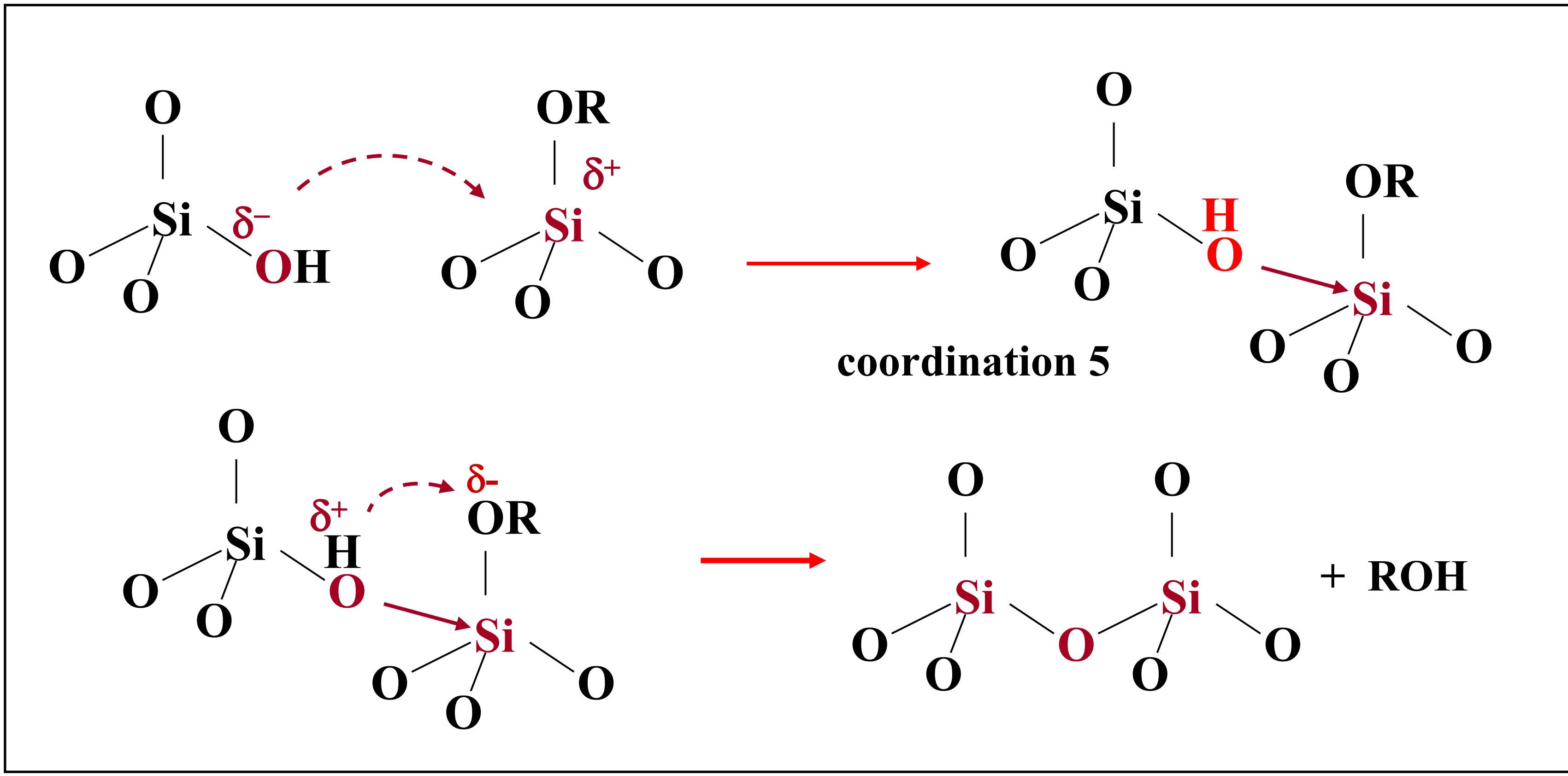
Nucleophilic substitution SN_2



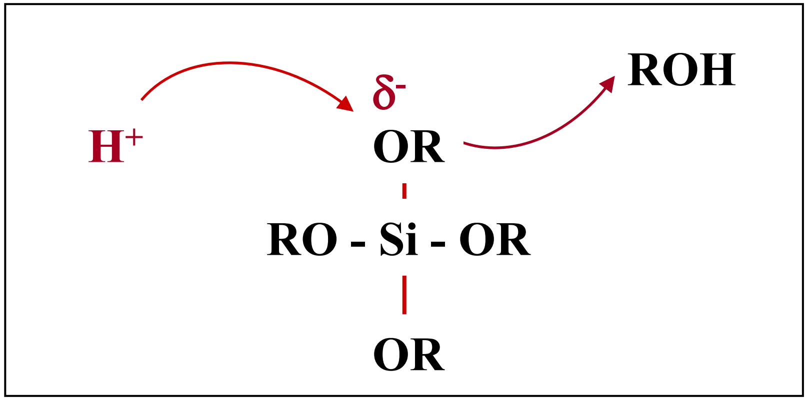
Polycondensation of silicon alkoxides $Si(OR)_4$

$$Si - O\boxed{R + HO} - Si \longrightarrow Si - O - Si + ROH$$

Nucleophilic substitution SN_2



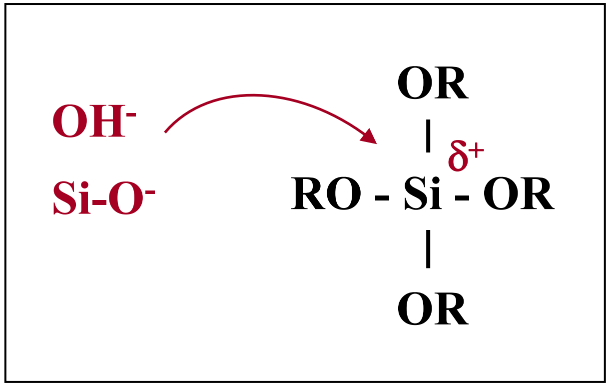
Acid catalysis (pH < 3)



protonation of Si-OH or Si-OR
that become better leaving groups

$H^+ > H_2O$

Base catalysis (pH > 3)



OH^- and Si-O^- better nucleophile
than H_2O or Si-OH

catalysis does not only speed up the reactions

it also controls the shape of the silica particles

Condensation

The condensation process is dynamic, and may be steered in the desired direction by adjusting the proper parameters.

Parameters which influences the condensation process:

- Type of precursor
- The ratio between alkoxide and water (RW)
- Type of catalyst used
- Type of solvent
- Temperature
- pH
- Relative and absolute concentrations of the reactants.

Example 1. Precursor

The stability and reactivity of the silicon alkoxides are influenced by a steric factor.

Bulky ligands slow down the hydrolysis:

Reactivity: $Si(OMe)_4 > Si(OEt)_4 > Si(OnPr)_4 > Si(OiPr)_4$

Example 2. PH

The water glass system

Three pH intervals:

$pH < 2$:
The species are positively charged and the reaction rate is proportional to $[H^+]$.

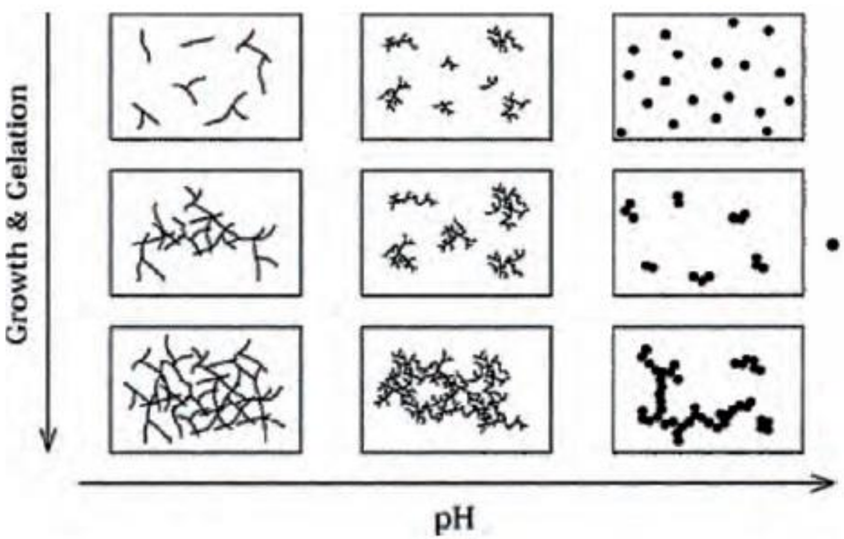
pH 2-7:
The reaction rate is proportional to $[OH^-]$.

$pH > 7$:
The re-dispersion rate is at maximum, and the solution consists of anionic species, which repels each other.

Isoelectronic point:
zero net charge
pH = 2.2 for silica

Example 3. Catalysis

- Acid-catalyzed
  - yield primarily linear or randomly branched polymer



- Base-catalyzed
  - yield highly branched clusters

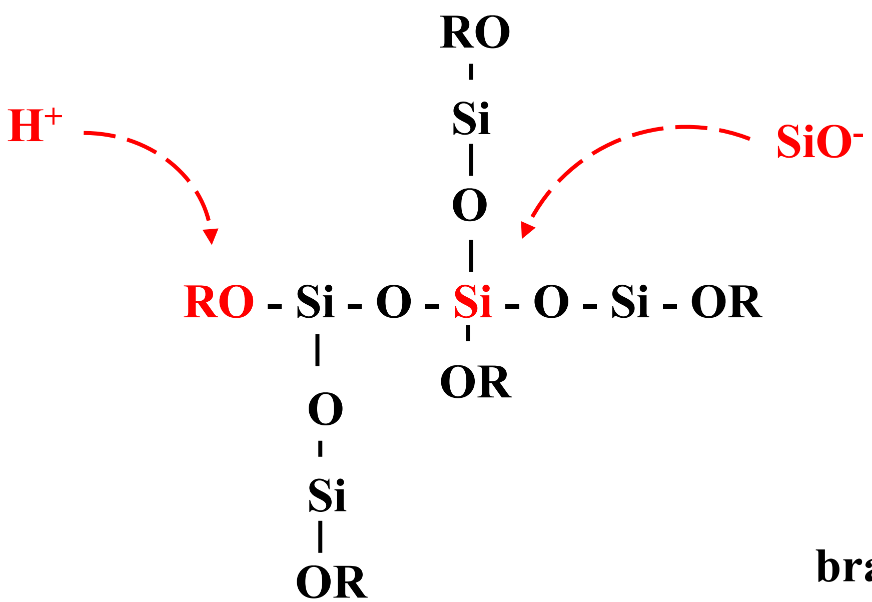
Catalysis

catalysis speeds up the reactions

And controls the shape of the silica particles

Acid catalysis

Base catalysis



end groups

↓

chain polymers

midle Si

↓

branched polymers

Catalysis controls the shape of silica particles

$$Si(OR)_4 + 4H_2O \xrightarrow{\text{hydrolysis}} Si(OH)_4 + 4ROH \xrightarrow{\text{condensation}} SiO_2 + 2H_2O$$

Acid catalysis (pH < 3)

fast hydrolysis

chain polymers

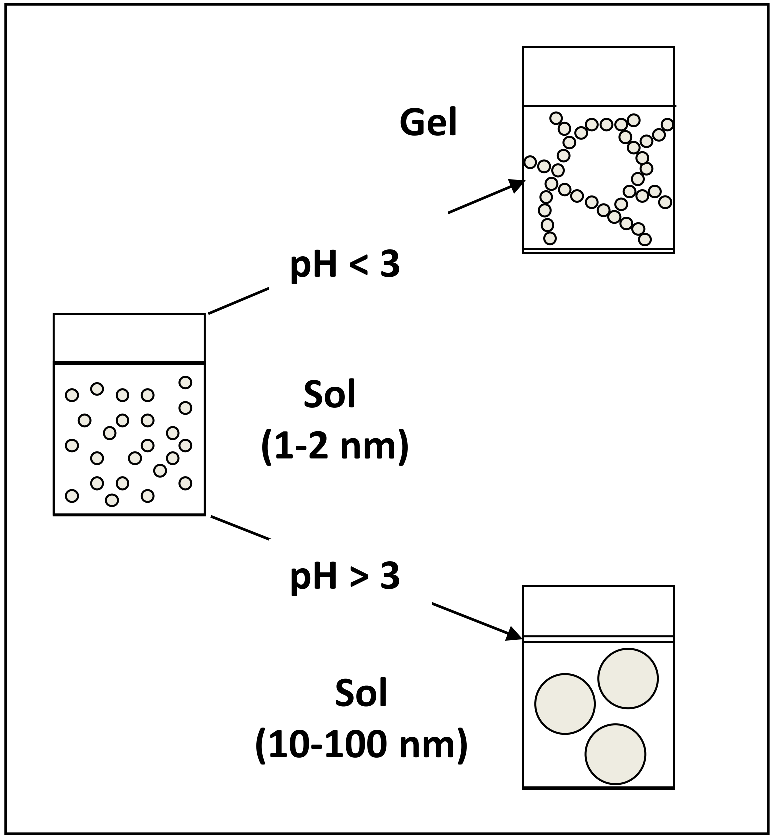
microporous gels (pores < 20Å)

Base catalysis (pH >3)

fast condensation

spherical particles (Stöber silica)

mesoporous gels (pores > 20Å)



Other metal alkoxides

$Si(OR)_4 < Sn(OR)_4 \sim Al(OR)_3 < Zr(OR)_4 < Ti(OR)_4$

- Larger and more electropositive metals are more susceptible to nucleophilic attack by water.
- The hydrolysis of most metal alkoxides is too rapid, leading to uncontrolled precipitation. Although the ratio of $H_2O/M(OR)_n$ may be tuned to control the hydrolysis rate, the nature of the metal alkoxide (e.g., altering the OR groups or metal coordination number) is the most powerful way to control the rate of hydrolysis.

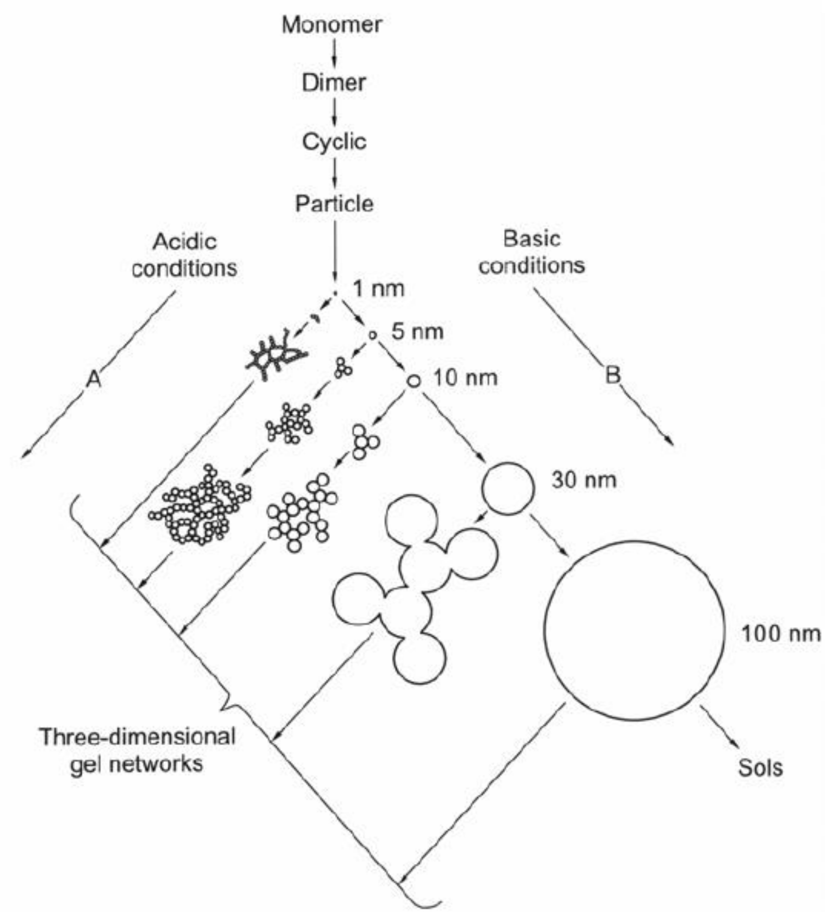
Gelation---Network formation

During reaction, objects will grow. However, a gel may not form.

As the sol aggregates the viscosity will increase until a gel is formed.

The sol-gel transition (gel-point) is reached when a continuous network is formed.

The gel-time is determined as the time when it is possible to turn the container upside-down. All fluid is kept in the gel, and the volume is maintained.



When the gel is formed, a large number of sol particles and clusters will still not have reacted. Ageing of the gel is therefore a very important stage in the process. The gel point is not a thermodynamic event.

Aging

As the viscosity rapidly increase, the solvent is “trapped” inside the gel.

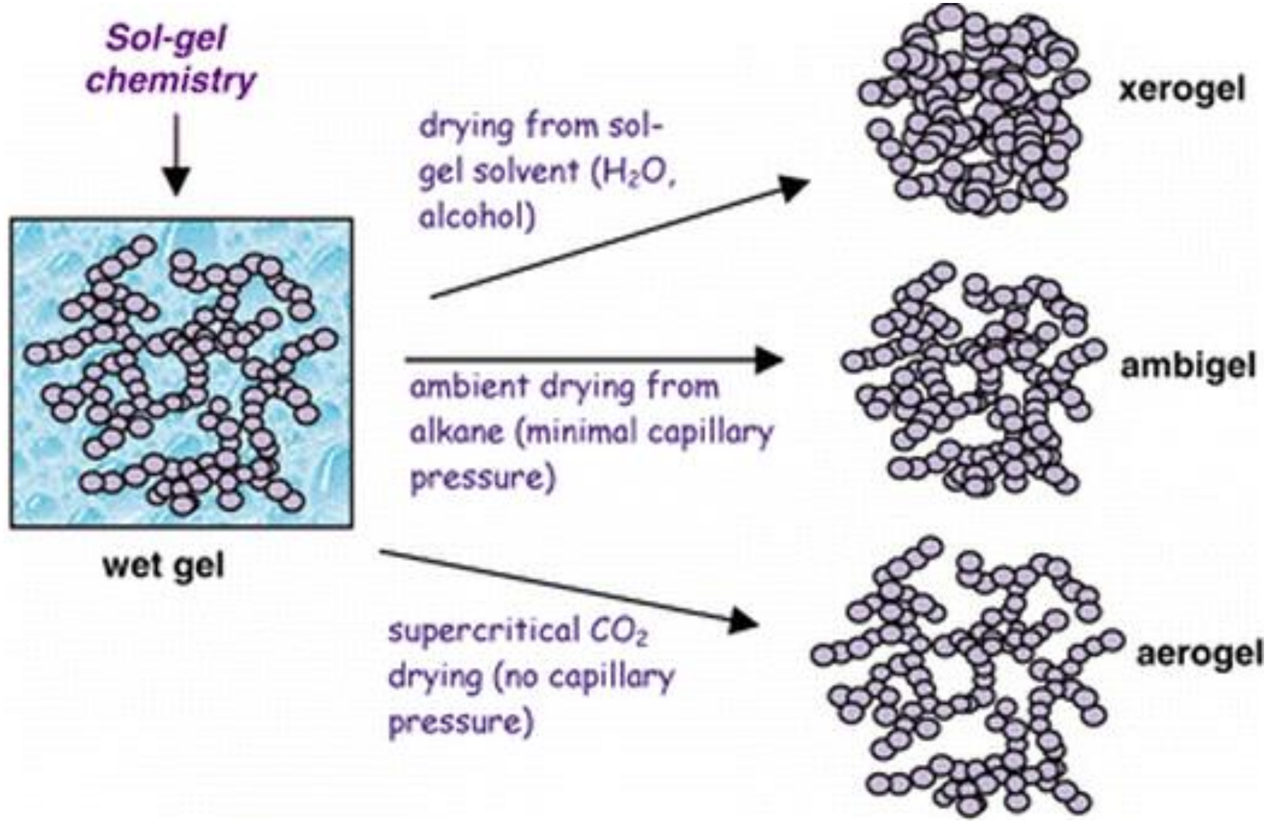
The structure may change considerably with time, depending on pH, temperature and solvent. The gel is still “alive”.

- The liquid phase still contains sol particles and agglomerates, which will continue to react, and will condense as the gel dries.

- The gel is originally flexible. Groups on neighboring branches will condense, making the gel even more viscous. This will squeeze out the liquid from the interior of the gel, end shrinkage occur. This process will continue as long as there is flexibility in the gel.

- Hydrolysis and condensation are reversible processes.

Drying



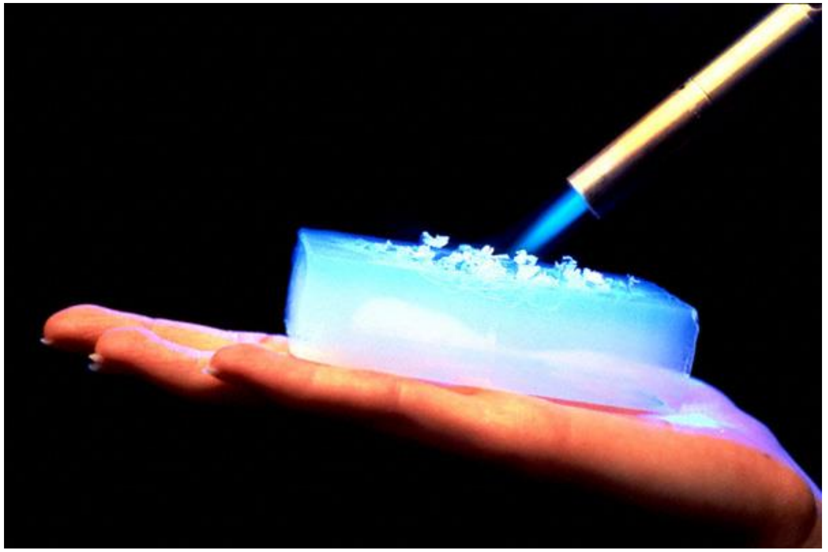
- If the solvent is removed through slow evaporation, a porous material known as a xerogel is formed.
- By contrast, if supercritical CO_2 is used to remove the solvent, a highly foam-like, porous, transparent material called an aerogel is formed.

Drying

One of the main problems in the preparation of bulk materials is avoiding cracking of the gel during drying, due to the stresses caused by the capillary forces associated with the gas-liquid interfaces. Fractures are initiated if these stress differences are greater than the tensile strength of the material.

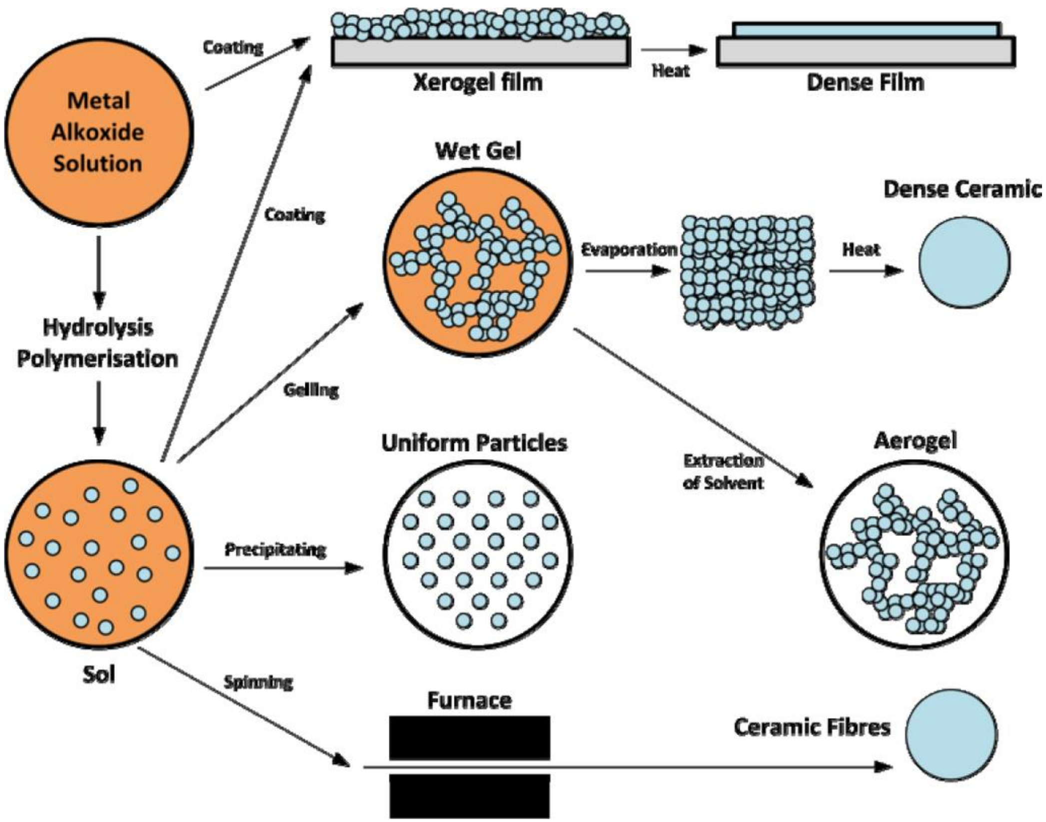
the most efficient way of neutralizing the undesired effects of surface tension is to suppress the liquid-vapor interface. This is achieved by treating the gel in an autoclave under supercritical conditions for the solvent, taking care that the path of the thermal treatment does not cross the equilibrium curve.

aerogel



- Silica aerogels consist of 99.8% air, and have a density of 1.9×10^{-3} mgL^{-1} and a thermal conductivity of 1.7×10^{-2} Wm^{-1}K^{-1}.
- The properties of aerogels afford a range of applications, among which include sound dampening, catalysis, desiccation, and thermal insulating (e.g., windows, refrigerators, walls).

Post-treatment



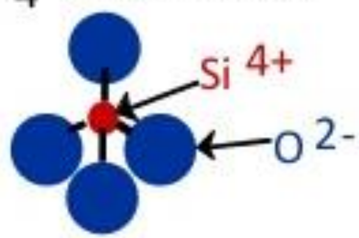
- Ultra-fine powders,
- Thin film coatings,
- Ceramic fibers,
- Microporous Inorganic,
- Membranes,
- Ceramics,
- Glasses,
- Extremely porous materials.

2.4.2 Glasses

- Glass is a hard material normally fragile and transparent. It is composed mainly of sand (silicates, SiO_2) and an alkali.
- These materials at high temperature (i.e. molten viscous state) fuse together; then they are cooled rapidly forming a rigid amorphous structure, however not having enough time to form a crystalline regular structure.

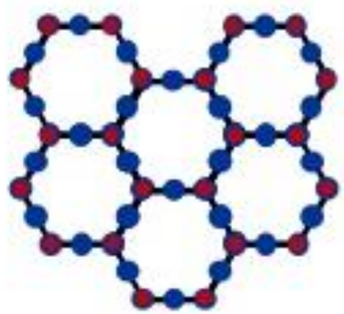
- Basic Unit:

SiO_4^{4-} tetrahedron



- Quartz is crystalline

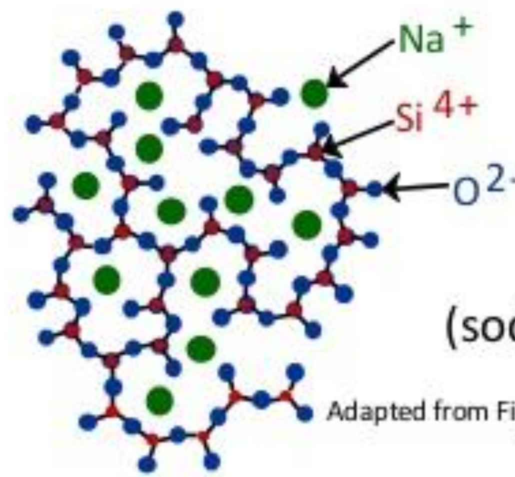
SiO_2:



- Glass is amorphous
- Amorphous structure

  occurs by adding impurities (Na^+,Mg^{2+},Ca^{2+}, Al^{3+})
- Impurities:

  interfere with formation of crystalline structure.



(soda glass)

Adapted from Fig. 12.11, *Callister, 7e*.

Ingredients

1. Sand (SiO_2 silica)

In its pure form it exists as a polymer, $(SiO_2)_n$.

2. Soda ash (sodium carbonate Na_2CO_3)

Normally SiO_2 softens up to 2000°C, where it starts to degrade (at 1713°C most of the molecules can already move freely). Adding soda will lower the melting point to 1000°C making it more manageable.

3. Limestone (calcium carbonate or $CaCO_3$) or dolomite ($MgCO_3$)

Also known as lime, calcium carbonate is found naturally as limestone, marble, or chalk.

The soda makes the glass water-soluble, soft and not very durable. Therefore lime is added increasing the hardness and chemical durability and providing insolubility of the materials.

Other materials and oxides can be added to increase properties (tinting, durability, etc.), produce different effects, colors, etc.

Properties

- Solid and hard material
- Disordered and amorphous structure
- Fragile and easily breakable into sharp pieces
- Transparent to visible light
- Inert and biologically inactive material.
- Glass is 100% recyclable and one of the safest packaging materials due to its composition and properties

These properties can be modified and changed by adding other compounds or heat treatment.

Types of glasses

- **Commercial glass or Soda-lime glass:**

-This is the most common commercial glass and less expensive. The composition of soda-lime glass is normally 60-75% silica, 12-18% soda, and 5-12% lime. A low percentage of other materials can be added for specific properties such as coloring.

- It has light transmission appropriate to be use in flat glass in windows;

- It has a smooth and nonporous surface that allows glass bottles and packaging glass to be easily cleaned;

- Soda-lime glass containers are virtually inert, resistant to chemical attack from aqueous solutions so they will not contaminate the contents inside or affect the taste.

-Whereas pure glass SiO_2 does not absorb UV light, soda-lime glass does not allow light at a wavelength of lower than 400 nm (UV light) to pass.

-The disadvantages of soda-lime glass is that is not resistant to high temperatures and sudden thermal changes. For example, everybody has experienced a glass breaking down when pouring liquid at high temperature, for example to make tea.

-Some of the use of soda-lime glass is primarily used for bottles, jars, everyday drinking glasses, and window glass.

• Lead glass:

-Lead glass is composed of 54-65% SiO_2, 18-38% lead oxide (PbO), 13-15% soda (Na_2O) or potash (K_2CO_3), and various other oxides. When the content of PbO is less than 18% is known as crystal glass.

- In moderate amounts lead increases durability;

- In high amounts it lowers the melting point and decreases the hardness giving a soft surface;

- In addition it has a high refractive index giving high brilliance glass.

-These two last properties make it appropriate for decorating purposes.

-Glass with high lead oxide contents (i.e. 65%) may be used as radiation shielding glass because lead absorb gamma rays and other forms of harmful radiation, for example, for nuclear industry.

-As with soda-lime glass, lead glass will not withstand high temperatures or sudden changes in temperature.

• Borosilicate glass:

-Borosilicate glass is mainly composed of silica (70-80%), boric oxide B_2O_3 (7-13%) and smaller amounts of the alkalis (sodium and potassium oxides) such as 4-8% of Na_2O and K_2O, and 2-7% aluminum oxide (Al_2O_3).

-Boron gives greater resistance to thermal changes and chemical corrosion.

-It is suitable for industrial chemical process plants, in laboratories, in the pharmaceutical industry, in bulbs for high-powered lamps, etc. Borosilicate glass is also used in the home for cooking plates and other heat-resistant products. It is used for domestic kitchens and chemistry laboratories, this is because it has greater resistance to thermal shock and allows for greater accuracy in laboratory measurements when heating and cooling experiments.

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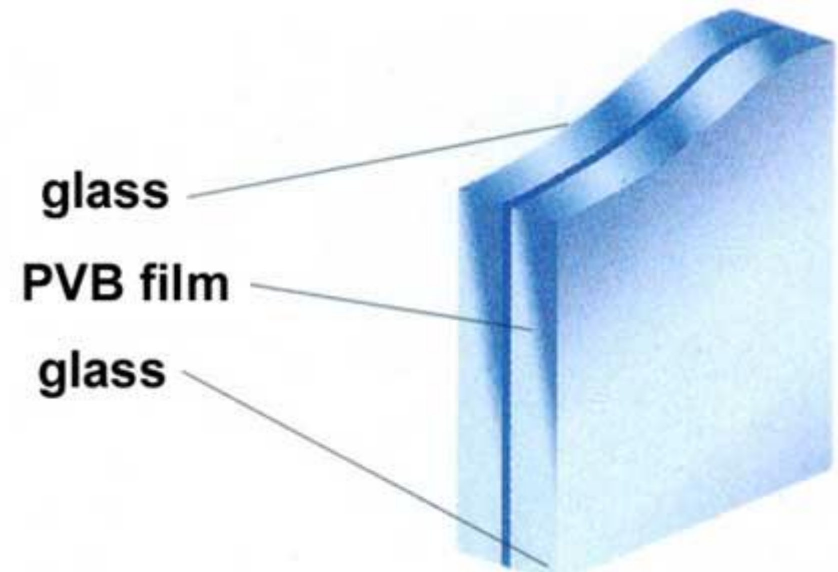
Solid-State Chemistry

- Safety glass, constructed of two pieces of plate glass join by a plastic to prevent the glass from scattering when broken.



- Fiberglass made from molten glass formed into continuous filaments that is used for fabrics or electrical insulation

- Foam glass made by trapping gas bubbles in glass to produce a spongy material for insulating purposes.

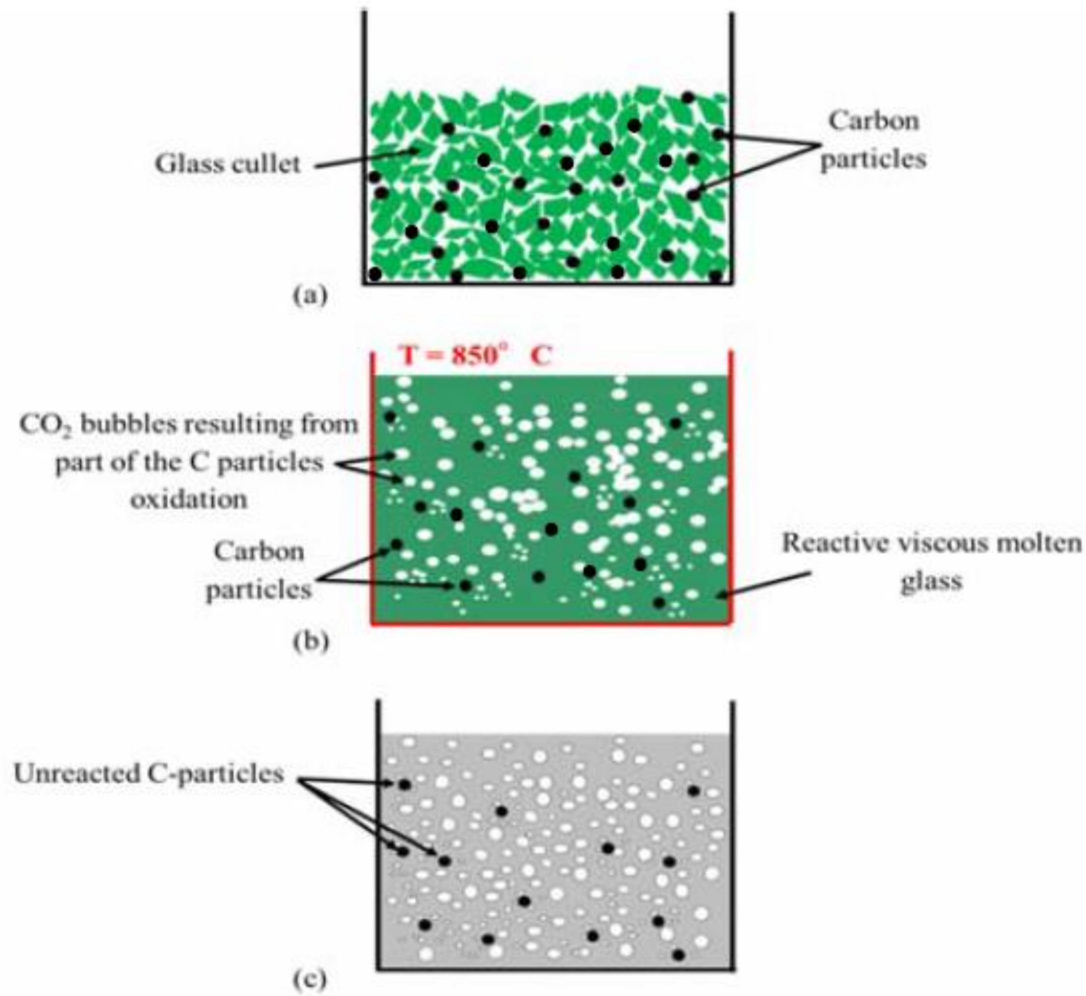
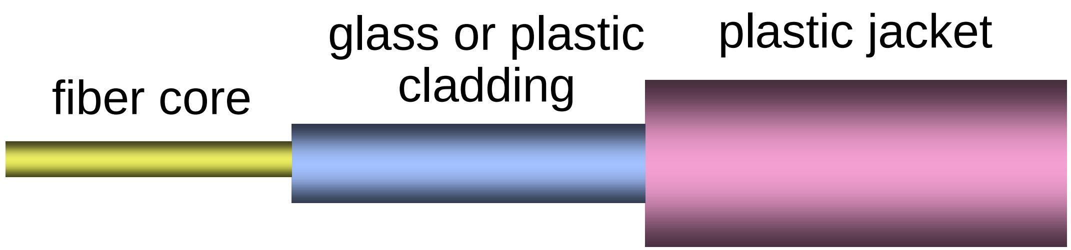

Glass cullet
Carbon particles
(a)
T = 850° C
CO2 bubbles resulting from part of the C particles oxidation
Carbon particles
Reactive viscous molten glass
(b)
Unreacted C-particles
(c)

Fig. 1

- **Optical fibers:**
  - ✓ You hear about fiber-optic cables whenever people talk about the telephone system, the cable TV system or the Internet.
  - ✓ They are also used in medical imaging and mechanical engineering inspection.
  - ✓ Fiber-optic lines are strands of optically pure glass as thin as a human hair that carry digital information over long distances.
  - ✓ They are arranged in bundles called optical cables and used to transmit light signals over long distances.

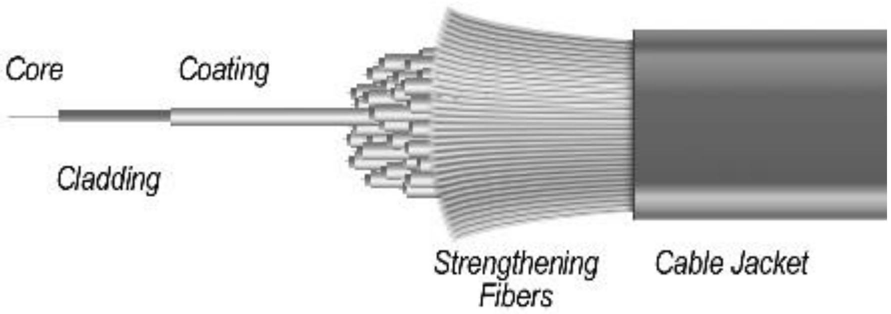


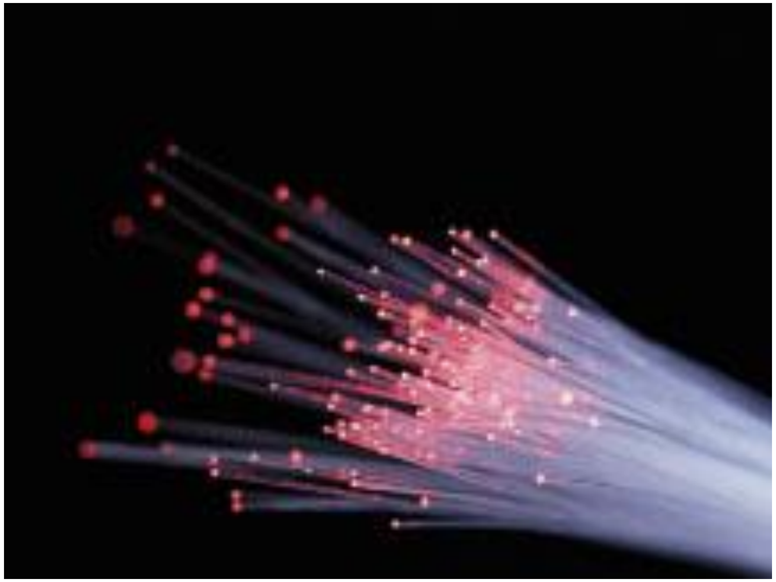
If you look closely at a single optical fiber, you will see that it has the following parts:

Core - Thin glass center of the fiber where the light travels

Cladding - Outer optical material surrounding the core that reflects the light back into the core

Buffer coating - Plastic coating that protects the fiber from damage and moisture

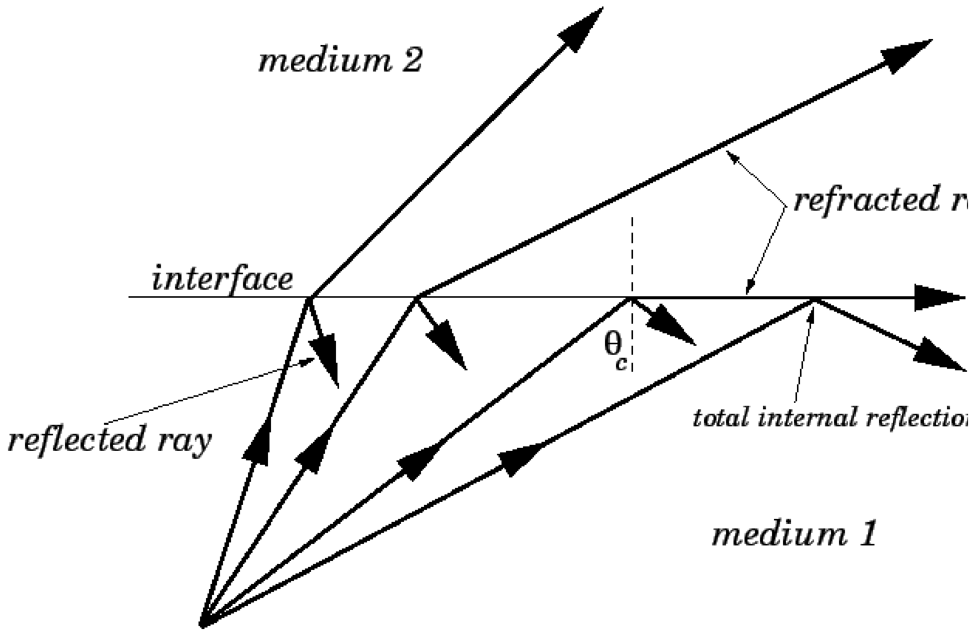

Core
Coating
Cladding
Strengthening Fibers
Cable Jacket

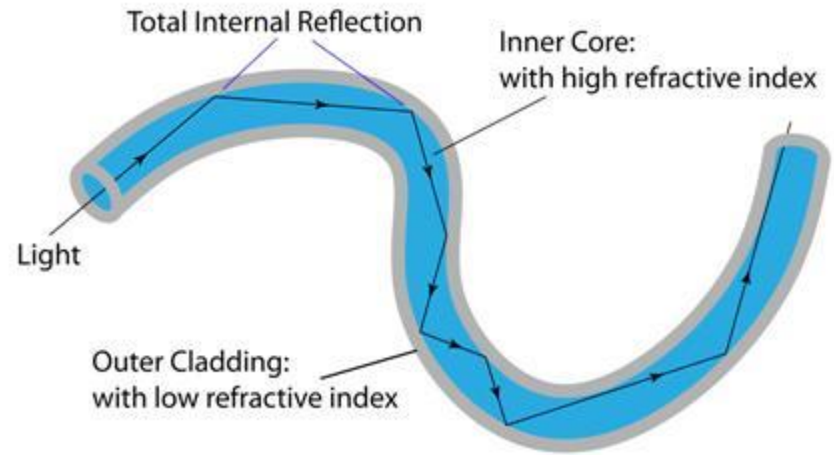


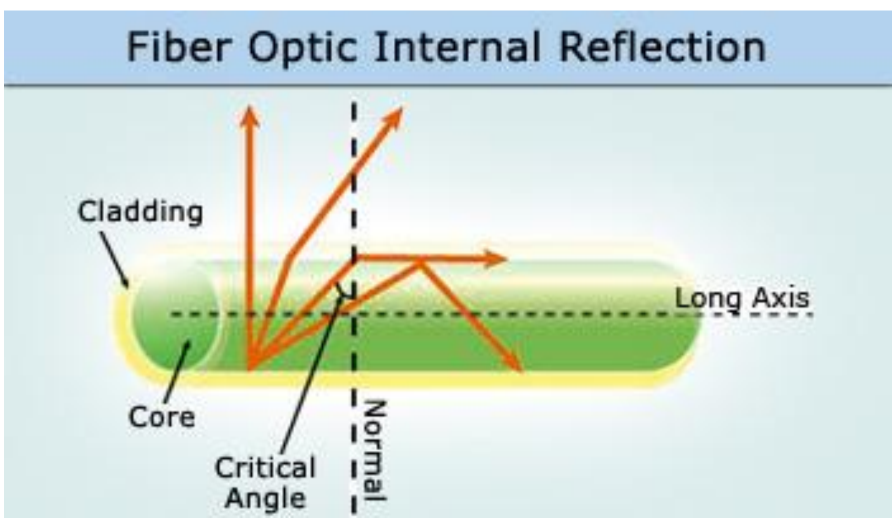
How Does an Optical Fiber Transmit Light?

- The light in a fiber-optic cable travels through the core (hallway) by constantly bouncing from the cladding (mirror-lined walls), a principle called **total internal reflection**.
- Because the cladding does not absorb any light from the core, the light wave can travel great distances.
- However, some of the light signal **degrades** within the fiber, mostly due to impurities in the glass. The extent that the signal degrades depends on the purity of the glass and the wavelength of the transmitted light

total internal reflection

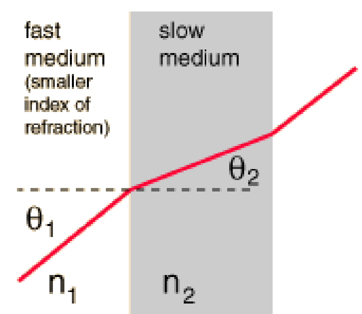






Snell's Law

$$\frac{n_1}{n_2} = \frac{\sin\theta_2}{\sin\theta_1}$$



Advantages

- Less expensive .
- Thinner
- Higher carrying capacity
- Less signal degradation
- Lightweight
- Flexible Medical imaging

    - in bronchoscopes, endoscopes, laparoscopes
- Mechanical imaging

    - inspecting mechanical welds in pipes and engines (in airplanes, rockets, space shuttles, cars)
- Plumbing

    - to inspect sewer lines

How Are Optical Fibers Made?

- Now that we know how fiber-optic systems work and why they are useful -- how do they make them? Optical fibers are made of extremely pure optical glass.

- We think of a glass window as transparent, but the thicker the glass gets, the less transparent it becomes due to impurities in the glass.

- However, the glass in an optical fiber has far fewer impurities than window-pane glass.

  One company's description of the quality of glass is as follows: If you were on top of an ocean that is miles of solid core optical fiber glass, you could see the bottom clearly.

Solid-State Chemistry

2.4.3 Cementitious Materials

The primary difference between concrete and cement is:

- Concrete is a composite material made of water, aggregate, and cement.
- Cement is a very fine powder made of limestone and other minerals, which absorbs water and acts as a binder to hold the concrete together.
- While cement is a construction material in its own right, concrete cannot be made without cement. The two terms often are incorrectly used interchangeably, but concrete and cement are distinctly separate products.

3000 BC – Egyptian Pyramids

A type of cement was used to hold together the limestone blocks of the great pyramids.

300 BC – 476 AD-Roman Architecture

The ancient Romans used Pozzolana ash that is remarkably close to modern cement to build many of their architectural marvels, such as the Colosseum, and the Pantheon.

1824-Portland Cement Invented

- There are many different types of cement, but the type most commonly used in construction is Portland cement.
- Joseph Aspdin of Britain developed the building material in the 1700s, when he found that adding clay to limestone and superheating the mixture allowed the resulting blend to set anywhere.
- Portland cement is a type of hydraulic cement, which means that when water is added, it starts a chemical reaction that is not dependent on how much water there is. This allows the cement to harden underwater and remain strong even in wet conditions.

Cement

- Cement is made from limestone, calcium, silicon, iron, and aluminum, among other ingredients.
- This mixture is heated in large kilns to about 2,700°F (1,482°C) to form a product known as clinkers, which roughly resemble marbles. These are ground into a powder and gypsum ($CaSO_4 \cdot 2H_2O$ – added to clinker to control the hardening rate) is added, creating the gray flour-like substance known as cement. When water is added to cement, it triggers a chemical process that allows it to harden.

Chemical Reactions Involved in the Manufacture of Cement

Following chemical reactions take place during the manufacture of cement:

1. First of all limestone decomposes into lime after getting heated to a high temperature.

$$\underset{\text{Calcium carbonate}}{CaCO_3} \xrightarrow{\text{Heat}} \underset{\text{Calcium oxide}}{CaO} + \underset{\text{Carbon dioxide}}{CO_2}$$

2. The lime produced in first reaction reacts with silicon dioxide to produce dicalcium silicate.

$$\underset{\text{Lime}}{2CaO} + \underset{\text{Silicon dioxide}}{SiO_2} \longrightarrow \underset{\text{Dicalcium silicate}}{2CaO.SiO_2}$$

3. Lime also reacts with silicon dioxide to produce one more compound tricalcium silicate.

$$\underset{\text{Lime}}{3CaO} + \underset{\text{Silicon dioxide}}{SiO_2} \longrightarrow \underset{\text{Tricalcium silicate}}{3CaO.SiO_2}$$

4. Lime also reacts with aluminum oxide to form tricalcium aluminate.

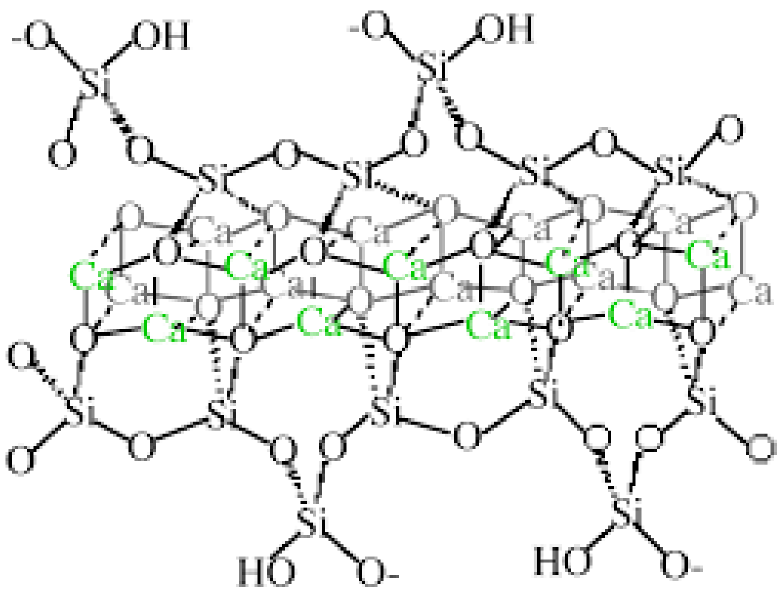
$$\underset{\text{Lime}}{3CaO} + \underset{\text{Aluminium oxide}}{Al_2O_3} \longrightarrow \underset{\text{Tricalcium aluminate}}{3CaO.Al_2O_3}$$

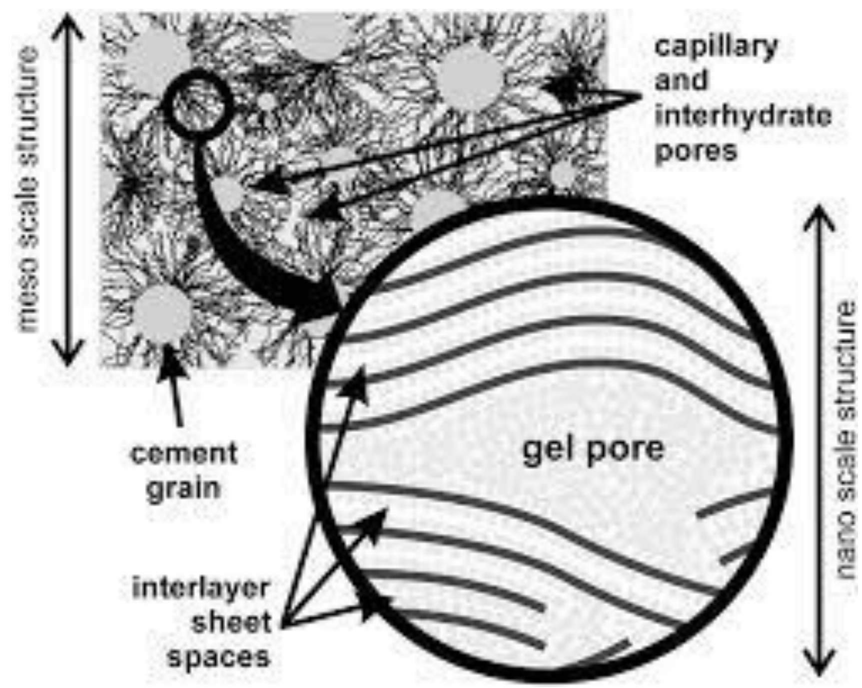
5. In the last step calcium oxide, aluminium oxide and ferric oxide react together to form cement.

$$\underset{\text{Calcium Oxide}}{4CaO} + \underset{\text{Aluminium oxide}}{Al_2O_3} + \underset{\text{Ferric oxide}}{Fe_2O_3} \longrightarrow \underset{\text{Cement}}{4CaO.Al_2O_3.Fe_2O_3}$$

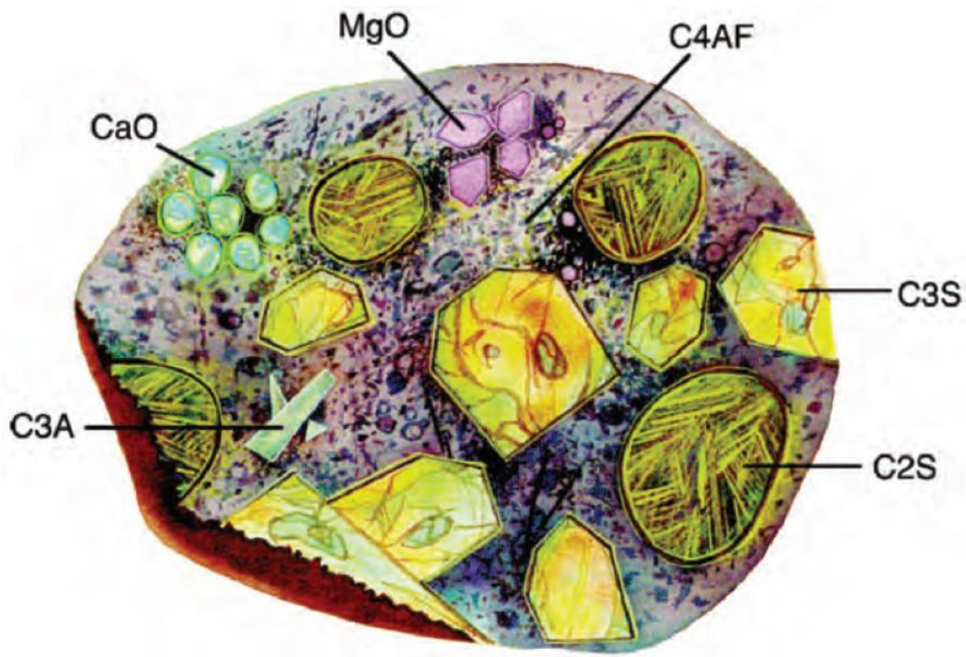
Hydration Reactions

2 ($3CaO{\bullet}SiO_2$) Tricalcium silicate	+ 11 H_2O Water	= $3CaO{\bullet}2SiO_2{\bullet}8H_2O$ Calcium silicate hydrate (C-S-H)	+ 3 ($CaO{\bullet}H_2O$) Calcium hydroxide
2 ($2CaO{\bullet}SiO_2$) Dicalcium silicate	+ 9 H_2O Water	= $3CaO{\bullet}2SiO_2{\bullet}8H_2O$ Calcium silicate hydrate (C-S-H)	+ $CaO{\bullet}H_2O$ Calcium hydroxide
$3CaO{\bullet}Al_2O_3$ Tricalcium aluminate	+ 3 ($CaO{\bullet}SO_3{\bullet}2H_2O$) Gypsum	+ 26 H_2O Water	= $6CaO{\bullet}Al_2O_3{\bullet}3SO_3{\bullet}32H_2O$ Ettringite
2 ($3CaO{\bullet}Al_2O_3$) Tricalcium aluminate	+ $6CaO{\bullet}Al_2O_3{\bullet}3SO_3{\bullet}32H_2O$ Ettringite	+ 4 H_2O Water	= 3 ($4CaO{\bullet}Al_2O_3{\bullet}SO_3{\bullet}12H_2O$) Calcium monosulfoaluminate
$3CaO{\bullet}Al_2O_3$ Tricalcium aluminate	+ $CaO{\bullet}H_2O$ Calcium hydroxide	+ 12 H_2O Water	= $4CaO{\bullet}Al_2O_3{\bullet}13H_2O$ Tetracalcium aluminate hydrate
$4CaO{\bullet}Al_2O_3{\bullet}Fe_2O_3$ Tetracalcium aluminoferrite	+ 10 H_2O Water	+ 2 ($CaO{\bullet}H_2O$) Calcium hydroxide	= $6CaO{\bullet}Al_2O_3{\bullet}Fe_2O_3{\bullet}12H_2O$ Calcium aluminoferrite hydrate



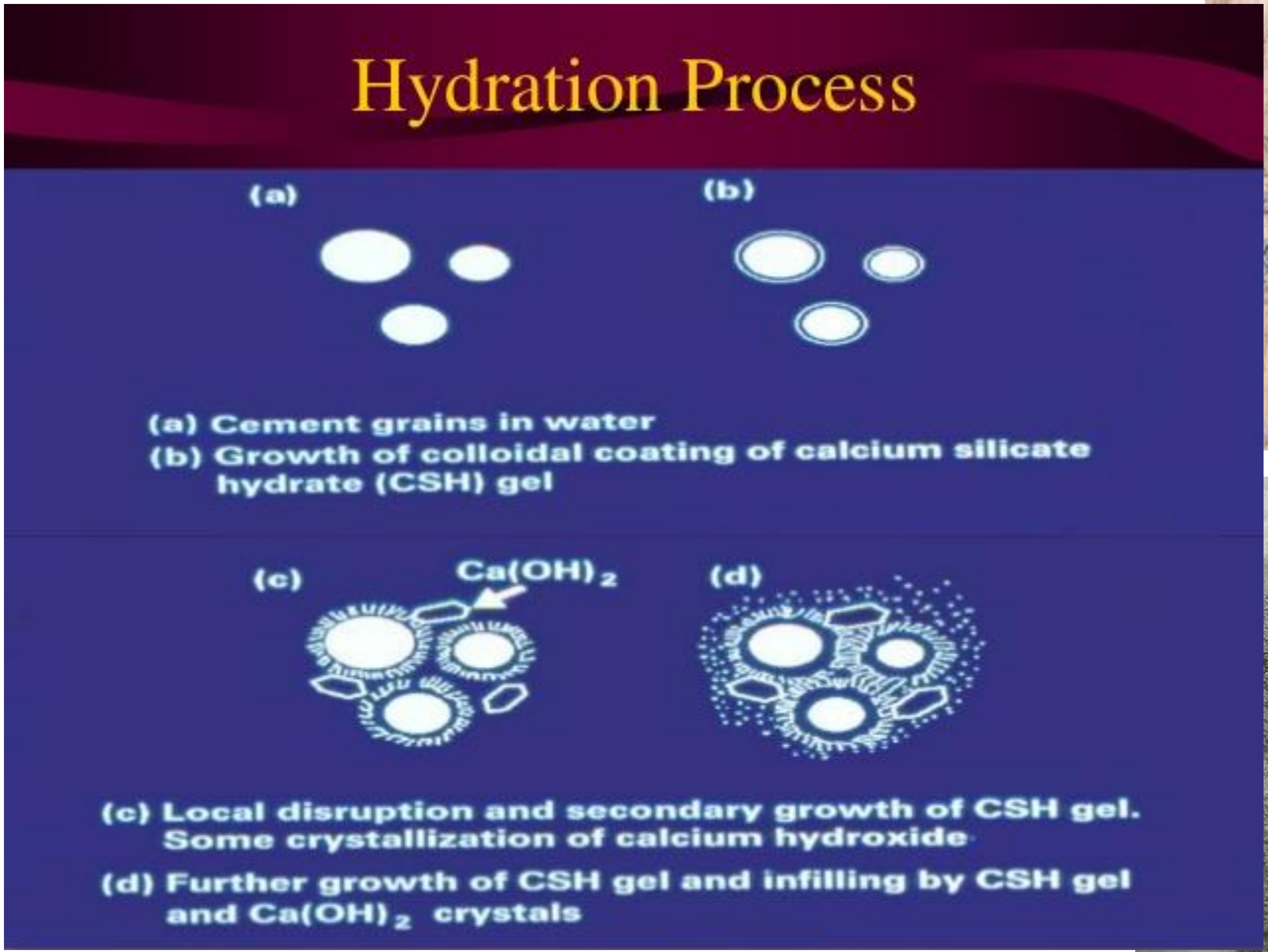


Calcium-silicate hydrate (C-S-H) gel



$3CaO.SiO_2$	C3S	50-70%
$2CaO.SiO_2$	C2S	15-30%
$3CaO.Al_2O_3$	C3A	5-10%
$4CaO.Al_nFe_{2-n}O_3$	C4AF	5-15%
Other additives	CaO or MgO	3-8%

- It is the hydration of the calcium silicate, aluminate, and aluminoferrite minerals that causes the hardening, or setting, of cement.
- The ratio of C3S to C2S helps to determine how fast the cement will set, with faster setting occurring with higher C3S contents.
- Lower C3A content promotes resistance to sulfates.
- Higher amounts of ferrite lead to slower hydration.
- The ferrite phase causes the brownish gray color in cements, so that “white cements” (i.e., those that are low in C4AF) are often used for aesthetic purposes.


Hydration Process
(a)
(b)
(a) Cement grains in water
(b) Growth of colloidal coating of calcium silicate hydrate (CSH) gel
(c)
Ca(OH)2
(d)
(c) Local disruption and secondary growth of CSH gel. Some crystallization of calcium hydroxide
(d) Further growth of CSH gel and infilling by CSH gel and Ca(OH)2 crystals

Concrete

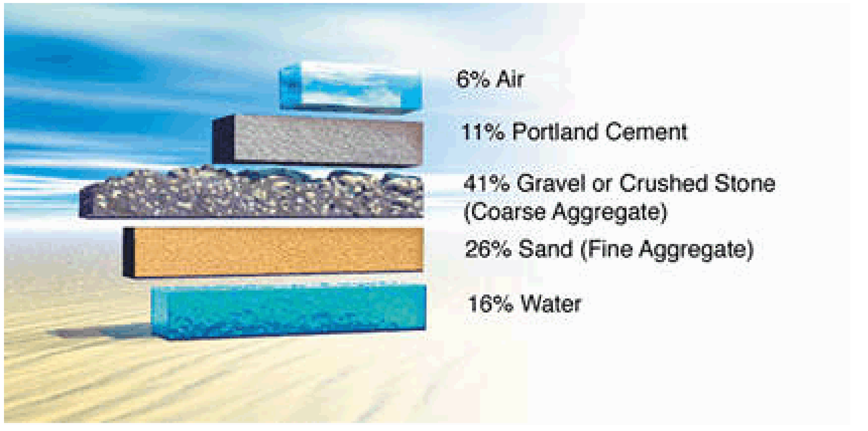
Concrete, is a masonry material that uses cement to bind together crushed stone, rock, and sand, also called aggregate.

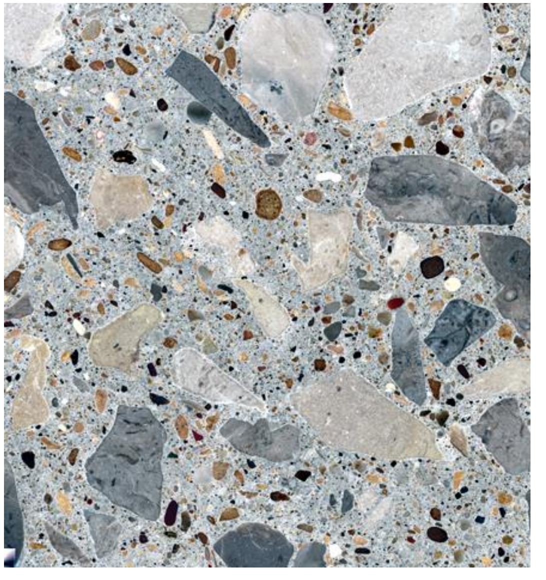
Cement makes up from 10% to 15% of the total mass of concrete; the exact proportions vary depending on the type of concrete being made.

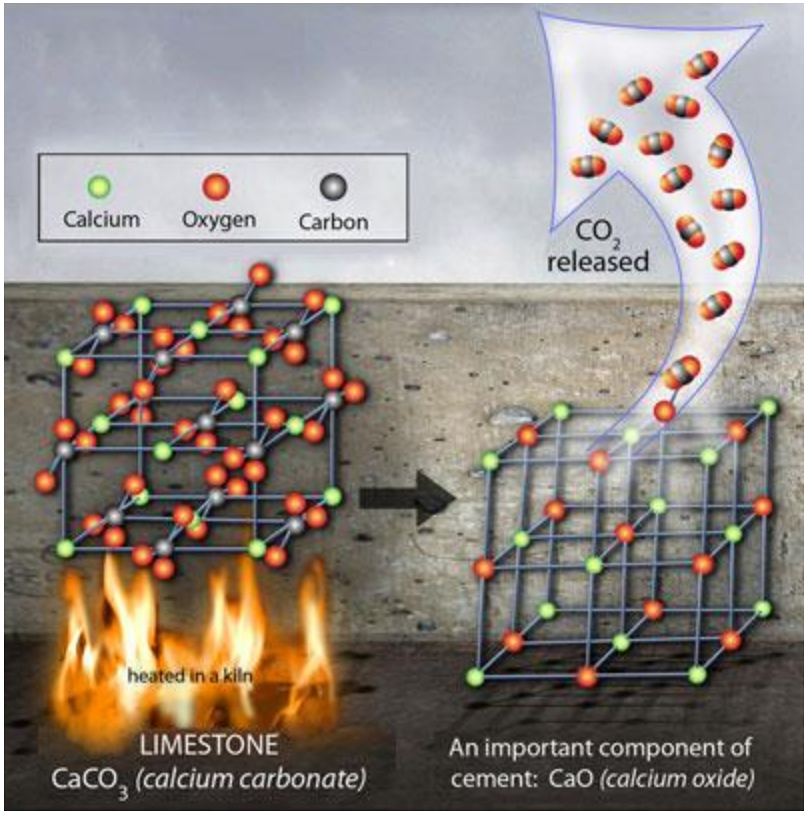
The aggregate and cement are mixed thoroughly with water, which starts the chemical reaction causing the cement to harden and set. Before this happens, the concrete mix can be poured into a mold so that it will harden in a specific shape, be it a block or a slab.

Concrete

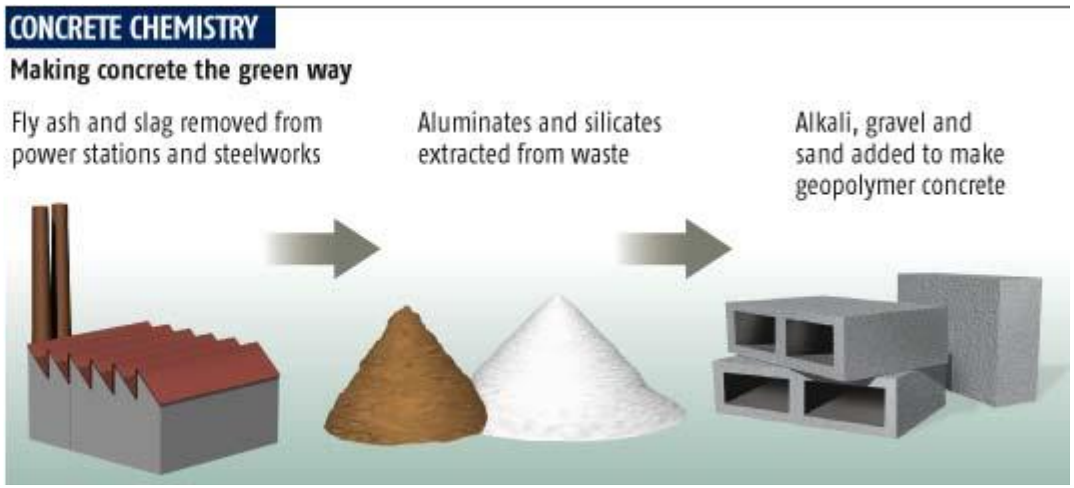
- A mixture of:
  - Portland Cement
  - Fine Aggregate
  - Coarse Aggregate
  - Water
  - Air
- Cement and water combine, changing from a moist, plastic consistency to a strong, durable rock-like construction material by means of a chemical reaction called "hydration"







The releasing of the greenhouse gas CO_2 causes environmental concerns over the world's increasing production of cement. It is estimated that Portland cement manufacturing accounts for over 10% of the world's total emission of CO_2.



There is an increasing focus on cements fabricated with fly ash (e.g., by-product from coal-fired power plants), that possess the same qualities as Portland cement clinker without the need for the calcination step.

- There are various types of concrete for different applications that are created by changing the proportions of the main ingredients.

- The mix design depends on the type of structure being built, how the concrete will be mixed and delivered, and how it will be placed to form the structure.

- Examples include:

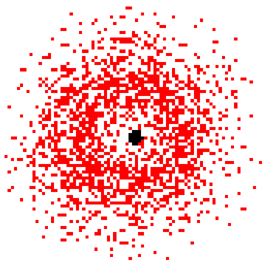
- Regular concrete
- Pre-Mixed concrete
- High-strength concrete
- Stamped concrete
- High-Performance concrete
- UHPC (Ultra-High Performance Concrete)
- Self-consolidating concretes
- Vacuum concretes
- Shotcrete
- Cellular concrete
- Roller-compacted concrete
- Glass concrete
- Asphalt concrete
- Rapid strength concrete
- Rubberized concrete
- Polymer concrete
- Geopolymer or Green concrete
- Limecrete
- Gypsum concrete
- Light-Transmitting Concrete

Review

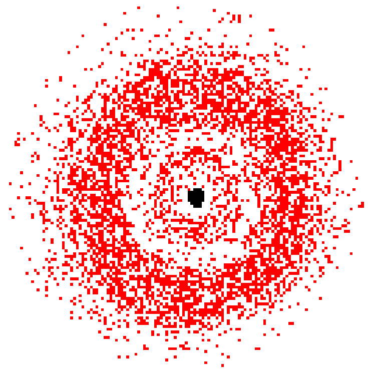
- Types of bonding in solids
  - -definition and properties
- Crystalline Solid
  - -Unit cells, Miller indices, Bravais unit cells.
  - -Lattices: close packing and interstitial sites.
  - -Symmetry elements
  - -Defects: definition and resultant properties.
  - -Properties
- Amorphous solid
  - -sol-gel: definition, reaction mechanism.
  - -glass: integration and properties.
  - -Cementitious materials: brief chemistry.

Optional Content

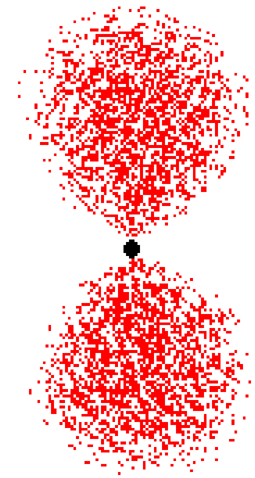
Atomic orbital and energy level



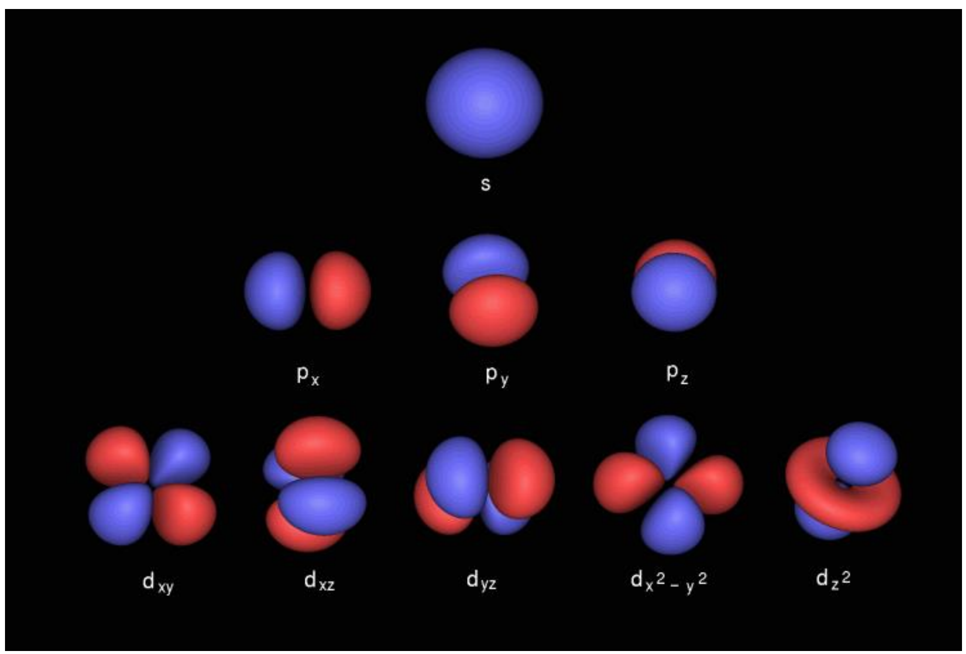
a 1s orbital



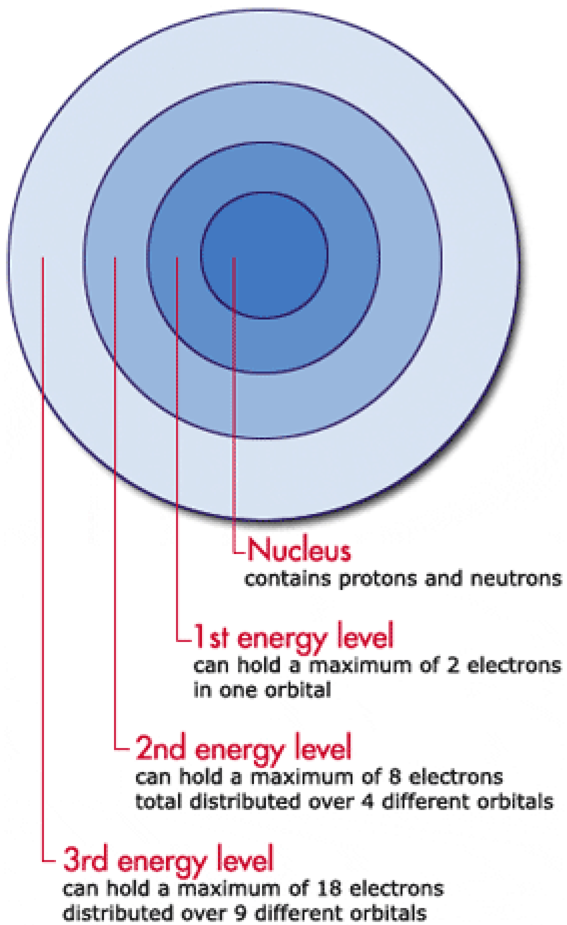
a 2s orbital

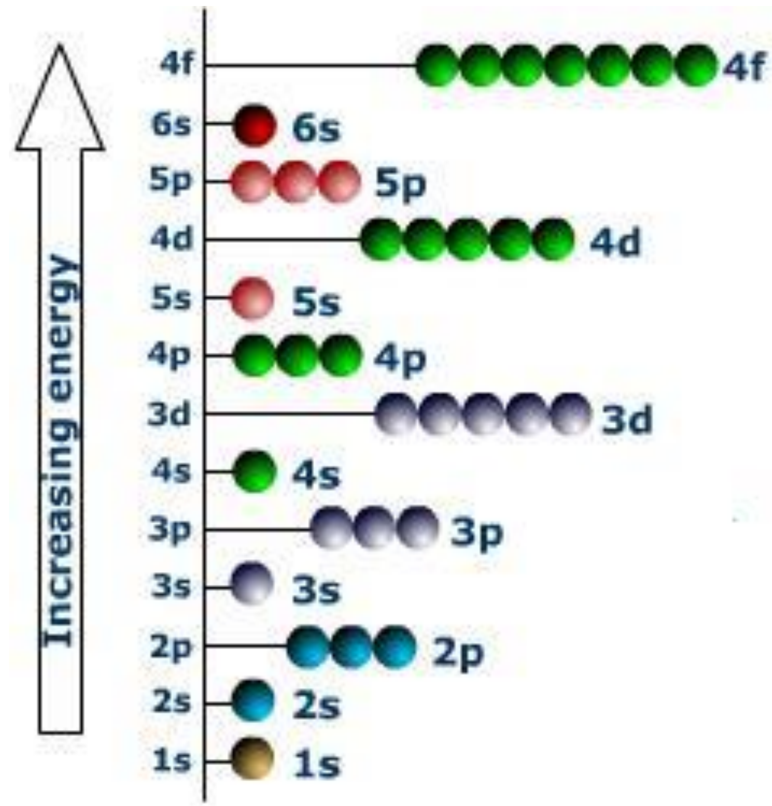


a p orbital

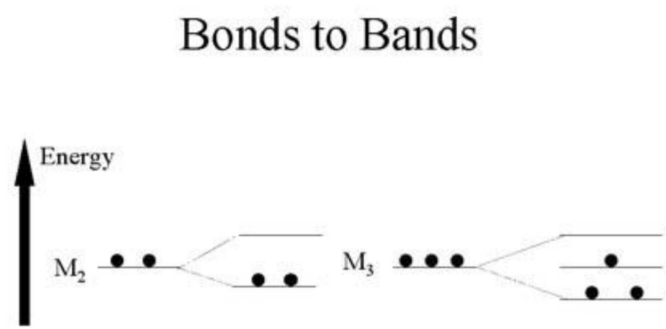


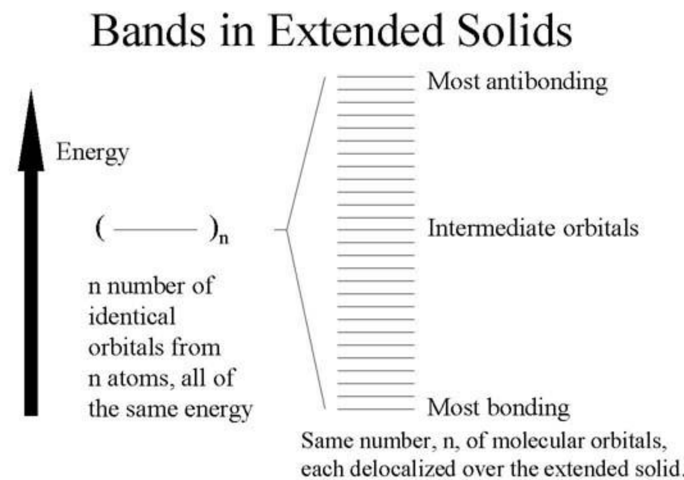
Hund's Rule

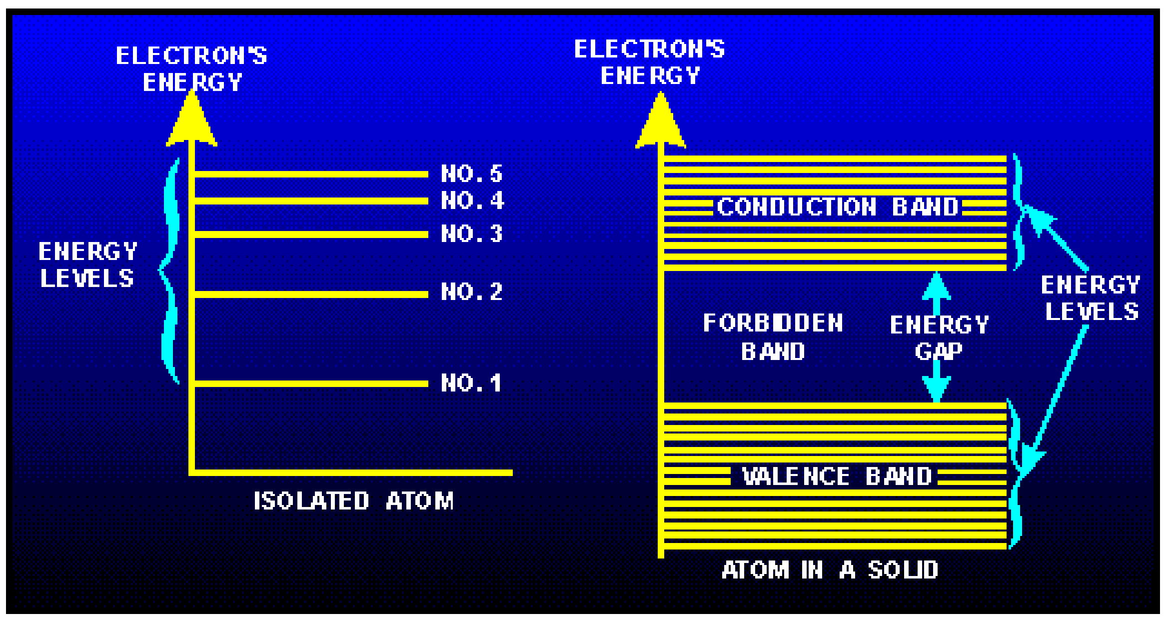

Nucleus
contains protons and neutrons
1st energy level
can hold a maximum of 2 electrons
in one orbital
2nd energy level
can hold a maximum of 8 electrons
total distributed over 4 different orbitals
3rd energy level
can hold a maximum of 18 electrons
distributed over 9 different orbitals


Increasing energy
4f
6s
5p
4d
5s
4p
3d
4s
3p
3s
2p
2s
1s

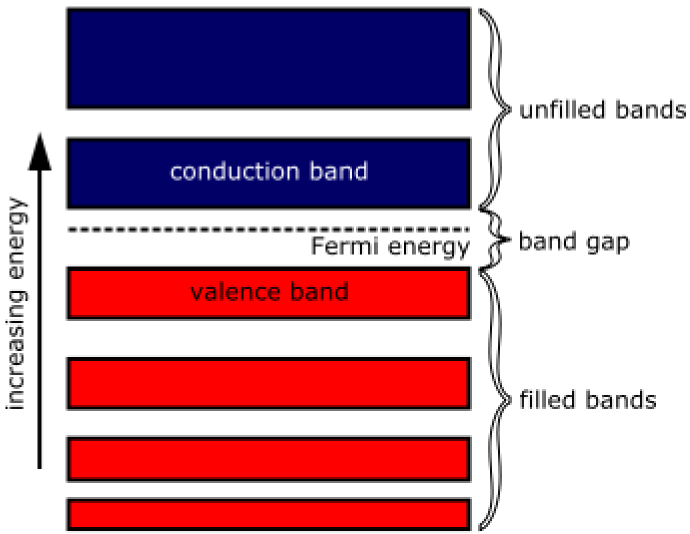
Solid Energy Band







Solid Energy Band



Highest occupied molecular orbitals (HOMOs)

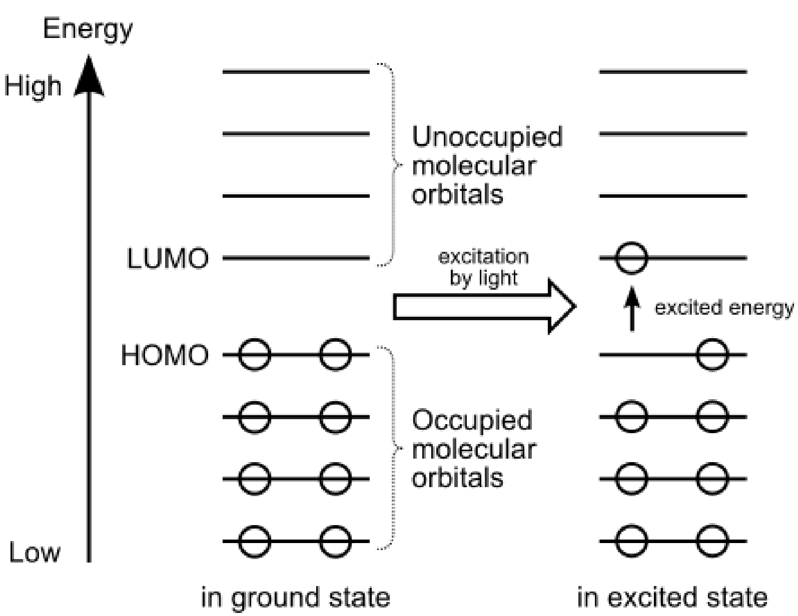
CONDUCTION BAND (unfilled band): Electrons in this band are easily removed by the application of external electric fields.

FORBIDDEN BAND or ENERGY GAP: Electrons are never found in this band, but may travel back and forth through it.

VALENCE BAND (electron-occupied band): is composed of a series of energy levels containing valence electrons, which are more tightly bound to the individual atom than the electrons in the conduction band. However, the electrons in the valence band can still be moved to the conduction band with the application of energy, usually thermal energy.

Solid Energy Band

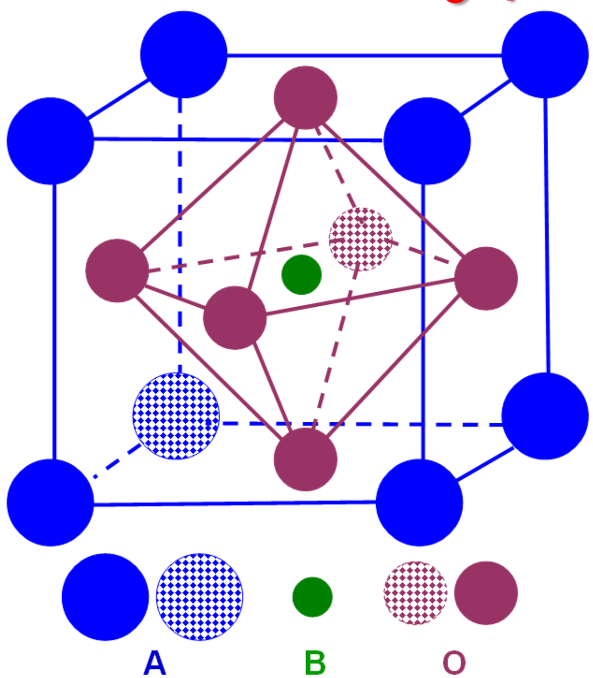
There are more bands below the valence band, but they are not important to the understanding of semiconductor theory and will not be discussed.

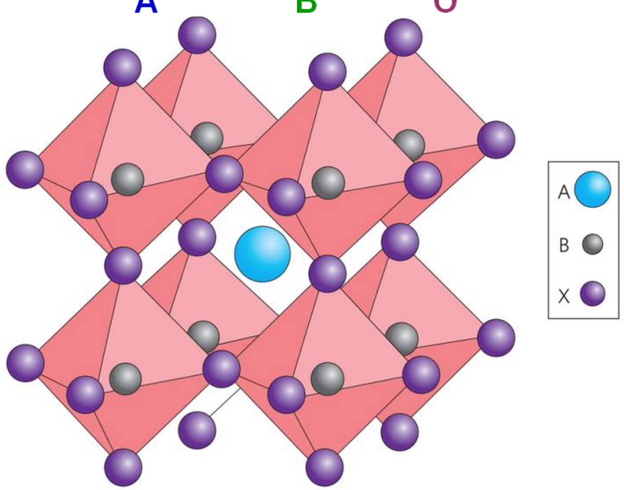


Lowest unoccupied molecular orbitals (LUMOs)

Highest occupied molecular orbitals (HOMOs)

Perovskite ABO_3 (A is a cation of larger size than B)





- A FCC arrangement of both oxide and the larger cation.
- The smaller cation occupies the octahedral hole at the position (½, ½, ½).
- High temperature super-conductors

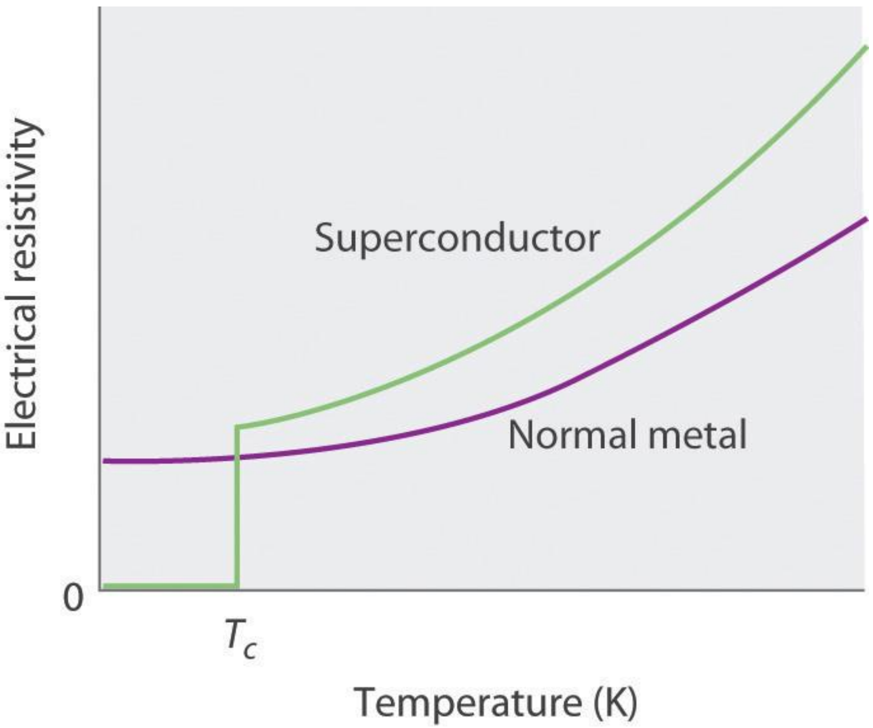
In the movie “Avatar” Hallelujah floating mountains

The **Hallelujah Mountains** (Na'vi name: ***Ayram alusìng*** meaning "Floating Mountains")[1] are floating islands that circulate slowly in the magnetic currents like icebergs at sea, scraping against each other and the towering mesa-like mountains of the region. On Pandora, huge outcroppings of unobtanium rip loose from the surface and float in the magnetic vortices due to the Meissner Effect

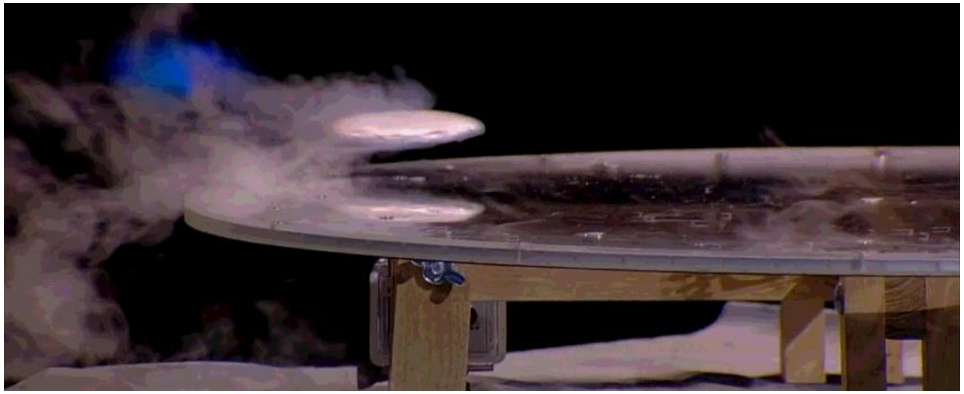
$$coductivity = \frac{1}{resistivity}$$

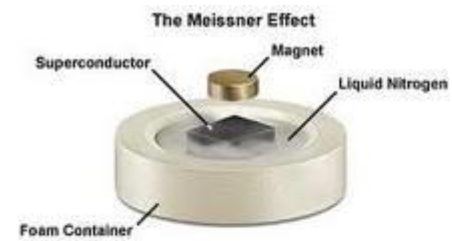
Superconductor is a material that show a resistivity of zero at temperatures approaching absolute zero.

Each of these superconductors has a characteristic superconducting transition temperature (Tc) at which its resistivity drops to zero.



H. Kamerlingh Onnes (1853–1926; Danish physicist, Nobel Prize in Physics, 1913)

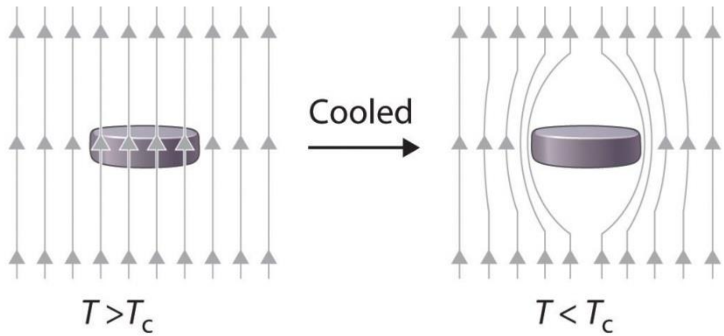




Meissner Effect: Exhibiting diamagnetic properties to the *total* exclusion of all magnetic fields. (Named for Walter Meissner.) This is a classic hallmark of superconductivity and can actually be used to levitate a strong rare-earth magnet.

Walter Hans. Meissner

Robert Ochsenfeld



(a)

(b)

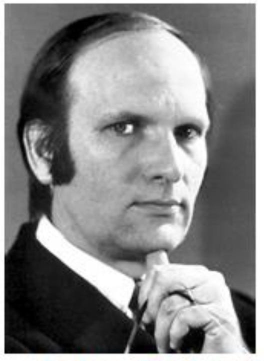
(a) Below its T_c, a superconductor completely expels magnetic lines of force from its interior. (b) In magnetic levitation, a small magnet "floats" over a disk of a high-temperature superconducting material ($YBa_2Cu_3O_{7-x}$) cooled in liquid nitrogen.

BCS Theory

In 1957, Bardeen, Cooper and Schrieffer (BCS) proposed a theory that explained the microscopic origins of superconductivity, and could quantitatively predict the properties of superconductors.

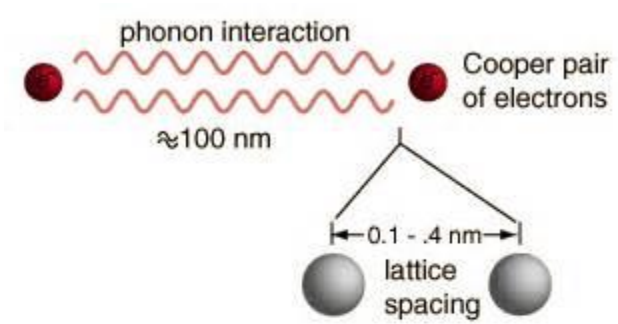
John Bardeen

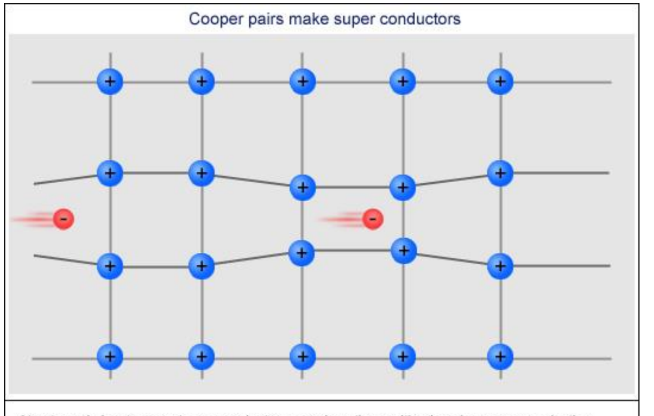
Leon Neil Cooper



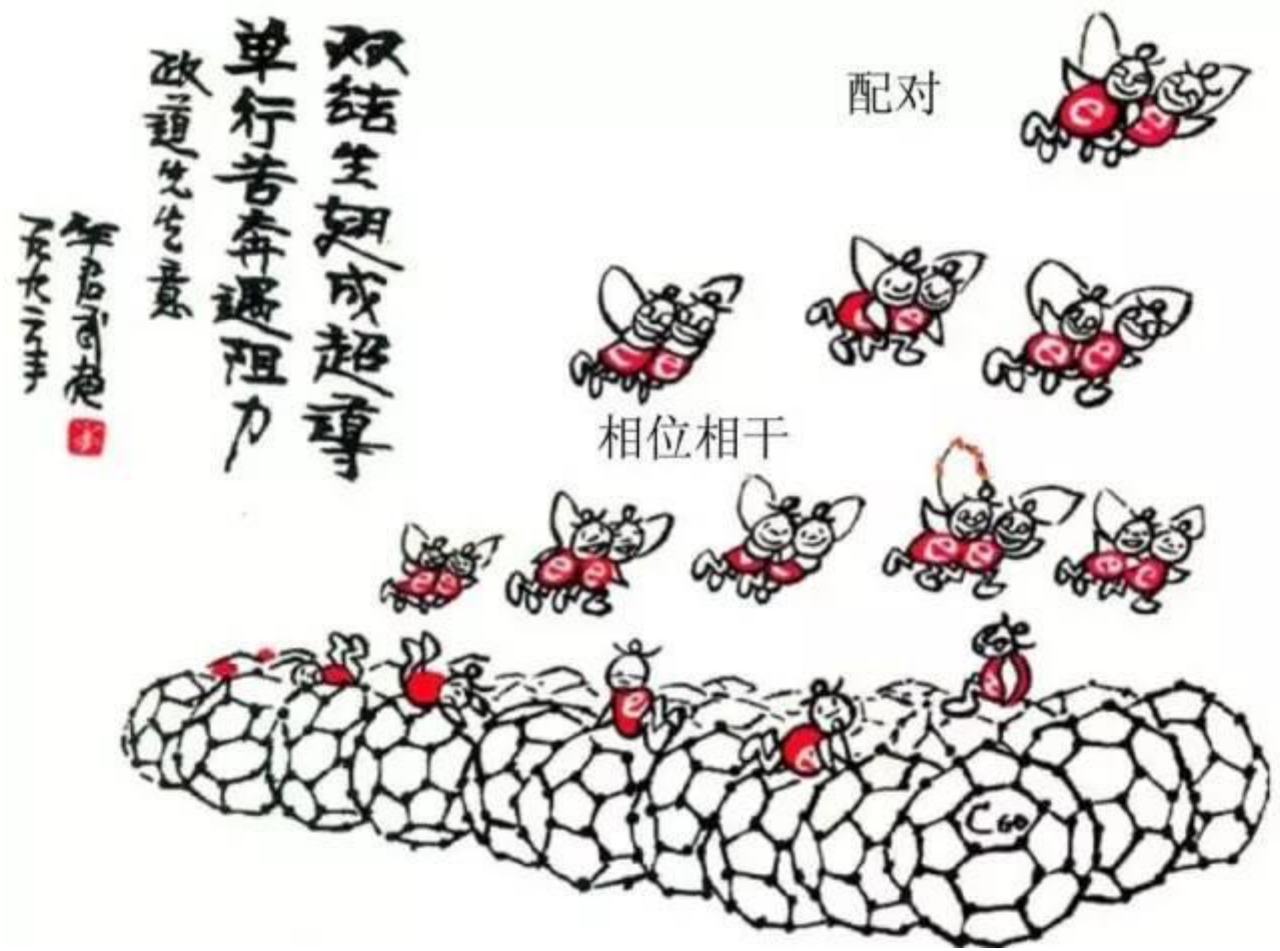
John Robert Schrieffer

Cooper pairs are formed by electron-phonon interactions: an electron in the cation lattice will distort the lattice around it, creating an area of greater positive charge density around itself. Another electron at some distance in the lattice is then attracted to this charge distortion (phonon) - the electron-phonon interaction. The electrons are thus indirectly attracted to each other and form a Cooper pair - an attraction between two electrons mediated by the lattice which creates a 'bound' state of the two electrons:





At extremely low temperatures, an electron can draw the positive ions in a superconducting material towards it. This movement of the ions creates a more positive region that attracts another electron to the area.



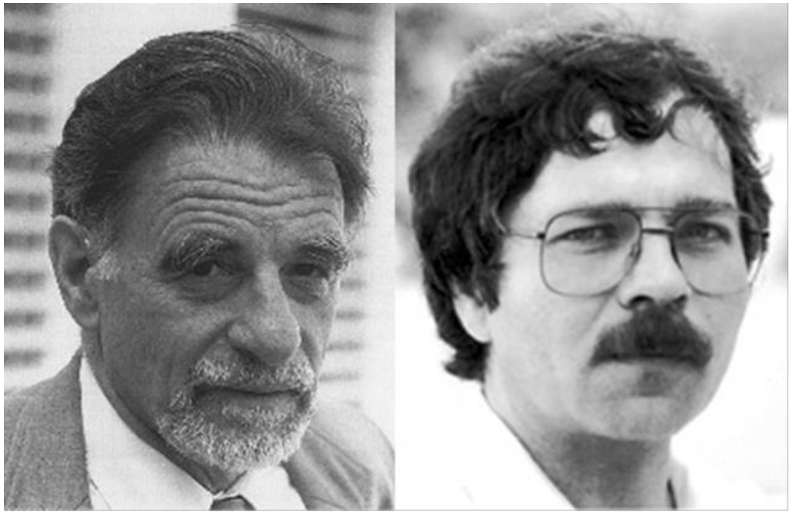
李政道先生提议的有关BCS超导机理漫画（单翅蜜蜂代表单个电子）题曰：“单行苦奔遇阻力，双结生翅成超导”，下面为蜂窝状的C60系列超导体。

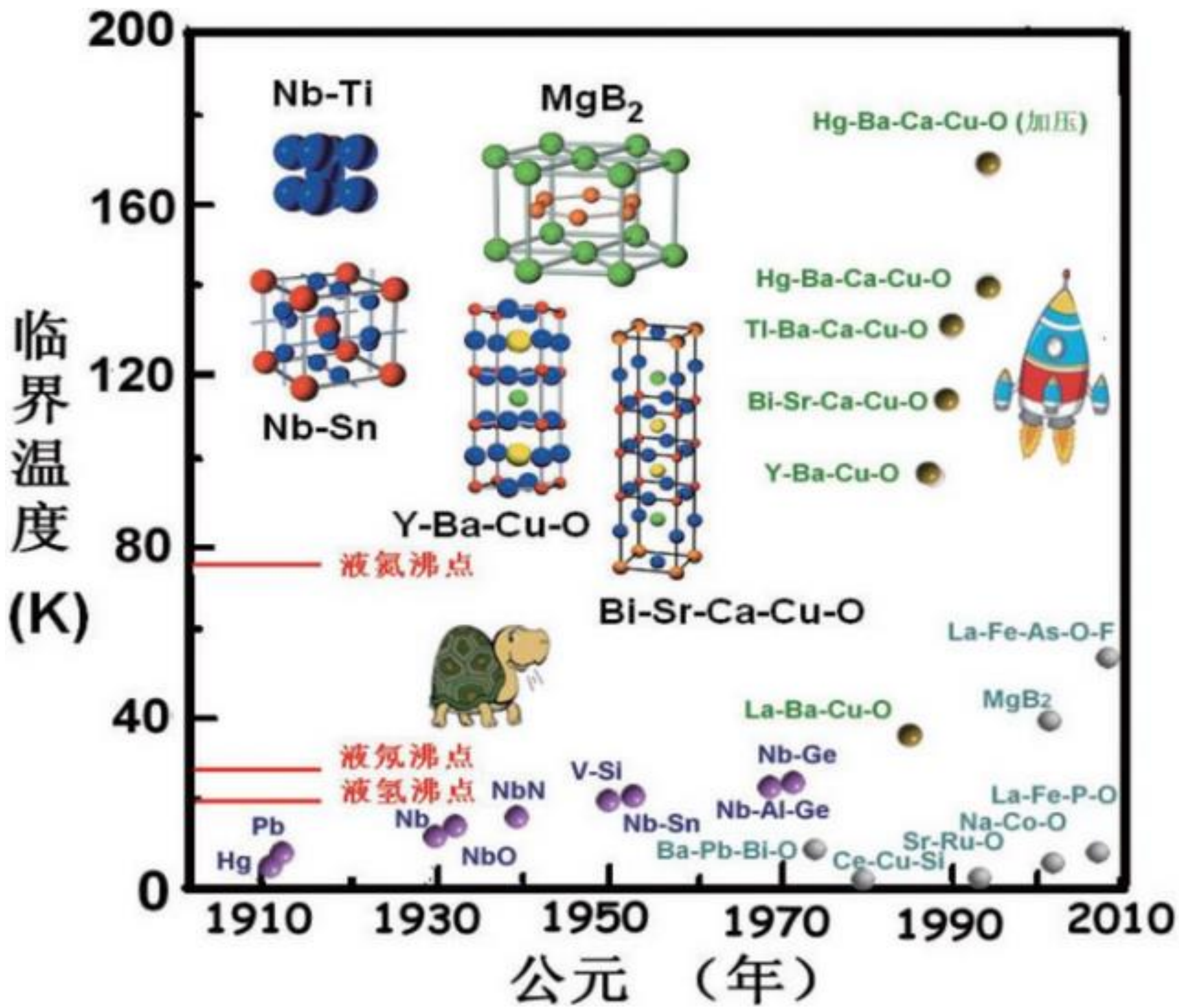
High Temperature Superconductor

T > 77 K

The temperature of liquid nitrogen, the least expensive cryogenic fluid.

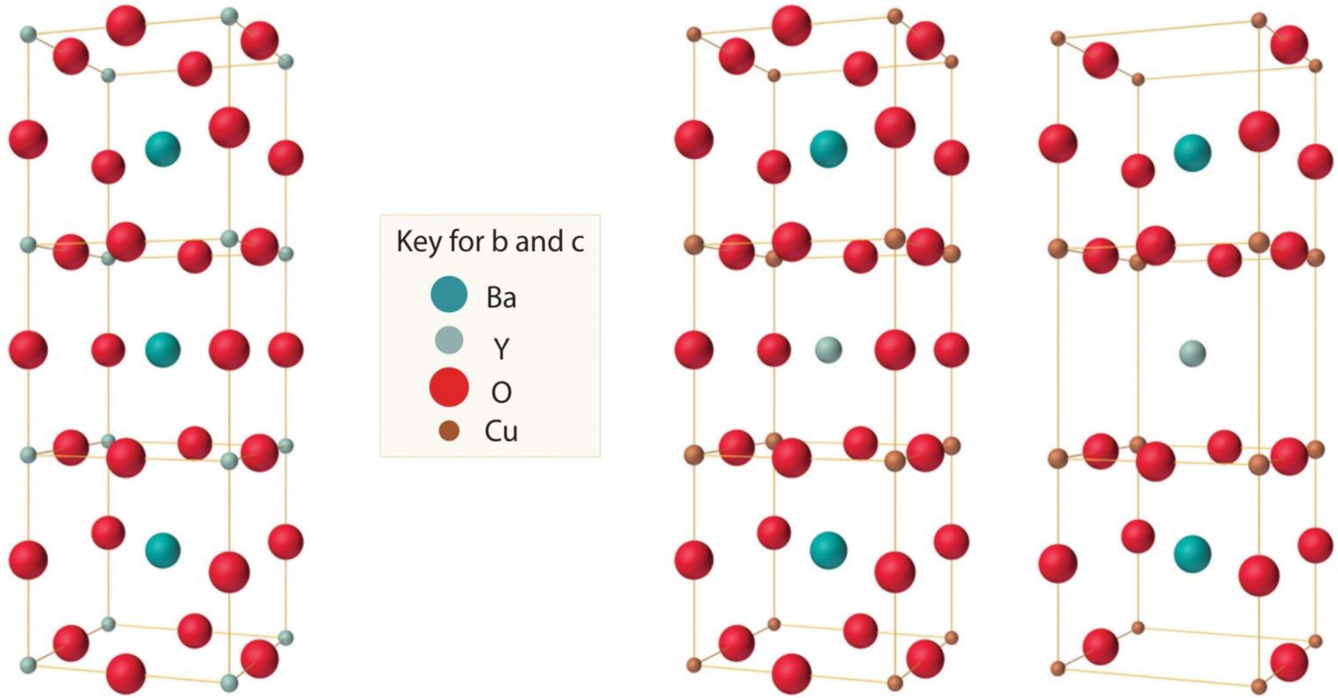
In 1986, Johannes G. Bednorz and Karl A. Müller, working for IBM in Zurich, showed that certain mixed-metal oxides containing La, Ba, and Cu exhibited superconductivity above 30 K.





200
160
120
80
40
0
临界温度 (K)
1910
1930
1950
1970
1990
2010
公元 （年）
Nb-Ti
MgB2
Nb-Sn
Y-Ba-Cu-O
Bi-Sr-Ca-Cu-O
Hg-Ba-Ca-Cu-O (加压)
Hg-Ba-Ca-Cu-O
Tl-Ba-Ca-Cu-O
Bi-Sr-Ca-Cu-O
Y-Ba-Cu-O
液氮沸点
液氖沸点
液氢沸点
La-Fe-As-O-F
MgB2
La-Ba-Cu-O
Nb-Ge
V-Si
NbN
Nb
Pb
Hg
NbO
Nb-Sn
Nb-Al-Ge
Ba-Pb-Bi-O
Ce-Cu-Si
Sr-Ru-O
Na-Co-O
La-Fe-P-O

The Relationship of the Structure of a Superconductor Consisting of Y-Ba-Cu-O to a Simple Perovskite Structure



(a) $CaTiO_3$ (b) $YBa_2Cu_3O_9$ (c) $YBa_2Cu_3O_7$

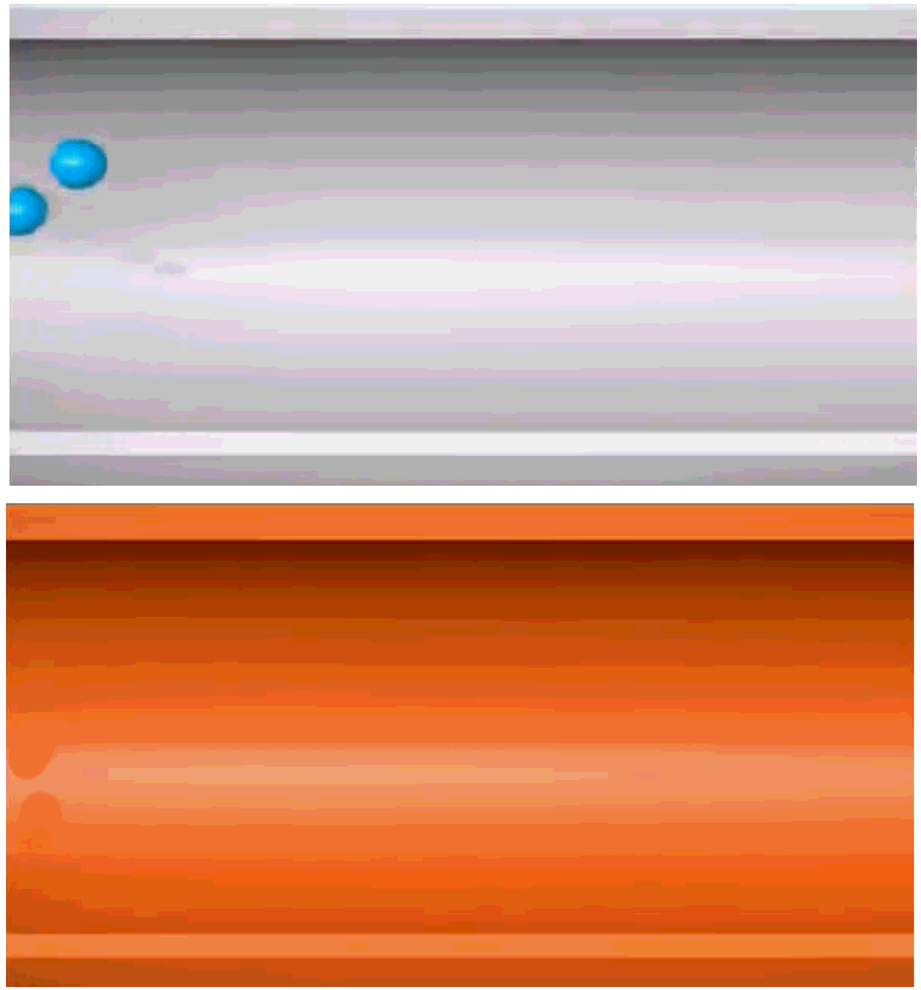
(a) Stacking three unit cells of the Ca-centered $CaTiO_3$ perovskite together with (b) replacement of all Ti atoms by Cu, replacement of Ca in the top and bottom cubes by Ba, and replacement of Ca in the central cube by Y gives a $YBa_2Cu_3O_9$ stoichiometry. (c) The removal of two oxygen atoms per unit cell gives the nominal $YBa_2Cu_3O_7$ stoichiometry of the superconducting material.

The only difference between the superconducting and insulating forms of the compound is that an O atom has been removed from between the Cu^{3+} ions, which destroys the chains of Cu atoms and leaves the Cu in the center of the unit cell as Cu^{+}. The chains of Cu atoms are crucial to the formation of the superconducting state.

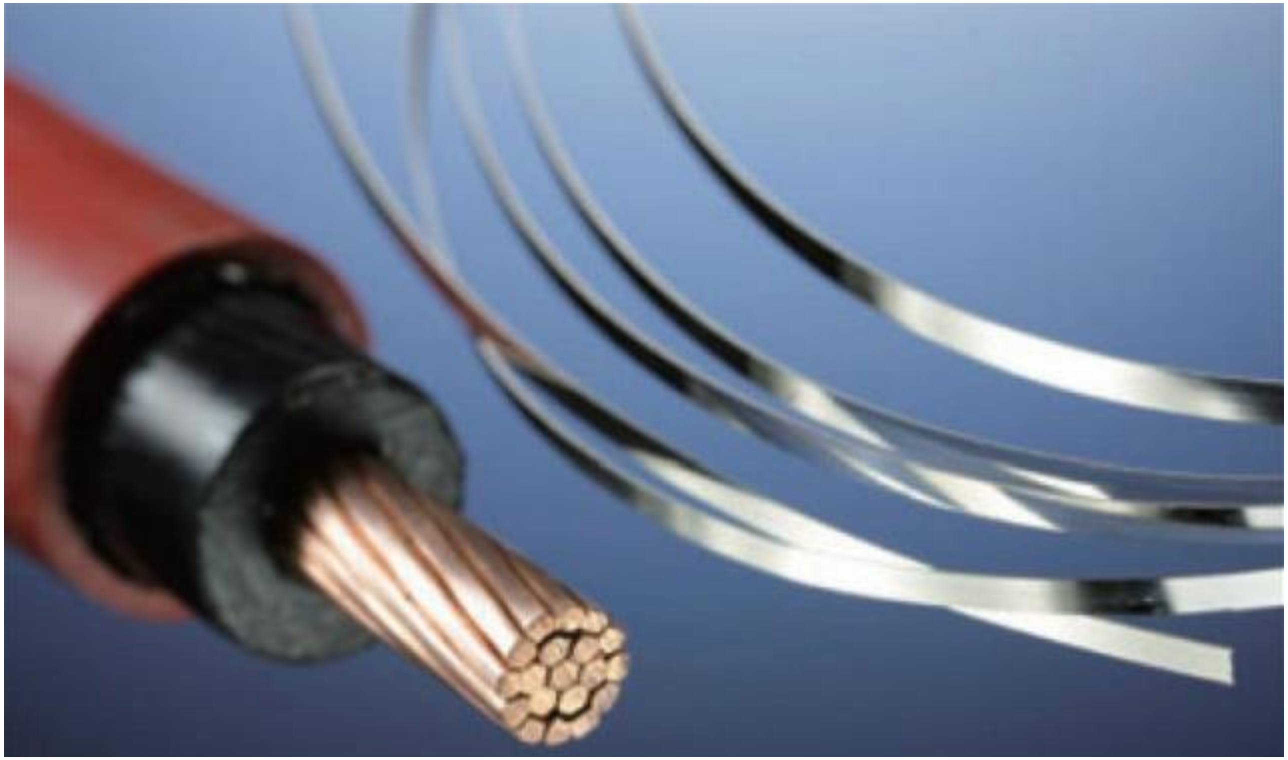
Compound	T_c (K)
$Ba(Pb_{1-x}Bi_x)O_3$	13.5
$(La_{2-x}Sr_x)CuO_4$	35
$YBa_2Cu_3O_{7-x}$	95
$Bi_2(Sr_{2-x}Ca_x)CuO_6$*	80
$Bi_2Ca_2Sr_2Cu_3O_{10}$*	110
$Tl_2Ba_2Ca_2Cu_3O_{10}$*	125
$HgBa_2Ca_2Cu_3O_8$*	133
K_3C_{60}	18
Rb_3C_{60}	30

***Nominal compositions only. Oxygen deficiencies or excesses are common in these compounds.**

零电阻特性——永不消逝的电流



超导体电子定向运动示意图超导体的零电阻特性在**电力输送**领域和**储能**领域都有着非常广阔的应用前景。



高温超导输电线和带材

超导的零电阻特性则可大大减少这部分损耗，超导技术应用于输电线路，人们得到的不仅是环保还有节能。

实际上超导体就是电力工业革命性的技术储备之一。目前采用常规金属合金制造的超导输电线已经得到了广泛的应用，而高温超导材料制作的超导输电线也即将投入市场。

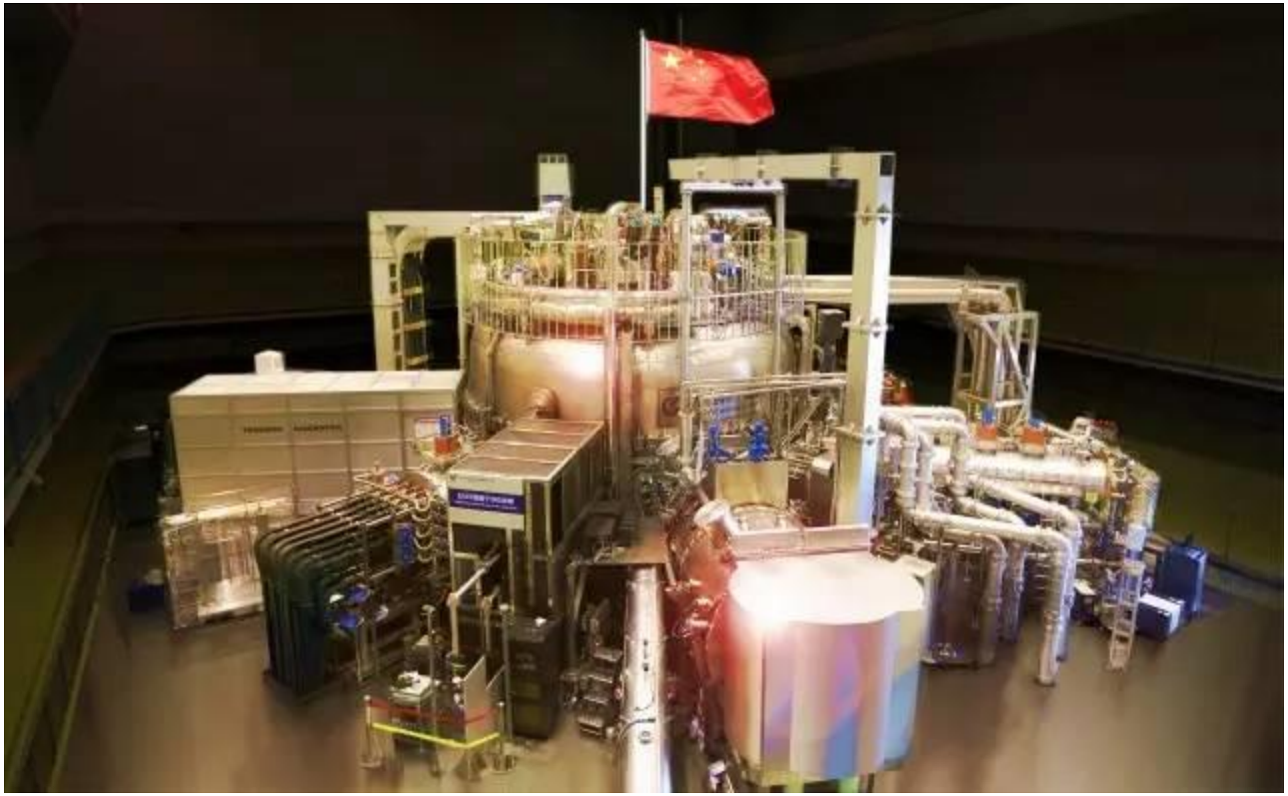
核磁共振成像仪 & 生物核磁共振脑成像

超导磁体所谓超导磁体是指采用超导线材或者带材绕制而成的用以产生磁场的超导线圈，超导磁体具有稳定性好、能耗低、磁场强等优势。

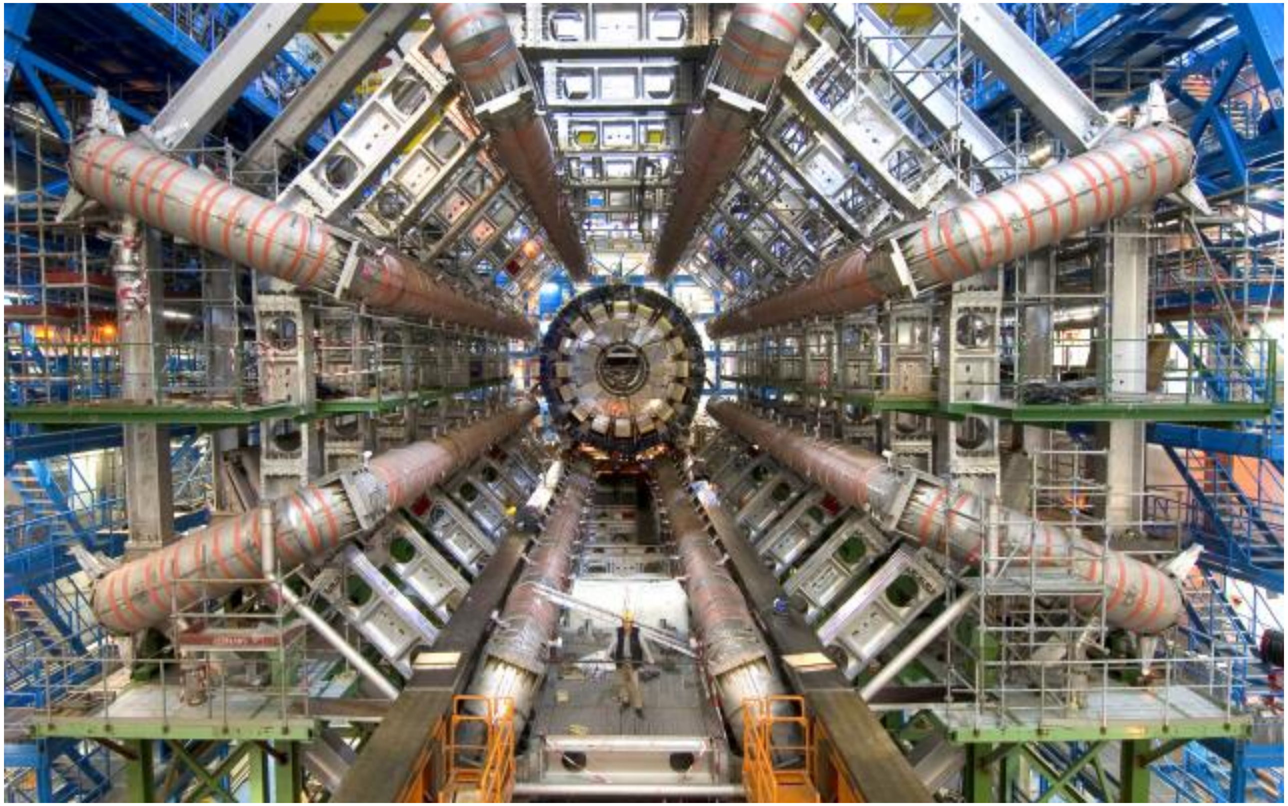
在生物学研究和临床医学上采用的高分辨核磁共振成像技术大都是采用超导磁体，其不仅体积小功耗低，还具有高场强高分辨率的特点。

磁悬浮列车

超导磁悬浮超导磁悬浮的原理是利用了超导体的完全抗磁性，由于超导体会将外部磁场的磁通线排出体外，因此当磁体靠近超导体时会受到很强的排斥力，当排斥力与重力抵消就实现了超导悬浮。



EAST全超导托卡马克



欧洲大型强子对撞机（LHC）

http://wenku.baidu.com/link?url=Cb7cLS5H8v1ChK0eGC5VDIuUdtevfxXkOiorfdGEdP4UozZyaUwB3DSpa6ack78FoSe_bdXDg6taFH-cYegQLvWkhDGrH0lkj4XvCemAl6O

超导

http://www.360doc.com/content/16/0314/13/9065871_542097111.shtml

http://quiz.thefullwiki.org/Close-packing_of_spheres